1 May 2003

TO: Pre–Installation Tab 088

SUBJECT: **DIRECTION 2294003, SIGNA Twin*Speed* PRE–INSTALLATION, REV 3**

Enclosed is *Direction 22294003, Signa TwinSpeed Pre–Installation,* Rev 3 revision. Table 1 below summarizes the major information incorporated in this revision. Changed content on individual pages is indicated by revision bar in the left hand margin.

TABLE 1
DIRECTION 2294003 REVISION 3 MAJOR CHANGES INCORPORATED

TAB	MAJOR INFORMATION CHANGES
Throughout manual	• Added BrainWave Option For Signa 1.5T Twin*Speed* with EXCITE Technology systems ONLY, information includes new revised heat output values.
1 SYSTEM CONFIGURATION	• Table 1–11 Note 2 changed M3043TA to M3043TC • Deleted Remote Keyboard Option for use with Operator Workspace PC LCD catalogs.
2 ROOM LAYOUTS	• Illustration 2–33 corrected Magnet Monitor vertical box and mounting flange dimensions.
4 SITE ENVIRONMENT	• Table 4–2 **bold** Note 2 to emphasize air cooling load average of maximum values. • Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC Table 4–7 Note 6 added Ethylene Glycol or Propylene Glycol wording. • Section 4–4–5 Water Cooled TGWC Requirements Table 4–8 updated Pressure Drop values, added note clarifying value for water and factor for 50/50, and Illustration 4–4 Water Flowrate/ Temperature requirement. • Section 4–7–3 Design Guidelines for acoustics added Structural Vibration Control information. • Section 4–8 Room Ventilation pressure equaliztion waveguide requirement added IEC regulation reference. •
5 POWER REQUIREMENTS	• Section 5–3–1 MDP and Illustration 5–1 clarified Emergency Off buttons provided with GE MDP option. • Illustration 5–1 added SPLICE on Run 296 BLK wire connection. • Illustration 5–3 external conduit metal requirement added 2002 NEC reference, also updated RF Filter metaliic conduit requirement and filter location for 2002 NEC requirement.
6 INTERCONNECT DATA	• Ilustration 6–1 added Equipment Room and Control Area delineating lines (requested by Installation Specialist). • Illustration 6–1, Tables 6–2 & 6–3 corrected cable Group 81 to connect between E02 and PP1 • Group 25 added missing Run 726 • Table 6–3 (multiple pages) added M3044TC to notes for increased usable length. • RF Transmit interconnects changed for new material & standard length cut at site. Added Group 72 from MR1 to MG2 for with RF Transmits (removed from Groups 25 & 45).
9 PREINSTALLATION	• Changed hospital to facility in several items.
10 TOOLS & TEST EQUIPMENT	• Deleted RF survey apparatus no longer available from GE Medical Systems.

UPDATE INSTRUCTIONS FOR YOUR DIRECTION 2294003 REV 1

Refer to Table 2 for instructions to properly update your Direction 2294003 Rev 2 with the enclosed Revision 3 pages.

TABLE 2
DIRECTION 2294003 REVISION 2 UPDATE INSTRUCTIONS

TAB	REMOVE PAGE(S)	INSERT PAGE(S)
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	i to ix	i to ix
1 SYSTEM CONFIGURATION	1–9/10 1–13/14	1–9/10 1–13/14
2 ROOM LAYOUT	2–1 to 2–71 (entire section)	2–1 to 2–75 (entire section)
4 SITE ENVIRONMENT	4–1 to 4–40 (entire section)	4–1 to 4–43 (entire section)
5 POWER REQUIREMENTS	5–1 to 5–18 (entire section)	5–1 to 5–18 (entire section)
6 INTERCONNECT DATA	6–1 to 6–43 (entire section)	6–1 to 6–45 (entire section)
8 SHIPPING & DELIVERY DATA	8–5/6	8–5/6
9 PREINSTALLATION	9–3/4 9–7/8	9–3/4 9–7/8
10 TOOLS & TEST EQUIPMENT	10–1 to 10–10 (entire section)	10–1 to 10–10 (entire section)

Debra C. Balis
MR SERVICE ENGINEERING

GE Medical Systems

Technical Publications

Direction 2294003
Revision 3

Signa® 1.5T Twin*Speed*
Pre–Installation

Architectural/Installation Planning Use Only

Operating Documentation

GE Medical Systems

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WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
Prelim A	Mar. 21, 2001	Initial Development Preliminary version. Information subject to change without notice
Rev 0	June 28, 2001	Initial Product Release
Rev 1	May 21, 2002	Added Signa TwinSpeed with EXCITE Technology system information throughout the document. Other major changes listed by section include the following **Sec 1:** Update power product offerings to E catalogs. **Sec 2:** Update TSCC & TGWC for revised E&W information; Added NEC 2002 Article references for MDP information; Added Network connection and telephone lines option to system connectivity requirements; Updated installed weight of Magnet, Enclosure, Body Coil, & Cryogens; Updated TAC Cabinet dimensions drawing. **Sec 4:** Added Outdoor TSCC & TGWC Operating Environment temperature & humidity specifications; Updated Water Cooled TSCC water cooling requirements; Clarified Magnet Room exhaust fan intake location requirement. **Sec 5:** Revised Voltage Transient specification; Updated Facility Ground Wire measurement techniques; Added NEC 2002 Article references. **Sec 6:** Updated miscellaneous system interconnects. **Sec 7:** Added illustration of Magnet Minimum Ceiling height area requirement; Clarified RF Shield Room vendor responsibility to supply torque specifications for all anchors, studs, and fastening hardware to meet the clamping force specified. **Sec 8:** Update magnet moving dimensions height. **Sec 9:** Updated references to Network connection and telephone lines option to system connectivity; Updated Cryogenic Vent connected to magnet prior to magnet ramp–up. **Sec 10:** Added new & deleted obsolete test equipment & part numbers. **Appendix C:** Added Ellis & Watts "Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*" document.
Rev 2	Nov. 19, 2002	Direction 2128126 updated to Rev 1. Changed SRF Cabinet to 1.5T SRFD II Cabinet information throughout the manual. **Sec 1:** Added M1090ZP & M1090ZN OW ONLY Long Cables; System Cooling & MDP selections Water Cooled TGWC added MDP turnkey solution choice to flowchart illustration; Added UPS for Magnet Monitor to Mobile magnet catalogs; For TSCC catalogs table added note for MDP required; Facility Options added full system UPS and Bypass panel options catalogs. **Sec 2:** Added Multiple MR System Sites shared Equipment Room requirements; Cradle extensions information added for TRM Coil replacement; Mounting requirements clarified for MDP top breaker highest position; Added UPS for Magnet Monitor to Mobile magnet catalogs; Revised System Monitoring & Support Connectivity Requirements for Magnet Monitor; Updated ACGD/PDU Cabinet weight & center of gravity values. **Sec 3:** Corrected reference typographical error; Magnetic Isogauss line plots added 40mT & 200mT required by IEC 60601 – 2 – 33 6.8.3 bb. **Sec 4:** Revised hearing protection requirement in the Magnet Room; Clarified water cooling required 365 days per year for Water Cooled TSCC & Water Cooled TGWC; Added metric value for recommend Magnet Room ventilation; Exhaust Fan illustrations added Room Ventilation Requirements/Recommendations reference; Added roof hatch delivery situation to Cryogenic Vent inspection CAUTION; Typical Cryogenic Vent Detail added height requirement for RF Shielded Room Contractor supplied vent pipe/waveguide in Magnet Room. **Sec 5:** Protective Disconnect Set–Up clarified power recommendation for Shield Cooler & TSCC; Added PDU feeder wiring requirement with full system UPS option; Updated Common Ground Stud location dimension in Magnet Room Grounding Requirements illustration. **Sec 6:** Added notes to identified RF Transmit & RF Receive cables; Updated miscellaneous system interconnects; Cable Groups 8 & 12 revised for Gradient Cable length reduction. **Sec 7:** References typo error corrected; Added use of Megger Insulation Tester to Electrical Isolation Measurement Method for Anchor Hardware Requirements for MR Equipment Inside RF Shielded Room. **Sec 8:** Added Transportation Environmental Conditions. **Sec 10:** Added Resistance Meter to table of Rigger/Customer Supplied Equipment. **Appendix D:** Added REQUIREMENTS FOR EQUIPMENT ROOM SHARED BY MULTIPLE MR SYSTEMS.
Rev 3	May 1, 2003	Added BrainWave Option For Signa 1.5T Twin*Speed* with EXCITE Technology systems ONLY, information includes new revised heat output values. **Sec 1:** Table 1–11 Note 2 changed M3043TA to M3043TC; Deleted Remote Keyboard Option for use with Operator Workspace PC LCD catalogs. **Sec 2:** Illustration 2–33 corrected Magnet Monitor vertical box and mounting flange dimensions. **Sec 4:** Table 4–2 **bold** Note 2 to emphasize air cooling load average for maximum value; Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC Table 4–7 Note 6 added Ethylene Glycol or Propylene Glycol wording. Section 4–4–5 Water Cooled TGWC Requirements Table 4–8 updated Pressure Drop values, added note clarifying value for water and factor for 50/50, and Illustration 4–4 Water Flowrate/Temperature requirement. Section 4–7–3 Design Guidelines for acoustics added Structural Vibration Control information. Section 4–8 Room Ventilation pressure equaliztion waveguide requirement added IEC regulation reference. Section 4–14–4 Transient Vibration specification added clarifications for measurements. **Sec 5:** Section 5–3–1 MDP and Illustration 5–1 clarified Emergency Off buttons provided with GE MDP option; Illustration 5–1 added SPLICE on Run 296 BLK wire connection & clarified Emergency OFF. Illustration 5–3 external conduit metal requirement added 2002 NEC reference, also updated RF Filter metaliic conduit requirement and filter location for 2002 NEC requirement. **Sec 6:** Ilustration 6–1 added Equipment Room and Control Area delinating lines (requested by Installation Specialist); Illustration 6–1, Tables 6–2 & 6–3 corrected cable Group 81 to connect between E02 and PP1Group 25 added missing Run 726; Table 6–3 (multiple pages) added M3044TC to notes for incresed usable length; RF Transmit interconnects changed for new material & standard length cut at site. Added Group 72 from MR1 to MG2 for with RF Transmits (removed from Groups 25 & 45). **Sec 9:** Changed hospital to facility in several items. **Sec 10:** Deleted RF survey apparatus no longer available from GE Medical Systems.

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GLOSSARY

ACGD – Abbreviation for Advanced Control Gradient Drivers.

CRYOGEN – A substance for producing low temperatures. Liquid helium is the cryogen used to cool the magnet to approximately 4 Kelvin (–269°C or –452°F).

CRYOSTAT – An apparatus maintaining a very low constant temperature. The cryostat consists of one concentric, cylindrical container housed in an outer vacuum tight vessel. The magnet and shim coils are mounted in the inner container. The container is filled with liquid helium. The shields surrounding the inner container are kept cold by a refrigeration device.

DEWAR – A container with an evacuated space between two highly reflective walls used to keep low temperature substances at near–constant temperatures. Liquid helium is usually stored and shipped in dewars.

EXCLUSION ZONE – Area where the magnetic flux density is greater than five gauss. Personnel with cardiac pacemakers, neurostimulators and other biostimulation devices must NOT enter this zone. Signs are posted outside the five gauss line alerting personnel of this requirement. Since the magnetic field is three–dimensional, signs are also posted on floors above and below the Magnet Room in which the five gauss line exists.

FERROUS MATERIAL – Any substance containing iron which is strongly attracted by a magnetic field.

GAUSS (G) – A unit of magnetic flux density. The earth's magnetic field strength is approximately one half gauss to one gauss depending on location. The internationally accepted unit is the tesla (1 Tesla = 10,000G and 1 milli Tesla = 10G).

GRADIENT – The amount and direction of the rate of change in space of the magnetic field strength. In the magnetic resonance system, gradient amplifiers and coils are used to vary the magnetic field strength in the x, y, and z planes.

HOMOGENEITY – Uniformity. The homogeneity of the static magnetic field is an important quality of the magnet.

ISOCENTER – Center of the imaging volume ideally located at the magnet center.

ISOGAUSS LINE – An imaginary line or a line on a field plot connecting identical magnetic field strength points.

MAGNETIC FIELD (H) – A condition in a region of space established by the presenece of a magnet and characterized by the presence of a detectable magnetic force at every point in the region. A magnetic field exists in the space around a magnet (or current carrying conductor) and can produce a magnetizing force on a body within it.

MAGNETIC RESONANCE (MR) – The absorption or emission of electromagnetic energy by nuclei in a static magnetic field, after excitation by a suitable radio frequency field.

MAGNETIC SHIELDING – Using material (e.g. steel) to redistribute a magnetic field , usually to reduce fringe fields.

QUENCH – Condition when a superconducting magnet becomes resistive thus rapidly boiling off liquid helium. The magnetic field reduces rapidly after a quench.

GLOSSARY (Continued)

RADIO FREQUENCY (RF) – Frequency intermediate between audio frequency and infrared frequencies. Used in magnetic resonance systems to excite nuclei to resonance. Typical frequency range for magnetic resonance systems is 5–130 Mhz.

RADIO FREQUENCY SHIELDING – Using material (e.g. copper, aluminium, or steel) to reduce interference from external radio frequencies. A radio frequency shielded room usually encloses the entire magnet room.

RESONANCE – A large amplitude vibration caused by a relative small periodic stimulus of the same or nearly the same period as the natural vibration period of the system. In magnetic resonance imaging, the radio frequency pulses are the periodic stimuli which are at the same vibration period as the hydrogen nuclei being imaged.

SECURITY ZONE – Area within the Magnet Room where the magnet is located. Signs are posted outside the Magnet Room warning personnel of the high magnetic field existing in the Magnet Room and the possibility of ferrous objects becoming dangerous projectiles within this zone.

SHIELD COOLER COLD HEAD – An external refrigeration device which maintains the shields inside the cryostat at a constant temperature.

SHIM COILS – Shim coils are used to provide auxiliary magnetic fields in order to compensate for inhomogeneities in the main magnetic field due to imperfections in the manufacturing of the magnet or affects of steel in the surrounding environment.

SHIMMING – Correction of inhomogeneity of the main magnetic field due to imperfections in the magnet or to the presence of external ferromagnetic objects.

SUPERCONDUCTING MAGNET – A magnet whose magnetic field originates from current flowing through a superconductor. Such a magnet is enclosed in a cryostat.

SUPERCONDUCTOR – A substance whose electrical resistance essentially disappears at temperatures near zero Kelvin. A commonly used superconductor in magnetic resonance imaging system magnets is niobium–titanium embedded in a copper matrix.

TESLA (T) – The internationally accepted unit of magnetic flux density. One tesla is equal to 10,000 gauss. One milli Tesla is equal to 10 gauss.

INTRODUCTION

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a magnetic resonance system. "Preinstallation work" is done to prepare the customer's premises for the installation of the products sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Medical System such as: the facility input power line to the power distribution unit, as well as emergency power lines and facility power lines to the the Magnet Room. The electrical contractor shall ring out and tag all wires at both ends. Color–coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non–electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Installation of RF shielding in Magnet Room and installation of equipment anchors.
- RF Screen Room including all openings (i.e. windows, doors, vents, etc.) need acoustic properties to meet local regulations and customer requirements.
- Site construction or renovation.
- Installation of structural reinforcements as required.
- Installation of selected magnetic shielding as required.
- Installation of water treatment equipment if necessary.
- Scheduling of riggers to move magnet (under GE Medical Systems direction) into its final location within the Magnet Room.

All work MUST be in compliance with national and local building and safety codes.

A structural steel analysis may be necessary during site evaluation for a magnetic resonance system. All site plans and preliminary concepts should be reviewed by GE Medical Systems MR Siting group prior to construction.

Unless specifically mentioned, GE Medical Systems does not provide or install: the facility input power lines to the Main Disconnect Panel and Power Distribution Unit or the power lines required in the Magnet Room, nor raised flooring, conduit, junction boxes, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, acoustic treatment to contain acoustic noise to the RF shielded room, or the RF shielded room illustrated in this document.

INTRODUCTION (Continued)

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Medical Systems personnel or third–party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE's field engineers, personnel of third–party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre–installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of pre–installation information.

Note

Direction 2223170, Signa MR/*i* & CV/*i* Pre–Installation contains the data and requirements for planning for a new Signa 1.0T & 1.5T MR/*i* & CV/*i* system.

Note

Direction 2304880, Signa 1.5T Twin*Speed* Upgrade Pre–Installation contains the data and requirements for planning for a Signa 1.5T Twin*Speed* system Upgrade.

Note

Direction 2142460, Signa Horizon Upgrade Pre–Installation contains the data and requirements for planning for a Signa Horizon system Upgrade.

Note

Direction 2223751, Signa MR/*i* & CV/*i* Upgrade Pre–Installation contains the data and requirements for planning for a Signa MR/*i* or CV/*i* system Upgrade with WideOpen Enclosure for a system with an LCC Magnet.

SECTION 1 – SYSTEM CONFIGURATION

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1–1 INTRODUCTION

Magnetic Resonance (MR) system Signa® 1.5T Twin*Speed* with EXCITE Technology and Signa® 1.5T Twin*Speed* are documented throughout this publication.

Note

There are some naming differences between the sales literature documents and the technical documentation. Listed below are the corresponding names. The Pre–Installation document uses the technical document names.

SALES LITERATURE NAME	=	TECHNICAL DOCUMENT NAME
Signa Infinity 1.5T Twin*Speed* with EXCITE Technology	=	Signa 1.5T Twin*Speed* with EXCITE Technology
Signa Infinity 1.5T Twin*Speed*	=	Signa 1.5T Twin*Speed*
CXK4 150 Magnet with K4 Technology	=	1.5T LCC (Active Shield) Magnet

Note

Throughout this document references to Signa Twin*Speed* apply to both Signa 1.5T Twin*Speed* with EXCITE Technology and Signa 1.5T Twin*Speed* except where explicitly documented as different.

1–2 BASIC SYSTEM

The basic system for fixed site operation consists of the following major equipment:

- System cooling equipment (See Illustration 1–1 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.)

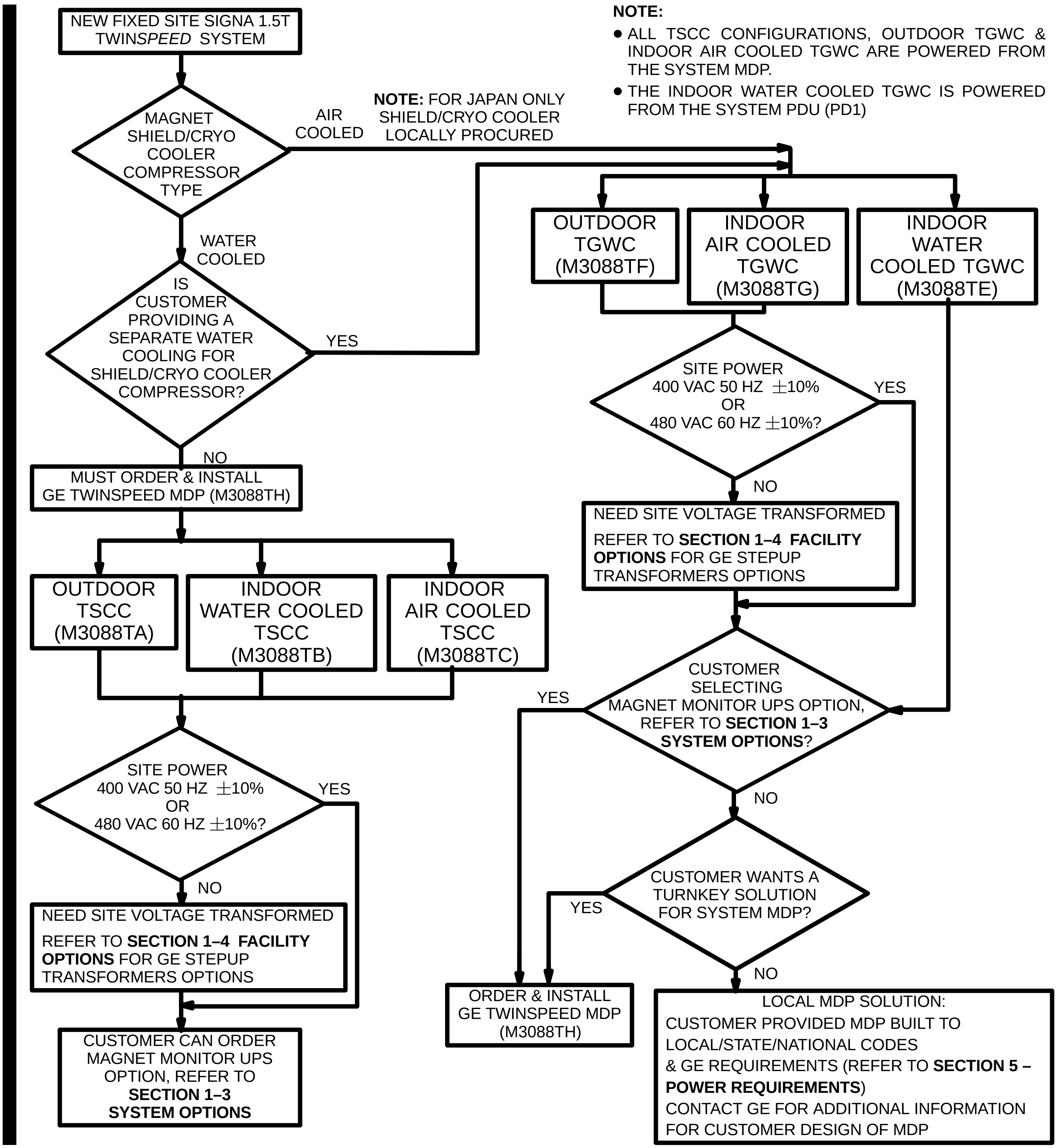
Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.

- 15 kilogauss (1.5T) LCC (Active Shield) Magnet & Magnet Enclosure
- Magnet and Magnet Accessories
- Shield/Cryo Cooler Compressor Cabinet
- TwinSpeed Gradient (TRM) and RF body coils and 1.5T Quad Head Coil
- Main Disconnect Panel (GE supplied or customer supplied, dependent on Illustration 1–1 flowchart decisions)
- System electronics cabinets: ACGD/PDU Cabinet, 1.5T SRFD II Cabinet, Twin Accessory Cabinet, and the System Cabinet
  - For Signa Twin*Speed* with EXCITE Technology the System Cabinet contains the Multi Generational Data Acquisition (MGD) chassis, Remote RF chassis, and Driver Module,
  - For Signa Twin*Speed* the System Cabinet contains the Integrated System Electronics (ISE) chassis and Transient Noise Suppression Module
- Operator Workspace (OW) equipment
- Patient Transport Table
- Patient accessories

The tables in this section list the catalogs for the hardware equipment which comprise the Signa 1.5T Twin*Speed* system. Illustrations 1–2 and 1–3 show the major equipment of the system. Refer to Section **1–2–1 System Cooling Equipment Selections** for explanation of available configurations.

1–2 BASIC SYSTEM (Continued)



FIXED SITE SYSTEM COOLING EQUIPMENT & MAIN DISCONNECT PANEL SELECTIONS
ILLUSTRATION 1–1

1–2 BASIC SYSTEM (Continued)

TABLE 1–1
SYSTEM ELECTRONICS MAJOR EQUIPMENT

CATALOG	DESCRIPTION
M3000TC Signa 1.5T Twin*Speed* with EXCITE Technology	• System Cabinet consisting of MGD chassis, Remote RF chassis, and Driver Module. • ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480 Volt, 50/60 Hz with power filter. • 1.5T SRFD II Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. • Twin Accessory Cabinet containing the Gradient Switch and the vacuum pump for Quiet Technology Magnet Enclosure. • Operator Workspace (OW) equipment: Octane 2 Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD panel, and chair. Note, refer to Tables 1–6, 1–7, and 1–8 for catalog choices required to complete Operator Workspace equipment. • Pneumatic Patient Alert System. • 1.5T Quad Head Coil • Patient Transport Table and cradle. • Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps. • Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.
M3000TA Signa 1.5T Twin*Speed*	• System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD), Phase Array hardware integrated, and with Fast Receiver. • ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480 Volt, 50/60 Hz with power filter. • 1.5T SRFD II Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. • Twin Accessory Cabinet containing the Gradient Switch and the vacuum pump for Quiet Technology Magnet Enclosure. • Operator Workspace (OW) equipment: Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD panel, and chair. Note, refer to Tables 1–6, 1–7, and 1–8 for catalog choices required to complete Operator Workspace equipment. • Pneumatic Patient Alert System. • 1.5T Quad Head Coil • Patient Transport Table and cradle. • Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps. • Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.

1–2 BASIC SYSTEM (Continued)

TABLE 1–2
1.5T MAGNET SUBSYSTEM (SELECT ONE)

CATALOG	DESCRIPTION
M3060TA	• 15 kilogauss (1.5T) LCC (Active Shield) Magnet for Fixed Site or Transportable/Relocatable Mobile (See Note 1). Includes Quiet Technology Enclosure partially installed on magnet, Magnet Rundown Unit, Magnet Monitor (performs cryogen meter function and magnet pressure control), and Global Modem. (**See Notes 2 & 3**)
M3061TA	• 15 kilogauss (1.5T) LCC (Active Shield) Magnet for True Mobile (See Note 1). Includes Quiet Technology Enclosure partially installed on magnet, Magnet Rundown Unit, Magnet Monitor (performs cryogen meter function and magnet pressure control), UPS for Magnet Monitor equipment, and Global Modem . (**See Note 2**).
Note	1 For Mobiles (Transportable, Relocatable, and True Mobile) contact your selected Van Manufacturer for site pre–installation, refer to Section 1–5, VAN MANUFACTURERS CONTACT INFORMATION. 2 **M1060JW Shield/Cryo Cooler Compressor (water cooled) must be ordered for LCC Magnet. The Shield/Cryo Cooler Compressor is not part of the LCC Magnet Catalogs but is required at magnet delivery. Refer to Section 5 POWER REQUIREMENTS and Section 4–4 Water Cooling Requirements for details of compressor 24 hour power and water cooling needs.** 3 If M1060TA Magnet is to be shipped by air then must order M1060SC CX Magnet Air Transport Crate (not required within Continental USA).

TABLE 1–3
LCC MAGNET 2.5 METER CEILING HEIGHT KIT (OPTIONAL)

CATALOG	DESCRIPTION
M1060SR	• LCC Magnet Low Ceiling Height (2.5 Meter) Siting Option **Note:** This option reduces the minimum required ceiling height for the Magnet Room to 2.5 meters from 2.67 meters. Check with GE Installation Specialist to determine if this is a required option.

TABLE 1–4
INTEGRATED BODY MODULE (SELECT ONE)

CATALOG	DESCRIPTION
M3085TA	• 1.5T Advanced 60 cm Twin Integrated Body Module (TRM)

TABLE 1–5
MAIN DISCONNECT PANEL (MDP) (SEE TABLE NOTE)

CATALOG	DESCRIPTION
M3088TH	• Signa Twin*Speed* Main Disconnect Panel
Note	1 See Illustration 1–1 flowchart to determine when GE MDP is to be utilized or customer supplied MDP which meets requirements can be considered.

TABLE 1–6
OPERATOR WORKSPACE MONITOR (SELECT ONE)

CATALOG	DESCRIPTION
M1000NT	• 18 inch LCD Color Monitor

1–2 BASIC SYSTEM (Continued)

TABLE 1–7
OPERATOR WORKSPACE TABLE (ONE REQUIRED)

CATALOG	DESCRIPTION
M1000MW	• Table for Operator Workspace *See Note 1*
Note 1 OW Table is an integral part of the regulatory approved system. OW Table provides mounting for several assemblies (e.g. OW Interface Module, OW Power Distribution Box, Modem, DASM) and cable routing for OW interconnects.	

TABLE 1–8
COUNTRY KITS (SELECT ONE)

CATALOG	DESCRIPTION
M1000MN	• English Keyboard
M1000MP	• French Keyboard
M1000MR	• German Keyboard
M1000MS	• Scandinavian Keyboard
M1000NH	• Italian Keyboard
M1000NJ	• Portuguese Keyboard
M1000NK	• Spanish Keyboard

TABLE 1–9
RECON PROCESSOR WITH BAM MEMORY (SELECT ONE)

CATALOG	DESCRIPTION
FOR SIGNA 1.5T TWIN*SPEED* WITH EXCITE TECHNOLOGY	
M3000RD	• EXCITE ReFlex100
M3000RE	• EXCITE ReFlex200
M3000RF	• EXCITE ReFlex400
FOR SIGNA 1.5T TWIN*SPEED*	
M3000RA	• ReFlex100 – Hex Power PC Processor with four 128 MB Bulk Acquisition Memory (BAM) boards (512 MB)

TABLE 1–10
FILMING INTERFACE (SEE NOTE 1 IN TABLE)

CATALOG	DESCRIPTION
M1000MJ	• Analog DASM
M1000MK	• Digital DASM
Note 1 Must choose one Filming DASM Board unless DICOM Print will be used exclusively for software filming to DICOM print peripheral devices.	

1–2 BASIC SYSTEM (Continued)

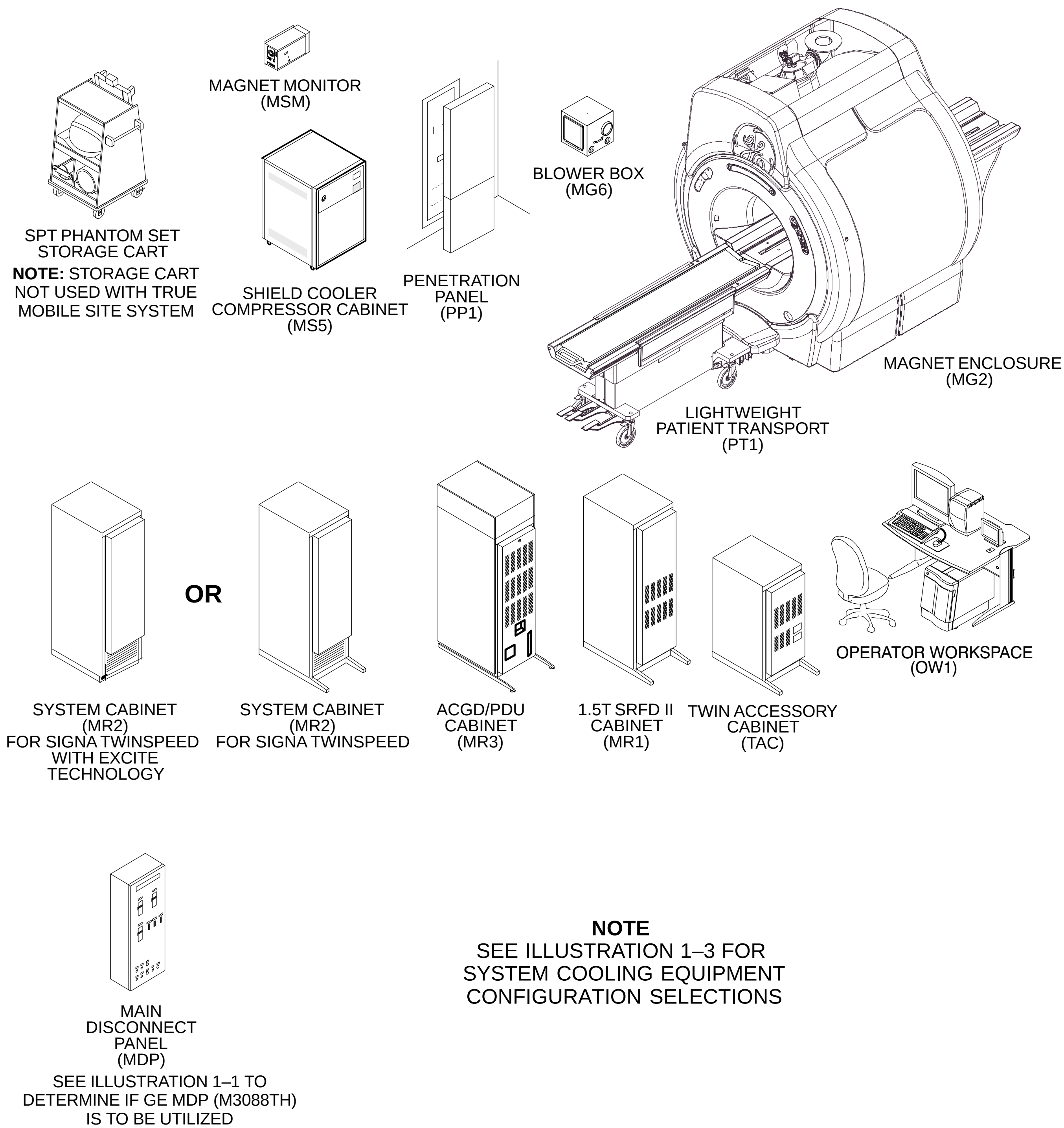
TABLE 1–11
SITE COLLECTOR KIT (SELECT ONE)

CATALOG	DESCRIPTION
FOR SIGNA 1.5T TWIN*SPEED* WITH EXCITE TECHNOLOGY	
M3043TC Refer to Notes 1 & 2	• LCC (Active Shield) Magnet Fixed Site Quiet Technology Enclosure • Penetration Panel with Phase Array hardware integrated and Penetration Panel Covers • Fixed Site system interconnect cables including Phased Array Option • Fixed Site Power Distribution Unit cables • SPT Phantom Set with Shipping/Storage Cart
M3053TC	• LCC (Active Shield) Magnet True Mobile Site Quiet Technology Enclosure • Penetration Panel with Phase Array hardware integrated • True Mobile Site system interconnect cables including Phased Array Option • True Mobile Site Power Distribution Unit cables • SPT Phantom Set
M3063TC	• LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Quiet Technology Enclosure • Penetration Panel with Phase Array hardware integrated • Transportable/Relocatable Mobile Site system interconnect cables including Phased Array Option • Transportable/Relocatable Mobile Site Power Distribution Unit cables • SPT Phantom Set with Shipping/Storage Cart
FOR SIGNA 1.5T TWIN*SPEED*	
M3043TA Refer to Notes 3 & 4	• LCC (Active Shield) Magnet Fixed Site Quiet Technology Enclosure • Penetration Panel with Phase Array hardware integrated and Penetration Panel Covers • Fixed Site system interconnect cables including Phased Array Option • Fixed Site Power Distribution Unit cables • SPT Phantom Set with Shipping/Storage Cart
M3053TA	• LCC (Active Shield) Magnet True Mobile Site Quiet Technology Enclosure • Penetration Panel with Phase Array hardware integrated • True Mobile Site system interconnect cables including Phased Array Option • True Mobile Site Power Distribution Unit cables • SPT Phantom Set
M3063TA	• LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Quiet Technology Enclosure • Penetration Panel with Phase Array hardware integrated • Transportable/Relocatable Mobile Site system interconnect cables including Phased Array Option • Transportable/Relocatable Mobile Site Power Distribution Unit cables • SPT Phantom Set with Shipping/Storage Cart

Note

1. EXCITE Technology Operator Workspace Long Cables (M1090ZP) is available to order only if the GE Installation Specialist determines the standard length cables (supplied in listed catalogs) for interconnects from OW to other system components will not work at the site. M1090ZP provides long length interconnects for OW to PDU, OW to System Cabinet, and OW to Penetration Panel ONLY.
2. Signa Twin*Speed* with EXCITE Technology Site Collector Kit with Long Cables (M3044TC) is available to order only if the GE Installation Specialist determines the standard length cables in M3043TC will not work at the site. M3044TC includes OW Long Cables with the long cables for the Equipment Room.
3. Operator Workspace Long Cables (M1090ZN) is available to order only if the GE Installation Specialist determines the standard length cables (supplied in listed catalogs) for interconnects from OW to other system components will not work at the site. M1090ZN provides long length interconnects for OW to PDU, OW to System Cabinet, and OW to Penetration Panel ONLY.
4. Signa Twin*Speed* Site Collector Kit with Long Cables (M3044TA) is available to order only if the GE Installation Specialist determines the standard length cables in M3043TA will not work at the site. M3044TA includes OW Long Cables with the long cables for the Equipment Room.

1–2 Basic System (Continued)



SIGNA 1.5T TWIN*SPEED* SYSTEM
ILLUSTRATION 1–2

1–2–1 System Cooling Equipment Selections

The Signa 1.5T Twin*Speed* with EXCITE Technology and the Signa 1.5T Twin*Speed* system requires water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. The MR Engineering Team together with Ellis & Watts International Inc. have designed several different water cooling cabinet options to optimize siting power requirements and monthly maintenance costs for the customer. The configuration options include:

- Twin System Cooling Cabinet (TSCC)
  TSCC units provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil located in the Magnet bore. See Table 1–12 for listing of TSCC offerings.

- Twin Gradient Water Cooler (TGWC)
  TGWC units provide water cooling for ONLY the Gradient Coil located in the Magnet bore. **When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet.** See Table 1–13 for listing of TGWC offerings.

Note

See Illustration 1–1 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.

TSCC Configurations

Refer to Table 1–12 for list of TSCC catalogs offerings.

TABLE 1–12
TWIN SYSTEM COOLING CABINET (TSCC) ***SEE TABLE NOTE 1***

CATALOG	DESCRIPTION
M3088TA	• Outdoor Air Cooled TSCC (Water–to–Air Chiller) 50/60 Hz for Gradient Coil Cooling & Shield/Cryo Cooler Compressor cooling *See Note 2* *TSCC installed external to building with Remote Control Panel installed in Equipment Room.* *The most cost effective for the customer.*
M3088TB	• Indoor Water Cooled TSCC (Water–to–Water Chiller) 50/60 Hz for Gradient Coil Cooling & Shield/Cryo Cooler Compressor cooling *See Note 3* *Installed in the Equipment Room, low additional Air Conditioning requirements.*
M3088TC	• Indoor Air Cooled TSCC (Water–to–Air Chiller) 50/60 Hz for Gradient Coil Cooling & Shield/Cryo Cooler Compressor cooling *Installed in the Equipment Room, increases Air Conditioning requirements.*
Note 1 The TSCC requires utilization of the GE MDP per Illustration 1–1. Therefore must order and install M3088TH TwinSpeed Main Disconnect Panel. 2 M3088TA is standard cooling equipment configuration of Transportable/Relocatable Mobile systems. 3 The water cooled configuration System Cooling Cabinets require customer supplied water for cooling. Refer to **Section 4–4–3 Water Cooled System Cooling Cabinet Configuration Requirements** for water cooling specifications.	

Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.

1–2–1 System Cooling Equipment Selections (Continued)

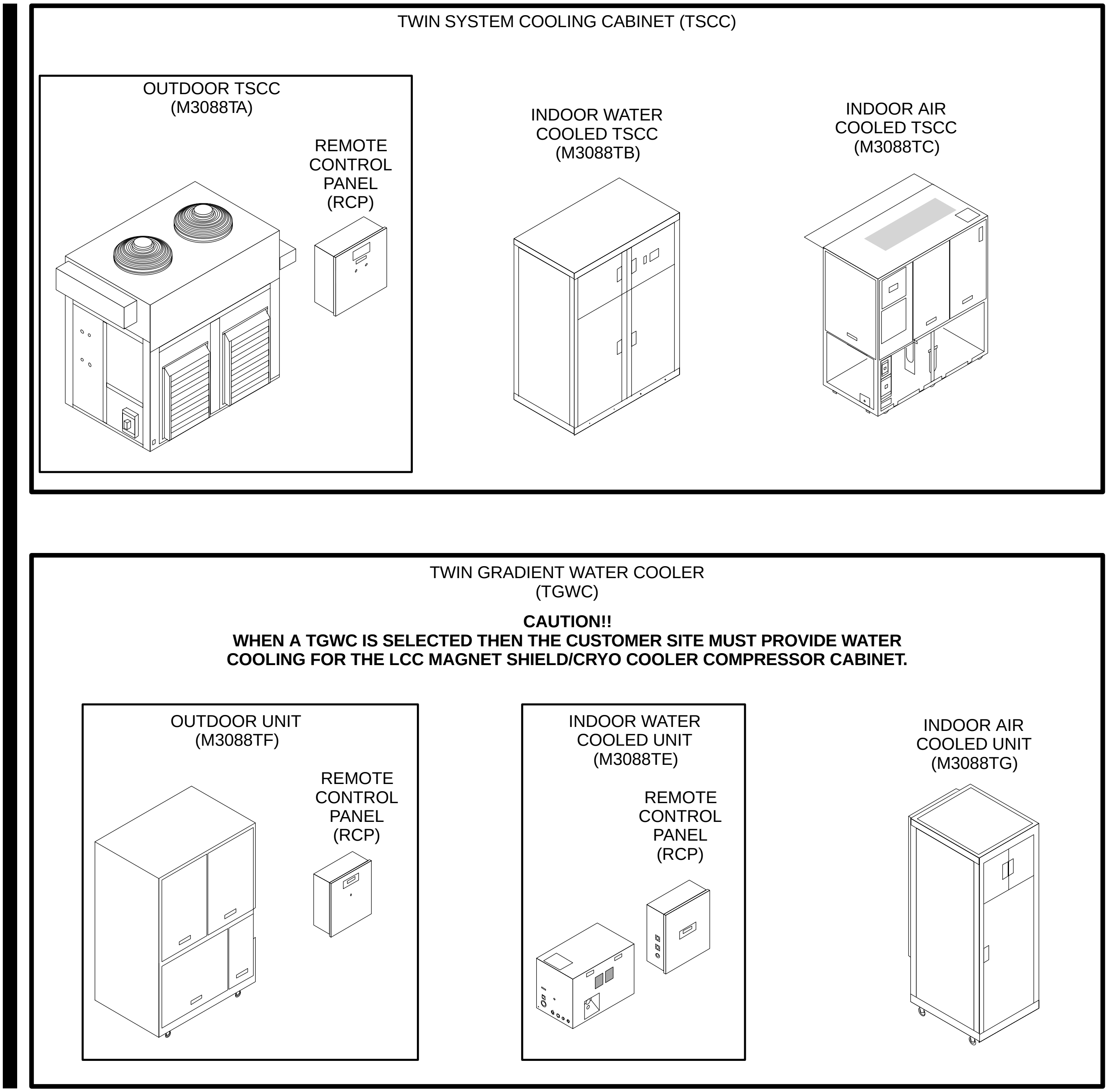
TGWC Configurations

Refer to Table 1–13 for list of TGWC offerings.

TABLE 1–13
TWIN GRADIENT WATER COOLER (TGWC) *See Note 1*

CATALOG	DESCRIPTION
M3088TF	• Outdoor Air Cooled TGWC (Water–to–Air Chiller) 50/60 for Gradient Coil Cooling *TGWC installed external to building with Remote Control Panel installed in Equipment Room.* *The most cost effective for the customer with separate cooling available for the Shield/Cryo Cooler Compressor.*
M3088TE	• Indoor Water Cooled TGWC (Water–to–Water Heat Exchanger) 50/60 Hz for Gradient Coil Cooling *See Note 2* *Unit and Remote Control Panel installed in the Equipment Room, low additional Air Conditioning requirements.*
M3088TG	• Indoor Air Cooled TGWC (Water–to–Air Chiller) 50/60 Hz for Gradient Coil Cooling *Installed in the Equipment Room, increases Air Conditioning requirements.*
Note	1 **When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet.** 2 The water cooled configuration of TGWC requires customer supplied water for cooling. Refer to **Section 4–4–3 Water Cooled System Cooling Cabinet Configuration Requirements** for water cooling specifications.

1–2–1 System Cooling Equipment Selections (Continued)



SYSTEM COOLING EQUIPMENT SELECTIONS
ILLUSTRATION 1–3

1–3 SYSTEM OPTIONS

This section lists options for the Signa 1.5T Twin*Speed* system which have site preparation impact.

- **For Signa 1.5T Twin*Speed* with EXCITE Technology systems ONLY:**
  EXCITE 8 Channel Option (M3033CA)
- 700 Volt Amp UPS for MR Magnet Monitor Equipment (E4504AG)

Note

The GE MDP (M3088TH) must be ordered and installed when the UPS for MR Magnet Monitor (E4504AG) is installed. The GE MDP provides Emergency Off control of the output of the UPS for system safety.

Note

The UPS for MR Magnet Monitor is standard equipment included in the Magnet for True Mobile (M3061TA).

- 2nd Order Resistive Shim Option for Signa 1.5T Twin*Speed* (M3080TA). The 2nd Order Shim Option correct inhomogeneities induced by the patient and restores the original homogeneity of the magent. The Resistive Shim Module is installed in the Twin Accessory Cabinet.
- Advantage Workstation (AW). Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation.*

Note

LCD monitor is recommended for the AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- VCR Interface Kit for Octane Computer (M1090TZ)
- Bar Code Reader (M1090PM) for use in conjunction with HIS/RIS software option.
- Additional Patient Transport table with cradle for WideOpen Enclosure (M1090TL).
- Multi–Nuclear Spectroscopy (MNS) option (M1033MD) provides MNS software and hardware. The hardware includes MNS Cabinet containing RF Broadband Amplifier, system interconnect cables, modification hardware for the basic 1.5T system, P31 Flexible Coil, and MR Spectroscopy Head Phantom.
- Advantage Windows Workstation
- **For Signa 1.5T Twin*Speed* with EXCITE Technology systems ONLY:** BrainWave Option (M1033BM) The BrainWave option provides an integrated set of software and hardware stimulus components for the Signa/AW RTIP application to produce brain activation images from MR Blood Oxygen Level Dependent (BOLD) scan data. Stimulus devices include audio (headphones), visual (glasses), and push–button response boxes held by the patient. Stimulus software resides on the AW (AW4.0 or later required) used by RTIP and on a new PC in the BrainWave Cabinet located in the Equipment Room.
- Other hard copy devices and patient accessories.
- Cryogen refill service.
- Various GE Medical Systems Group service contracts.

1–4 FACILITY OPTIONS

- The GE MDP, all configurations of TSCC, the Outdoor TGWC, and the Indoor Air Cooled TGWC require high voltage power. If system input voltage configuration does not meet specifications in **SECTION 5 – POWER REQUIREMENTS** then customer provided transformer may be needed. Refer to Table 1–14 for low voltage step up transformer options.

TABLE 1–14
LOW VOLTAGE (200 or 208 VOLTS) STEP UP TRANSFORMER OPTIONS

CATALOG See Note 1	**DESCRIPTION**
E4500AS	• 150 KVA 208 to 480Y277 Volt, 60 Hz transformer
R4500BD	• 150 KVA 200 to 400Y230 Volt, 50/60 Hz transformer
Note 1 A step–up transformer is required if the site input voltage configuration does not meet specifications in **SECTION 5 – POWER REQUIREMENTS**. The GE Main Disconnect Panel (MDP) & Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC) configurations require high voltage power.	

- Direct current (DC) lighting controller for the magnet room:
  - E4503AD 20 Amp Maximum Constant Lighting Level System, surface/semi–flush mount
  - E4503AF 20 Amp Maximum Variable Lighting Level System, surface/semi–flush mount
  - E4503AW 28 Amp Maximum Constant Lighting Level System, surface/semi–flush mount
  - E4503AY 28 Amp Maximum Variable Lighting Level System, surface/semi–flush mount.

Note

DC Lighting Controller 50 Hz designs are available by special order, for more information contact:

Dwight Gilbert,GE Supply	414–527–6638	Dwight.Gilbert@supply.ge.com
Lou Hernandez, GE Industrial Systems	262–797–4910	lou.hernandez.indsys.ge.com

- Signature 5000 Series 3 UPS 100KVA (E4502FB) provides reliable, clean, constant voltage power for the complete MR system. The use of a full system UPS enables the system imaging to be completed after loss of supply power and allows for saving of valuable data and orderly system shutdown. Also recommend installing a 100KVA UPS Bypass Panel (E4504CG) which feeds power to the UPS in the normal mode and enables the imaging system to operate when the UPS is in manual bypass mode for routine servicing of the UPS.

- Signa System Seismic Anchorage Service (R4390JA).

Note

Magnet Seismic anchoring is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems. Refer to Section 7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM for details.

- Oxygen Monitor Kit (M1060KM) which includes Oxygen Monitor and Remote Oxygen Monitor Module.

Note

The Oxygen Monitor does not bear a CE monogram and therefore may not be acceptable in all European countries.

1–5 VAN MANUFACTURERS CONTACT INFORMATION

Pre–Installation requirements may vary for specific Mobile configuration (True Mobile, Transportable, or Relocatable) and between Van Manufacturers. Listed below is contact information for GE Van Manufacturers.

- Ellis & Watts
  4400 Glen Willow Lake Lane — Telephone: 513–752–9000
  Batavia, OH 45103 — FAX: 513–752–4983
  USA

- PDC Facilities
  700 Walnut Ridge Drive, PO Box 900 — Telephone: 262–367–7700
  Hartland, WI 53029–0900 — FAX: 262–367–7744
  USA

- SMIT Mobile Equipment
  Buys Ballotstraat 6
  3261 La Oud–Beijerland — Telephone: (31) 186–6–14322
  Holland — FAX: (31) 186–6–19367

- AK Specialty Vehicles
  16745 South Lathrop Avenue — Telephone: 708–596–5066
  Harvey, IL 60426 — FAX: 708–596–2480
  USA
  AK Specialty Vehicles is also the contact for used Calumet Coach trailers.

SECTION 2 – ROOM LAYOUTS

TABLE OF CONTENTS

TABLE OF CONTENTS

2–1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on certain medical implants (including cardiac pacemakers, neurostimulators, and biostimulation devices) and the environmental effect (motors,steel, etc.) on the field homogeneity. The maximum magnetic field in which the equipment can be located is listed in Table 2–1. Selected magnetic shielding of some devices and equipment is possible but must be handled on an individual basis. Refer to Section 6, INTERCONNECTION DATA, for cable length considerations.

The RF shielded room (Magnet Room) is unique in that the room must be shielded from outside radio frequency interference. This is done by enclosing the room with metal walls, floors and ceiling. These shielding requirements impose special considerations which are addressed in Section 7, RF SHIELDED ROOM.

The Magnet Room can be magnetically shielded to reduce the magnet fringe field or to shield the magnet from the effects of the external environment. Refer to Section 3–4, MAGNETIC SHIELDING, for magnet shielding considerations.

TABLE 2–1
PROXIMITY LIMITS

GAUSS (mT) LIMIT See Notes 1 & 2	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS See Note 4	• Nuclear cameras	
1 GAUSS (0.1mT) OR LESS See Note 4	• Positron Emission Tomography scanner • Linear Accelerator • Cyclotrons • Accurate Measuring scale • Image intensifiers • Color TV	• Video display (color, B/W, monochrome) • CT scanner • Ultrasound • Lithotriptor • Electron microscope • Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS See Note 4	• Power transformers • Main electrical distribution transformers • Moving steel equipment such as: – Vehicular traffic – Fork lift trucks – Dumb waiters – Electric transport carts	– Loading dock (truck traffic) – Elevators – Escalators – Helicopters (See Note 3)
5 GAUSS (0.5mT) OR LESS	• Cardiac pacemakers • Neurostimulators	• Biostimulation devices
10 GAUSS (1mT) OR LESS	• Magnetic tapes and floppy discs • Hard copy imagers • Line printers • Video Cassette Recorder (VCR) • Film processor • Credit cards, watches, and clocks • Telephone switching station • Water cooling equipment • HVAC equipment	• Major mechanical equipment room • Large steel equipment such as: – Emergency generators – Commercial laundry equipment – Food preparation area – Air conditioning chiller – Fuel storage tanks – Motors greater than 5 horsepower • X–ray tubes
30 GAUSS (3mT) OR LESS	• System Cabinet • Twin Accessory Cabinet	• TSCC equipment • TGWC equipment
50 GAUSS (5mT) OR LESS	• Octane Computer • Operator Workspace Cabinet • LCD Color Monitor for OW (See note 5) • 1.5T SRFD II Cabinet • ACGD/PDU Cabinet • Main Disconnect Device	• Magnet Monitor • Telephones • Metal detector for screening • LCD Color Monitor for Advantage Workstation • BrainWave Audio Console • BrainWave Cabinet
100 GAUSS (10mT) OR LESS	• S/C Power Supply • Service Tool Magnet Power Supply Cabinet • Service Tool Shim Power Supply Cabinet	• Shield/Cryo Cooler Compressor Cabinet • Oxygen Monitor • Pneumatic Patient Alert Control Box
200 GAUSS (20mT) OR LESS	• Penetration Panel • Blower Box	• Magnet Rundown Unit • Remote Oxygen Sensor Module
800 GAUSS (80 mT) OR LESS	• BrainWave Cart	

Note 1 Refer to SECTION 3, MAGNETIC FIELD CONSIDERATIONS, for magnet field plots.

2 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable.

3 Verify operating limits with provider of helicopter service

4 The 0.5 gauss (0.05 mT), 1 gauss (0.1 mT), and 3 gauss (0.3 mT) or less proximity limits apply to ALL magnets. For additional Active Shield magnet proximity limits refer to **Section 4–13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS**.

5 If gauss limit is more than indicated for OW with LCD Color Monitor then contact GE to determine impacts on OW equipment.

2–2 ROOM SIZES

Table 2–2 contains minimum room dimensions necessary for an MR suite. Table 2–2 also contains issues which are created by reduction in service access, operator access, and equipment space.

TABLE 2–2
SIGNA TWIN*SPEED* SYSTEM CONFIGURATION MINIMUM ROOM INSIDE CLEAR SPACE DIMENSIONS

<table>
<tr><th rowspan="2">SYSTEM CONFIGURATION</th><th colspan="2">EQUIPMENT ROOM MINIMUM VALUES
See Note 6</th><th colspan="3">MAGNET ROOM MINIMUM VALUES
See Note 4</th><th colspan="2">CONTROL ROOM MINIMUM VALUES
See Note 6</th><th rowspan="2">TOTAL SYSTEM AREA
ft^2 (m^2)</th></tr>
<tr><th>W x D
ft–in.
(m)</th><th>AREA
ft^2 (m^2)</th><th>W x D
ft–in.
(m)</th><th>AREA
ft^2
(m^2)</th><th>CEILING HEIGHT
ft–in. (m)</th><th>W x D
ft–in.
(m)</th><th>ARE
A
ft^2
(m^2)</th></tr>
<tr><td>System with
Outdoor TSCC</td><td>8–9 x 13–0
(2.67 x 3.96)</td><td>113.75
(10.57)</td><td rowspan="6">10–11.27 X 19–7.1
(3.334 x 5.972)
See Note 1, 2, 3, 5</td><td rowspan="6">214.3
(19.91)</td><td rowspan="6">8–9
(2.67)
See Note 8</td><td rowspan="6">5* x 7
(1.52* x 2.13)</td><td rowspan="6">35
(3.24)</td><td>363.05
(33.72)</td></tr>
<tr><td>System with
Indoor Water Cooled TSCC</td><td>8–9 x 18–0
(2.67 x 5.49)</td><td>157.50
(14.65)</td><td>406.80
(37.81)</td></tr>
<tr><td>System with
Indoor Air Cooled TSCC</td><td>8–9 x 18–6
(2.67 x 5.64)</td><td>161.88
(15.06)</td><td>411.18
(38.21)</td></tr>
<tr><td>System with
Outdoor TGWC</td><td>8–9 x 13–0
(2.67 x 3.96)
See Note 7</td><td>113.75
(10.57)
See Note 7</td><td>363.05
(33.72)
See Note 7</td></tr>
<tr><td>System with
Indoor Water Cooled TGWC</td><td>8–3 x 14–8
(2.51 x 4.47)
See Note 7</td><td>121.00
(11.22)
See Note 7</td><td>370.30
(34.37)
See Note 7</td></tr>
<tr><td>System with
Indoor Air Cooled TGWC</td><td>8–9 x 15–4
(2.67 x 4.67)
See Note 7</td><td>134.17
(12.47)
See Note 7</td><td>383.47
(35.62)
See Note 7</td></tr>
</table>

Note

1 Must locate magnet in the side–to–side center of magnetic room shield.

2 Minimum dimensions dependent on the magnetic field containment requested and dimensions of magnetic shield design.

3 Room dimensions in front of LCC Magnet MUST allow for Gradient Coil Assembly and split bridge installation/servicing. LCC Magnets MUST USE special Gradient Coil Replacement Tool Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 107.75 in. (2737 mm) clear space in front of magnet. Note split bridge servicing requires 77.5 in. (1969 mm) clear space in front of the magnet.

4 Absolute Minimum Magnet Room dimensions will result in limited operator clearances and increased Magnet Service time. Illustration 2–8shows only 1.5 ft (0.46 m) clearance at end of Patient Transport.

5 Room dimensions do not contain 5 gauss line to room.

6 Minimum Equipment Room and Control Room dimensions do not permit placement of air conditioning units in room. Nor do they permit space for any optional equipment such as Advantage Workstation option, Laser Camera, etc.

7 When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. Minimum Equipment Room for system with TGWC does not permit placement of equipment which will provide water cooling for Shield/Cryo Cooler Compressor.

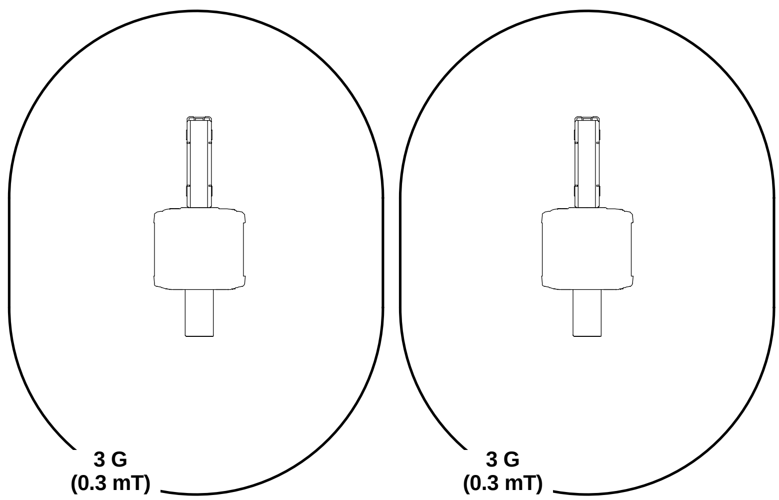
8 Magnet Room minimum ceiling height is 8 ft 2.5 in. (2.50 meter) when Low Ceiling Height Siting Option (M1060SR) is installed.

* Width is dependent on Magnet Room door location and customer's approval of limited space available for operator.

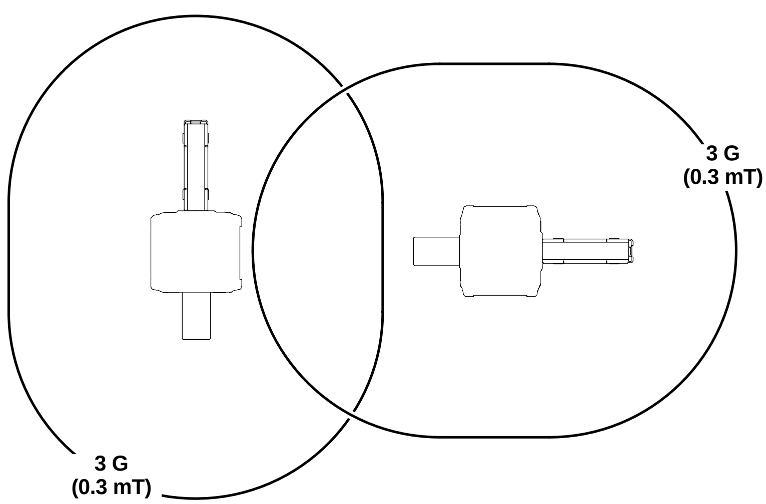
2–3 MULTIPLE MR SYSTEMS SITE

2–3–1 Two Magnet Site Layout

For two magnet installations interaction can occur between the magnetic fields. For 2 magnets not to interact at all (including when bringing magnet to field) the 3 gauss lines of each magnet must not intersect. If the 3 gauss lines intersect but remain outside each magnet's cryostat there will be interaction between the magnets when bringing to field. The orientation of the magnets is irrelevant. Consult the MR Siting & Shielding group for closer proximity of magnets.



NO INTERACTION



INTERACTION WHEN BRINGING MAGNET TO FIELD

TWO MAGNET INSTALLATION
ILLUSTRATION 2–1

2–3–2 Equipment Room Shared By Multiple MR Systems

When the Equipment Room is shared by two or more MR systems of the same field strength there is a potential for cross–talk of RF energy between the MR systems. RF cross–talk may cause noise artifacts in images. Proper planning and installation of the multiple systems in the shared Equipment Room can reduce the potential for cross–talk.

- The RF Screen of the Magnet Room for each system needs to meet the RF Attenuation specifications in **Section 7–1 RF SHIELDED ROOM SPECIFICATION**.
- The Equipment Room design, layout, and installation must meet the requirements of **APPENDIX D – REQUIREMENTS FOR EQUIPMENT ROOM SHARED BY MULTIPLE MR SYSTEMS** for:
  - Relative location of certain equipment cabinets of one MR system relative to other MR system(s) equipment cabinets.
  - Separation of the Penetration Panel of one MR system relative to the Penetration Panel of other MR system(s).
  - Separation of interconnects of one MR system relative to interconnects other system(s) and the managing of any excess RF Receive and RF Transmit cables for each MR System.

2–4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY

Table 2–3 lists minimum actual clearance opening dimensions for doors and hallways required by Signa equipment. Installation or replacement of components listed in Table 2–4 must be taken into consideration when determining hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for Signa Component shipping dimensions.

TABLE 2–3
MINIMUM HALLWAY/DOOR DIMENSIONS

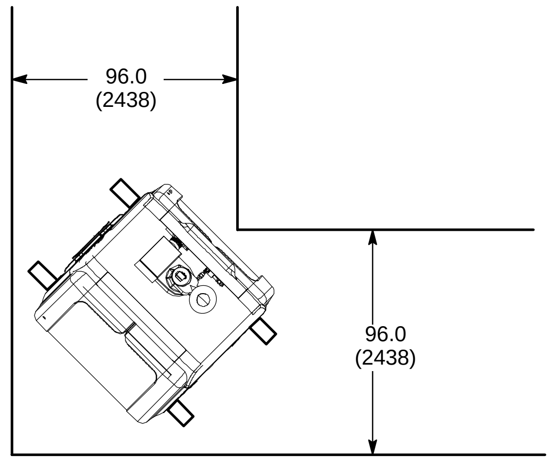
COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Operator Workspace Table	32 (813)	80 (2032)	
Indoor Water Cooled TSCC [Model LC18MT/WC]	36 (914)	80 (2032)	
Indoor Air Cooled TSCC [Model LC20MT]	36 (914)	80 (2032)	
Equipment Cabinets	36 (914)	80 (2032)	Includes Indoor Water Cooled TGWC [Model LTL10] or Indoor Air Cooled TGWC [Model LC10M]
Cryogen delivery route and Storage Room	43 (1092)	80 (2032)	Width requirements due to size of 500 liter dewars. Width and height requirements vary dependent on the dewars used. Check with cryogen supplier.
Magnet	Refer to Note 1	Refer to Note 2	Refer to Table 2–4 for uncrated magnet dimensions.
RF Room Door	Refer to Section 7, RF SHIELDED ROOM	Refer to Section 7, RF SHIELDED ROOM	

Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension.

1 Minimum width depends on access route to removable panels of RF shielded room wall. For straight path (i.e. no bends or turns) recommended to allow 6 in. (153 mm) on both sides of magnet. Appropriate calculations must be performed if turns exist along proposed magnet delivery route. Illustration 2–2 shows dimensions for 90° turn.

2 Final dimension is dependent on rigger equipment used, refer to SECTION 8, SHIPPING AND DELIVERY DATA.

2–4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY(continued)



NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

MAGNET MINIMUM DOOR/HALLWAY DIMENSIONS 90° TURN
ILLUSTRATION 2–2

TABLE 2–4
COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
Magnet (uncrated)	Refer to comments.	Refer to comments.	Refer to Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
Split Bridge (LCC Magnet Enclosure)	40 (18)	21.5 x 77.3, x 7 (546 x 1969 x 18)	Room dimensions in front of LCC Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement, See Note 3.
Replacement RF Body Coil	155 (70)	30 x 30 x 60 (762 x 762 x 1524)	Replacement coil is shipped in a protective case. Weight & dimensions are for coil & case.
Replacement Gradient (TRM) Coil Assembly on a Shipping Cradle/Cart	See Note 1	35.5 x 99.84 x 55.88 (902 x 2536 x 1420) See Note 2 & 3	Initial TRM Gradient Coil Assembly is shipped installed in the Magnet. Shipping/installation cart is used to install replacement coil assembly only. Refer to Section 2–4–1, Gradient Coil Assembly, Cradle, & Cart Requirements.
Gradient (TRM) Coil Replacement Tool Kit Crate for LCC Magnets	750 (340)	30 x 86 x 28 (762 x 2184 x 711)	See Note 3.

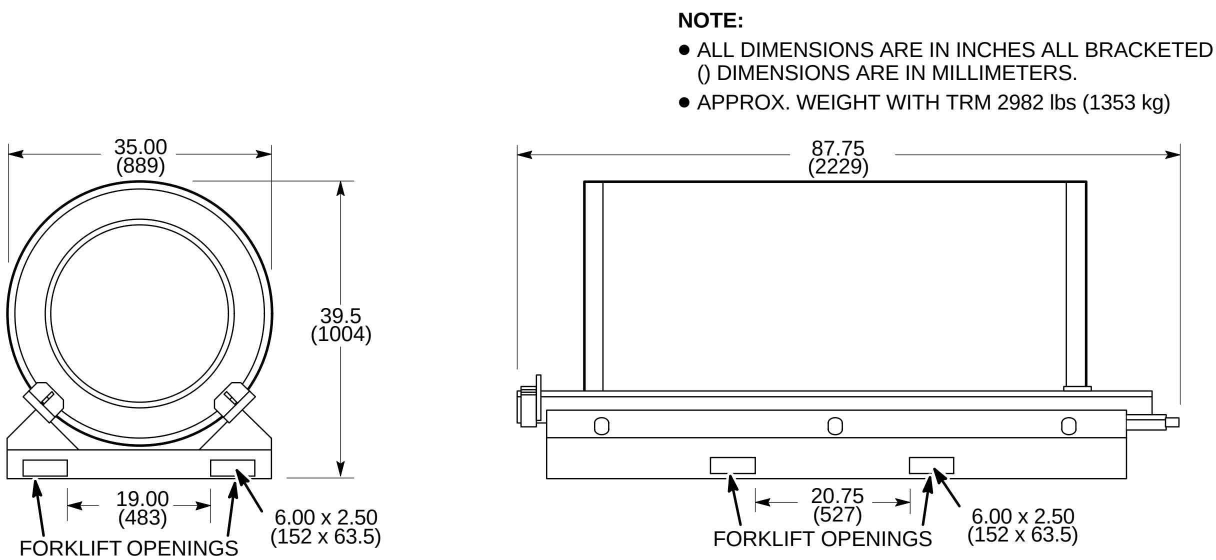
Note 1 The replacement TRM Gradient Coil Assembly weight is approximately 2850 lbs (1293 kg), the shipping cradle is 132 lbs (60 kg), and the Gradient Coil Assembly shipping/installation cart weighs 855 lbs (389 kg). Therefore total shipping weight is 3837 lbs (1742 kg). The coil assembly outside diameter x length dimensions are 35.0 x 74.6 in. (888 x 1895 mm).

2 Gradient (TRM) Coil Assembly and shipping cart dimensions are with cart in lowest position. Cart can be adjusted to maximum height of 61.88 in. (1572 mm).

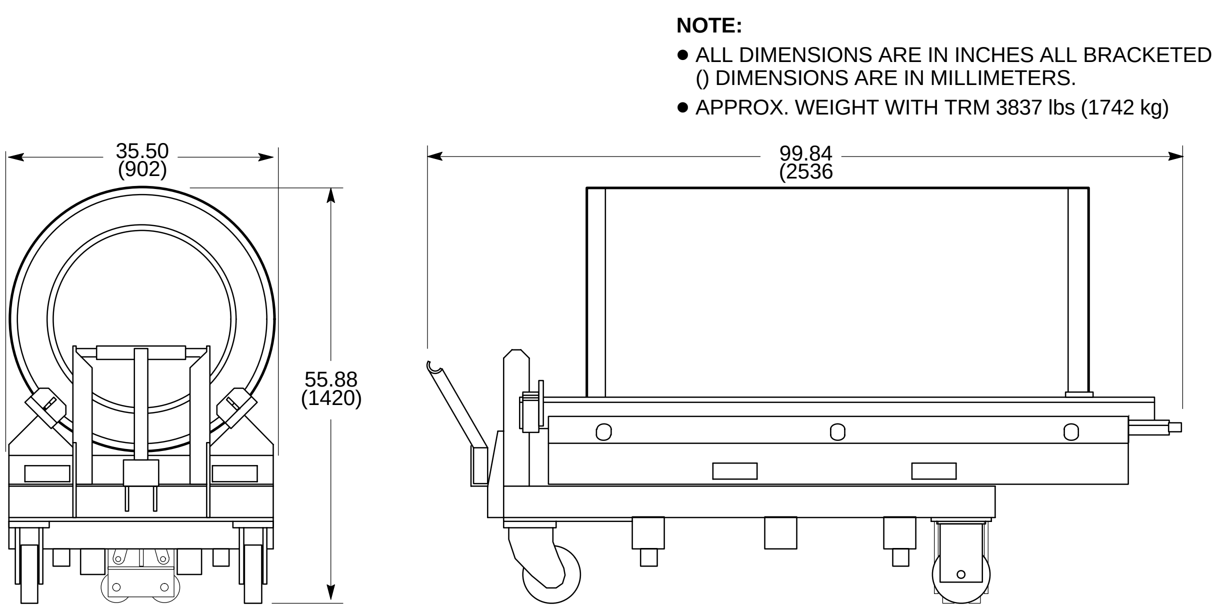
3 LCC Magnets MUST USE GE Service Tool Gradient Coil Replacement Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 107.75 in. (2737 mm) clear space in front of magnet.

2–4–1 Gradient Coil Assembly, Cradle, & Cart Requirements

Initial TRM Gradient Coil Assembly is shipped installed in the Magnet. Shipping/installation cart is used to install replacement coil assembly only. The Gradient Coil Assembly will be delivered on an aluminium (re–useable) cradle. A forklift or crane/hoist rated for 4000 lbs (1818 kg) may be required to position the coil/cradle assembly onto the cart for installing the coil into the magnet, see Illustrations 2–3 and 2–4.



REPLACEMENT GRADIENT (TRM) COIL ASSEMBLY & CRADLE
ILLUSTRATION 2–3



SHORT ROOM REPLACEMENT GRADIENT (TRM) COIL ASSEMBLY & CRADLE ON CART
ILLUSTRATION 2–4

2–5 CABLING CONSIDERATIONS

Several different methods for running cables are listed below and the customer should carefully consider the advantages and disadvantages of each.

Care must be taken to protect interconnecting cords and cables from physical damage (including water). Branch circuit conductors must be enclosed in metal raceway or metal wireway when concealed or when installed under access flooring.

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

Note

Only non–magnetic metal material (e.g. aluminum) can be used when routing cables in the Magnet Room.

2–5–1 Floor Duct

Recessed floor duct has advantages when used within a single room or two adjacent rooms. Floor duct combines a neat functional appearance with accessibility and room for expansion. The disadvantage is the amount of work required to install it, which is generally prohibitive in old installations. Floor ducts can be used in the Magnet Room, however, they must meet the requirements in Section 7, RF SHIELDED ROOM.

2–5–2 Raceway

Raceways offer some unique advantages when routing cables. It is very practical to use in existing structures since it is surface–mounted. There is no problem with pre–terminated cables since the entire raceway system can be opened. Raceway systems are relatively easy to expand as compared to other means of routing cables. However, surface–mounted raceways are not recommended for routing cables within the Magnet Room due to the number/size of cables and the trip hazard of the raceway.

2–5–3 Access Flooring

Access flooring (i.e. removable panel flooring) is recommended for use in both the Equipment and Magnet Rooms due to the number and size of cables in the system. Cable accessibility and ease of alteration are just a few advantages of using access flooring. Floor duct with dividers placed above the Magnet Room floor but beneath the access flooring is a convenient method of separating electrical lines from water lines. However, if the area under the access flooring is used for an air plenum, cables may have to be in raceway depending on local and national codes. **Note**, the Signa system interconnecting cables in the Equipment room are FT4 or equivalent rated, not plenum rated.

2–5–4 Conduit

Conduit has some important restrictions when used with a MR system. The primary problem is that the majority of cables used are pre–terminated, which greatly simplifies interconnection, but makes cabling difficult because of the added dimensions of the connectors. As a consequence, conduit size must allow for the dimension of the connectors and the possibility of additional cables being added as the system is upgraded in the future. Always size the conduit to allow the cable to pass through with all other cables already in the conduit. Conduit should not be used for running the main GE MR system cables in the Magnet Room due to the number and size of conduits needed.

Note

MR personnel must have an unobstructed path from the patient table to the area directly behind the rear pedestal. Therefore cable routing methods must not interfere with this pathway.

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes.

2–6 FLOORING

The finished floor in the Magnet Room should be water resistant and a type flooring to minimize the possibility of a static discharge. Use of a access floor with covering to minimize static discharge is recommended in the Equipment Room and the portion of the Magnet Room designated for cable routing. For safety purposes in the Magnet Room, it is required that the access flooring be made of aluminum. Depending on local and national codes, the area under the access floor may possibly be used as an air conditioning plenum. If the area under the access floor is to be used as an air plenum and for cable routing, 10 in. (254mm) of clear space from the underside of the access floor to the permanent floor is recommended. Cabling, plumbing (water lines), etc. routed under the access floor may affect air flow and needs to be considered if used as an air conditioning plenum. Also check local and national codes for fire protection requirements under access floor. **Note**, the Signa system interconnecting cables in the Equipment room are FT4 or equivalent rated, not plenum rated.

Ensure that the access flooring, if used, can support the equipment and any transport device needed to move the equipment.

If carpeting is used, it should either be anti–static carpeting or treated with an anti–static solution. Carpeting is not recommended in the Magnet Room or cryogen storage area as well as along the dewar delivery route due to possible problems with docking the patient table or moving the dewars.

Information on RF shielded room floor requirements can be found in Section 7, RF SHIELDED ROOM.

2–7 System Cooling Equipment

The Signa Twin*Speed* system requires water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. Refer to Section 1–2–1 System Cooling Equipment Selections for listing of catalog

- Twin System Cooling Cabinet (TSCC) which provides water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil located in the Magnet bore. Refer to Section 2−7−1 TSCC Siting Considerations for details.
- Twin Gradient Water Cooler (TGWC) which provides water cooling for ONLY the Gradient Coil located in the Magnet bore. When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. Refer to Section 2−7−2 TGWC Siting Considerations for details.

Note
True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.

2–7–1 TSCC Configurations Siting Considerations

CAUTION

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival. Therefore the Twin System Cooling Cabinet (TSCC) and Main Disconnect Panel (MDP) must be installed and operational prior to magnet arrival.

Note
Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TSCC is available in three fixed site configuration options:

- Outdoor Air Cooled TSCC [Model LC–20MT/O] has the TSCC unit located outside the building and a Remote Control Panel (RCP) located in the MR system Equipment Room; see Illustration 2–14 for Outdoor Air Cooled TSCC and Illustration 2–16 for the Remote Control Panel.
- Indoor Water Cooled TSCC [MODEL LC–18MT/WC]; see Illustration 2–17 and refer to Section 4–4–3 Water Cooled TSCC Requirements for water cooling requirements.
- Indoor Air Cooled TSCC [MODEL LC–20MT]; see Illustration 2–19.

Note
Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet. Refer to Section 4–4–2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation.

2–7–1 TSCC Configurations Siting Considerations (Continued)

TSCC & System Interconnects/Separation Limitations

Location of the Outdoor TSCC must meet the following limitations for the water lines (customer provided copper line and E&W/GE provided flexible tubing combined).

- Outdoor TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- Outdoor TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 160 ft (48.8 m) water line length.
- The Outdoor TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 100 ft (30.5 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.
- The Outdoor TSCC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total interconnect length.

Location of the Indoor Water Cooled TSCC or Indoor Air Cooled TSCC must meet the following limitations for the water lines (E&W/GE provided flexible tubing).

- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 60 ft (18.3 m) water line length.
- The Indoor Water Cooled TSCC or Indoor Air Cooled TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 15 ft (4.6 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.

The TSCC is powered from the system Main Disconnect Panel (MDP) via customer supplied wiring. Refer to **Section 6, INTERCONNECT DATA**, for additional interconnects information and requirements.

2–7–1 TSCC Configurations Siting Considerations (Continued)

Responsibility For Installation Tasks For TSCC Equipment

The TSCC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2–5 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2–5
TSCC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG				RESPONSIBILITY		
OUTDOOR (LC20MT/O)	**INDOOR WATER COOLED (LC18MT/W)**	**INDOOR AIR COOLED (LC20MT)**	**INSTALLATION TASKS FOR TSCC EQUIPMENT** See Notes 1, 2, & 3	**CUSTOMER CONTRACTOR**	**GE SERVICE**	**E&W SERVICE PROVIDER**
			WHEN TSCC IS AT SITE & PRIOR TO MAGNET DELIVERY			
X	X	X	Unload TSCC from truck.	X		
X			Move TSCC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TSCC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TSCC to Equipment Room and mount. (Use seismic tie down brackets & vibration isolators where required by law)	X		
X	X		Install customer water supply and return lines to TSCC	X		
X	X	X	Connect customer supplied power wiring from MDP to TSCC.	X		
X			Install power and control cable between TSCC & RCP	X		
			WHEN MAGNET IS DELIVERED/INSTALLED			
X			Install water cooling lines from customer supplied/installed ball values & hose barbs on copper lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
	X	X	Install water cooling lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
X	X	X	Fill TSCC with Propylene Glycol and startup unit		X	
X	X	X	See **APPENDIX C Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*** and review for completeness. Send email request for initial performance verification to service@elliswatts.com. Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
X	X	X	Verify proper operation per E&W Chiller Startup Check List			X

Note
1. Refer to Ellis & Watts *Technical Publication 470 Pre installation/Installation/Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Outdoor TSCC information and details.
2. Refer to Ellis & Watts *Technical Publication 471 Pre installation/Installation/Operating LC 18MT/ Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Indoor Water Cooled TSCC information and details.
3. Refer to Ellis & Watts *Technical Publication 472 Pre installation/Installation/Operating LC 20MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Indoor Air Cooled TSCC information and details.

2–7–2 TGWC Configurations Siting Considerations

CAUTION

When a TGWC is selected for the system then the customer site must still provide cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet; refer to Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC

Note

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TGWC is available in three fixed site configuration options:

- Outdoor Air Cooled TGWC [MODEL LC–10MO] has the TGWC unit located outside the building and a Remote Control Panel (RCP) is located in the MR system Equipment Room; see Illustration 2–21 for Outdoor Air Cooled TGWC and Illustration 2–23 for the RCP.
- Indoor Water Cooled TGWC [MODEL LTL–10] has the TGWC unit and a Remote Control Panel both located in the MR system Equipment Room; see Illustration 2–24 and refer to **Section 4–4–5 Water Cooled TGWC Requirements** for water cooling requirements.
- Indoor Air Cooled TGWC [MODEL LC–10M]; see Illustration 2–26.

TGWC & System Interconnects/Separation Limitations

Location of the Outdoor TGWC must meet the following limitations for the water lines (customer provided copper line and GE provided flexible tubing combined).

- Outdoor TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- The Outdoor TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 100 ft (30.5 m) above or below the Gradient Coil.
- The Outdoor TGWC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total interconnect length.

Location of the Indoor Water Cooled TGWC or Indoor Air Cooled TGWC must meet the following limitations for the water lines (GE provided flexible tubing).

- Indoor Water Cooled TGWC or Indoor Air Cooled TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- The Indoor Water Cooled TGWC or Indoor Air Cooled TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

The Outdoor TGWC and Indoor Air Cooled TGWC are powered from the Main Disconnect Panel (MDP) via customer supplied wiring. The Indoor Water Cooled TGWC is powered from the MR system PDU. Refer to Section 6, INTERCONNECT DATA, for additional interconnects information and requirements.

2–7–2 TGWC Configurations Siting Considerations (Continued)

Responsibility For Installation Tasks For TGWC Equipment

The TGWC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2–6 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2–6
TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG			INSTALLATION TASKS FOR TGWC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
OUTDOOR (LC10MT/O)	INDOOR WATER COOLED (LC10)	INDOOR AIR COOLED (LC10MT)		CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
			WHEN TGWC IS AT SITE & PRIOR TO MAGNET DELIVERY			
X	X	X	Unload TGWC from truck.	X		
X			Move TGWC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TGWC and Remote Control Panel to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TGWC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
X	X		Install customer water supply and return lines to TGWC	X		
X		X	Connect customer supplied power wiring from MDP to TGWC.	X		
X	X		Install power and control cable between TGWC & RCP	X		
			WHEN MAGNET IS DELIVERED/INSTALLED			
X			Install water cooling lines from customer supplied/installed ball values & hose barbs on copper lines from TGWC to Gradient Coil. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
	X	X	Install water cooling lines from TGWC to Gradient Coil. Refer to **Section 6 – INTERCONNECT DATA** for details of interconnects.		X	
X		X	Fill TGWC with Propylene Glycol and startup		X	
X		X	Verify proper operation per E&W Chiller Startup Check List			X

Note
1. Refer to Ellis & Watts *Technical Publication 474 Pre installation/Installation/Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Outdoor TGWC information and details.
2. Refer to Ellis & Watts *Technical Publication 469 Pre installation/Installation/Operating LTL10 Chilled Water System for GE Twin Magnet MRI Equipment* for additional Indoor Water Cooled TGWC information and details.
3. Refer to Ellis & Watts *Technical Publication 473 Pre installation/Installation/Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Indoor Air Cooled TGWC information and details.

(Continued)

2–7–2 TGWC Configurations Siting Considerations (Continued)

TABLE 2–6 (Continued)
TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG				RESPONSIBILITY		
OUTDOOR (LC10MT/O)	**INDOOR WATER COOLED (LC10)**	**INDOOR AIR COOLED (LC10MT)**	**INSTALLATION TASKS FOR TGWC EQUIPMENT** See Notes 1, 2, & 3	**CUSTOMER CONTRACTOR**	**GE SERVICE**	**E&W SERVICE PROVIDER**
			WHEN MR PDU IS DELIVERED/INSTALLED			
	X		Connect E&W supplied power cable from PDU to TGWC.		X	
	X		Fill TGWC with Propylene Glycol, startup and verify operation		X	
X	X	X	See **APPENDIX C Prerequisites for Scheduled Performance Verification of Chillers For Signa Twin*Speed*** and review for completeness. Send email request for initial performance verification to service@elliswatts.com. Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	

Note
1. Refer to Ellis & Watts *Technical Publication 474 Pre installation/Installation/Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Outdoor TGWC information and details.
2. Refer to Ellis & Watts *Technical Publication 469 Pre installation/Installation/Operating LTL – 10 Chilled Water System for GE Twin Magnet MRI Equipment* for additional Indoor Water Cooled TGWC information and details.
3. Refer to Ellis & Watts *Technical Publication 473 Pre installation/Installation/Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional Indoor Air Cooled TGWC information and details.

2–8 SPECIAL SITING CONSIDERATIONS

The following system accessories or options have special siting concerns which need to be considered.

2–8–1 Blower Box (MG6)

The Blower Box (MG6) provides cooling air for the Patient Comfort Module in the Magnet Enclosure. The Blower Box will be mounted within the RF Shielded Room and connect to the Patient Comfort Module by 4.0 inch (101.6 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or a support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room.** The flexible vinyl air duct routes from the Blower Box through the Magnet Enclosure Rear Pedestal cable access and connects to the Patient Comfort Module in the Magnet Enclosure. Refer to Illustrations 2–27 for Blower Box and mounting dimensions.

Note

Blower Box mounting requires customer supplied hardware (ie. lag bolts, screws, etc.) appropriate for the surface on which the box will be mounted.

2–8–2 Pneumatic Patient Alert (PA1)

The Pneumatic Patient Alert system is a stand alone system that will allow the Patient to contact the Operator even when the intercom volume is turned down. The Control Box is to be located near the Operator Workspace. The Control Box audible and visual alarm will be activated by the patient squeeze bulb which is located on the Magnet Enclosure and connected by pneumatic tubing through the Penetration Panel to the Control Box. The Control Box should be mounted with consideration for ease of use by operator, remaining within sight of operator, and within 5 ft (1.5 m) of an electrical outlet. The Control Box can be powered from an outlet on the Operator Workspace. Refer to Illustration 2–42 for Control Box mounting dimensions.

2–8–3 SPT Phantom Set Shipping/Storage Cabinet

System Performance Test provides the customer and GE Service with a means to quickly verify whether critical parameters affecting image quality are within specifications. The test uses a set of phantoms and a nesting plate for proper positioning of the phantoms on the Patient Table. The phantom set and nesting plate are provided in a cabinet which protects the pieces during shipment and storage at site. The cabinet is not magnetic therefore it can be stored inside the Magnet Room if so desired and moved to the Patient Table for ease of positioning the phantoms. See Illustration 2–44 for dimensions information.

2–8–4 Oxygen Monitor (OM1) (Optional)

The optional Oxygen Monitor should be mounted near the Operator Workspace. The Oxygen Monitor alarm will be activated by the remote oxygen sensor located in the Magnet Room. All cellular telephones, even if not in use, should be kept at least 20 feet (6.1 meters) away from the Oxygen Monitor to prevent possible false trips of Oxygen Monitor alarms.

2–8–5 Spectroscopy for 1.5T Magnet Only (Optional)

Multi–Nuclear Spectroscopy (MNS)

The MNS option provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet (contains the Broadband RF Amplifier) and several module kits to modify the SRF Cabinet, Penetration Panel in the Magnet Room, and system cabling. Refer to **Section 6–8, SPECTROSCOPY INTERCONNECTIONS** cable length information and other cable data.

Hydrogen Only (PROBE SV/Q) Spectroscopy

No additional siting impacts.

2–8–6 Advantage Workstation Option Additional Hardware Features

The Advantage Workstation (AW) option can be utilized with cardiac and neurovascular software offerings. Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation* for more information.

The customer must determine the desired location for the AW relative to the imaging system. The following factors should be considered in determining AW location.

- If the customer is purchasing RTIP (neurovascular) option, then the customer may want to observe the experiment real time while operating the Signa system versus reviewing the data at a later time. If the former, then consideration should be made to co–locate the AW near the system Operator Workspace (OW).

Note

LCD monitor is recommended for the AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- If the customer is going to perform cardiac studies only, then co–location is not necessary at this time. However, co–location of the AW may be recommended for future applications.

2–8–7 Magnet Monitor

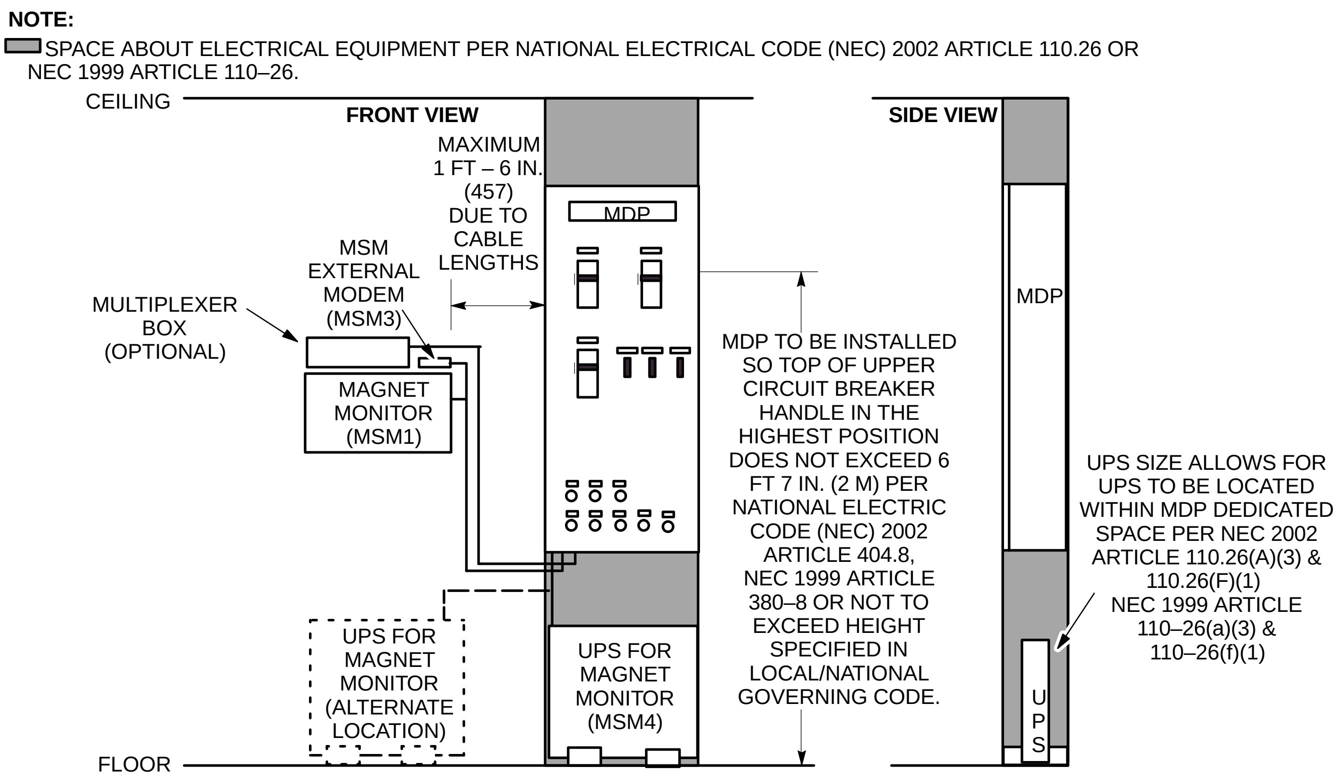
The Magnet Monitor performs the functions of a cryogen meter and LCC Magnet pressure control with readout display capability on the unit The Magnet Monitor also monitors vacuum level of the Quiet Technology Magnet Enclosure and the TSCC/TGWC operation parameters. The Magnet Monitor with its modem allows for remote monitoring during system warranty period or available as part of a GE Service contract. A dedicated direct–distance–dialing voice–grade telephone line with access located near the Magnet Monitor in the Equipment Room for remote monitoring is required, refer to **Section 2–8–8 System Monitoring & Support Connectivity Requirements** for additional information.

Magnet Monitor remote monitoring operation is valuable to maximize proper uninterrupted magnet operation. The Magnet Monitor is powered from the Main Disconnect Panel (MDP) and must be mounted near the MDP, see Illustration 2–5. The Magnet Monitor should be mounted approximately 60 in. (1524 mm) above the floor in the Equipment Room but outside the 10 gauss zone. Refer to **Section 6 – INTERCONNECT DATA** for details of Magnet Monitor, optional UPS, and system interconnects.

An optional small Uninterruptible Power System (UPS) is available for the Magnet Monitor. This optional UPS provides power to the Magnet Monitor equipment via the MDP. In the event of facility power outage, the UPS will maintain Magnet Monitor operation for sufficient time to communicate status via the remote monitoring.

Note

The UPS for the Magnet Monitor equipment is provided as part of the Magnet subsystem for True Mobile configurations.



MAGNET MONITOR (MSM) & ASSOCIATED EQUIPMENT, UPS FOR MAGNET MONITOR, AND MDP LOCATIONS
ILLUSTRATION 2–5

2–8–8 System Monitoring & Support Connectivity Requirements

One of the system monitoring and support connectivity configurations listed in Table 2–7 must be provided for system installation and serviceability purposes. The network connection (if selected configuration) and telephone lines are to be provided and paid for by the customer.

TABLE 2–7
SYSTEM MONITORING & SUPPORT CONNECTIVITY REQUIREMENTS

CONFIGURATION	PHONE LINE	USE/LOCATION
Network connection & telephone lines	One voice–grade telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	Two Network connection with a static IP Addresses **See Note 1**	One access located near the Operator Workspace (OW) in the Control Room.
		One access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. This Ethernet connection must not lose power when the MR system is shutdown.
Multiple telephone lines	One voice–grade telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct–distance–dialing voice–grade line (data line)	Access located near the Operator Workspace (OW) in the Control Room. **See Note 2 & 3**
	One line must be a dedicated direct–distance–dialing voice–grade line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. **See Note 2 & 3**

Note 1 **For North America & South America:** One Internet accessible Virtual Private Network Connection with a static IP Address is required. **For Europe:** One ISDN Network Connection with a static IP Address is required.

2 A dedicated direct–distance–dialing voice–grade telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase.
46–328475P1 4 Line Phone Multiplexer box; 115 VAC input power
46–328475P3 4 Line Phone Multiplexer box; 220 VAC input power
If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table.

3 If a Multiplexer Box is used then the Magnet Monitor **MUST** be Channel 1 to allow for call out after a power outage.

2–8–9 BrainWave Option For Signa 1.5T Twin*Speed* with EXCITE Technology Systems ONLY

The BrainWave option provides an integrated set of software and hardware stimulus components for the Signa/AW RTIP application to produce brain activation images from MR Blood Oxygen Level Dependent (BOLD) scan data. Stimulus devices include audio (headphones), visual (goggles), and push–button response box held by patient. Stimulus software resides on the AW (AW4.0 or later required) used by RTIP and on a PC in the BrainWave Cabinet located in the Equipment Room.

The BrainWave option consists of the following major items:

- BrainWave Audio Console
  Used to select the sound media and communicate to/from the patient. The Audio Console is to be located on the Signa system Operator Workspace Table.

- BrainWave Cabinet
  Located in the Equipment Room, the BrainWave Cabinet contains the following BrainWave equipment: Stimulus PC, LCD Display, keyboard, and mouse for the Stimulus PC, Video Interface, Graphic (audio) Equalizer, and Response Interface Module.

- BrainWave Cart
  Located in the Magnet Room, the BrainWave Cart holds the Visual Display Assembly consisting of the Visual Projector, Image Guides, and Glasses which are factory connected together as one assembly. The color LCD Visual Projector receives video signals from the Video Interface in the BrainWave Cabinet, converts the signals to optical images, and then optically transmits the images to the left/right Patient Glasses via two flexible Image Guides. The Glasses are mounted to the inside of the Signa Head Coil with the Glasses Mount Assembly that provides for proper viewing of images by patients.

- Interconnect cables
  The Audio Console is powered from the Operator Workspace. The BrainWave Cabinet is powered from the Signa System Cabinet. There are interconnects from Audio Console to the Signa System Cabinet and Penetration Panel, from System Cabinet to the Penetration Panel to the BrainWave Cart via the Magnet Enclosure. Refer to **Section 6, INTERCONNECT DATA**, for additional interconnects information and requirements.

2–9 ARCHITECTURAL REMINDERS

1. Pay attention to isogauss limits, not only for placement of equipment in rooms, but also for isogauss limits with respect to outside environment.

2. The customer is responsible for establishing protocols to warn persons with cardiac pacemakers, neurostimulators, and biostimulation devices of the potential danger of entering magnetic fields greater than 5 gauss (exclusion zone).

3. Due to the periodic cryogen servicing of the magnet, consideration must be given to the delivery route of the cryogens to the Magnet Room. The service route should be level and therefore steps or steep ramps must be avoided. Maximum acceptable incline along dewar delivery route is 1:12 (5°).

3A. Cryogen dewars must not be stored within the Magnet Room due to the safety issues of seismic considerations, spillage effects, fire hazards and explosive effects of compressed gas. Also the magnetic field inhomogeneity is affected by the physical shape of the non–magnetic dewars. If the magnet is shimmed with no dewars in the Magnet Room, the magnetic field inhomogeneity will slightly change once dewars are moved into the room (or vise versa). All dewars must be stored outside of the Magnet Room and more than 10 ft (3.05 m) from isocenter of the magnet in all directions.

3B. Means must be provided to secure gas cylinders used for cryogen transfills in an upright position using a removable chain or strap. This is to prevent the cylinders from falling, which may cause injury or damage.

4. If elevators are to be used along cryogen delivery route, verify that elevator dimensions and weight capacity is sufficient to handle the cryogen dewars. Also, elevator must be dedicated with restricted access during cryogen transport (will not allow stops between initial start and final floor destination).

5. The operator seated at the Operator Workspace should have an unobstructed view of the patient on the transport table when table is docked to the magnet.

6. It is recommended that the Magnet Room viewing window be of fine mesh screening material (as opposed to a “honeycomb–type pattern”) for better visibility of the patient from the Operator Workspace.

7. Operators in Magnet Room must have easy access to the scan control switches located on both front side panels of the magnet enclosure.

8. A patient preparation/emergency area should be located near the Magnet Room and direct patient access must be available from the Magnet Room to the patient preparation/emergency area.

9. Customer provided and paid for telephone lines must be supplied for system installation and serviceability purposes per **Section 2–8–8 System Monitoring & Support Connectivity Requirements**.

10. A lockable storage cabinet can be provided and maintained by GE Medical Systems Service for storage of GE Medical Systems service documentation/tools. Cabinet to be approximately 36 in. (914 mm) wide, 18 in. (457 mm) depth, and 72 in. (1829 mm) high.

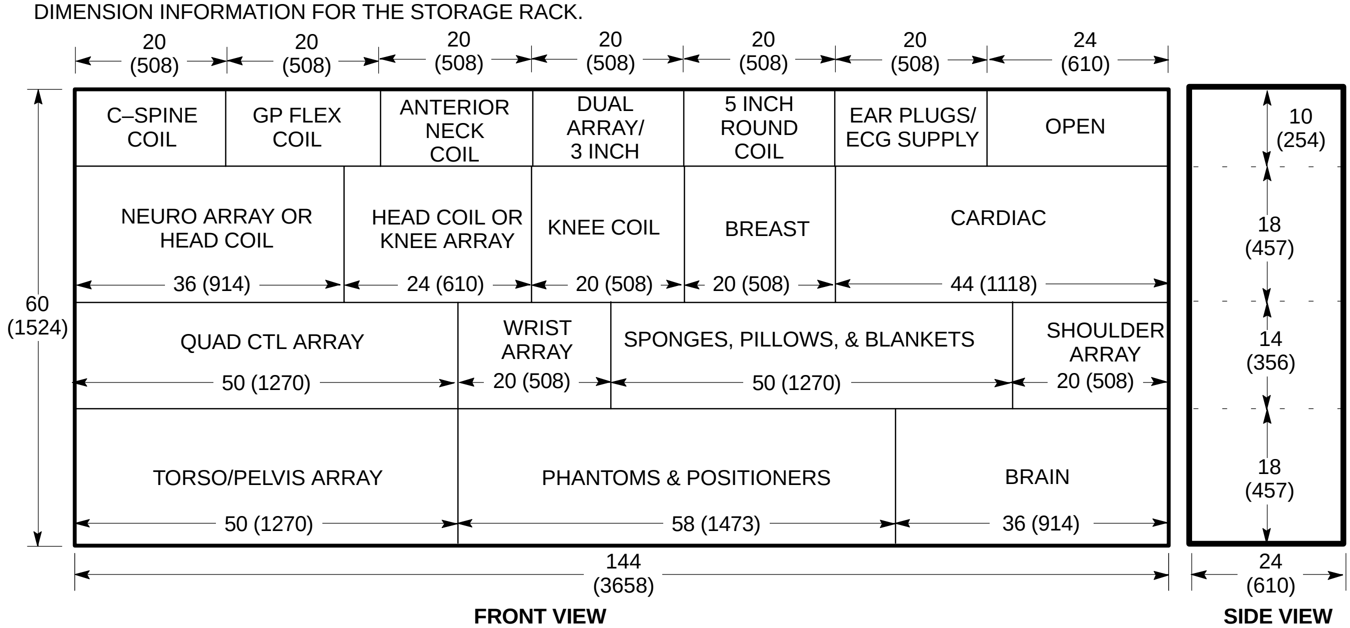
2–9 ARCHITECTURAL REMINDERS (Continued)

11. Corrosive chemicals must not be stored or used in the Equipment Room. These include chemicals used for film processor storage tanks, processor chemical recovery systems, etc. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control area.

12. Storage space for system accessories and supplies should be planned for and included in room layout drawings. Illustrations 2–6 and 2–7 show a suggestion for a shelf/cabinet arrangement.

13. Recommend protecting floors while moving heavy pieces of equipment (e.g. ACGD/PDU Cabinet, TRM Body Coil Assembly, TSCC, TGWC).

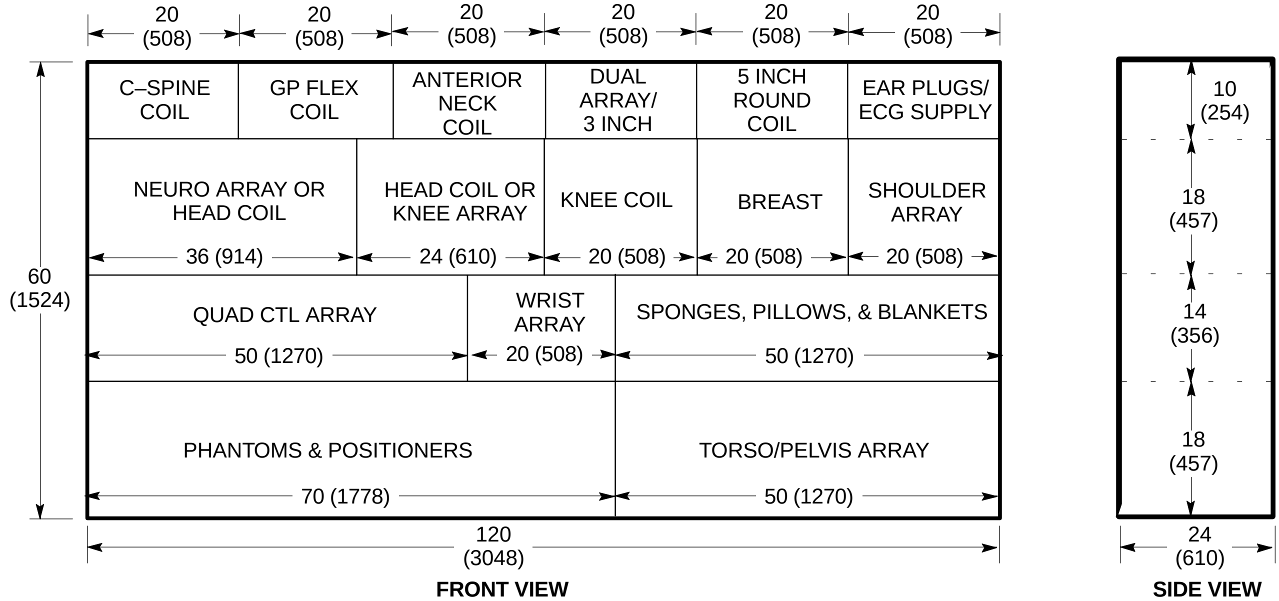
NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- PERIPHERAL VASCULAR ARRAY IS NOT INCLUDED IN THE BELOW STORAGE SHELF/CABINET. CONTACT GE MEDICAL SYSTEMS ACCESSORIES FOR PV ARRAY STORAGE RACK OPTION AND DIMENSION INFORMATION FOR THE STORAGE RACK.



SUGGESTION FOR ACCESSORIES & SUPPLIES STORAGE SHELF/CABINET FOR SIGNA TWIN*SPEED* WITH EXCITE TECHNOLOGY
ILLUSTRATION 2–6

2–9 ARCHITECTURAL REMINDERS (Continued)

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- PERIPHERAL VASCULAR ARRAY IS NOT INCLUDED IN THE BELOW STORAGE SHELF/CABINET. CONTACT GE MEDICAL SYSTEMS ACCESSORIES FOR PV ARRAY STORAGE RACK OPTION AND DIMENSION INFORMATION FOR THE STORAGE RACK.



SUGGESTION FOR ACCESSORIES & SUPPLIES STORAGE SHELF/CABINET FOR SIGNA TWIN*SPEED*
ILLUSTRATION 2–7

2–10 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for the MR system. Listed in Table 2–8 are the weights, floor loading, and normal mounting methods for MR components.

TABLE 2–8
FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
LCC Magnet, Quiet Technology Enclosure, and Coils	See Notes 1 & 3 & Refer to Section 2–10–1	83.27 x 117.67 x 101.96 (2115 x 2989 x 2587.21)	See Illustration 2–11 See Illustration 7–1.	Magnet & Table Dock Asm. Resting on base, bolted to floor with RF Screen Room vendor supplied/ installed bolts & anchors.
Patient Table (See Note 4)	630 (286)	27 x 89 x 38 (686 x 2261 x 965)	Rectangular base 23 x 65 (584 x 1651). Four casters 8 (203) dia. x 1.5 (38) wide.	Mobile.
Blower Box	20 (9)	14.5 x 14 x 13.7 (368 x 356 x 348)	Square base 14.5 x 14 (368 x 356)	Anchor to floor, shelf, or wall.
1.5T SRFD II Cabinet	495 (225)	23.46 x 45.5 x 71.06 (596 x 1156 x 1805)	Rectangular base 22 x 33.5 (559 x 851). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 Twin*Speed* with EXCITE Technology System Cabinet	468 (213)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 Twin*Speed* System Cabinet	494 (225)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
ACGD/PDU Cabinet	2136 (969)	23.5 x 37.25 x 85.6 (597 x 927 x 2174) See Note 5	Rectangular base 33.5 x 22 (851 x 559). Four casters each 3.0 (76.2) dia. x 1.82 (46.2) wide.	Set on floor on casters.
Twin Accessory Cabinet	600 (273)	23.32 x 38.75 x 50.13 (592 x 984 x 1273)	Rectangular base 22 x 30 (584 x 762). Four casters 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor and rest on casters.
Shield/Cryo Cooler Compressor Cabinet	275 (125)	17.72 x 19.7 x 26.93 (450 x 500 x 684)	Rectangular base 17.72 x 19.7 (450 x 500). Four casters 3 (76) dia. x .75 (19) wide.	Set on floor and rest on casters.
Twin System Cooling Cabinet (TSCC) See Note 6				
Outdoor Air Cooled	1200 (544)	74.83 x 45.5 x 66.88 (1901 x 1156 x 1699)	See Illustration 2–14.	Bolted to mounting pad external to building.
Indoor Water Cooled	1000 (454)	55.4 x 33.59 x 73 (1407 x 358 x 1854)	See Illustration 2–17.	Set on floor and rest on casters.
Indoor Air Cooled	1410 (640)	65 x 32.5 x 71.26 (1651 x 825.5 x 1810)	See Illustration 2–19.	Set on floor and rest on casters.
Twin Gradient Water Cooler (TGWC) See Note 6				
Outdoor Air Cooled	780 (354)	49.5 x 34 x 76.2 (1257 x 864 x 1936)	See Illustration 2–21.	Bolted to mounting pad external to building.
Indoor Water Cooled	175 (79)	33.94 x 19.66 x 23.67 (86.21 x 49.94 x 60.13)	See Illustration 2–24.	Set on floor and rest on 4 casters.
Indoor Air Cooled	1290 (585)	28.4 x 33 x 73 (721 x 838 x 1854)	See Illustration 2–26.	Set on floor and rest on casters & front pads on anti–tip legs

(Continued)

2–10 FLOOR LOADING AND WEIGHTS (Continued)

TABLE 2–8 (Continued)
FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
Operator Workspace Table with LCD Color Monitor (See Note 7)	175 (80)	54 x 43 x 52 (1372 x 1092 x 1321)	See Illustration 2–37.	Set on floor: Table on leveling pads & Cabinet rest on casters. Anchor Table to floor per Section 2–10–3.
Operator Workspace Cabinet (See Note 7)	192 (87)	18.5 x 29 x 26 (470 x 737 x 660)	See Illustration 2–38.	Set on floor and rest on casters.
UPS for Magnet Monitor*	34 (15)	19.4 x 3.5 x 17 (494 x 89 x 432)	See Illustration 2–45	Set on floor on 2 mounting brackets.
Advantage Workstation*	Refer to Direction *2261309–100 Advantage Workstation 3.1 Ultra60 Pre–Installation.* Also refer to **Section 2–8–6** ***Advantage Workstation Option Additional Hardware Features*** to determine location of workstation.			
BrainWave Cabinet*	387 (176)	23.5 x 23 x 69 (597 x 584 x 1753)	Rectangular base 23 x 22 (584 x 559). Four leveling pads each 1.5 (38) dia.	Casters for location. Set on floor on leveling pads.
BrainWave Cart*	71 (32)	18 x 36 x 44.5 (457 x 914 x 1130)	Rectangular base 16.5 x 22.5 (419 x 572). Four casters each 2.875 (73) dia. x 2.0 (50.8) wide.	Set on floor on casters.
MNS Cabinet*	430 (195)	23.56 x 42 x 70.7 (598 x 1067 x 1797)	Rectangular base 33.5 x 22 (851 x 559). Four leveling pads each 1.5 (38) dia.	Casters for location. Set on floor on four leveling pads.
500 Liter Helium Dewar**	812 (369)	42 dia. x 67.5 high (1067 x 1715)	Mobile.	Four casters.
250 Liter Helium Dewar**	519 (236)	36 dia. x 64.0 high (914 x 1626)	Mobile.	Four casters.

Note 1 Weight of 1.5T LCC (Active Shield) magnet with Quiet Technology Enclosure, RF/Gradient (TRM) Body Coil, and cryogens is 12,826 lbs (5818 kg).

2 Consult a structural engineer on method of calculating proper weight/unit area for floor loading.

3 Refer to **Section 2–4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY** for Gradient Coil Assembly replacement weight and dimension requirements.

4 630 lbs (286 kg) includes 350 lbs (159 kg) patient.

5 ACGD/PDU Cabinet height with Fan Module removed is 74.8 inches (1900 mm).

6 System Cooling Cabinet is available in the following configuration options (must choose one):
TSCC (Provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil).
- Outdoor Air Cooled TSCC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room
- Indoor Water Cooled TSCC located in or near the Signa system Equipment Room
- Indoor Air Cooled TSCC located in or near the Signa system Equipment Room.
- True Mobile system equipment, contact Van Manufacturer for details.

TGWC (Provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor, refer to Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.
- Outdoor Air Cooled TGWC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room.
- Indoor Water Cooled TGWC and Remote Control Panel (RCP) wall mounted both located in the Signa system Equipment Room
- Indoor Air Cooled TGWC located in or near the Signa system Equipment Room.

7 The Operator Workspace Cabinet minimum area location is under the Workspace Table, see in Illustration 2–37. An alternate location is possible with the Operator Workspace Cabinet located against the right side of the Workspace Table.

* Optional Equipment.

** Dewar specifications may vary. Check with cryogen supplier for exact weight and dimensions.

2–10–1 Magnet Loading Considerations

In addition to the weight of the riggers equipment, special consideration must be given to the weight of the magnet along the delivery route. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for the shipping weight of the magnet (i.e. magnet shipped with liquid cryogens and RF/Gradient coil inside magnet bore and with partial enclosure installed). Structural reinforcement may be required along the magnet delivery route. It is required that a structural engineering analysis be performed on the Magnet Room floor and delivery route to determine its load bearing capacity.

During magnet installation, leveling plates or shims MUST be installed between magnet feet and the floor to compensate for variation within floor levelness. Refer to Section 7–6–4 FLOORS for levelness requirements. For LCC (Active Shield) Magnet, GE Medical Systems supplied aluminum shims are installed to complete contact between entire surface of each of the four magnet feet and the floor.

2–10–2 Anchoring And Seismic Considerations

It is the responsibility of the customer to obtain any and all approvals necessary for the construction of equipment support and seismic anchoring.

The center of gravity for MR system components are given for use in seismic calculations. If the MR cabinets are required by code to be anchored, refer to seismic drawings available on request from your local GEMS Installation Specialist. For the TSCC and TGWC units refer to Table 2–9 for mounting information.

TABLE 2–9
SYSTEM COOLING EQUIPMENT MOUNTING INFORMATION

SYSTEM COOLING EQUIPMENT	MOUNTING ILLUSTRATION	COMMENTS (See Note 1)
Outdoor TSCC unit	2–14 & 2–15	Unit has pre–drilled holes that should be used for mounting which should be used to aid in preventing the cabinet from shifting during a seismic event.
Indoor Water Cooled TSCC	2–17 & 2–18	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.
Indoor Air Cooled TSCC	2–19 & 2–20	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.
Outdoor TGWC	2–21 & 2–22	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event.
Indoor Water Cooled TGWC	2–24	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.
Indoor Air Cooled TGWC	2–26	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.

Note 1 Sites requiring seismic anchoring by code should have the site architect and engineer review the response spectra and/or Uniforn Builders Code (UBC) for their location and then contact Ellis & Watts International for assistance in seismic planning of the site.
Contact: Ellis & Watts International Inc.
4400 Glen Willow Lake Lane Telephone: 513–752–9000
Batavia, OH 45103 USA FAX: 513–752–4983

2–10–2 Anchoring And Seismic Considerations (Continued)

Magnet Mounting

It is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems. Refer to Table 2–10 for magnet mounting information and **Section 7–7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM**.

TABLE 2–10
MAGNET MOUNTING INFORMATION

MAGNET	MOUNTING ILLUSTRATION	COMMENTS
1.5T LCC (Active Shield)	2–11	The magnet must be bolted to the floor, refer to Section 4–15, Vibration. Bolt hole openings in the magnet base are to be used to anchor the magnet.

2–10–3 Operator Workspace Mounting Requirements

The Operator Workspace Table sets on the floor with the Workspace Cabinet positioned under the Table towards the right side, see Illustration 2–37. Note, the Workspace Table with the Color Host Monitor must be bolted to the floor for safe use of the table and equipment. The Color Host Monitor can be bolted to the Workspace Table to prevent the monitor being moved within the magnetic field.

2–11 COMPONENT DIMENSIONS

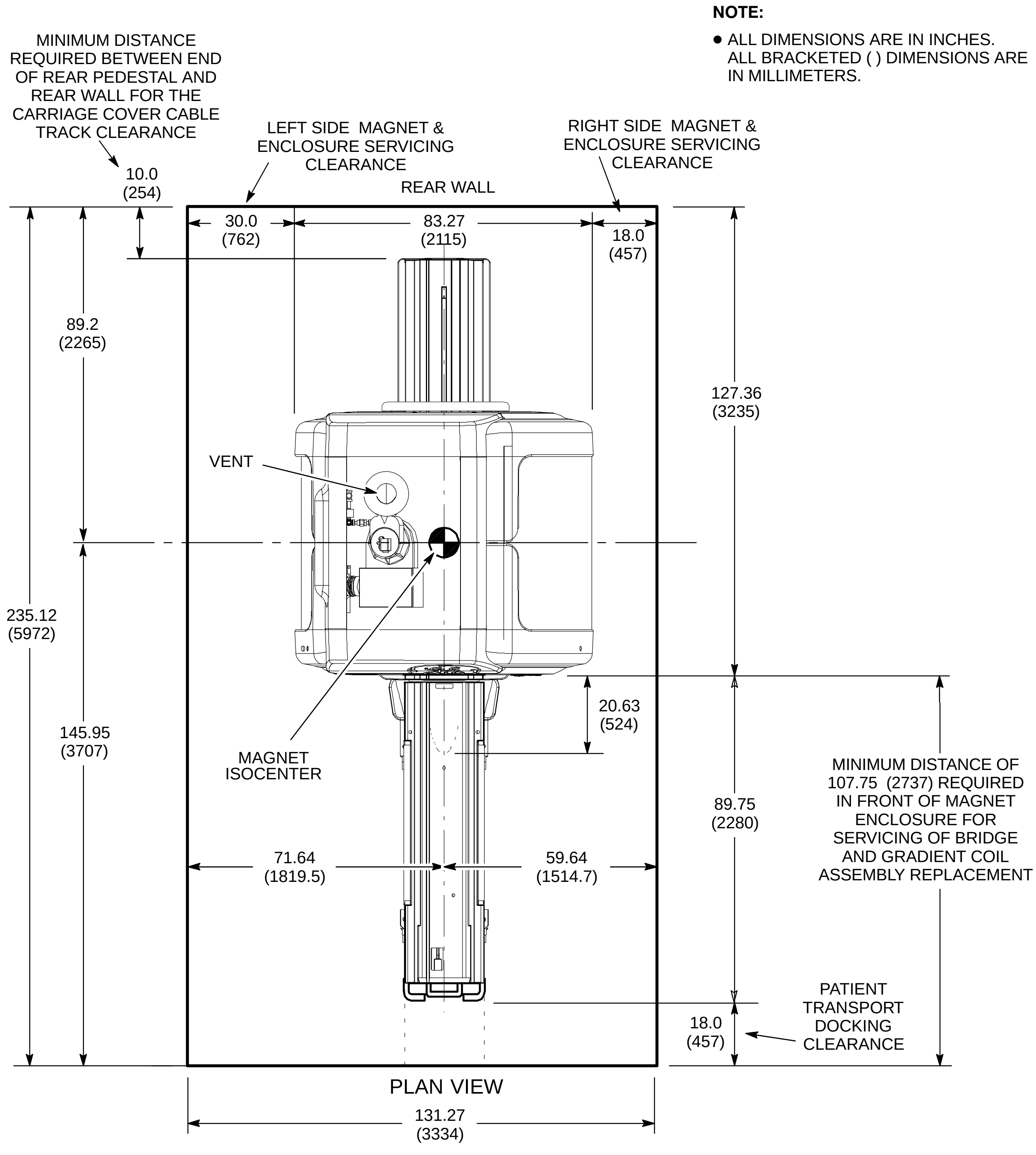
To assist in completing your room layout, refer to Table 2–11 for list of component Illustrations.

TABLE 2–11
MR SYSTEM COMPONENT ILLUSTRATIONS LIST

ILLUSTRATION NAME	ILLUSTRATION NUMBER
SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET (MINIMUM SERVICE AREA)	2–8
SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS)	2–9
SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS	2–10
1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN	2–11
SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)	2–12
SIGNA TWIN*SPEED* MAIN DISCONNECT PANEL (MDP)	2–13
OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2–14
OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING	2–15
REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O]	2–16
INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC–18MT/WC] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2–17
INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING	2–18
INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2–19
INDOOR AIR COOLED TSCC MOUNTING	2–20
OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10MTO] (TGWC) FOR GRADIENT COIL COOLING WATER ONLY	2–21
OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO] MOUNTING	2–22
REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO]	2–23
INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL 10] (TGWC) FOR GRADIENT COIL COOLING WATER ONLY	2–24
REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL 10]	2–25
INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC) FOR GRADIENT COIL COOLING WATER ONLY	2–26
BLOWER BOX (MG4)	2–27
1.5T SRFD II CABINET (MR1)	2–28
ACGD/PDU CABINET (MR3)	2–29
SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED* WITH EXCITE TECHNOLOGY	2–30
SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED*	2–31
TWIN ACCESSORY CABINET (TAC)	2–32
MAGNET MONITOR (MSM1)	2–33
PATIENT TRANSPORT TABLE (PT1)	2–34
PENETRATION PANEL (PP1)	2–35
PENETRATION PANEL COVER	2–36
OPERATOR WORKSPACE (OW1)	2–37
OPERATOR WORKSPACE CABINET (OW1 A2)	2–38
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR OPTION	2–39
(Continued)	

2–11 COMPONENT DIMENSIONS (Continued)

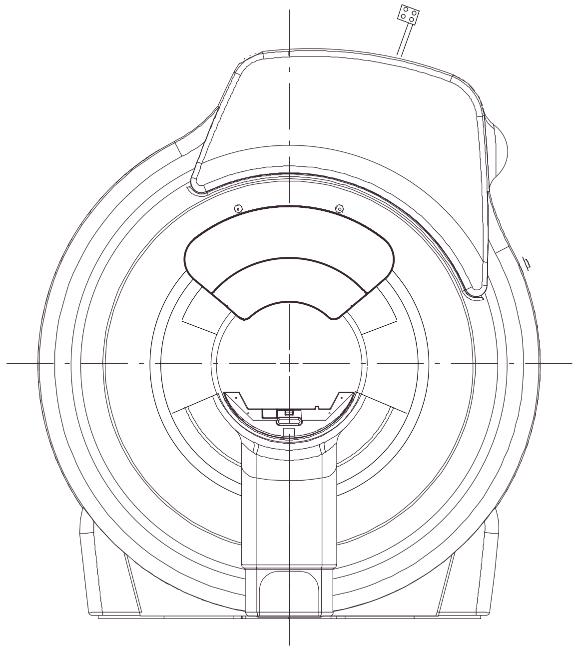
TABLE 2–11
MR SYSTEM COMPONENT ILLUSTRATIONS LIST (Continued)



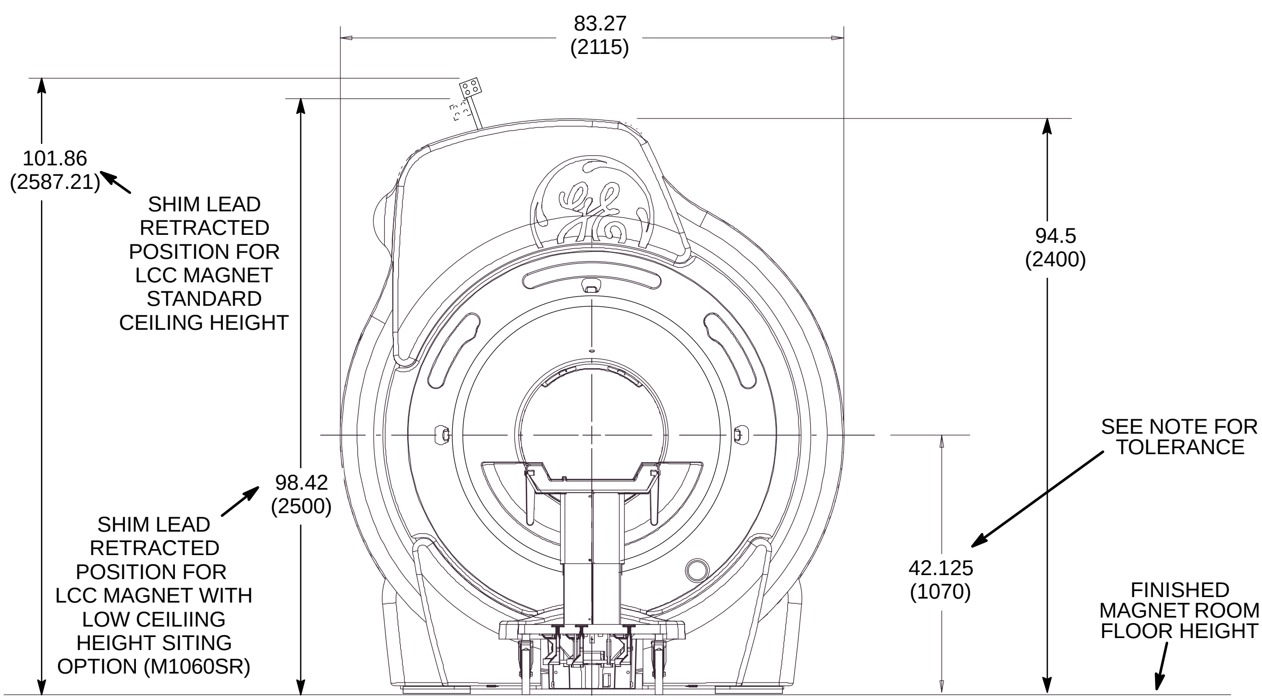
SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET (MINIMUM SERVICE AREA)
ILLUSTRATION 2–8

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- FINISHED FLOOR TO MAGNET CENTER LINE HEIGHT DIMENSION MUST BE 42.125±0.25 in. (1070±6.35 mm) TO ALLOW PATIENT TABLE TO PROPERLY DOCK TO THE ENCLOSURE.



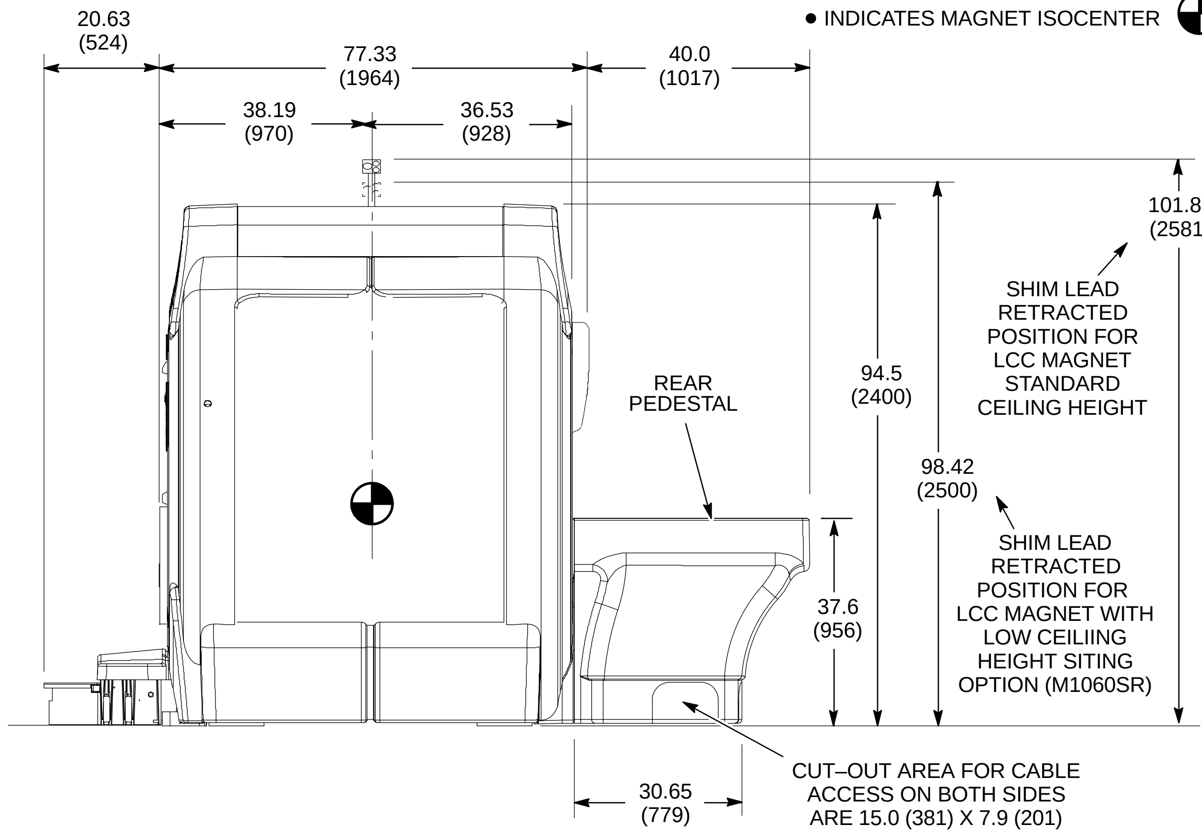
REAR VIEW

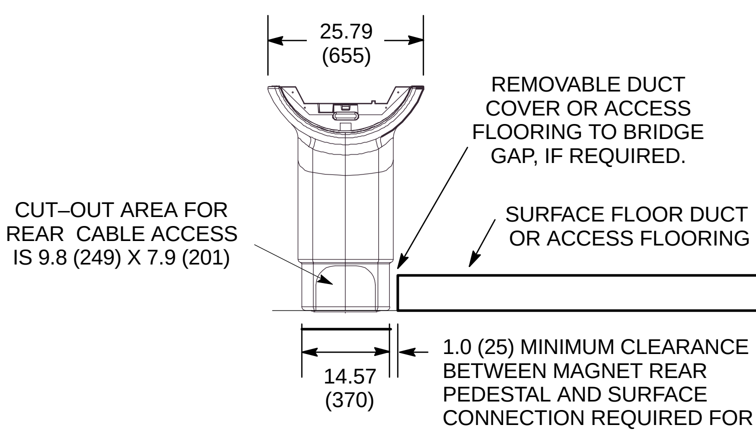


FRONT VIEW

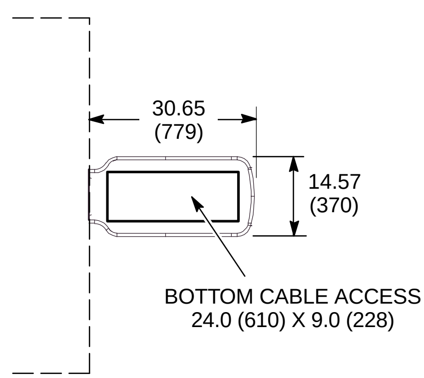
SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS)
ILLUSTRATION 2–9

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER





REAR PEDESTAL (REAR VIEW)

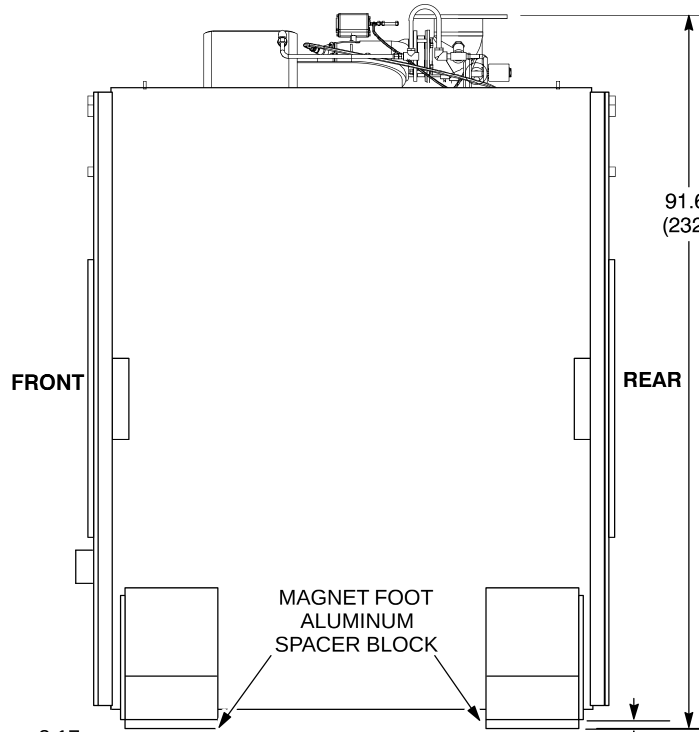


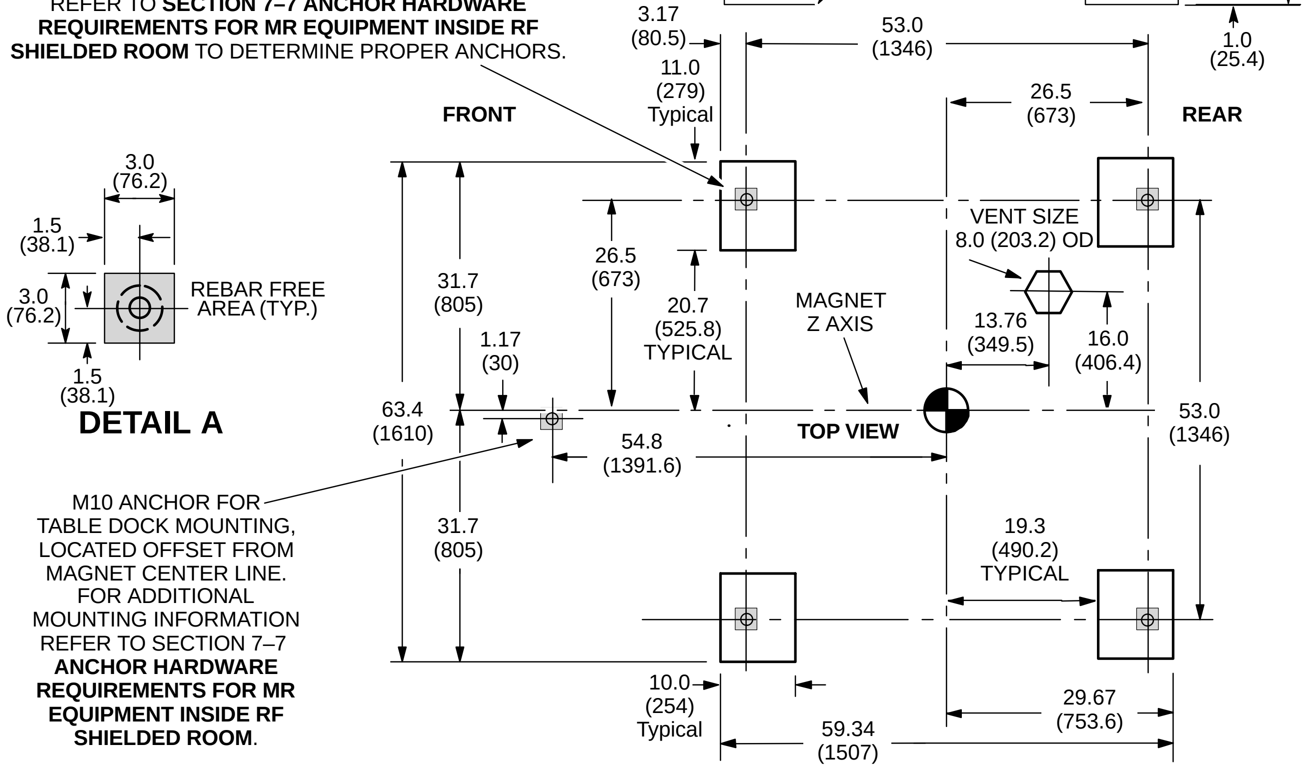
REAR PEDESTAL (FLOOR CONTACT VIEW)

SIGNA TWIN*SPEED* 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS
ILLUSTRATION 2–10

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8–1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH TRM 12,826 lbs (5818 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **SECTION 8 SHIPPING AND DELIVERY DATA**

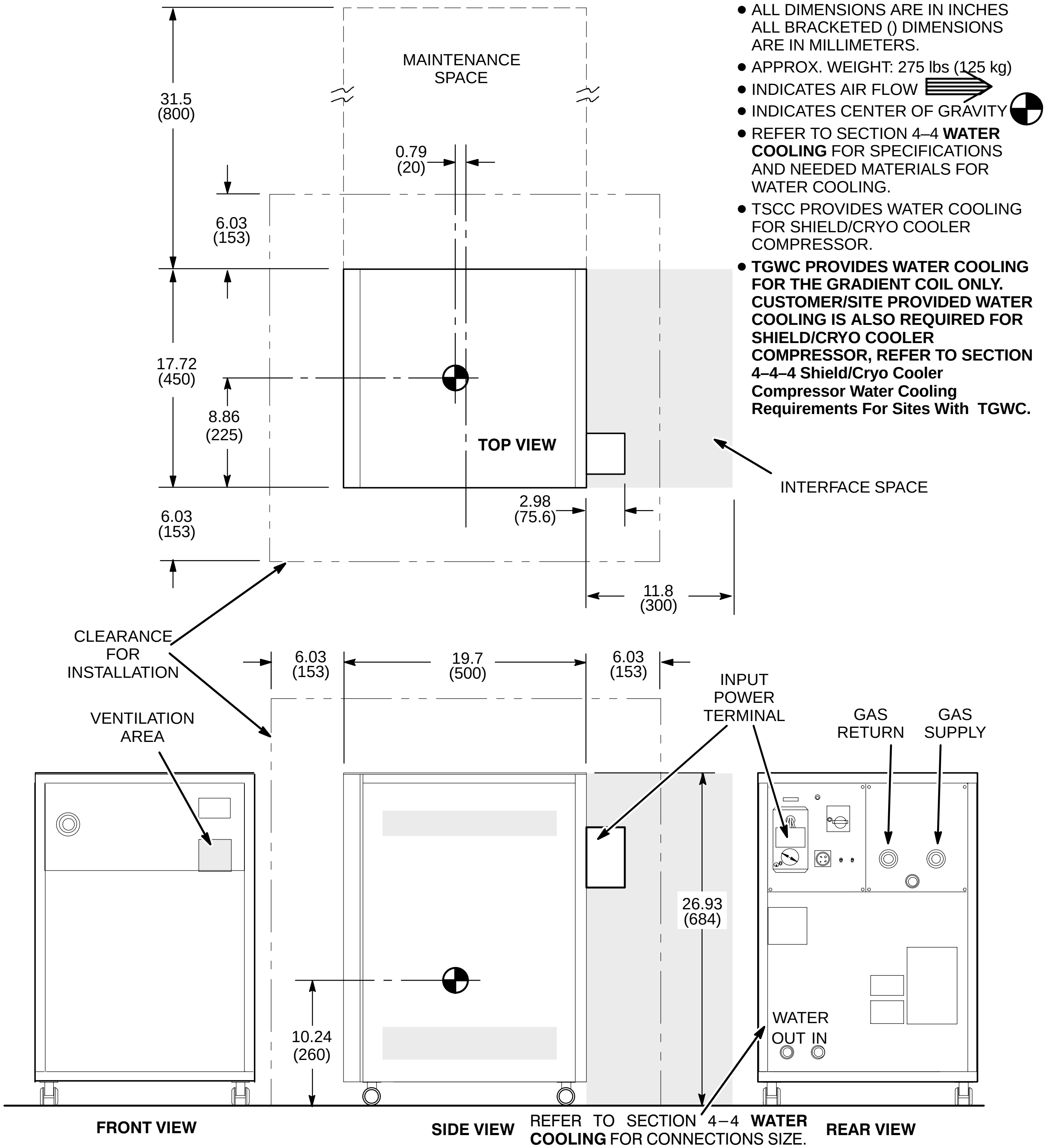




1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN
ILLUSTRATION 2–11

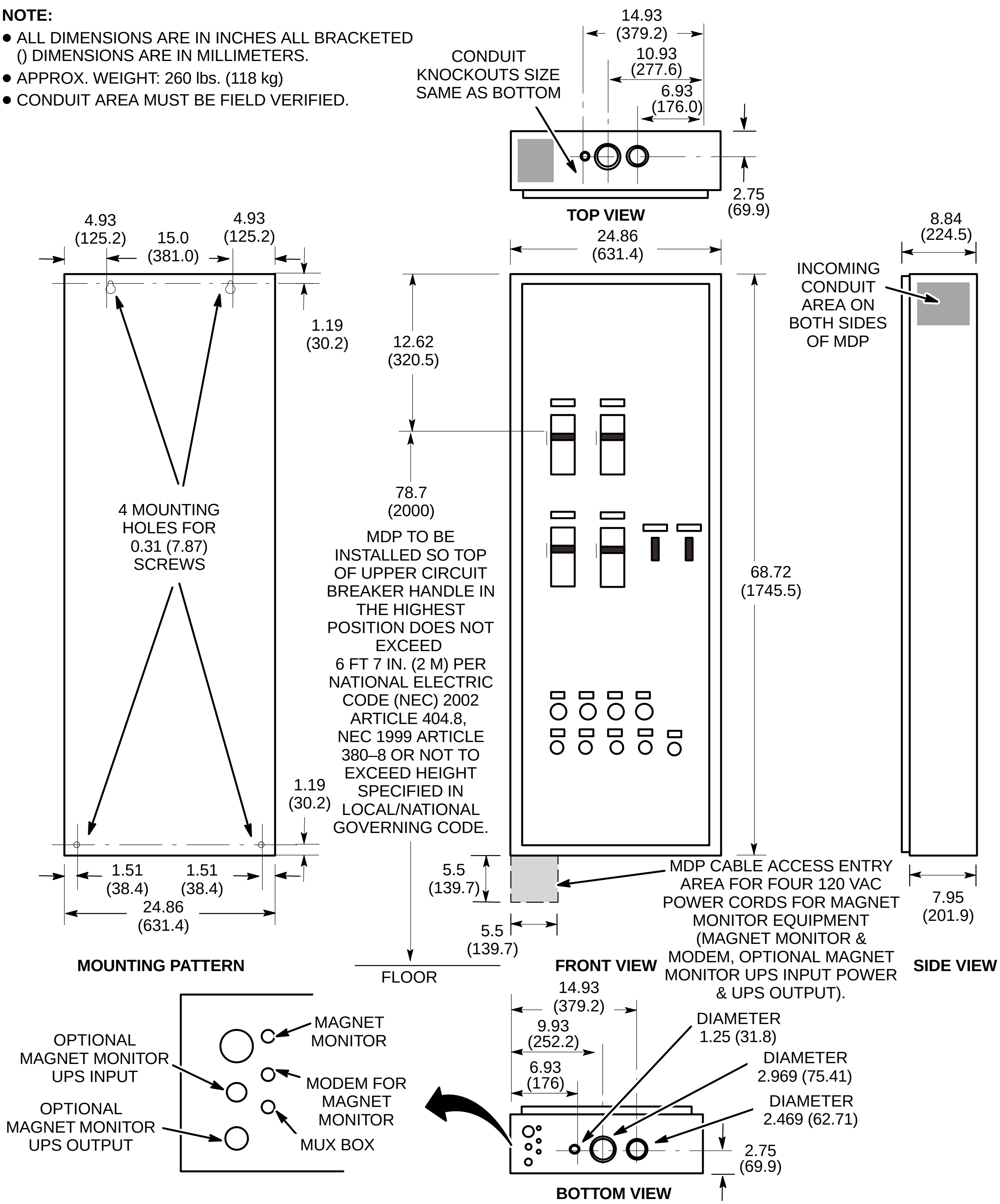
NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 275 lbs (125 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- REFER TO SECTION 4–4 **WATER COOLING** FOR SPECIFICATIONS AND NEEDED MATERIALS FOR WATER COOLING.
- TSCC PROVIDES WATER COOLING FOR SHIELD/CRYO COOLER COMPRESSOR.
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)
ILLUSTRATION 2–12

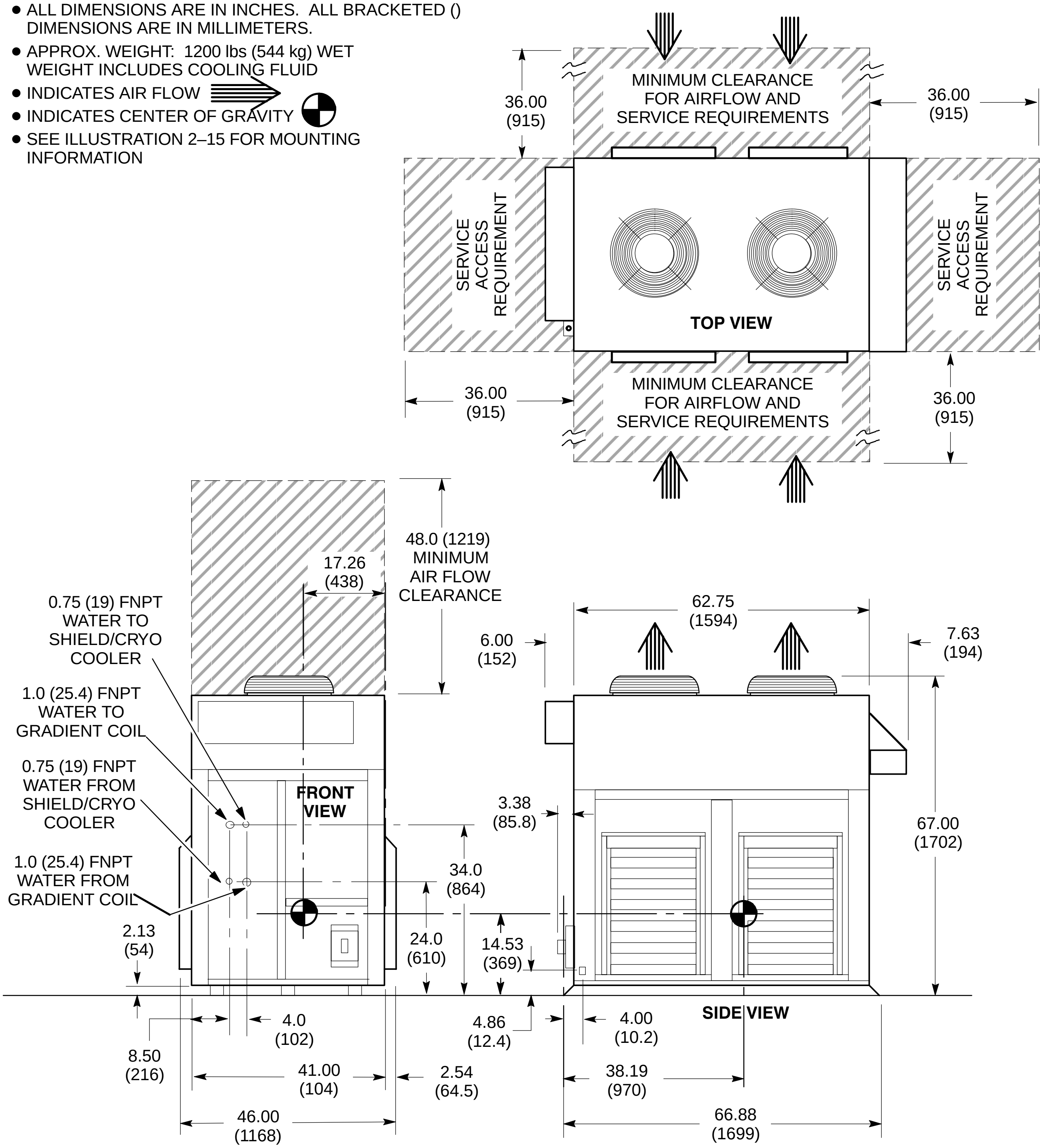
NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 260 lbs. (118 kg)
- CONDUIT AREA MUST BE FIELD VERIFIED.



SIGNA TWIN*SPEED* MAIN DISCONNECT PANEL (MDP) M3088TH
ILLUSTRATION 2–13

NOTE:

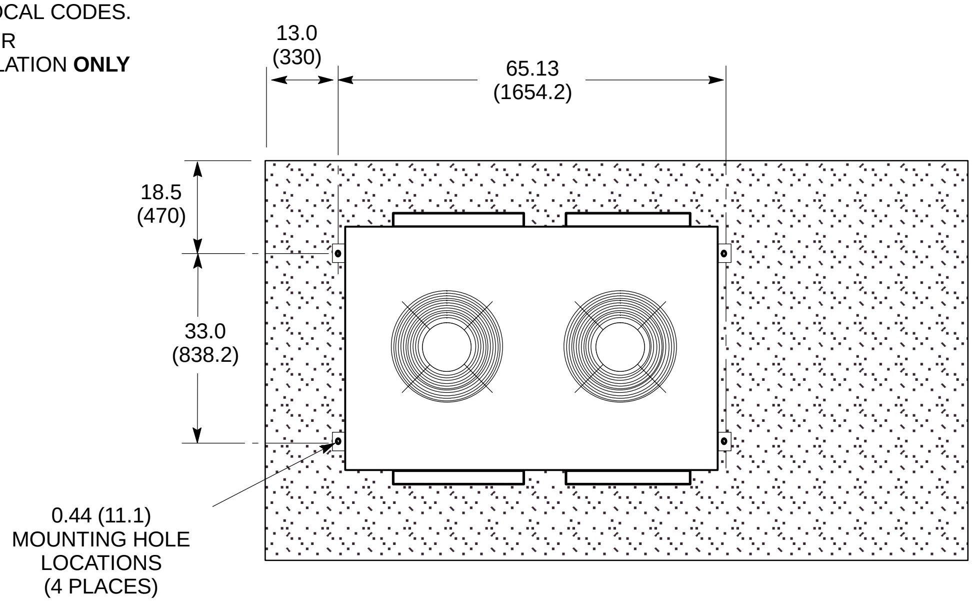
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1200 lbs (544 kg) WET WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- SEE ILLUSTRATION 2–15 FOR MOUNTING INFORMATION

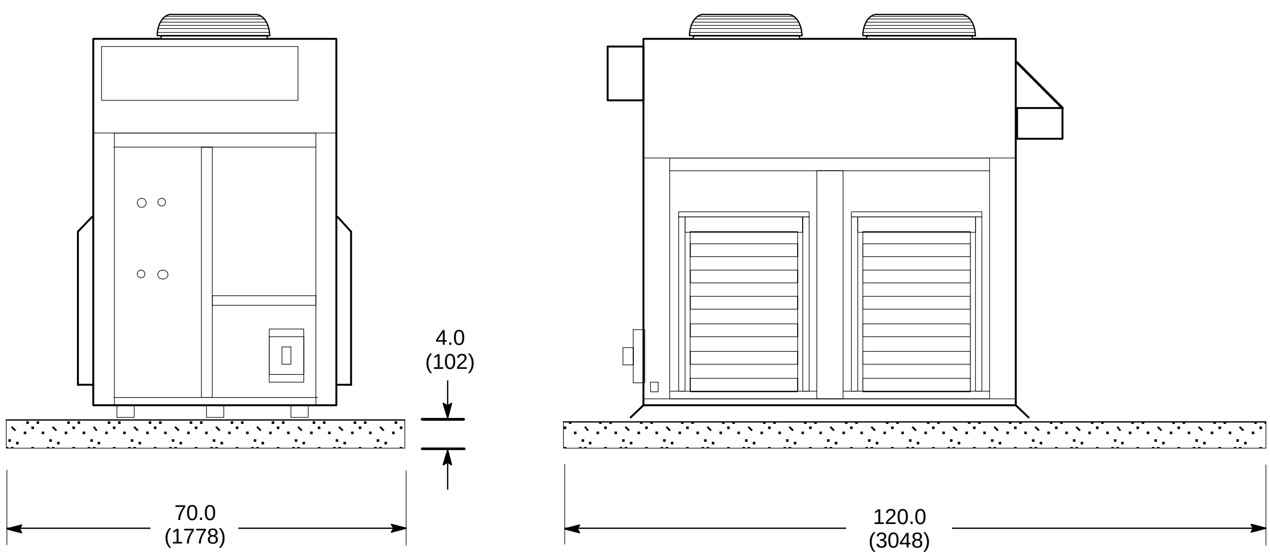


OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2–14

NOTE:

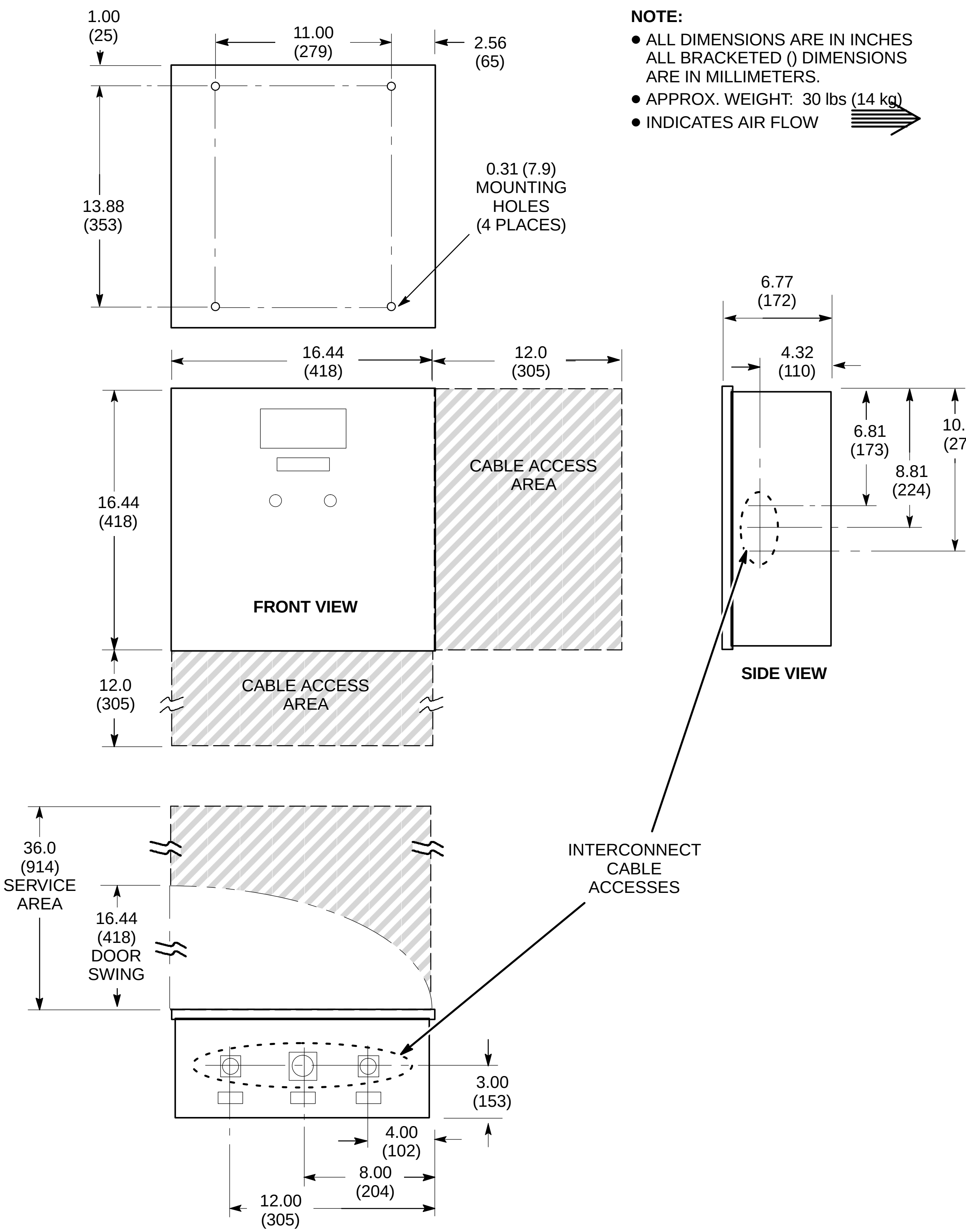
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON–GRADE (GROUND) INSTALLATION **ONLY**





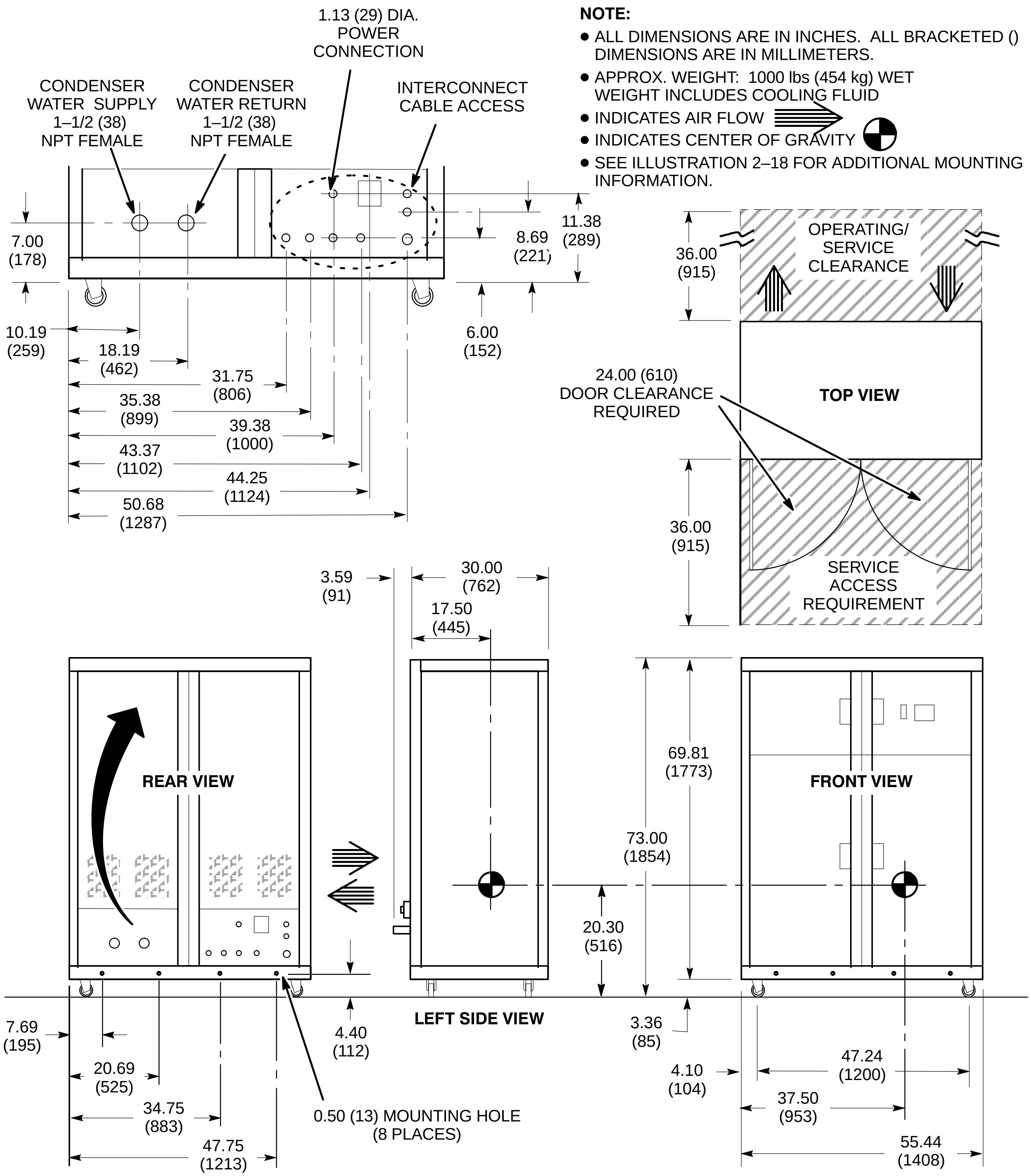
FRONT VIEW **LEFT SIDE VIEW**

OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING
ILLUSTRATION 2–15



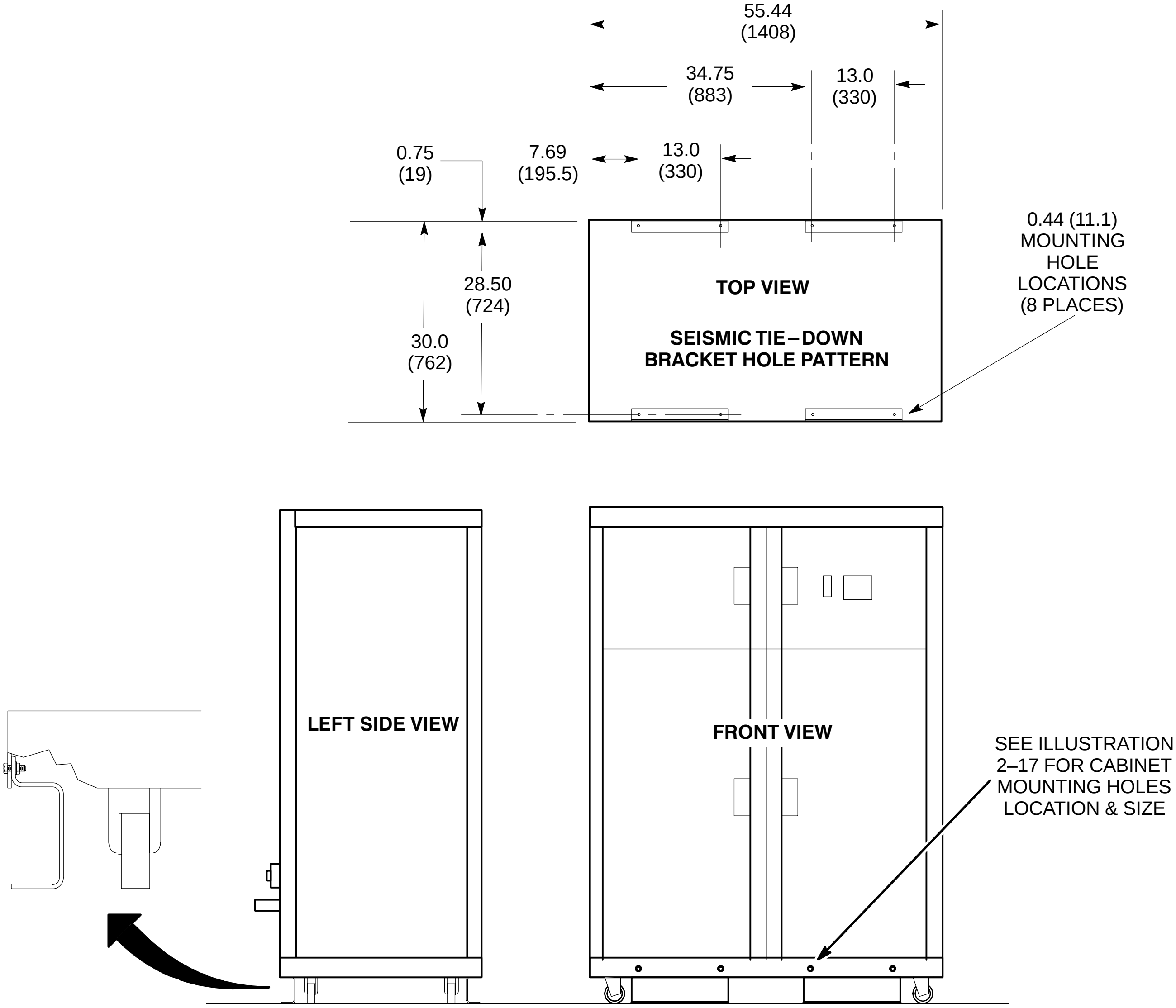
REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O]
ILLUSTRATION 2–16

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1000 lbs (454 kg) WET WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- SEE ILLUSTRATION 2–18 FOR ADDITIONAL MOUNTING INFORMATION.



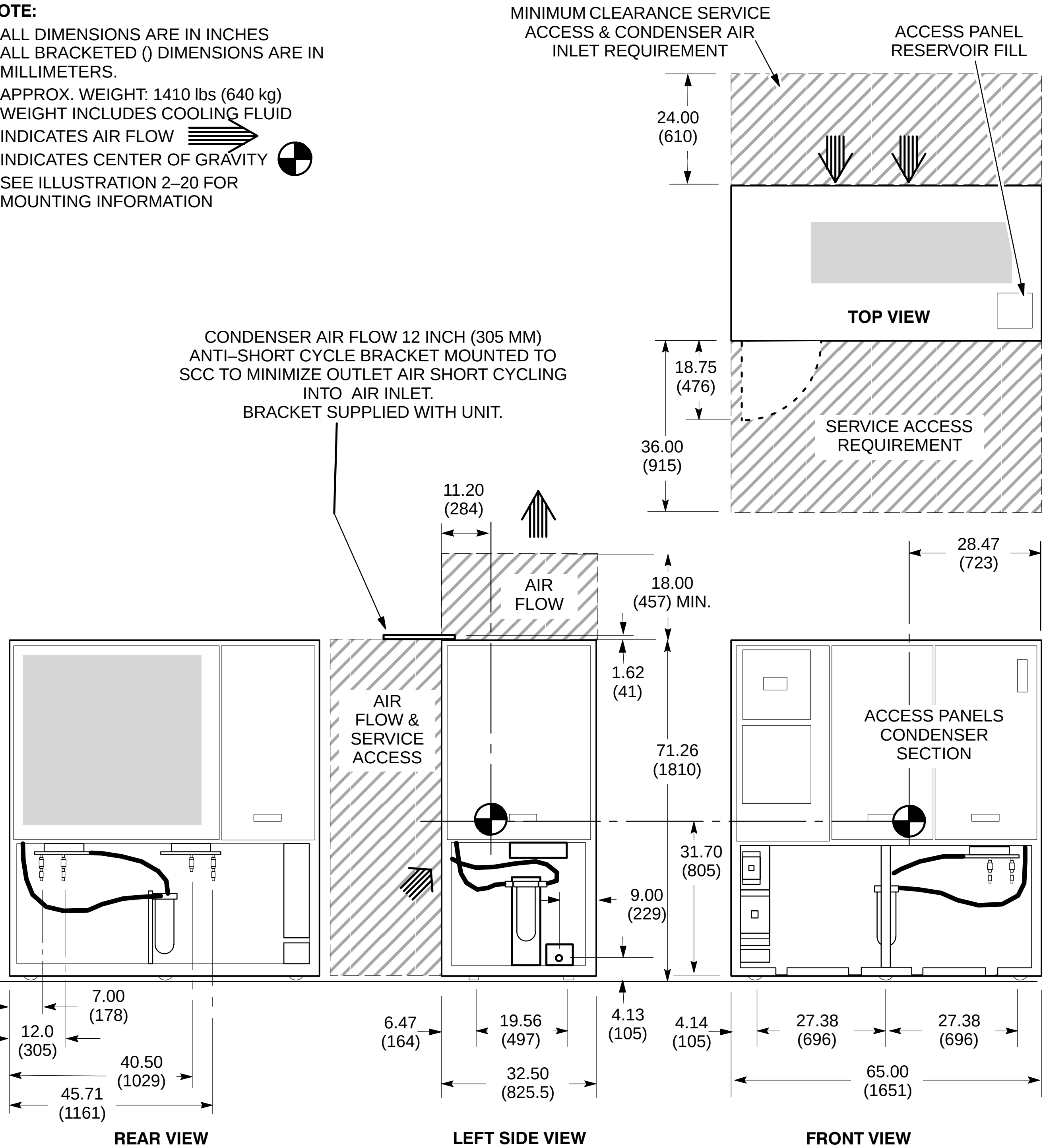
INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 18MT/WC] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2–17

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



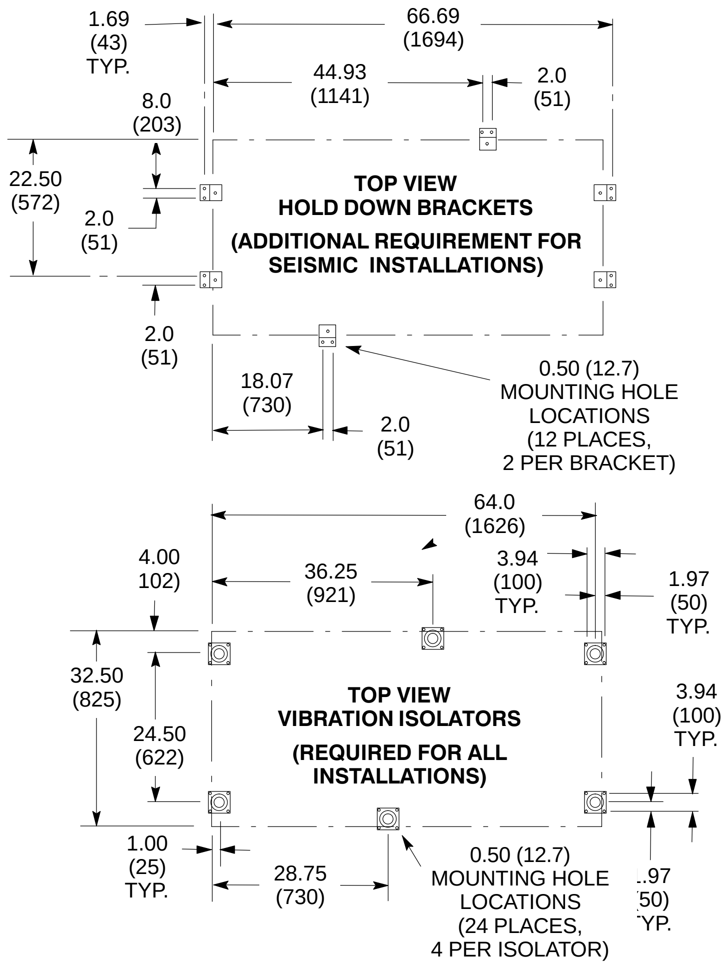
INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING
ILLUSTRATION 2–18

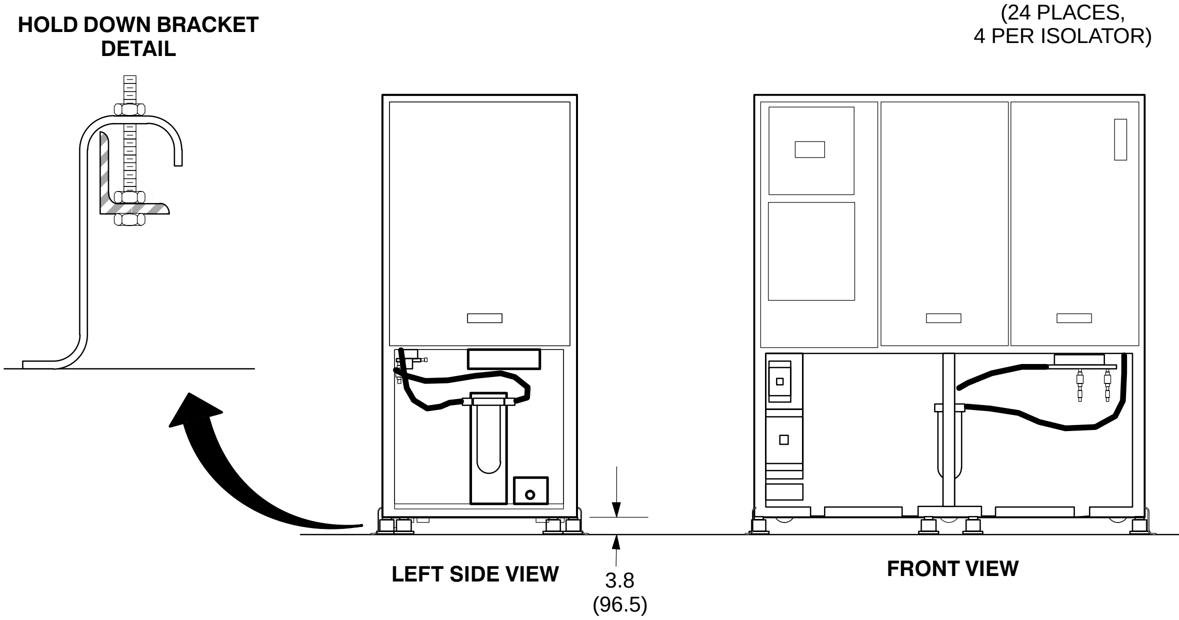
NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1410 lbs (640 kg)
  WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- SEE ILLUSTRATION 2–20 FOR MOUNTING INFORMATION



INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2–19

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

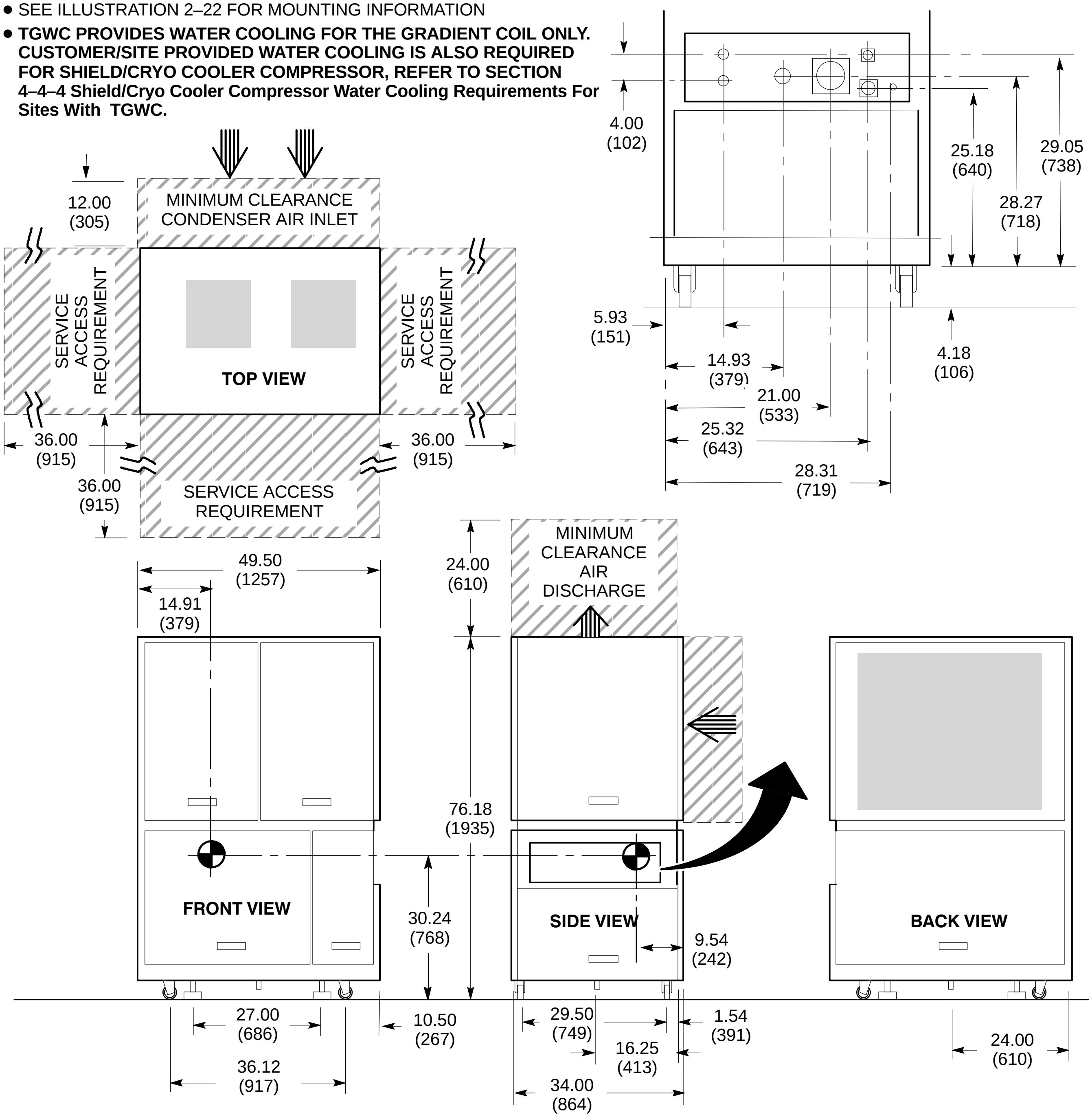




INDOOR AIR COOLED TSCC MOUNTING
ILLUSTRATION 2–20

NOTE:

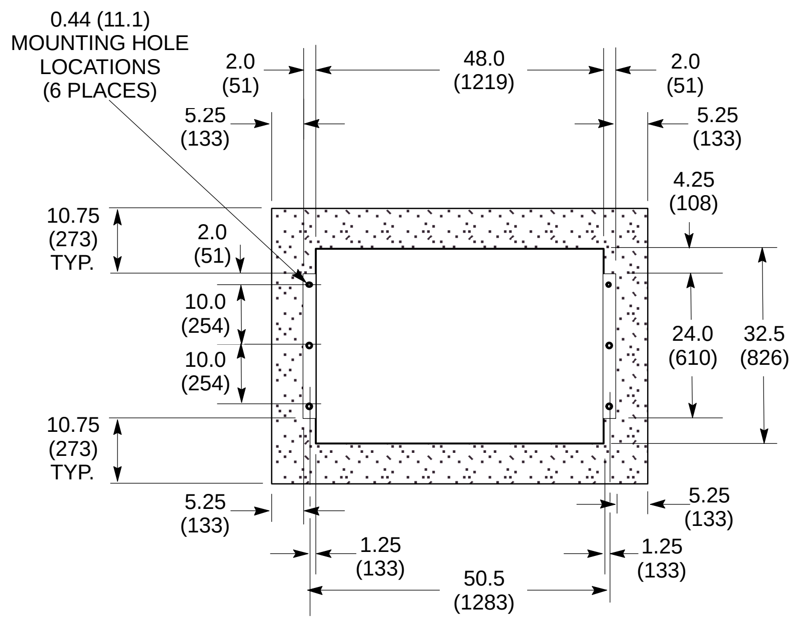
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 780 lbs (345 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- SEE ILLUSTRATION 2–22 FOR MOUNTING INFORMATION
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



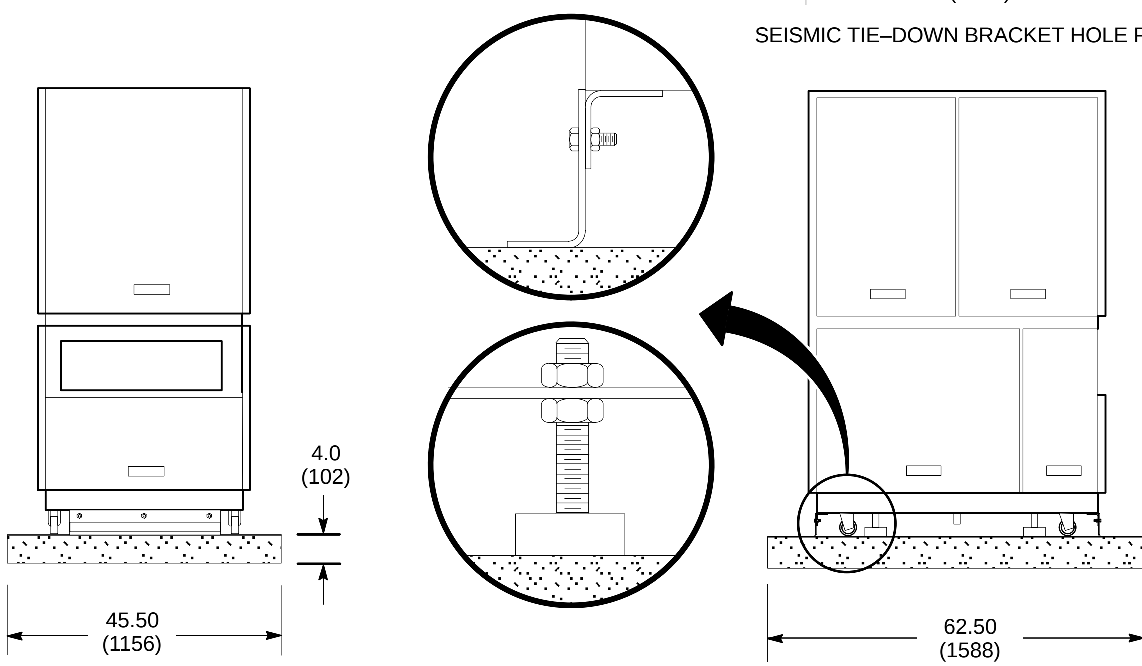
OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10MTO] (TGWC)
FOR GRADIENT COIL COOLING WATER ONLY
ILLUSTRATION 2–21

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON–GRADE (GROUND) INSTALLATION **ONLY**

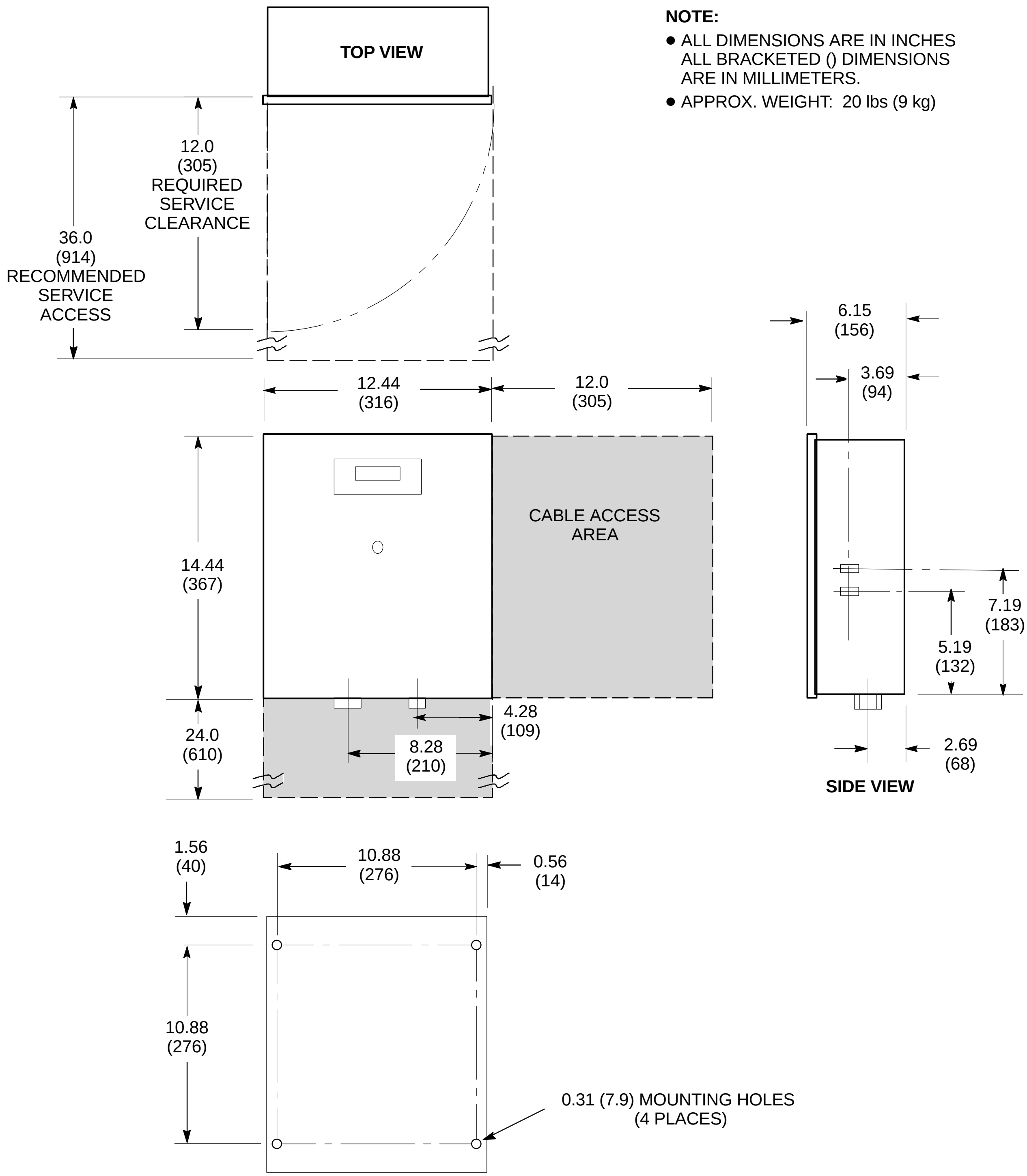


SEISMIC TIE–DOWN BRACKET HOLE PATTERN



RIGHT SIDE VIEW **FRONT VIEW**

OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO] MOUNTING
ILLUSTRATION 2–22

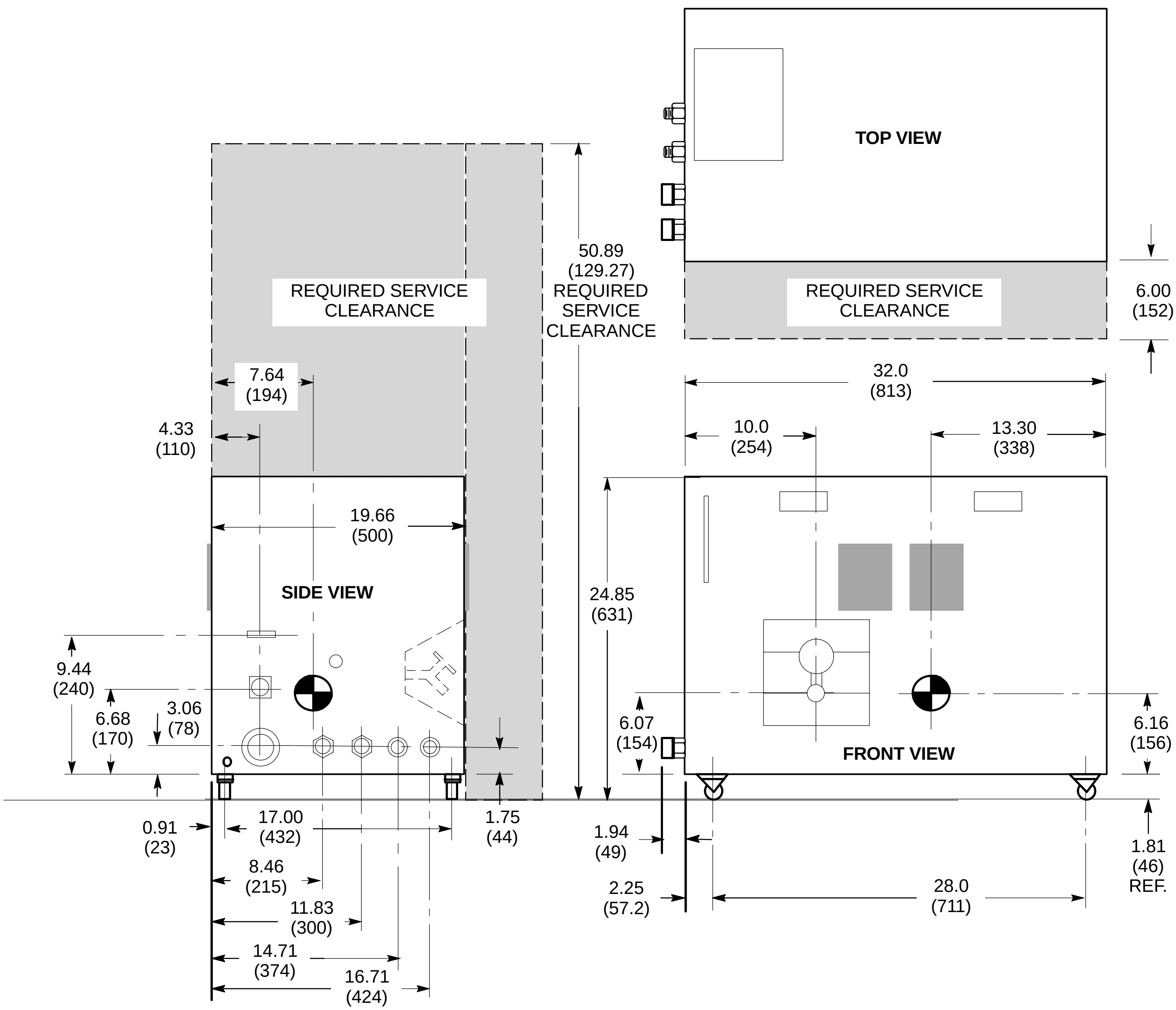


NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS
  ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)

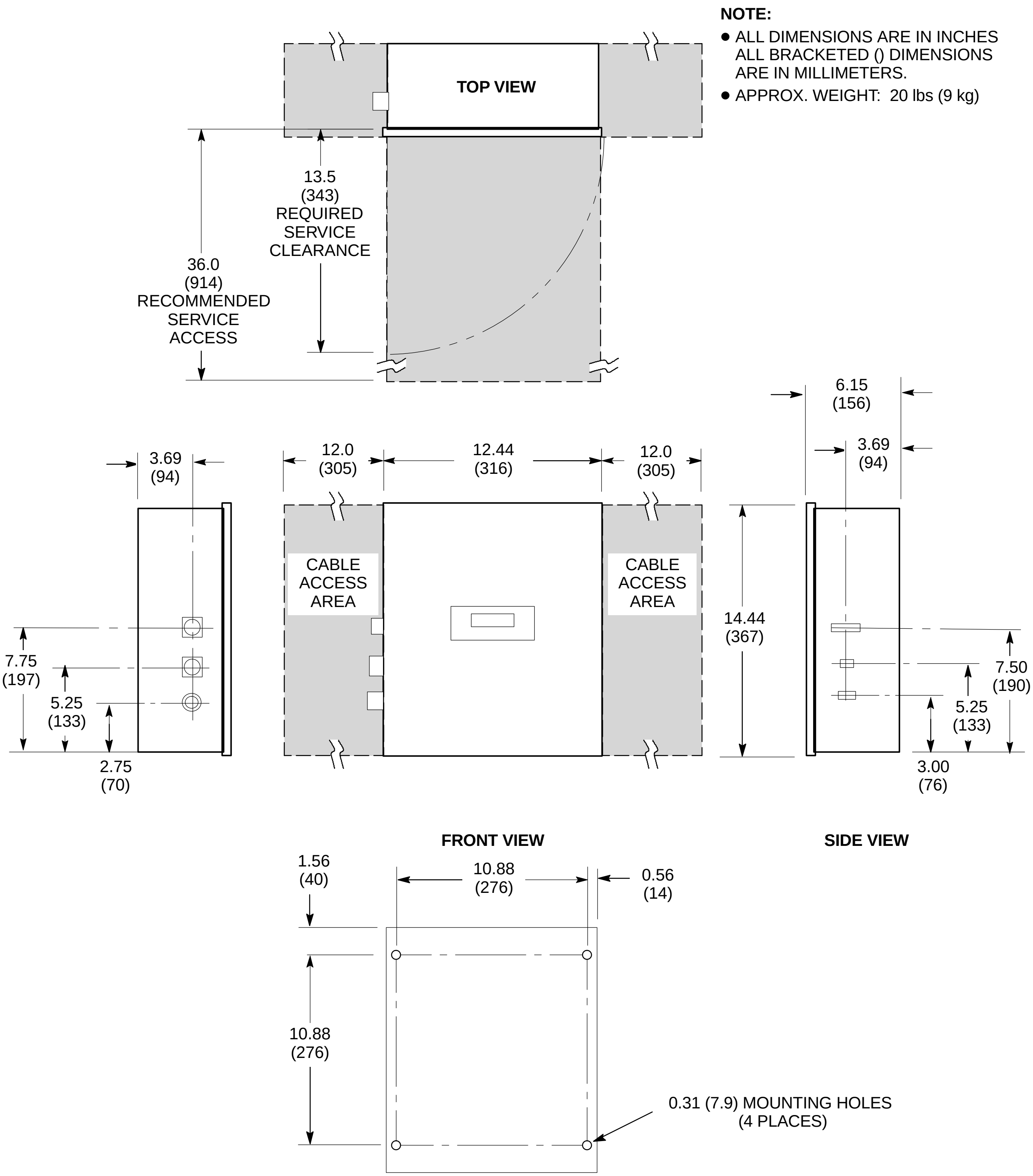
REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO]
ILLUSTRATION 2–23

NOTE:

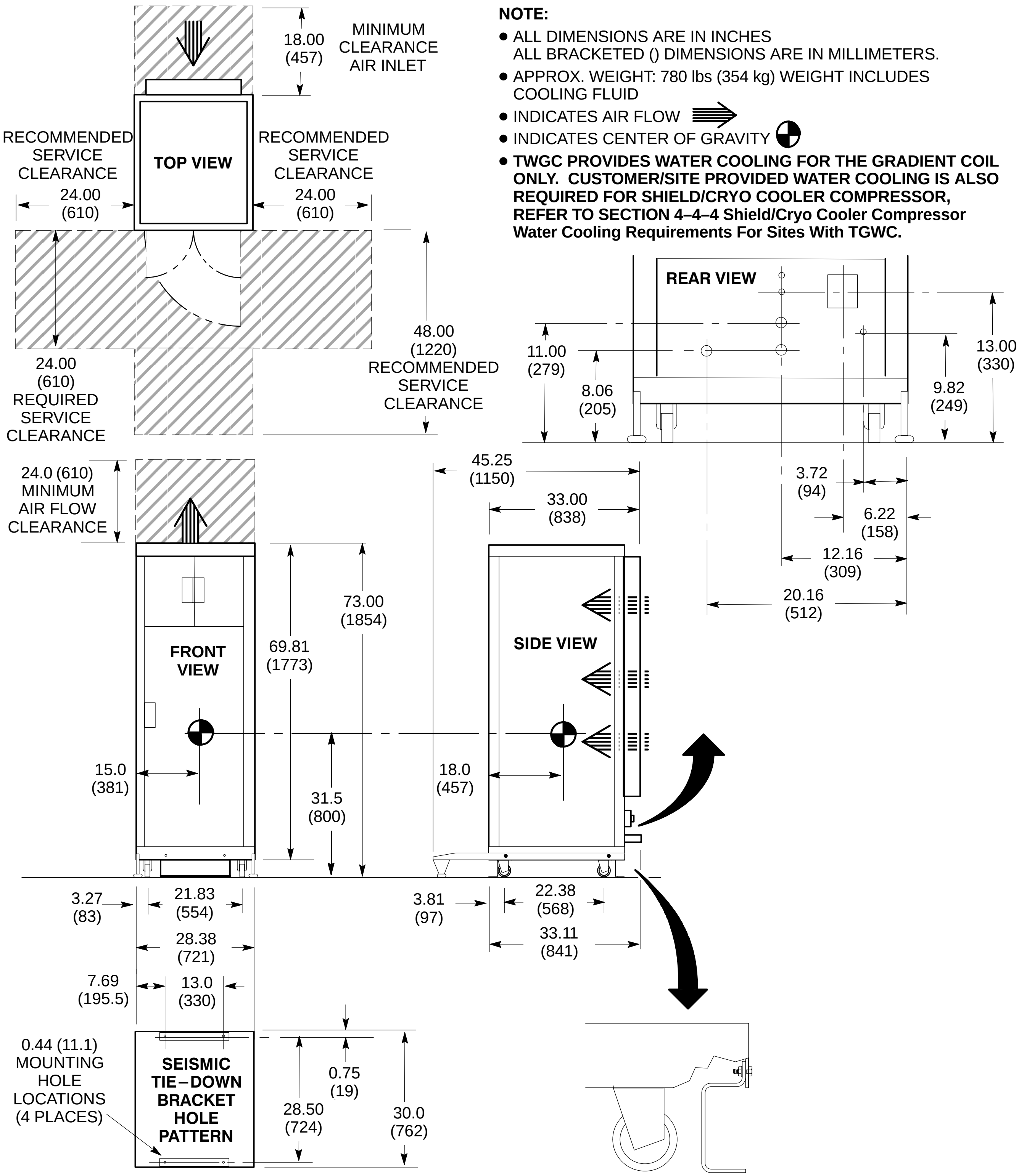
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: TGWC 175 lbs (79 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL 10] (TGWC)
FOR GRADIENT COIL COOLING WATER ONLY
ILLUSTRATION 2–24

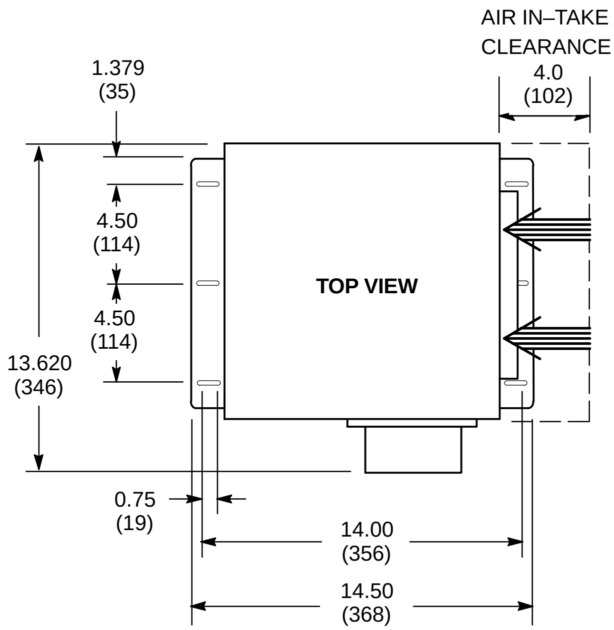


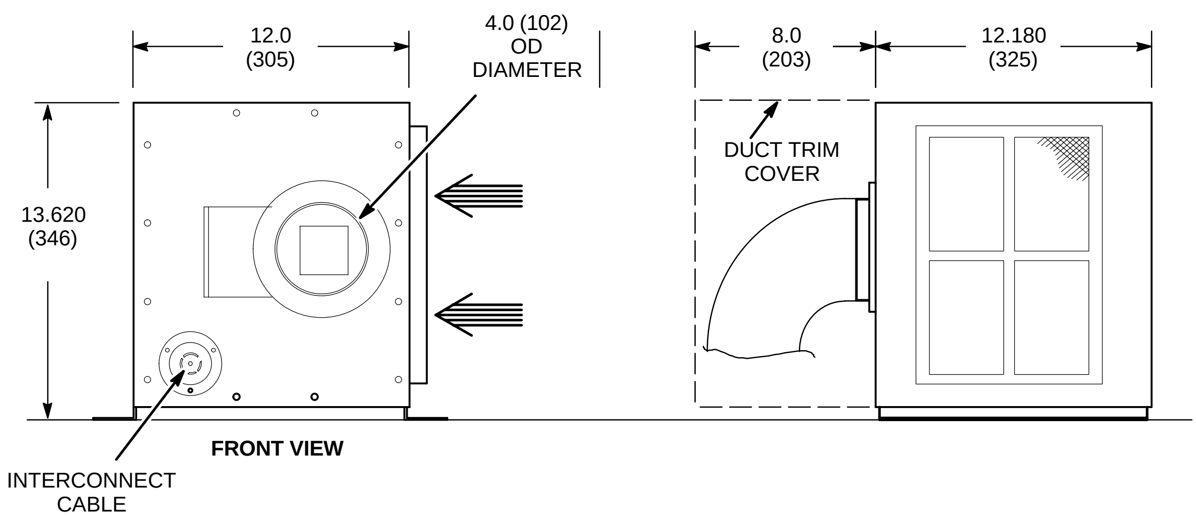
REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL 10]
ILLUSTRATION 2–25



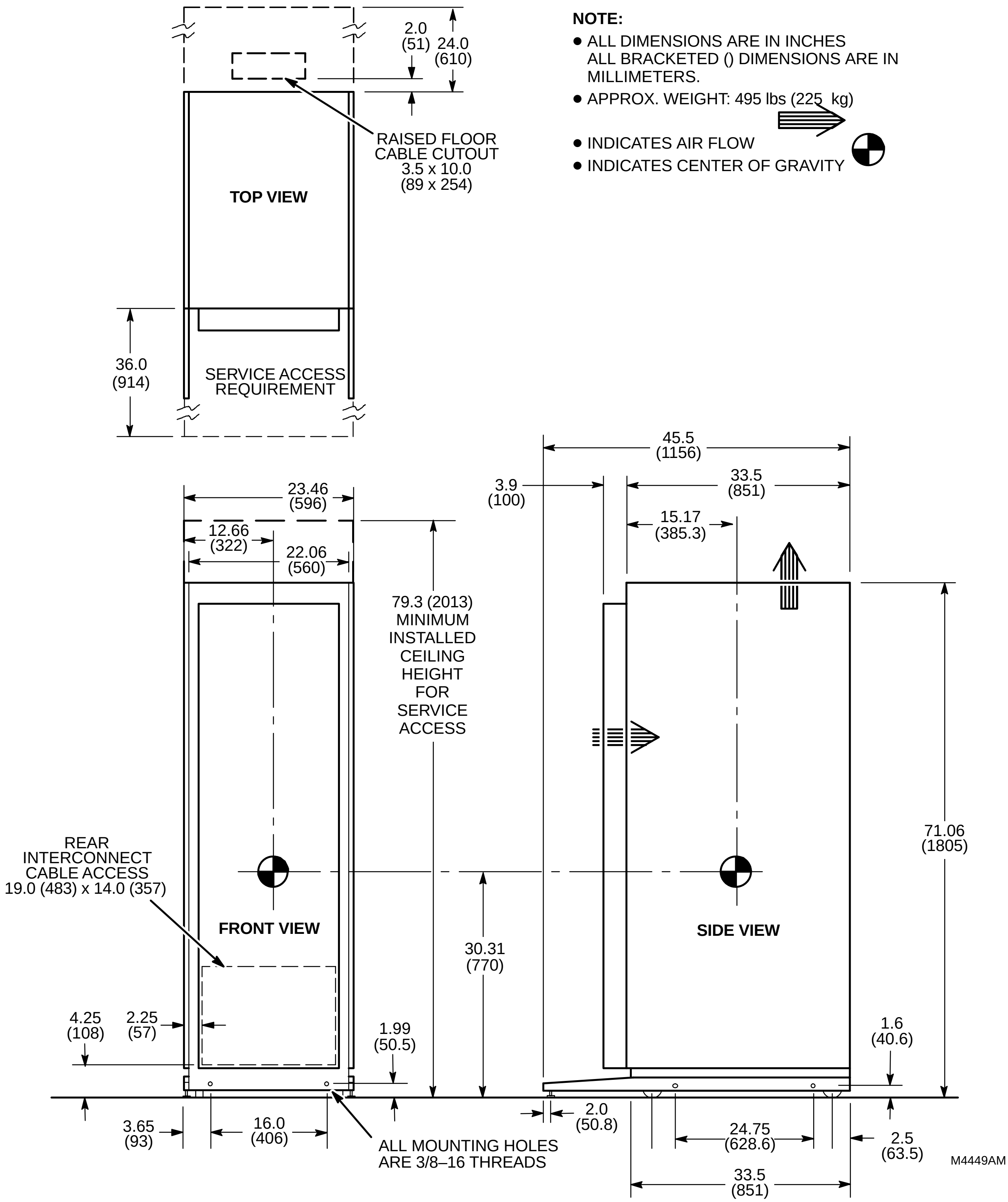
INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC)
FOR GRADIENT COIL COOLING WATER ONLY
ILLUSTRATION 2–26

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)
- BLOWER BOX MUST BE ANCHORED DUE FERROUS COMPONENTS.
- INDICATES AIR FLOW





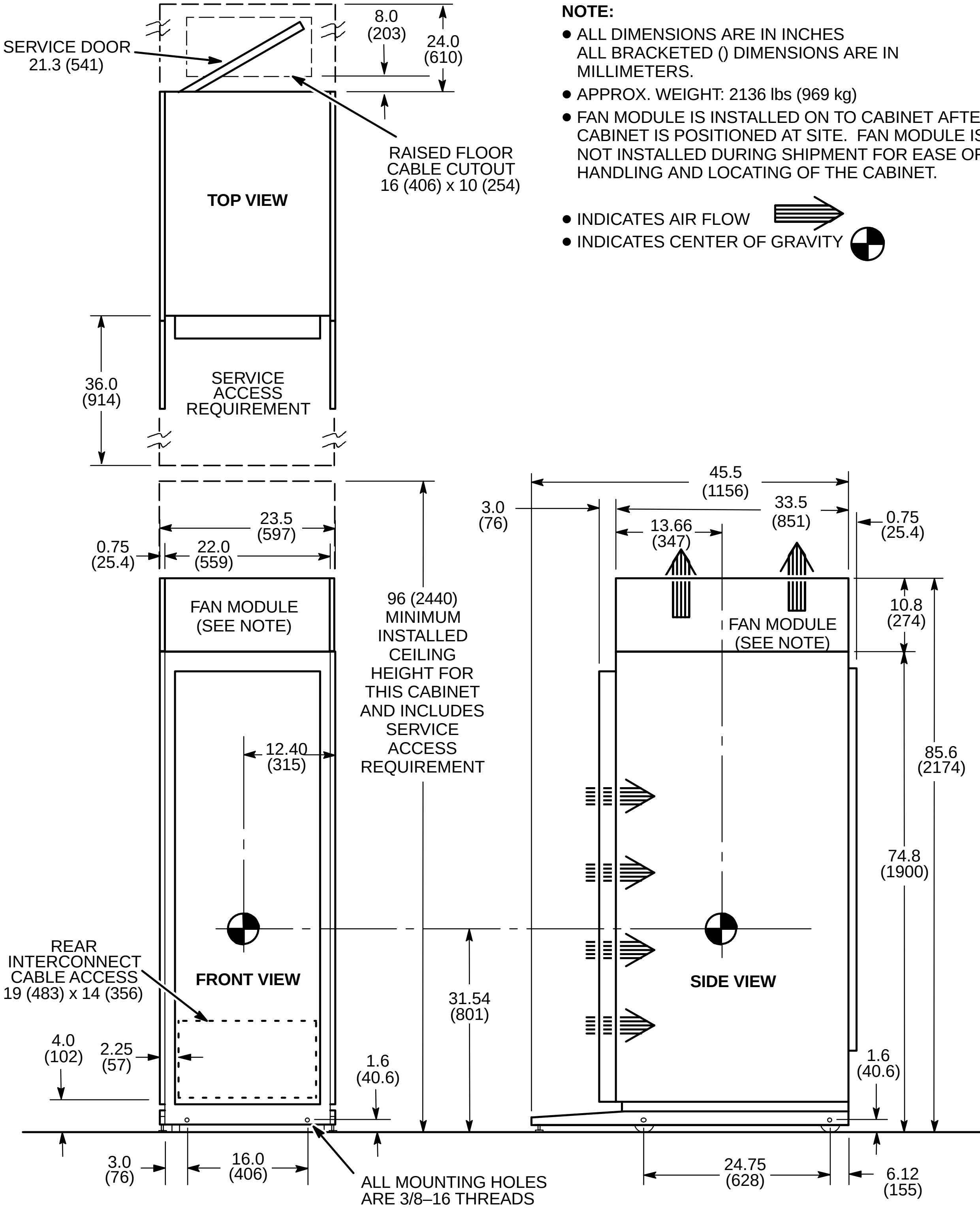
BLOWER BOX (MG6)
ILLUSTRATION 2–27



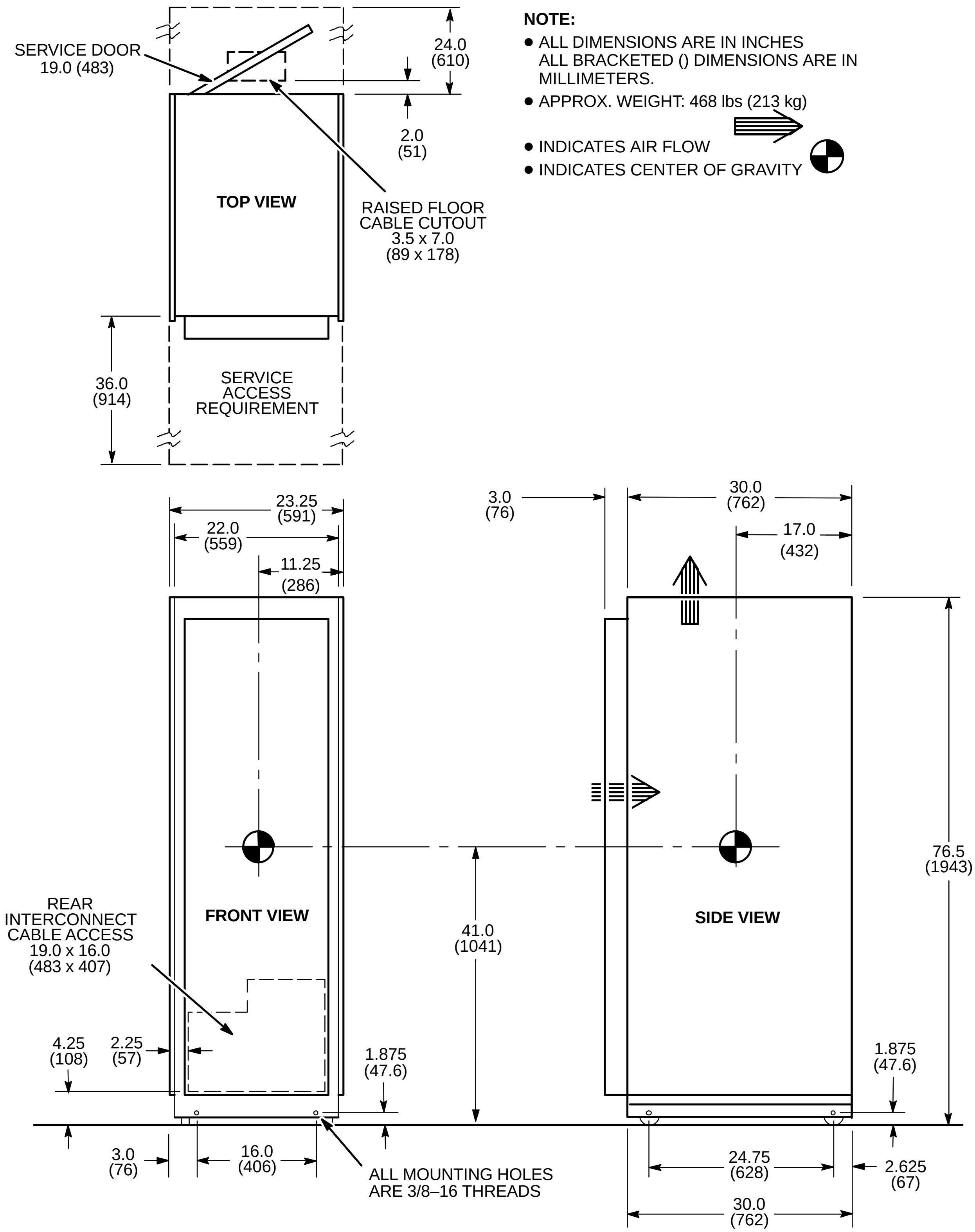
1.5T SRFD II CABINET (MR1)
ILLUSTRATION 2–28

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 2136 lbs (969 kg)
- FAN MODULE IS INSTALLED ON TO CABINET AFTER CABINET IS POSITIONED AT SITE. FAN MODULE IS NOT INSTALLED DURING SHIPMENT FOR EASE OF HANDLING AND LOCATING OF THE CABINET.
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY

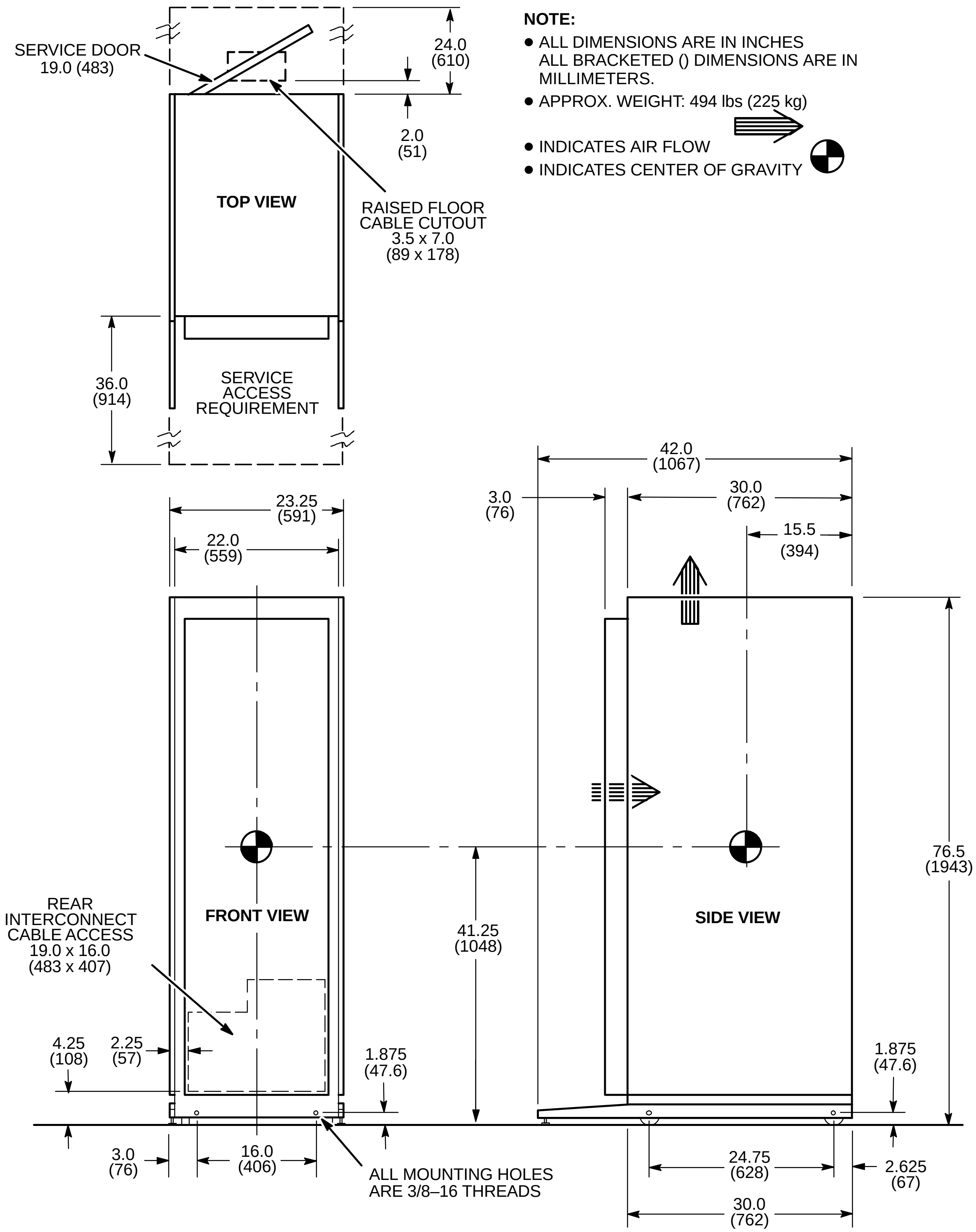


ACGD/PDU CABINET (MR3)
ILLUSTRATION 2–29

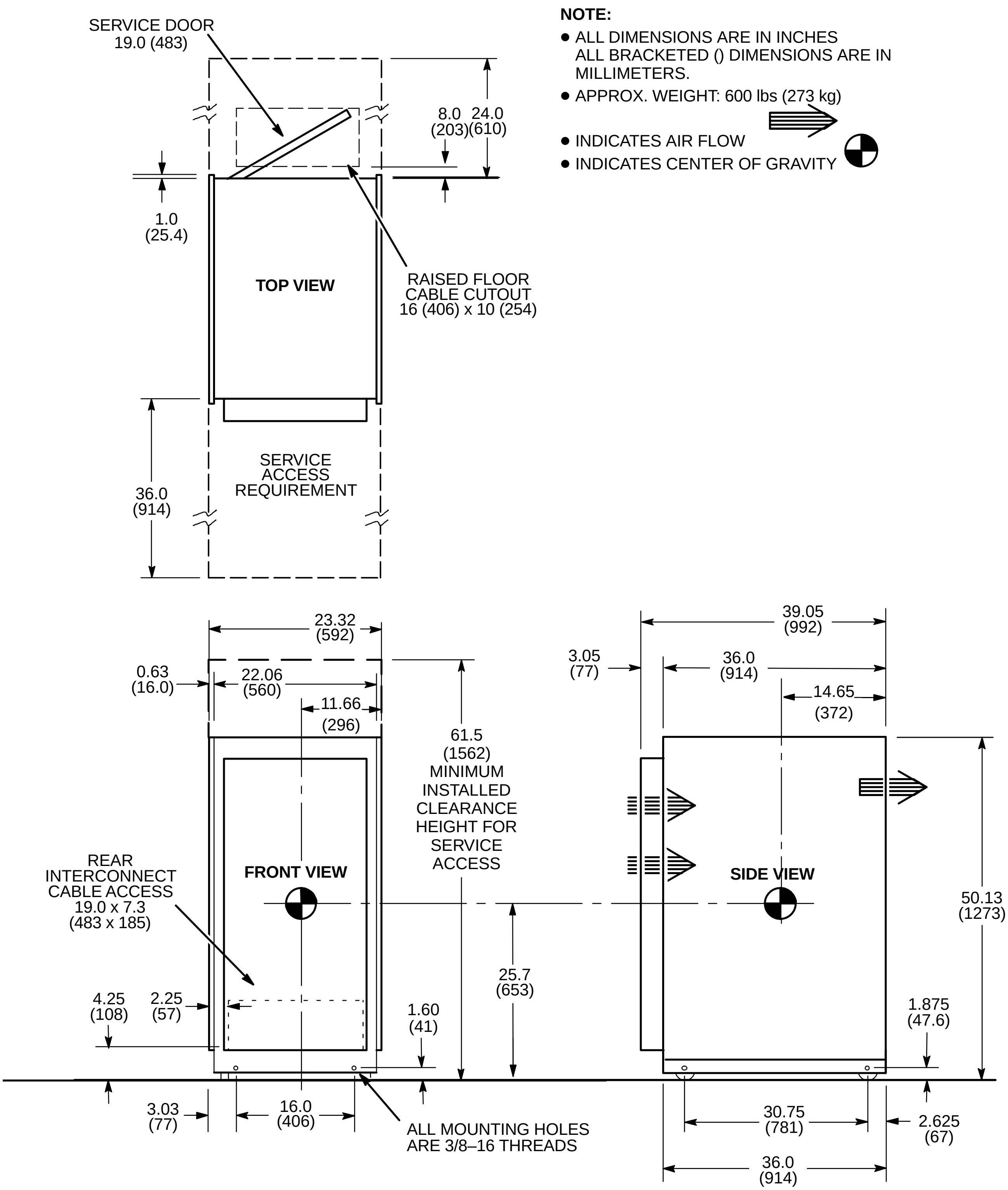


SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY

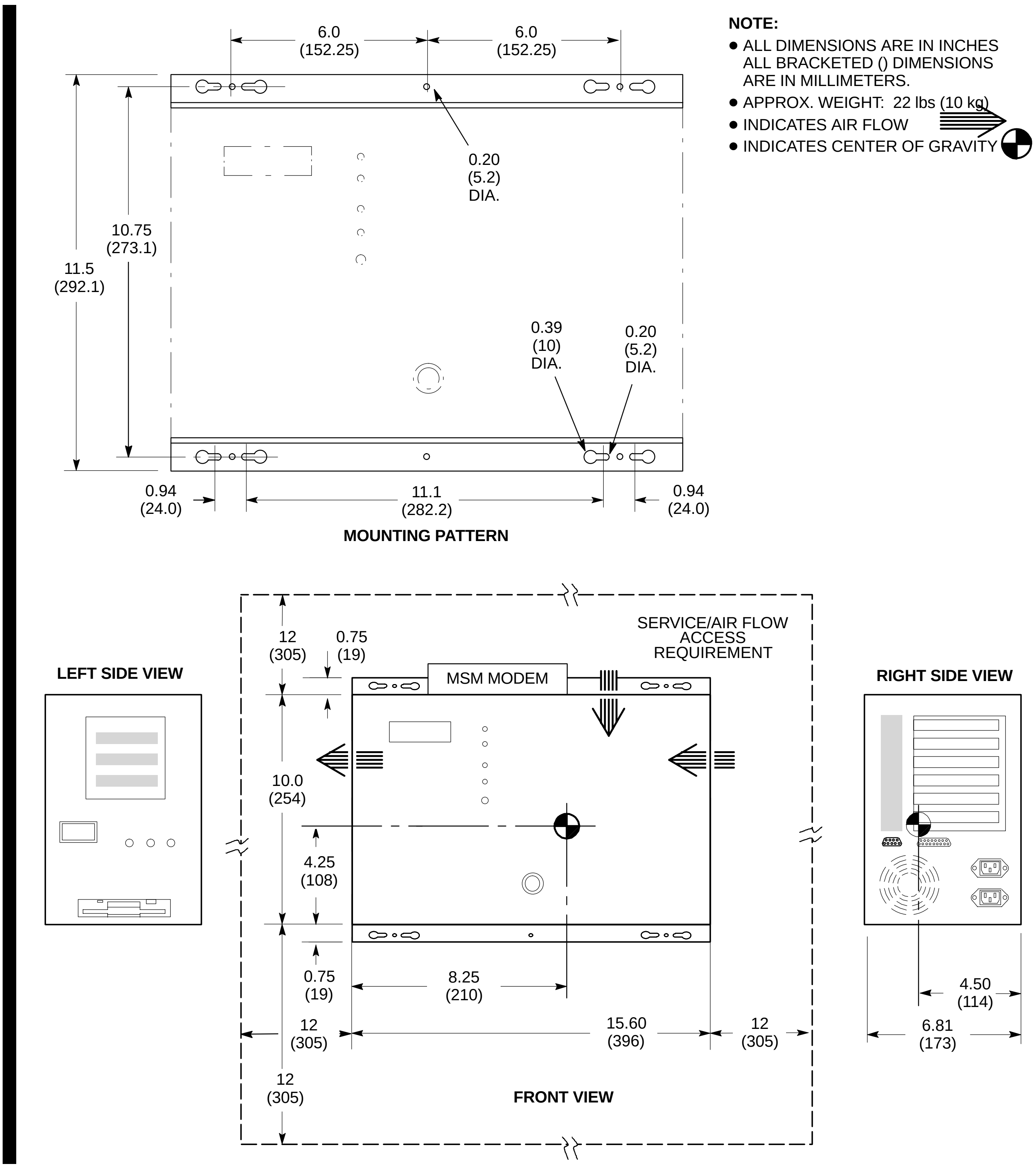
ILLUSTRATION 2–30



SYSTEM CABINET (MR2) FOR SIGNA TWIN*SPEED*
ILLUSTRATION 2–31

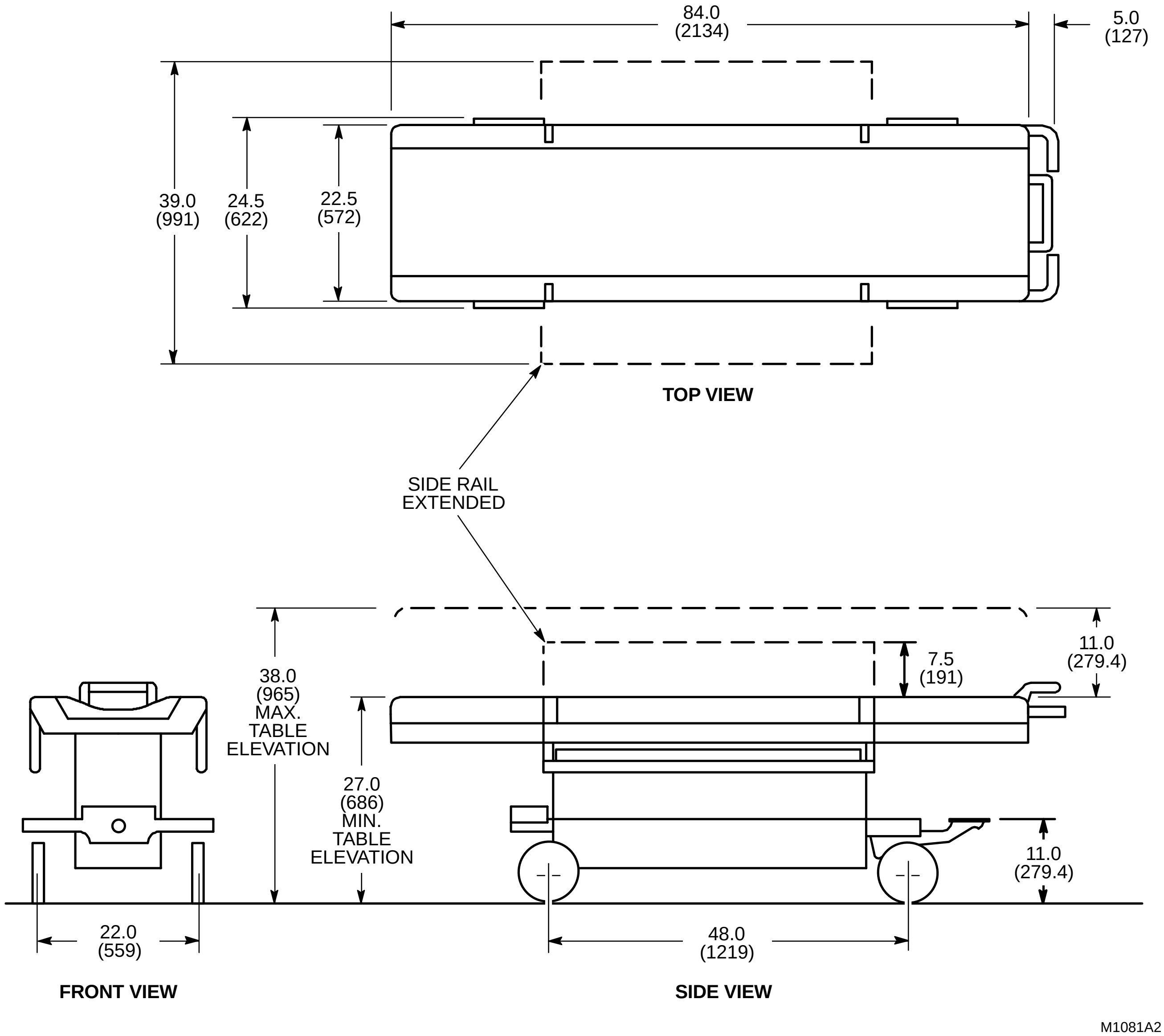


TWIN ACCESSORY CABINET (TAC)
ILLUSTRATION 2–32



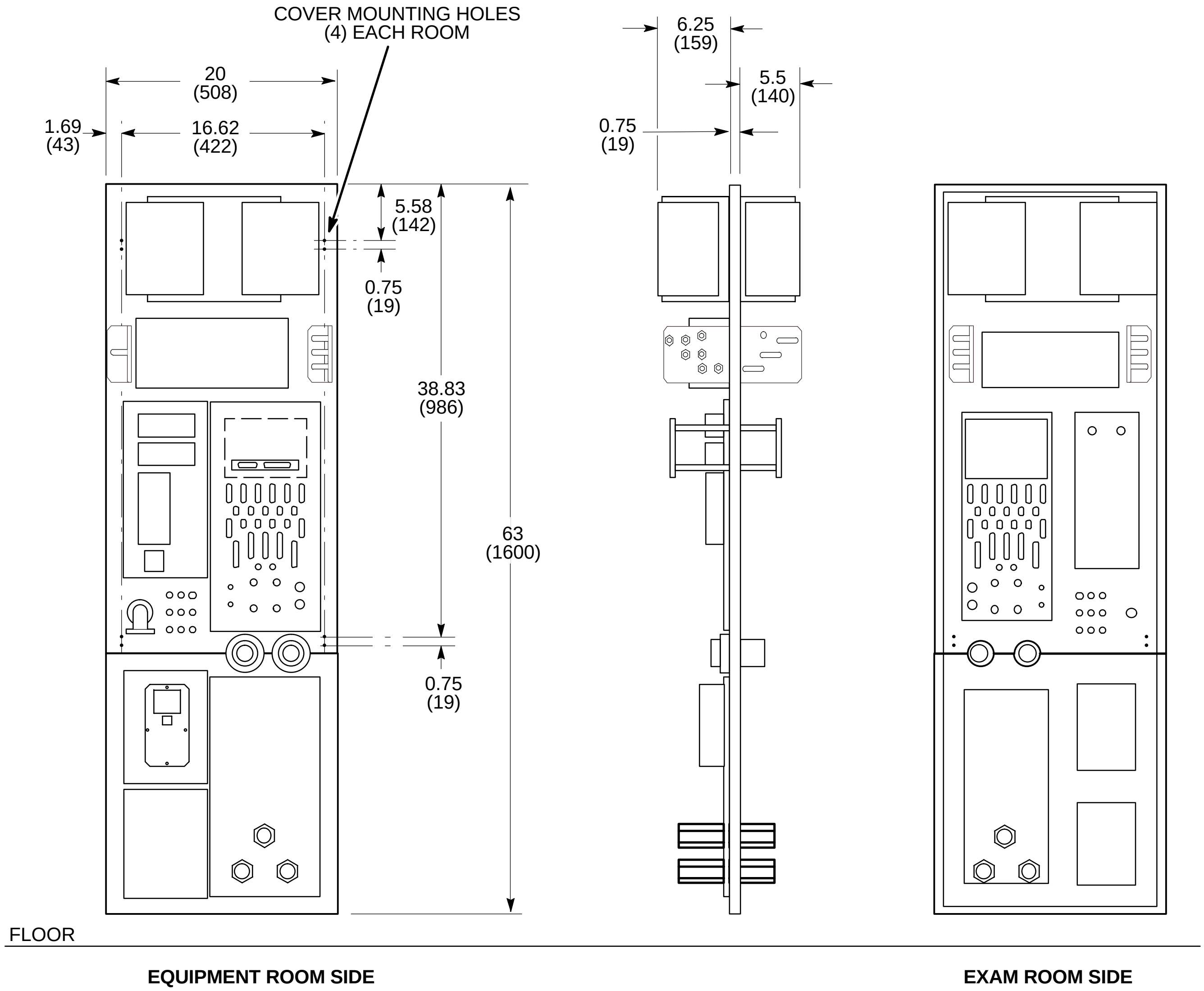
MAGNET MONITOR (MSM1)
ILLUSTRATION 2–33

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- WEIGHT: 280 lbs (127 kg)



PATIENT TRANSPORT TABLE (PT1)
ILLUSTRATION 2–34

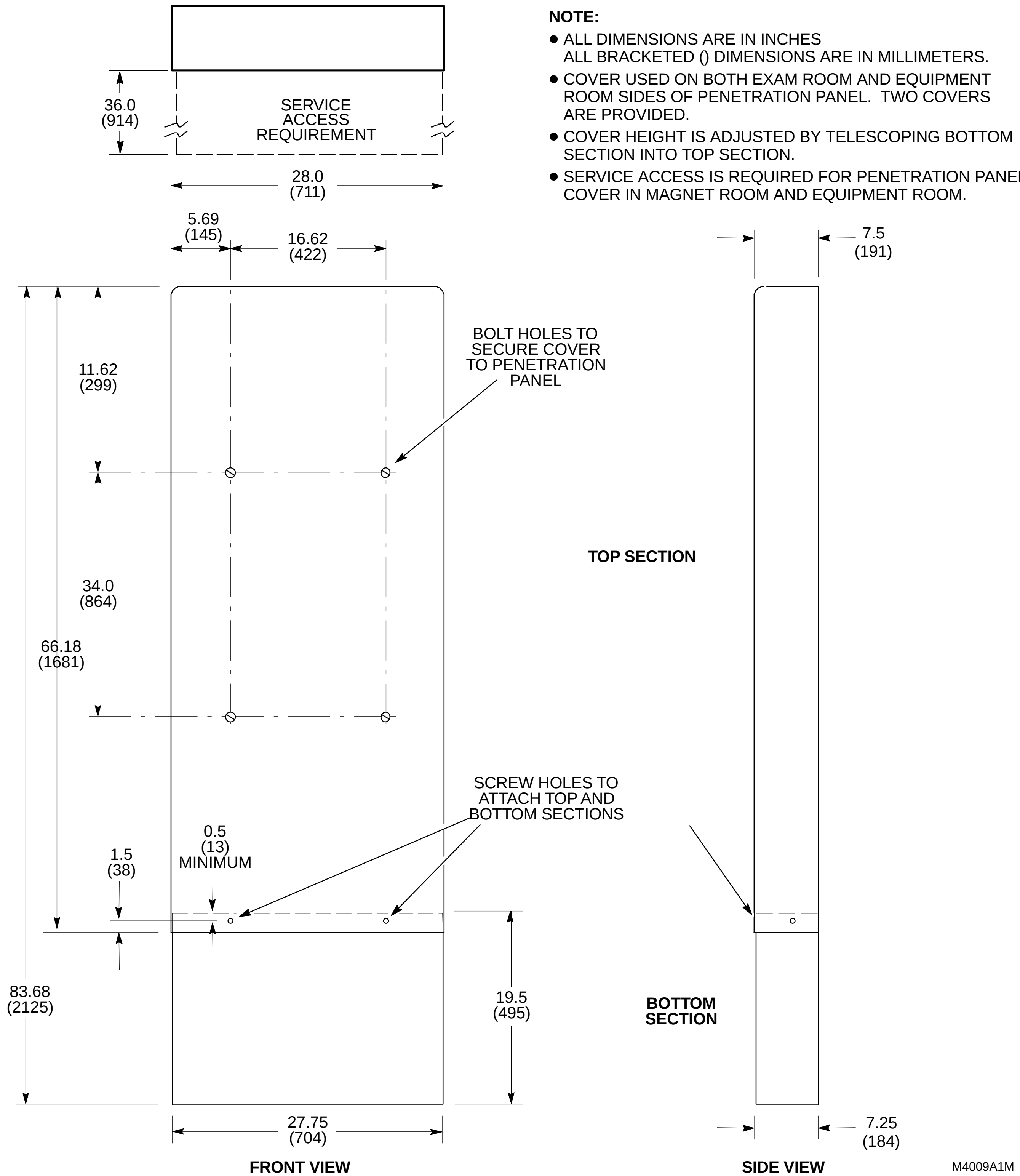
NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 52 lbs (23.6 kg)



PENETRATION PANEL (PP1)
ILLUSTRATION 2–35

NOTE:

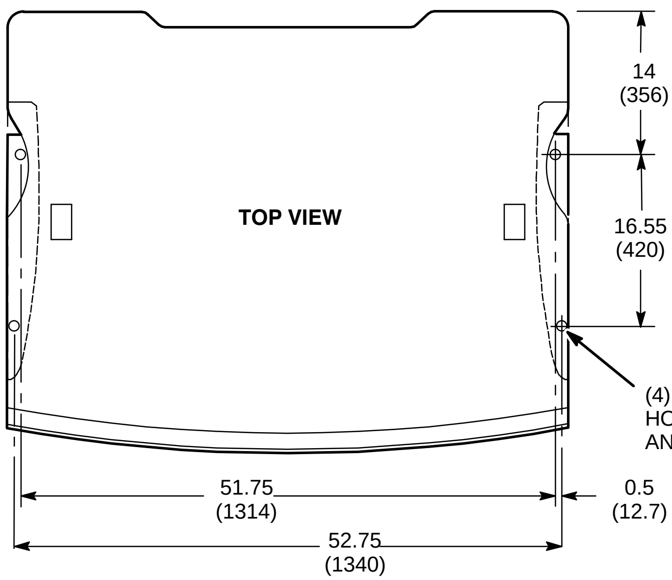
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- COVER USED ON BOTH EXAM ROOM AND EQUIPMENT ROOM SIDES OF PENETRATION PANEL. TWO COVERS ARE PROVIDED.
- COVER HEIGHT IS ADJUSTED BY TELESCOPING BOTTOM SECTION INTO TOP SECTION.
- SERVICE ACCESS IS REQUIRED FOR PENETRATION PANEL COVER IN MAGNET ROOM AND EQUIPMENT ROOM.

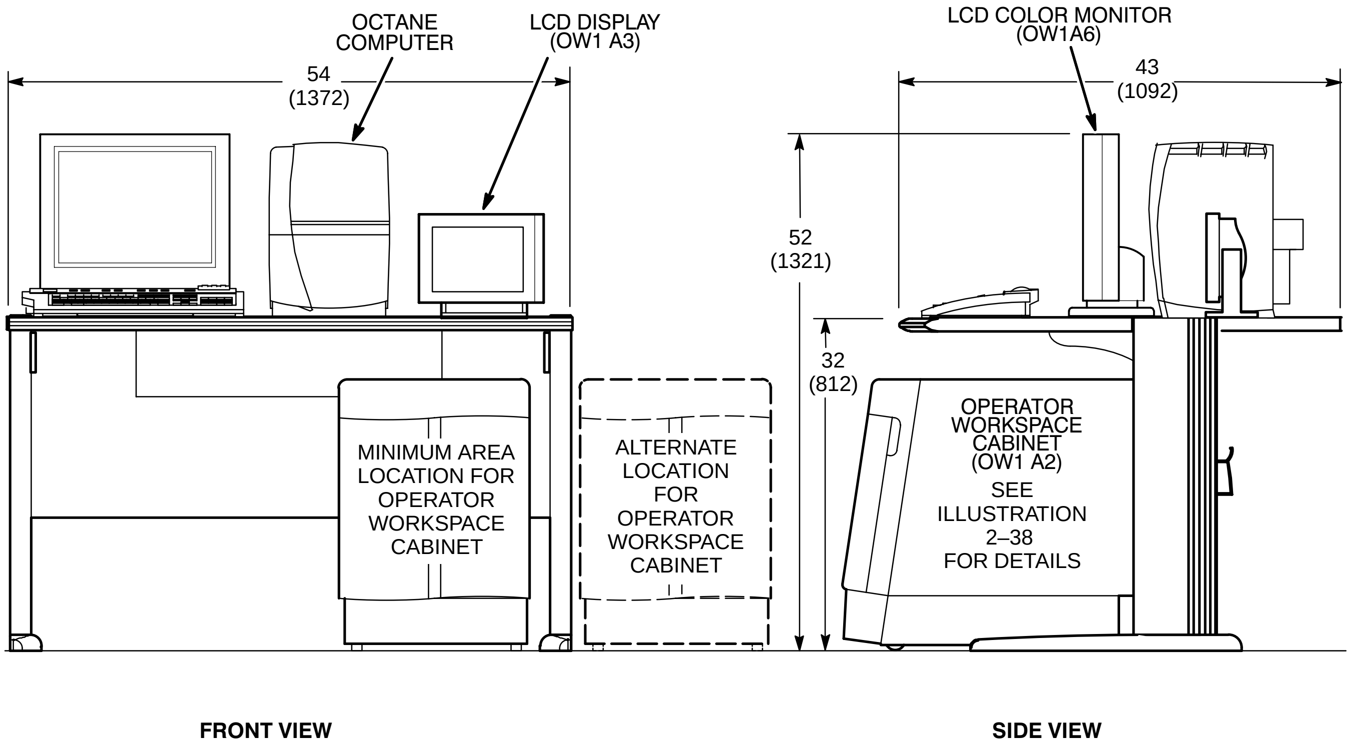


PENETRATION PANEL COVER
ILLUSTRATION 2–36

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ASSEMBLIES WHICH MOUNT TO UNDERSIDE OF TABLE AND IN TABLE CABLE TRAY ARE NOT SHOWN.
- OW TABLE IS AN INTEGRAL PART OF THE REGULATORY APPROVED SYSTEM. OW TABLE PROVIDES MOUNTING FOR SEVERAL ASSEMBLIES (E.G. OW INTERFACE MODULE, OW POWER DISTRIBUTION BOX, MODEM, DASM) AND CABLE ROUTING FOR OW INTERCONNECTS. DIMENSIONS OF EQUIPMENT LOCATED ON TOP OF OW TABLE ARE PROVIDED IN ILLUSTRATIONS 2–39 THROUGH 2–41 OW TABLE TOP LAYOUT PLANNING PURPOSES.

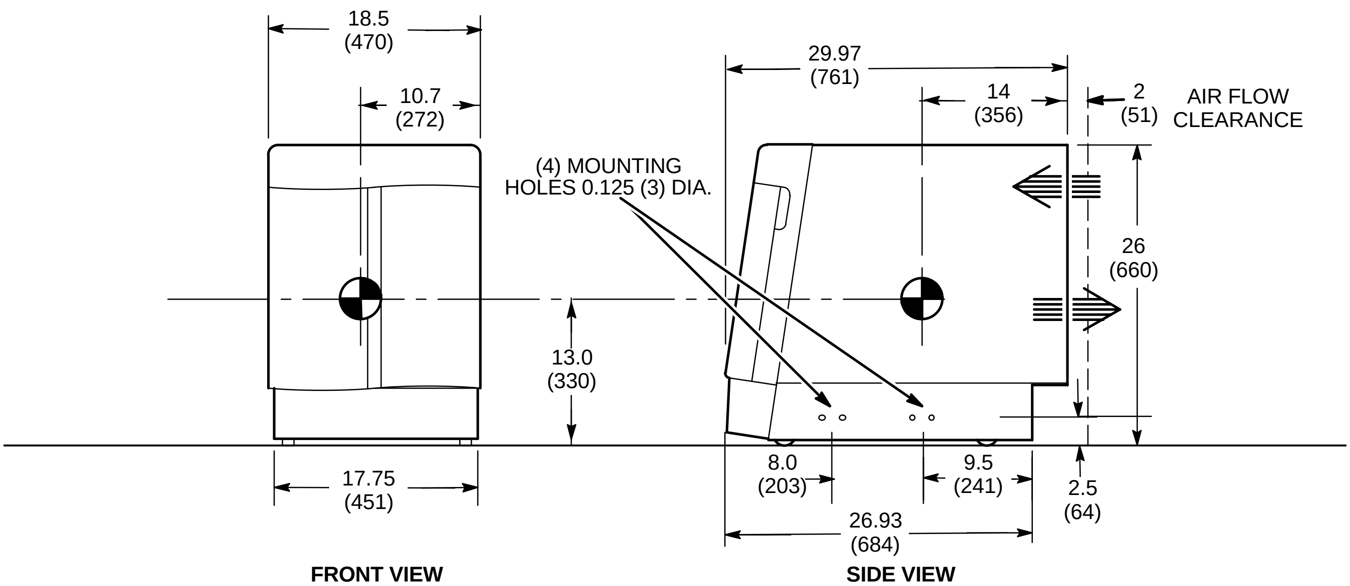




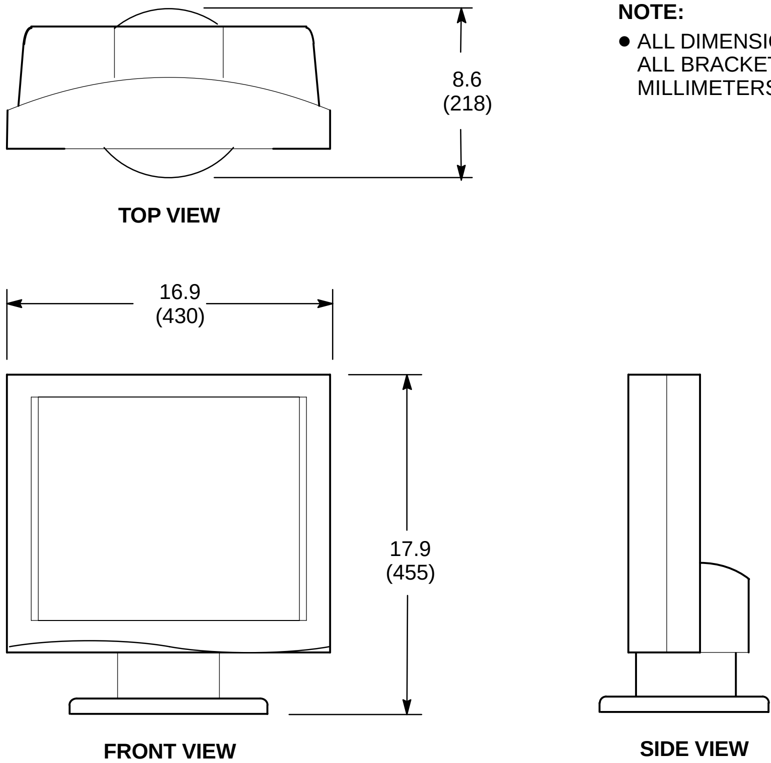
OPERATOR WORKSPACE (OW1)
ILLUSTRATION 2–37

NOTE:

- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 192 lbs (87 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



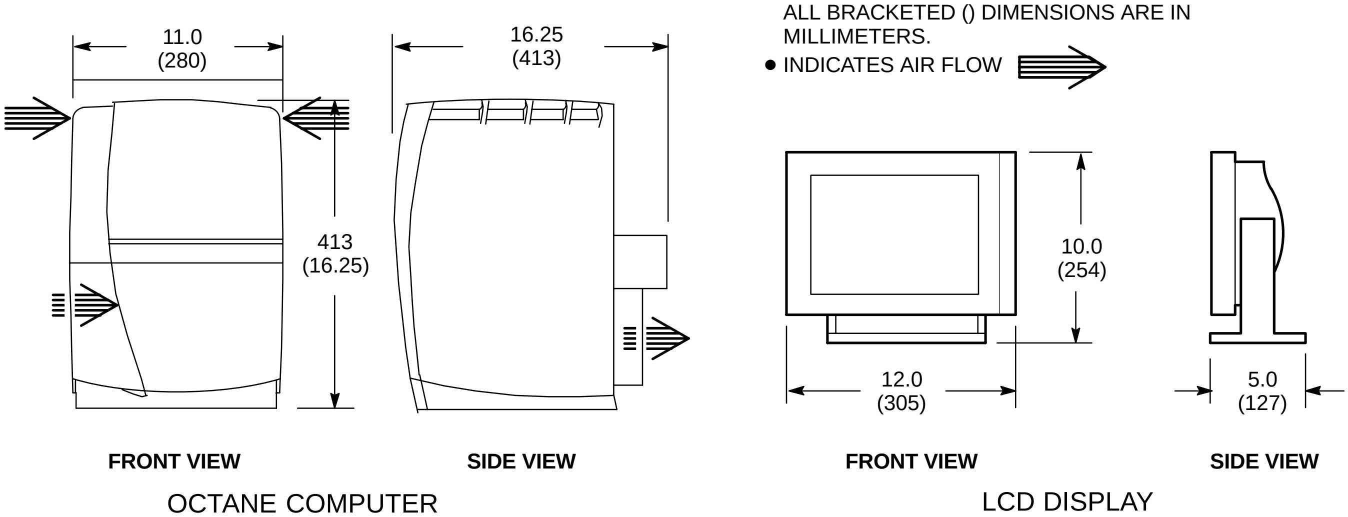
OPERATOR WORKSPACE CABINET (OW1 A2)
ILLUSTRATION 2–38



NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

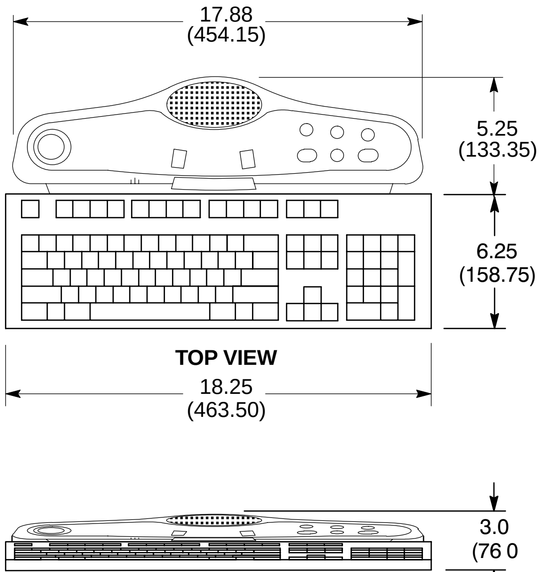
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR
ILLUSTRATION 2–39

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- INDICATES AIR FLOW

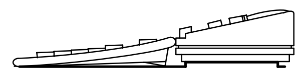


OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER
ILLUSTRATION 2–40

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



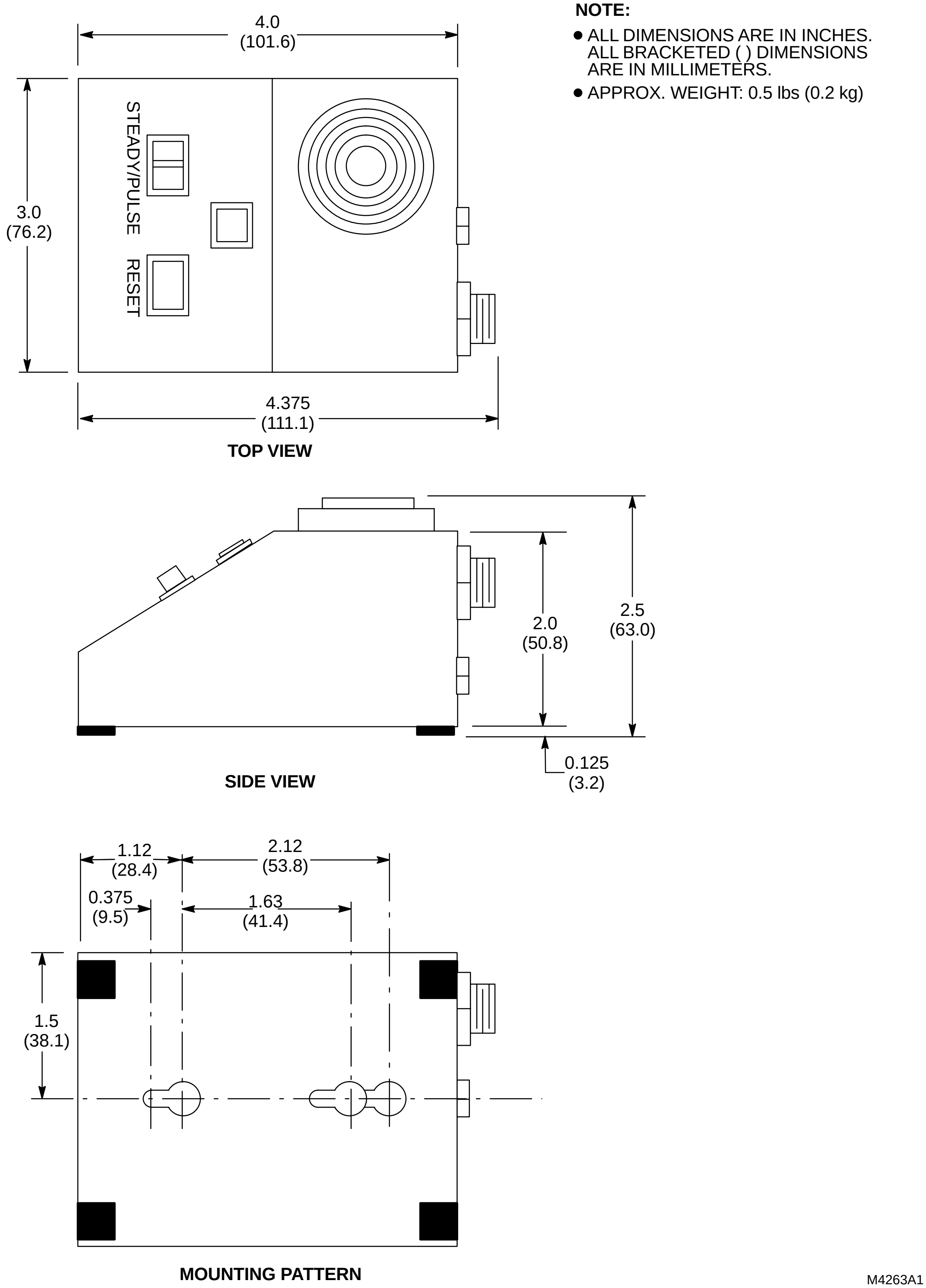
FRONT VIEW



SIDE VIEW

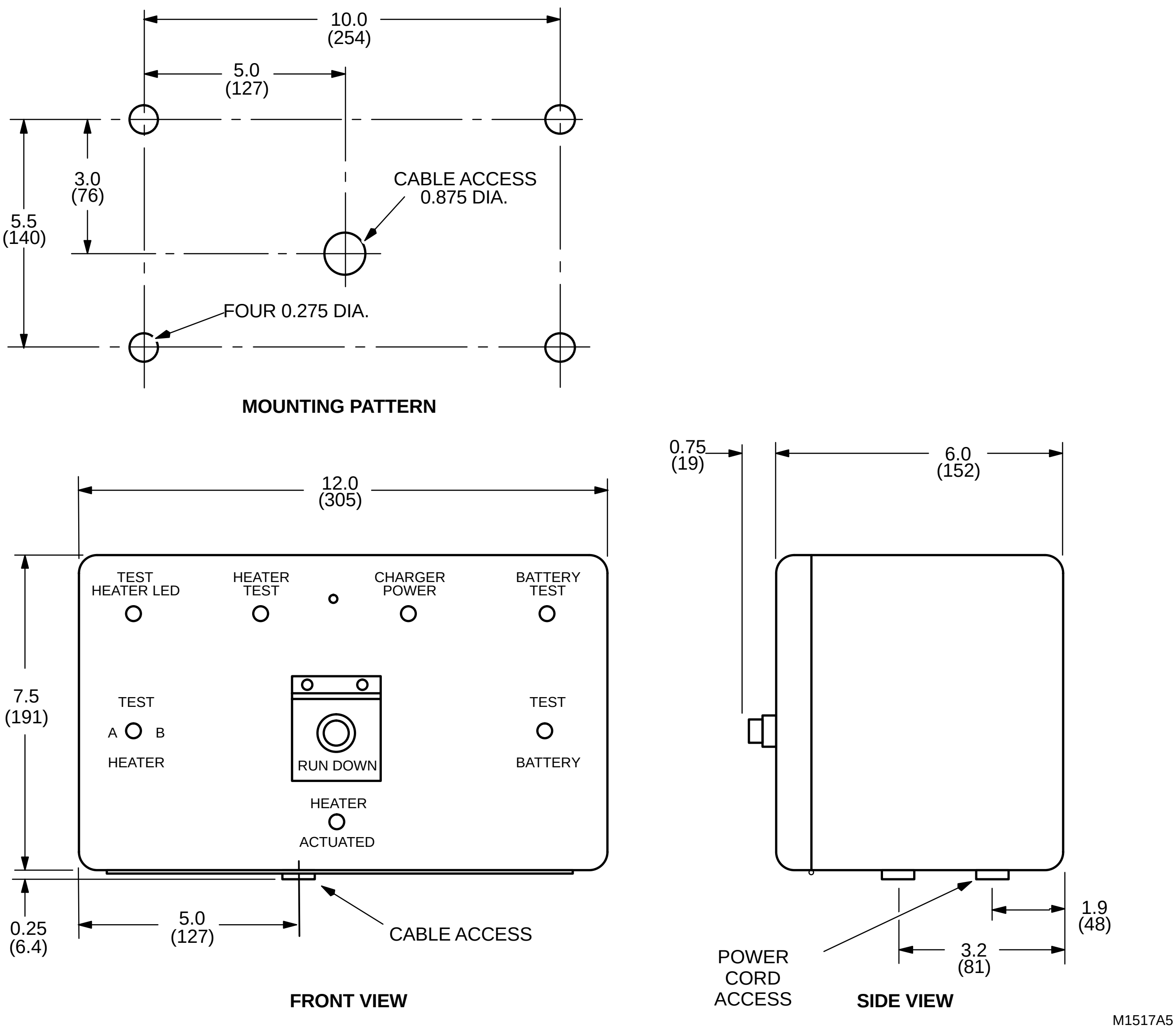
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – KEYBOARD
ILLUSTRATION 2–41

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 0.5 lbs (0.2 kg)



PNEUMATIC PATIENT ALERT CONTROL BOX (PA1)
ILLUSTRATION 2–42

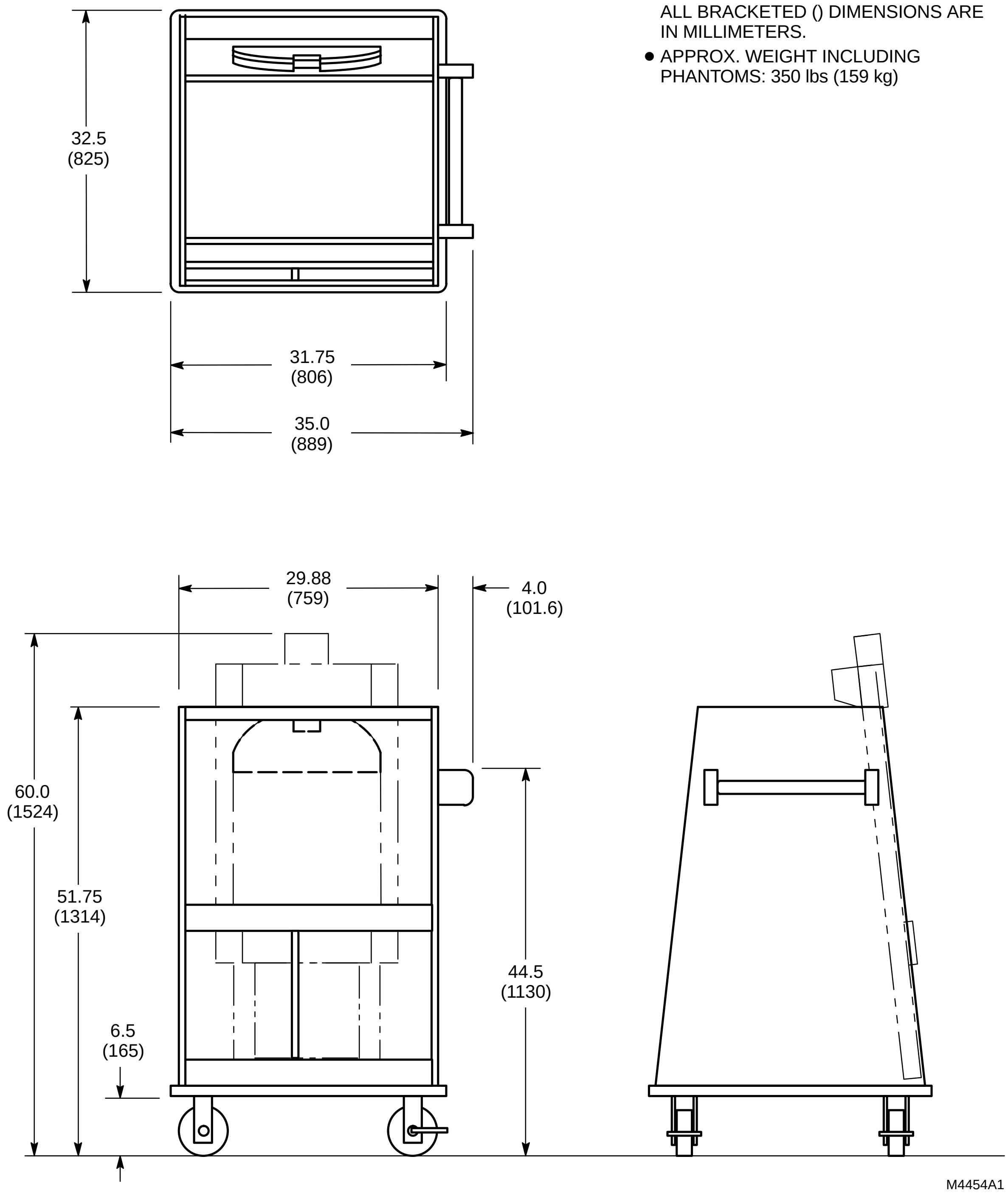
NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 8.8 lbs. (4kg)



MAGNET RUNDOWN UNIT (MS4)
ILLUSTRATION 2–43

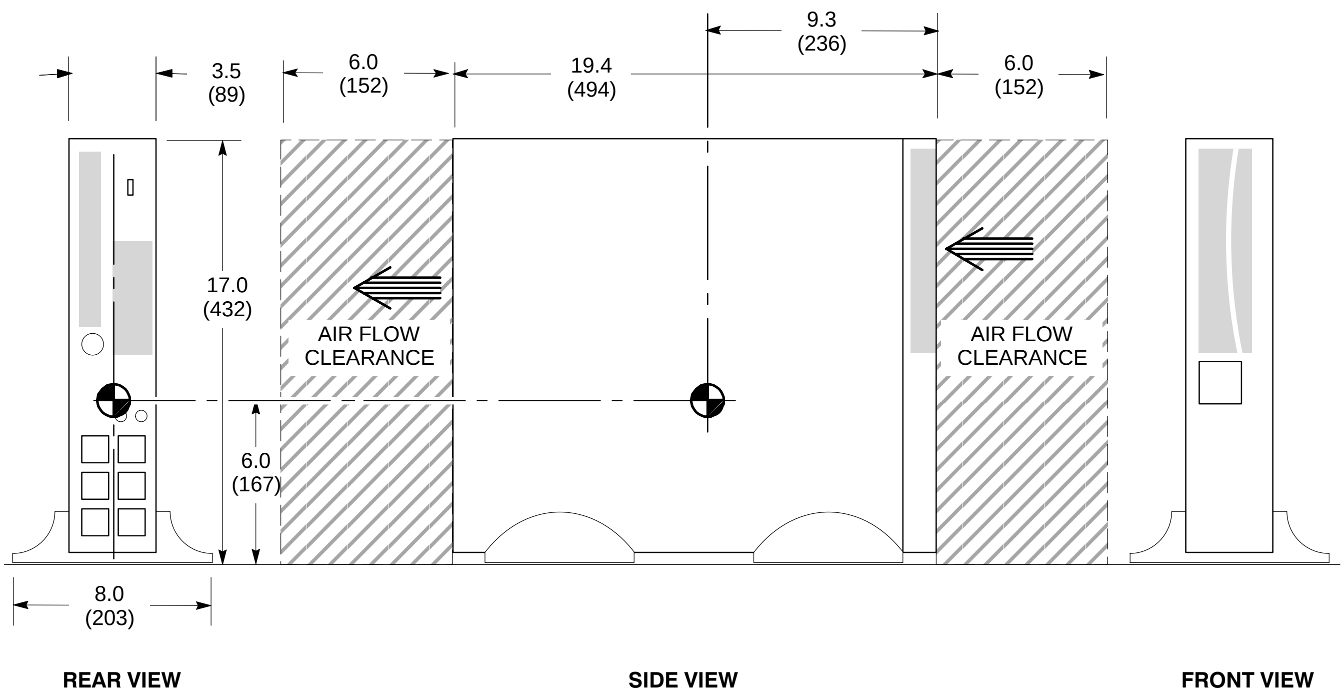
NOTE:

- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT INCLUDING PHANTOMS: 350 lbs (159 kg)



SPT PHANTOM SET SHIPPING/STORAGE CART
ILLUSTRATION 2–44

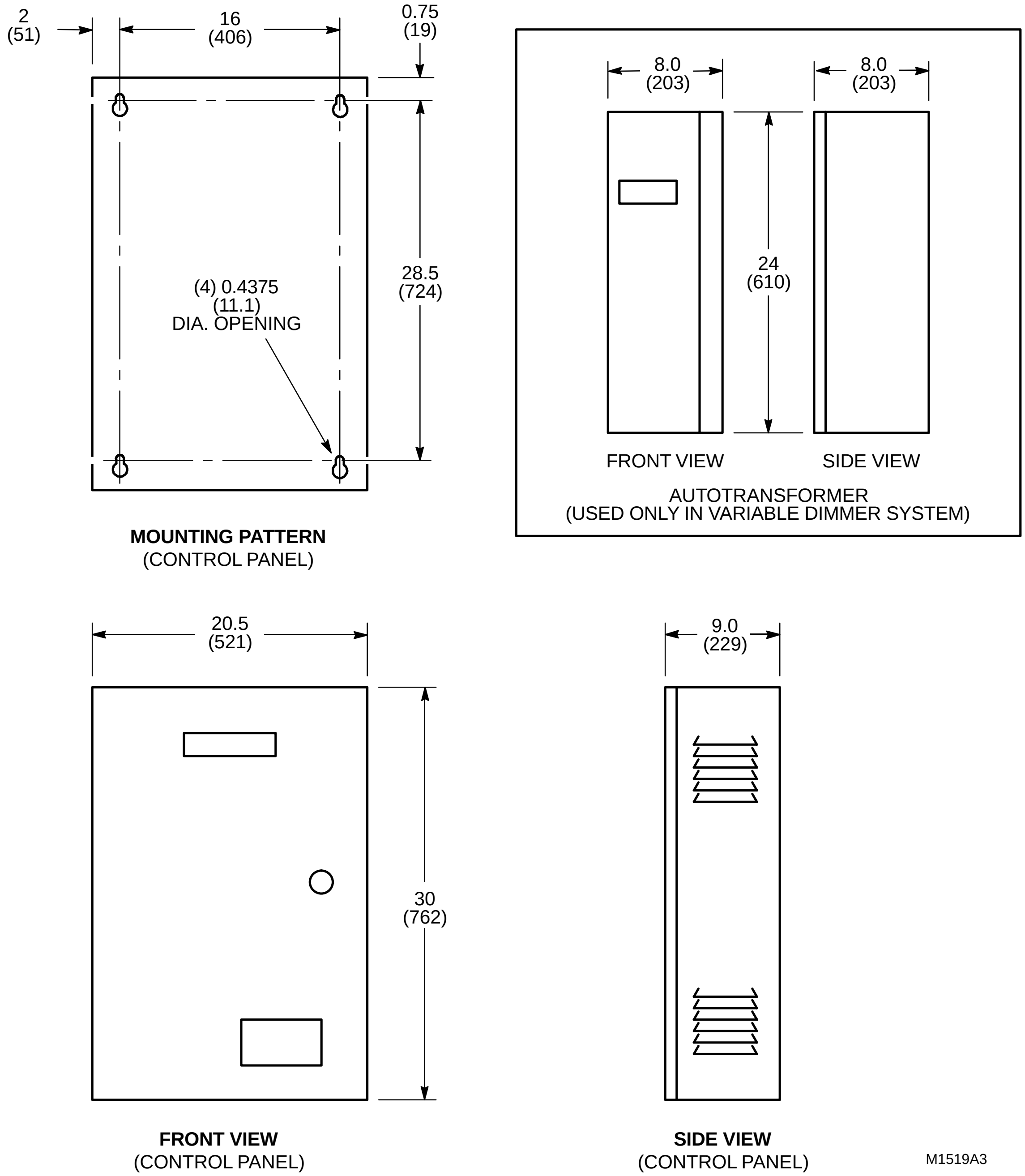
NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 50 lbs (23 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



UPS FOR MAGNET MONITOR – STANDARD FOR TRUE MOBILE CONFIGURATION, OPTIONAL FOR OTHER CONFIGURATIONS
ILLUSTRATION 2–45

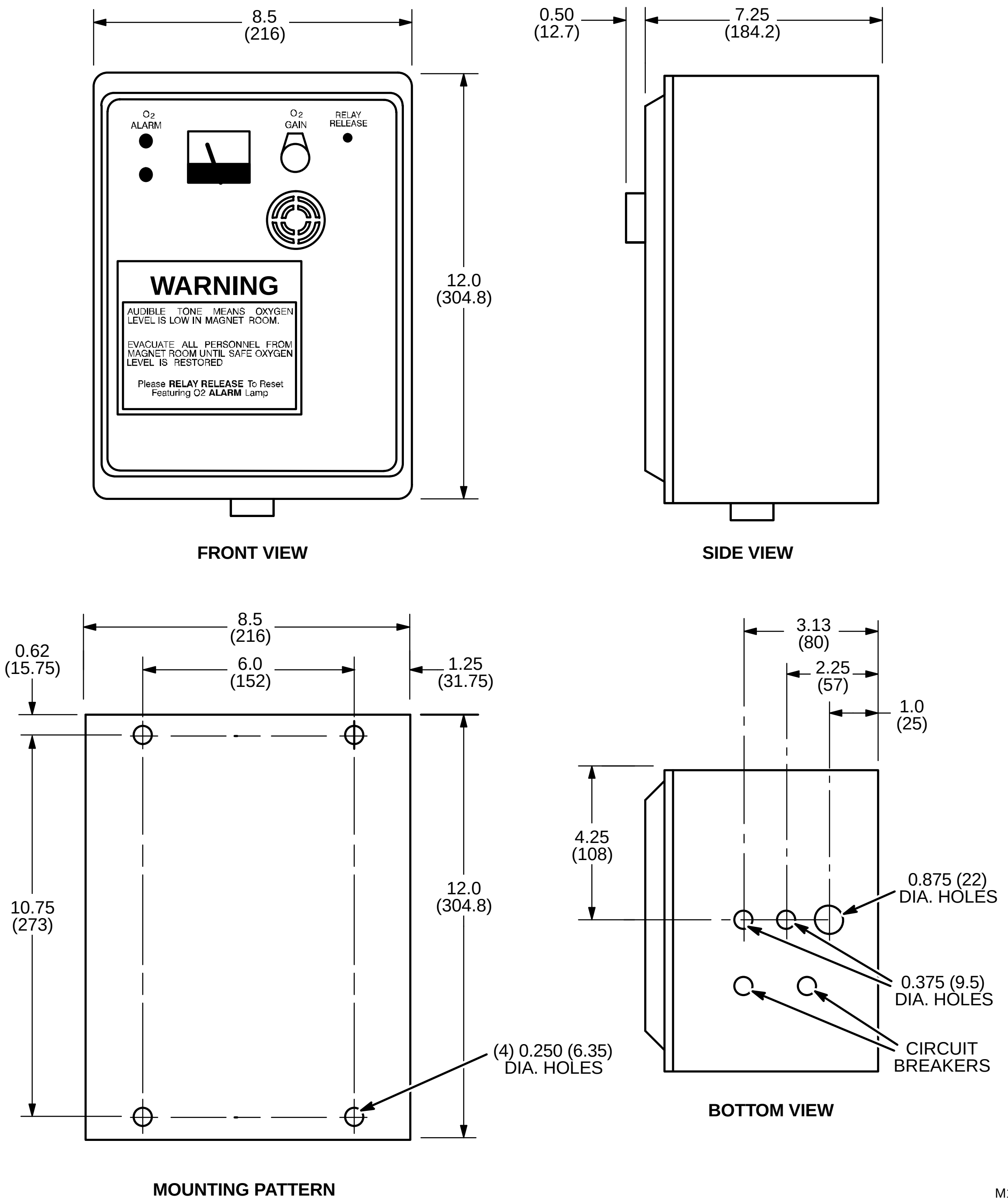
NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHTS:
  CONTROL PANEL: 155 lbs (70 kg)
  AUTOTRANSFORMER: 60 lbs (27 kg)



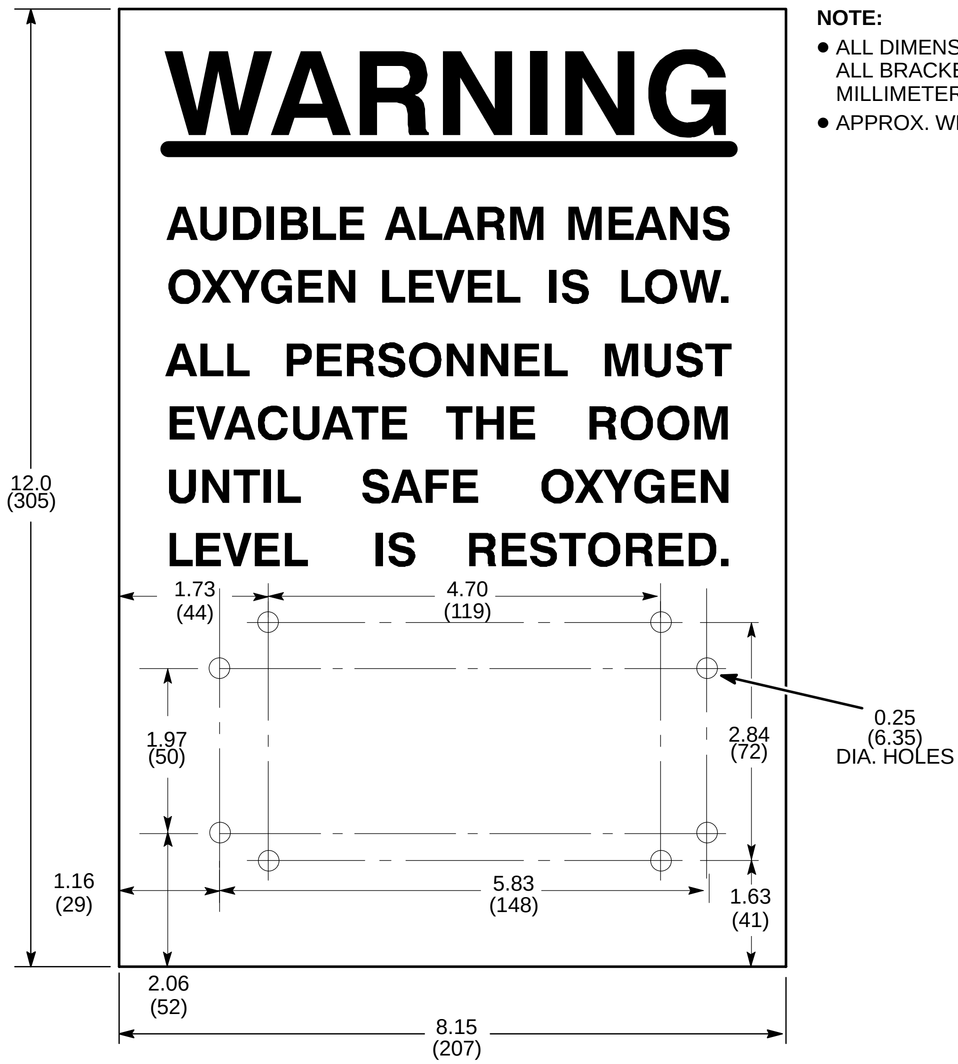
DC LIGHTING CONTROLLER – OPTIONAL
ILLUSTRATION 2–46

NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 9 lbs (4.1 kg)

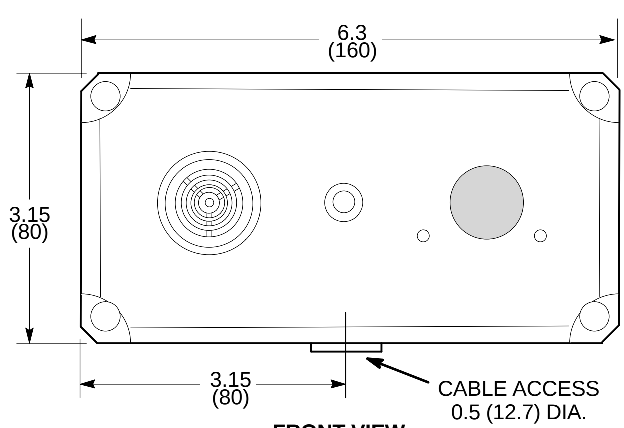


OXYGEN MONITOR (OM1) – OPTIONAL
ILLUSTRATION 2–47

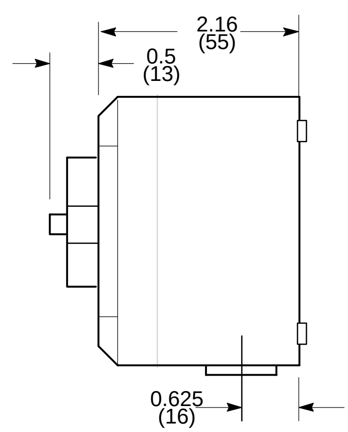
NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT OF MODULE: 1.5 lbs (0.68 kg)



MOUNTING PATTERN



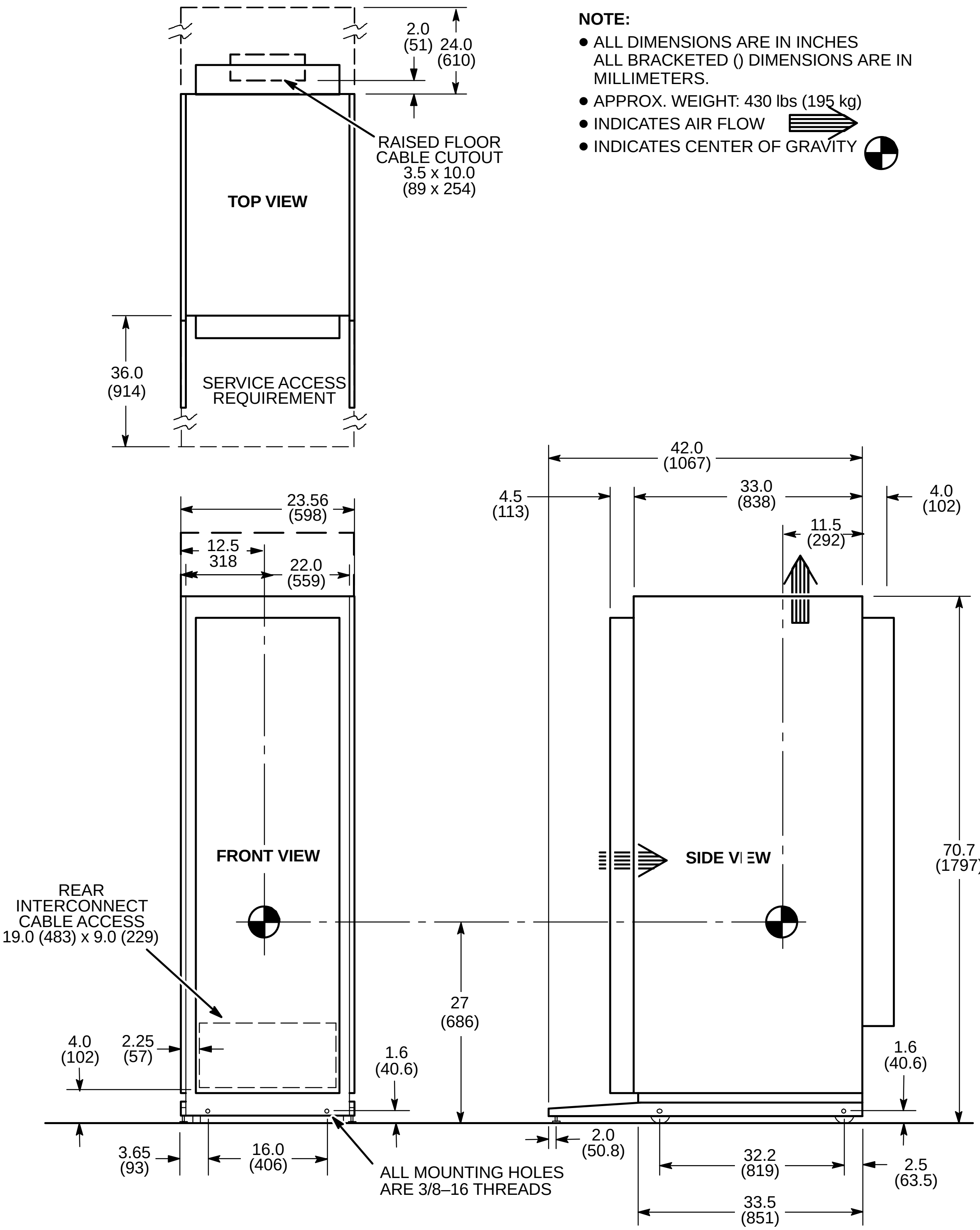
FRONT VIEW



SIDE VIEW

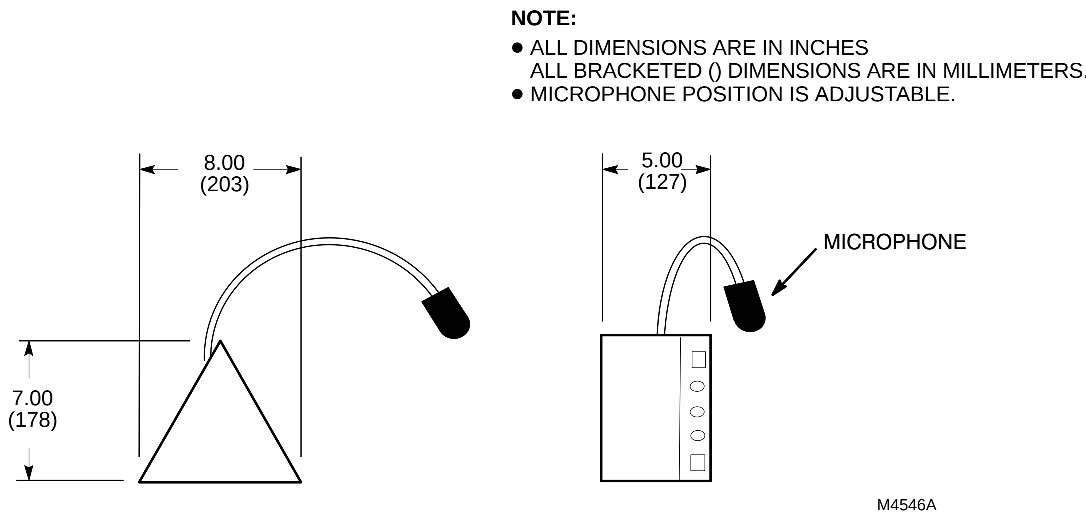
M3346A1M

REMOTE OXYGEN SENSOR MODULE (OM3) – OPTIONAL
ILLUSTRATION 2–48

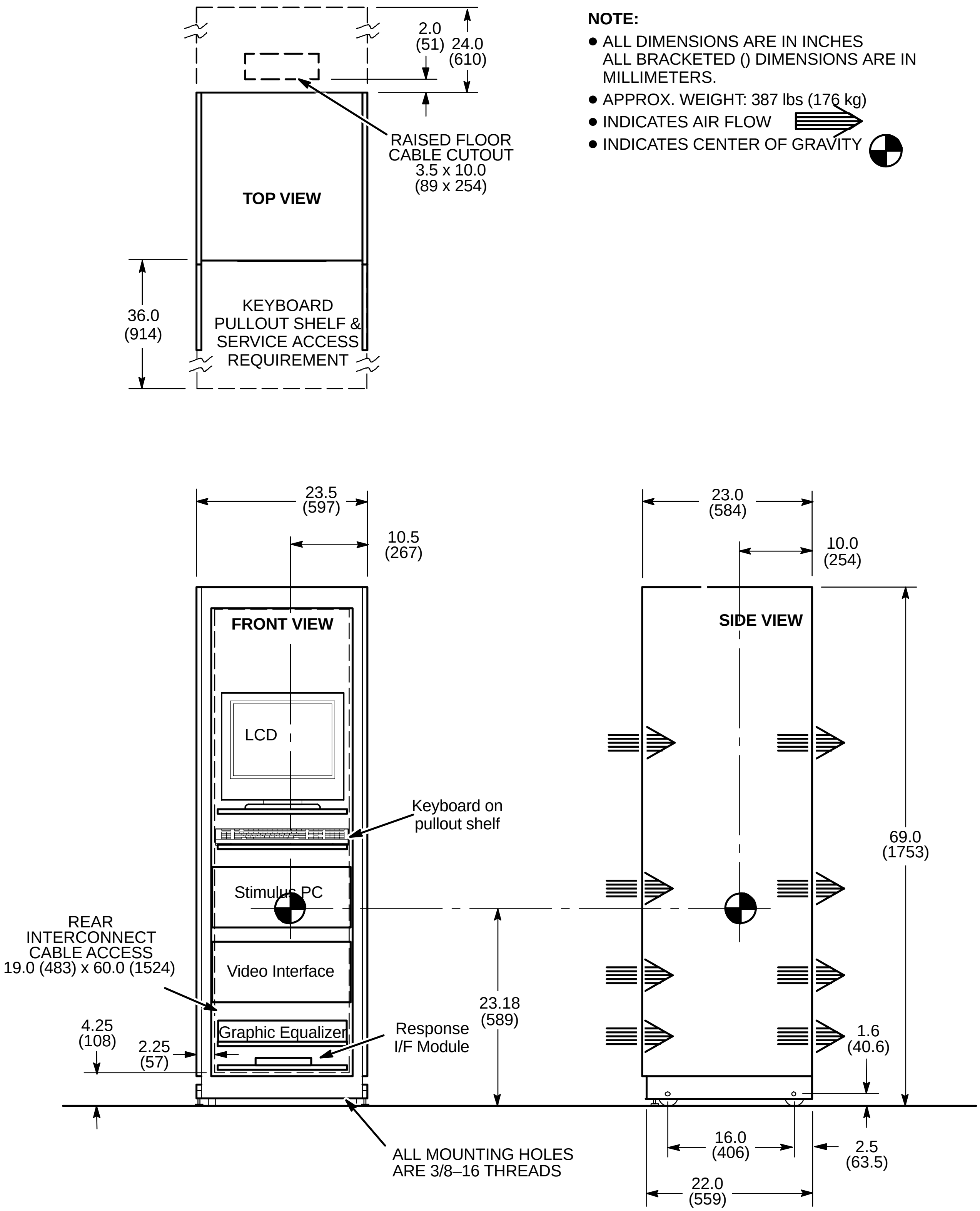


MNS CABINET (MNS) – 1.5T OPTION

ILLUSTRATION 2–49



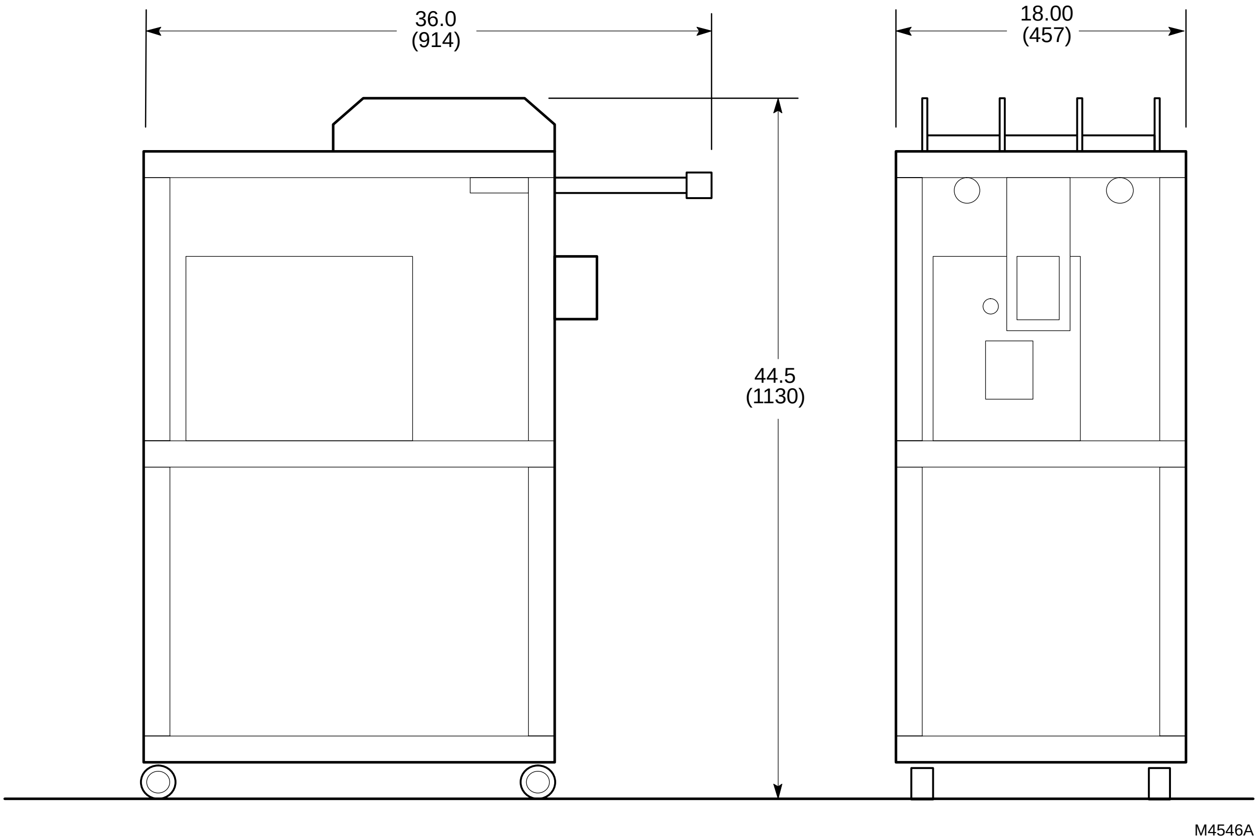
BRAINWAVE AUDIO CONSOLE – BRAINWAVE OPTION
ILLUSTRATION 2–50



BRAINWAVE CABINET – BRAINWAVE OPTION

ILLUSTRATION 2–51

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 71 lbs (32 kg)



BRAINWAVE CART – BRAINWAVE OPTION
ILLUSTRATION 2–52

SECTION 3 – MAGNETIC FIELD CONSIDERATIONS

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3–1 INTRODUCTION

The static magnetic field is three–dimensional and extends into space above and below the magnet as well as to the surrounding space on the same level. Objects within this three–dimensional space can be affected by the magnetic field or can affect the magnetic field, refer to Table 2–1 PROXIMITY LIMITS in Section 2 – ROOM LAYOUTS. Therefore all ferromagnetic material within this three–dimensional magnetic field must be thoroughly examined to ensure it is not significantly affected by nor affects the magnetic field.

3–2 HOMOGENEITY REQUIREMENTS

Structural steel within the static magnetic field of an unshielded magnet has a definite impact on the homogeneity or uniformity of the field. The magnet's field homogeneity is an important criteria that impacts image quality of the magnet. Homogeneity requirements for imaging are just as stringent as the homogeneity requirements for chemical shift analysis (spectroscopy).

3–3 STRUCTURAL STEEL EVALUATION OF PROPOSED SITES

Structural steel near of the magnet (especially unshielded and actively shielded magnets) causes perturbations in the magnetic field within the imaging region of the magnet. This may degrade the homogeneity of the magnet and thus system performance. An evaluation of the effects of structural steel on the magnet is required in some instances, refer to Section 4–14–1, Floors.

Therefore, the customer must provide information indicating mass and location of all iron and steel within an 8 feet (2.5 meter) radius of the 1.5T (Actively Shielded) magnet isocenter. This 8 foot (2.5 meter) radius is shown as the shaded region in Illustrations 3–1 and 3–2. This includes iron below the magnet such as sewer pipes, floor beams and any steel rebar in the concrete floor or structural members. Any structural steel required for the installation of the magnet at the particular site (i.e. floor reinforcement) must also be indicated.

If the steel in close proximity to the magnet exceeds the limits found in Section 4–14–1 Floors, one of the following actions may be taken:

- Choose an alternate site.
- Redesign steel structure.
- Request a steel analysis by the MR Siting & Shielding group.
- Install external passive shims. This is iron securely placed on the magnet which compensates for the effect of nearby structural steel. The MR Siting & Shielding group will assist in determining if this is a feasible alternative. Note, use of this technique is greatly limited for Active Shield magnets.
- Install magnet internal passive shims. These shims are iron plates placed within the magnet. The MR Siting & Shielding group will assist in determining if this is a feasible alternative. Note, use of this technique is greatly limited for most magnets other than for minor steel disturbances.
- Install magnetic shielding (refer to Section 3–4, MAGNETIC SHIELDING). If containment of the stray magnetic field is required, shielding will also usually mask the effects of nearby structural steel.

3–4 MAGNETIC SHIELDING

Magnetic shielding is used to reduce the fringe field around the magnet.

Room Shield

Room magnetic shielding generally consists of iron plates in the room walls, floor, and ceiling. Special consideration should be given when selecting a magnet site location due to the expense and effort required to provide magnetic shielding.

Designing a magnetic shield requires a comprehensive computer analysis which predicts the effect the shield will have on the magnetic field as well as the effect of the shield on the homogeneity of the magnet. The structural capacity of the site and space availability are important factors in the design of the shield. The MR Siting & Shielding Group has the capability to design magnetic shields which meet a broad range of site requirements.

3–5 MAGNETIC FIELD

Illustrations 3–1 and 3–2 are the fringe field plots for the 1.5T Active Shield magnet. These plots illustrate the three–dimensional area of magnetic field without the influence of any nearby ferrous objects or the earth's ambient magnetic field. Actual magnetic field intensity at given locations will vary from these plots due to the following effects:

- Ferrous materials used in building construction which will become permanently magnetized when in close proximity to the MR generated magnetic field.
- Earth's magnetic field – about 0.5 gauss in strength and unidirectional.

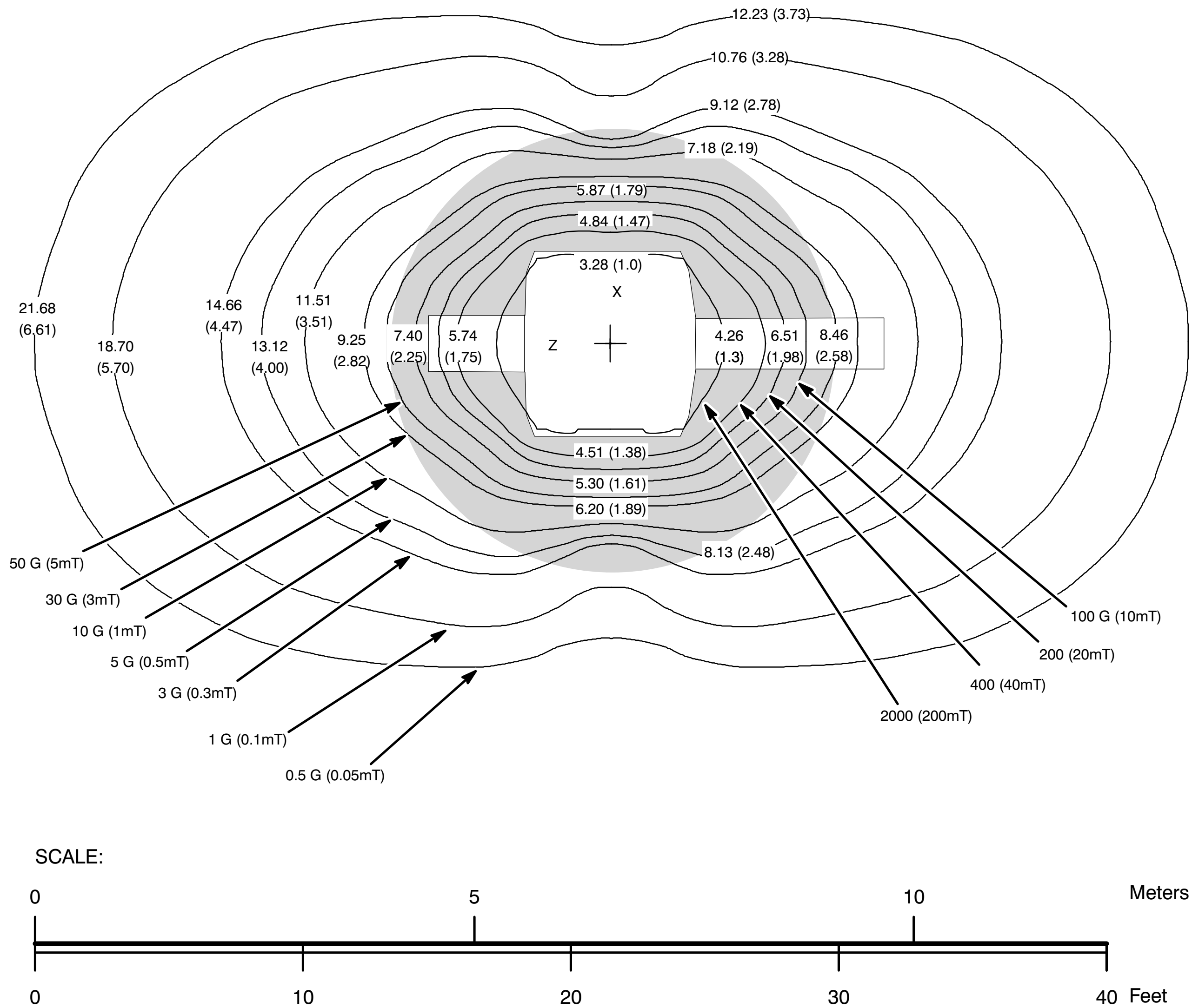
Therefore, these plots are only approximations of actual field intensities found at points surrounding the magnet. These plots should be used as an aid in reviewing the location of MR and hospital equipment and services (i.e. elevators, vehicular traffic, computer monitors, etc.). Refer to Section 2, ROOM LAYOUTS, for the sensitivities of various equipment within the magnetic field.

3–6 Exclusion Zone For 1.5T LCC (Active Shield) Magnets

The five gauss exclusion zone for cardiac pacemakers, neurostimulators, and other biostimulation devices is shown in Illustration 3–1 and 3–2 for 1.5T LCC (Active Shield) magnet. It should be noted the vertical views for the various magnetic field plots show 12 ft (3.66 m) between floors for reference. If the distance between floors is other than 12 ft (3.66 m), appropriate corrections must be made.

The interaction of the main magnet coils and the cancellation coils results in the effective shielding for the active shield magnet. Certain kinds of magnet quenches can actually cause a very short magnetic field transient resulting in the 5 gauss (0.5mT) field expanding for 2 seconds or less as noted in Illustrations 3–1 and 3–2 for 1.5T LCC (Active Shield) magnet. It should be noted that normal rampdowns or Magnet Rundown Unit (MS4) initiated quenches WILL NOT cause the magnetic field to expand.

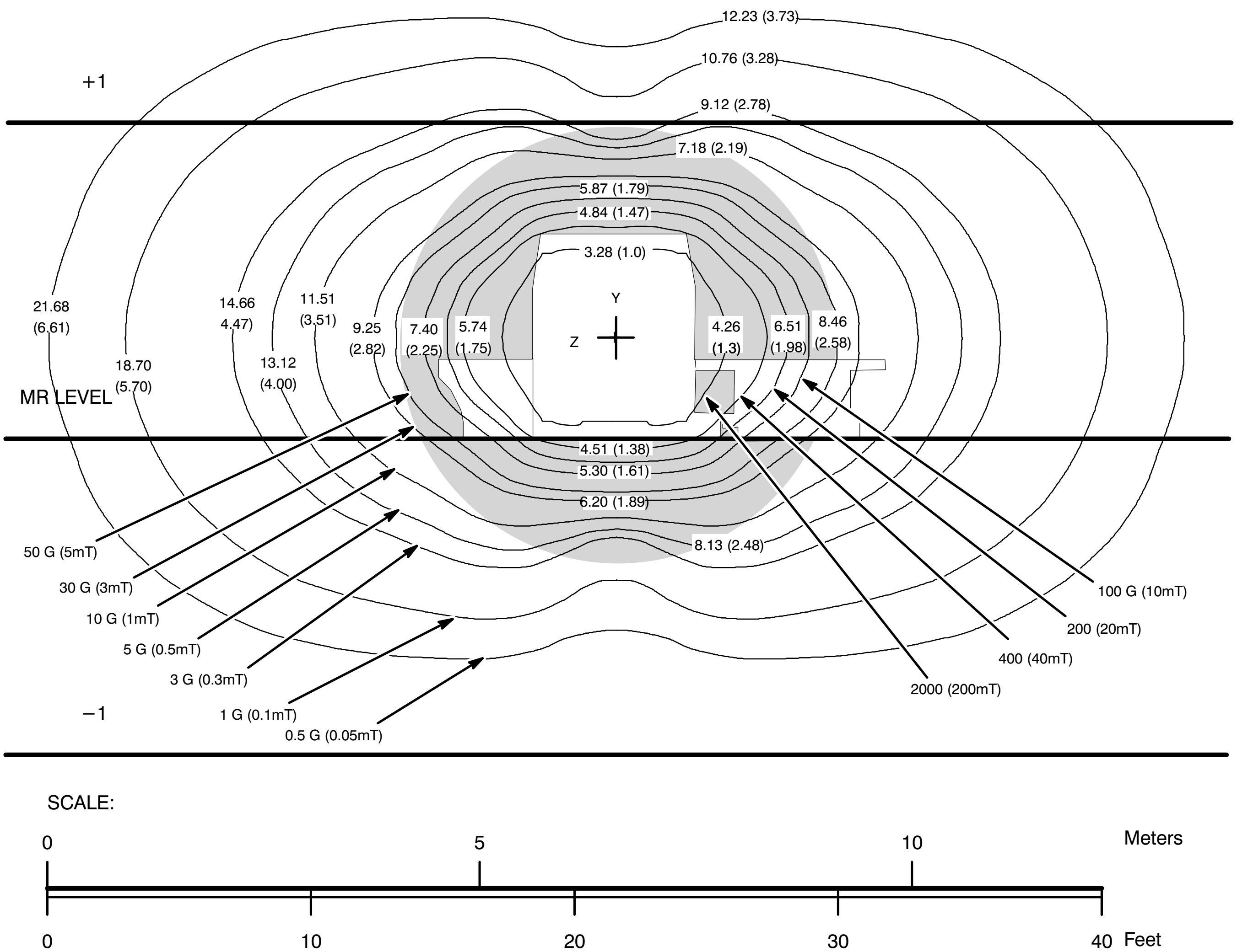
It is recommended every site consider the event of a quench and plan accordingly (such as placing 5 gauss (0.5mT) warning signs at the expanded locations).



NOTE:

- MEASURED MAGNETIC FLUX DENSITY WILL VARY FROM PLOT DUE TO FACTORS SUCH AS CONCENTRATING EFFECTS OF NEARBY FERROUS OBJECTS AND AMBIENT FIELDS, INCLUDING EARTH'S MAGNETIC FIELD.
- POTENTIAL EXISTS UNDER FAULT CONDITIONS THAT THE 5 GAUSS LINE MAY EXPAND TO 16.40 ft (5.0 m) RADIALLY AND 22.96 ft (7.0 m) AXIALLY FOR 2 SECONDS OR LESS.

1.5 TESLA LCC (ACTIVE SHIELD) MAGNETIC ISOGAUSS LINE PLOT
MR CENTER LEVEL
ILLUSTRATION 3–1



NOTE:

- 12 ft (3.66 m) BETWEEN FLOORS.
- MEASURED MAGNETIC FLUX DENSITY WILL VARY FROM PLOT DUE TO FACTORS SUCH AS CONCENTRATING EFFECTS OF NEARBY FERROUS OBJECTS AND AMBIENT FIELDS, INCLUDING EARTH'S MAGNETIC FIELD.
- POTENTIAL EXISTS UNDER FAULT CONDITIONS THAT THE 5 GAUSS LINE MAY EXPAND TO 16.40 ft (5.0 m) RADIALLY AND 22.96 ft (7.0 m) AXIALLY FOR 2 SECONDS OR LESS.

1.5 TESLA LCC (ACTIVE SHIELD) MAGNETIC ISOGAUSS LINE PLOT
VERTICAL VIEW
ILLUSTRATION 3–2

SECTION 4 – SITE ENVIRONMENT

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4–1 INTRODUCTION

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections. The environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding these restrictions. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

4–2 TEMPERATURE AND HUMIDITY SPECIFICATIONS

4–2–1 System Suite

Note

If these temperature and humidity specifications are not strictly adhered to, failures of the ACGD modules of the ACGD/PDU Cabinet may occur.

Use the specifications listed in Table 4–1 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmental controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

To help prevent a patient from feeling uncomfortably warm during a scan, make sure the magnet room temperature does not exceed 69.8°F (21°C) maximum.

TABLE 4–1
TEMPERATURE AND HUMIDITY SPECIFICATIONS

AREA	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	HUMIDITY (%)	HUMIDITY CHANGE (%/Hr)	MAX. ROOM GRADIENT °F (°C)
Equipment Room at Inlet to Equipment	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)**
Magnet Room	59–69.8 (15–21)	5 (3)	30–60*	5	5 (3)
Operator's Control Room	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)

Note * Non–condensing humidity with 50% nominal at 65°F (18.3°C).
** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.

4–2–2 Outdoor TSCC & TGWC Operating Environment

The Outdoor TSCC and Outdoor TGWC are designed to be located external to the building and operate in environments meeting the following specifications.

- Operating Ambient Temperature: –22°F (–30°C) to 122°F (50°C)
- Operating Humidity: 5–100%

4–3 AIR COOLING REQUIREMENTS

Table 4–2 contains the heat output of the equipment listed in the typical site location. These values do not include people, lights and non–MR equipment. Actual site average values will vary depending on system use (ie. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room.

Requirements

Physical placement of the air conditioning equipment (compressor, etc.) is an important factor due to the homogeneous field requirements of the magnet. Therefore, it is important this equipment be located outside the 10 gauss line. Refer to Section 3, MAGNETIC FIELD CONSIDERATIONS, for plot of gauss lines.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4–2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal and Blower Box air intake for patient cooling.

Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner. Due to the large variation in heat loads, the compressors should be equipped with unloaders or hot gas bypass to prevent moisture stripping of the evaporator coils.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch high raised flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets.

4–3 AIR COOLING REQUIREMENTS (Continued)

TABLE 4–2
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	MAGNET ROOM *See Note 1 below*		EQUIPMENT ROOM *See Note 2 below*		OPERATOR/CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
RF/Gradient Body Coil Assembly, Magnet Enclosure Equipment	8189	2400				
Patient Blower Box	1366	400				
Main Disconnect			900	264		
1.5T SRFD II Cabinet			18,423	5400		
ACGD/PDU Cabinet			34,120	10,000		
System Cabinet			6140	1800		
Twin Accessory Cabinet			2354	690		
TSCC Configurations for Magnet & Body Coil water cooling:						
• Outdoor Air Cooled Remote Control Panel (See Note 3 & 5)			34	10		
• Water Cooled (See Note 3 & 6)			12,000	3517		
• Indoor Air Cooled (See Note 3)			120,000	35,140		
TGWC Configurations for only Body Coil water cooling (requires site cooling for Shield Cooler Compressor):						
• Outdoor Air Cooled Remote Control Panel (See Note 4 & 5)			34	10		
• Water Cooled TGWC (See Note 4 & 6) RCP for water cooled TGWC			10,240 34	3000 10		
• Indoor Air Cooled (See Note 4)			61,000	17,880		
Shield/Cryo Cooler Compressor			Heat dissipation to air neglible			
Magnet Monitor			205	60		
Operator Workspace (See Note 7) with LCD Color Monitor					4095	1200
Magnet Monitor UPS* & Modem			450	132		
*MNS Cabinet			2730	800		

Note
- * Optional equipment.
- † Maximum heat output is defined for temperature and humidity as defined in Table 4–1.
1. Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room specifications as listed in Section 4–2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Blower Box intake which contain a patient cooling fan.
2. **FOR EQUIPMENT ROOM ONLY: The air cooling load averaged over a working day (~12 hours) is typically 1/2 of the maximum value.**
3. The TSCC provides water cooling for the Shield/Cryo Cooler Compressor Cabinet (MS5) and the Gradient Coil.
4. The TGWC provides water cooling for the Gradient Coil only. **When a TGWC is selected then the customer/site must still provide water cooling for the Shield/Cryo Cooler Compressor Cabinet.**
5. The Outdoor Air Cooled TSCC and Outdoor Air Cooled TGWC are designed to be installed external to building and exhaust heat to outside air. A Remote Control Box is located in the Equipment Room.
6. The water cooled TSCC or TGWC require customer provided chilled water cooling in addition to the air cooling, refer to Section 4–4 Water Cooling Requirements for details.
7. Operator Workspace equipment includes the following: LCD Color Monitor, Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, Keyboard, and interface modules mounted to Workspace Table

(Continued)

4–3 AIR COOLING REQUIREMENTS (Continued)

TABLE 4–3
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	MAGNET ROOM *See Note 1 below*		EQUIPMENT ROOM *See Note 2 below*		OPERATOR/CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
*Advantage Workstation Option					See Note 8	
*BrainWave Audio Console					136	40
*BrainWave Cabinet			5118	1500		
*BrainWave Cart	1638	480				

Note * Optional equipment.
8 Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation.*

4–4 WATER COOLING REQUIREMENTS

4–4–1 Gradient Coil Temporary Backup Water Cooling Recommendation

There are no options available to support temporary backup water cooling for the TRM Gradient Coil. Water cooling must be provided by the cooling cabinet (TSCC or TGWC) supplied with the system to prevent contamination/damage to the coil and for proper image quality.

4–4–2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation

Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet (MS5). The backup cooling design can utilize open loop city water only as temporary backup during loss of the closed loop water cooling from the TSCC or customer provided water cooling to the Shield/Cryo Cooler Compressor. Long term open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching that can take place which will eventually contribute to failure. Water system capacity must be selected to make sure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Cooler Compressor Cabinet, refer to Tables 4–4 and 4–5 and Illustration 4–1 for water cooling specifications.

CAUTION

Switching the Shield/Cryo Cooler Compressor inlet/outlet cooling from the TSCC to a temporary water backup supply will result in approxiamately 1.5 gallons (5.5 liters) of 50% mixture of Dowfrost HD and de–ionized water being discharged. This discharge may have site impacts due to local regulatory codes. Make sure to understand and follow local regulatory requirements when designing and implementing a temporary backup water system. The design of the change over equipment from TSCC to city water and vise–a–versa must not allow contamination of the closed loop system in the TSCC.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

Note

Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

TABLE 4–4

LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de–ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.

4–4–2 Shield/Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation (Continued)

TABLE 4–5
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPER-ATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOM-MENDED FLOW RATE gal/min (liter/min) See Note 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERATURE RISE ∆°F (∆°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28)	Minimum 29 (200) Maximum 100 (690)	Minimum 1.1 (4) See Notes 2 & 3	7.5 (0.5) [52] at minimum flow rate See Note 5	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation	28,320* (8300) for 60 Hz operation
			Maximum 2.6 (10) See Notes 2 & 3	47 (3.2) [324] at maximum flow rate See Note 5	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation	20,728 (6100) for 50 Hz operation	23222* (6700) for 50 Hz operation

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The backup cooling system design must have allowances for pressure/temperature changes due to distance located from the Shield/Cryo Cooler Compressor.

1 Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.

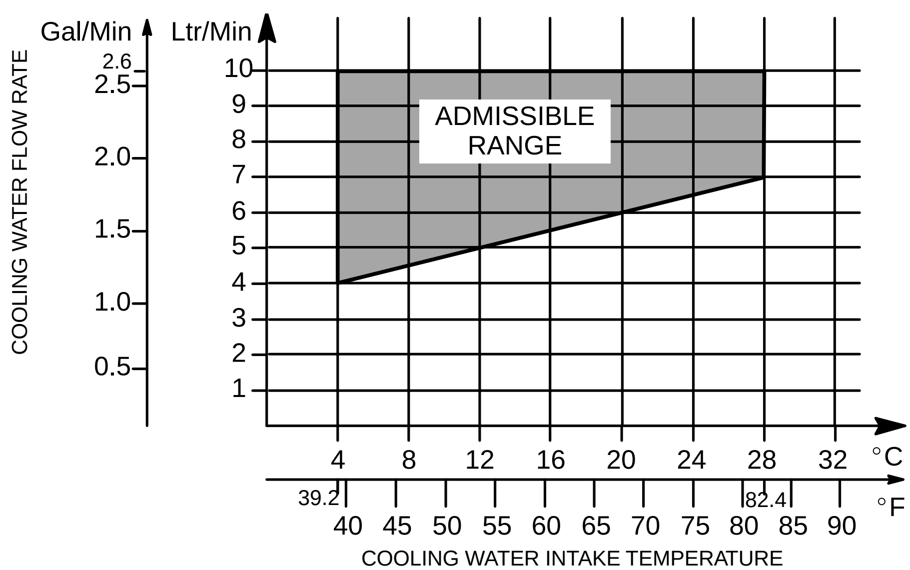
2 Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4–1 for graphic water temperature and flow rate admissible range.

3 Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.

4 Water flow rate and temperature rise value are based on water.

5 Pressure drop values based on new system, may rise due to calcification.

6 Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.



SHIELD/CRYO COOLER COMPRESSOR REQUIREMENTS FOR TEMPORARY BACKUP WATER COOLING
ILLUSTRATION 4–1

4–4–3 Water Cooled TSCC Requirements

The water cooled configuration of the Twin System Cooling Cabinet (TSCC) contains a dedicated, closed loop, liquid–to–liquid water chiller system providing water cooling for the Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil. The water cooled TSCC configuration requires customer provided water cooling, refer to Table 4–6 and Illustration 4–2 for cooling water requirements. The water cooled TSCC has 1–1/2 inch (38 mm) NPT female connections at the lower back of the TSCC unit for customer water cooling connection. Refer to Illustration 2–17 for water supply and return connection location details. The water supply and return lines must be cleared of debris (solder, dirt, etc.) before connecting to the TSCC, refer to **APPENDIX C Prerequisites for Scheduled Start–up of Chillers For Signa Twin*Speed***.

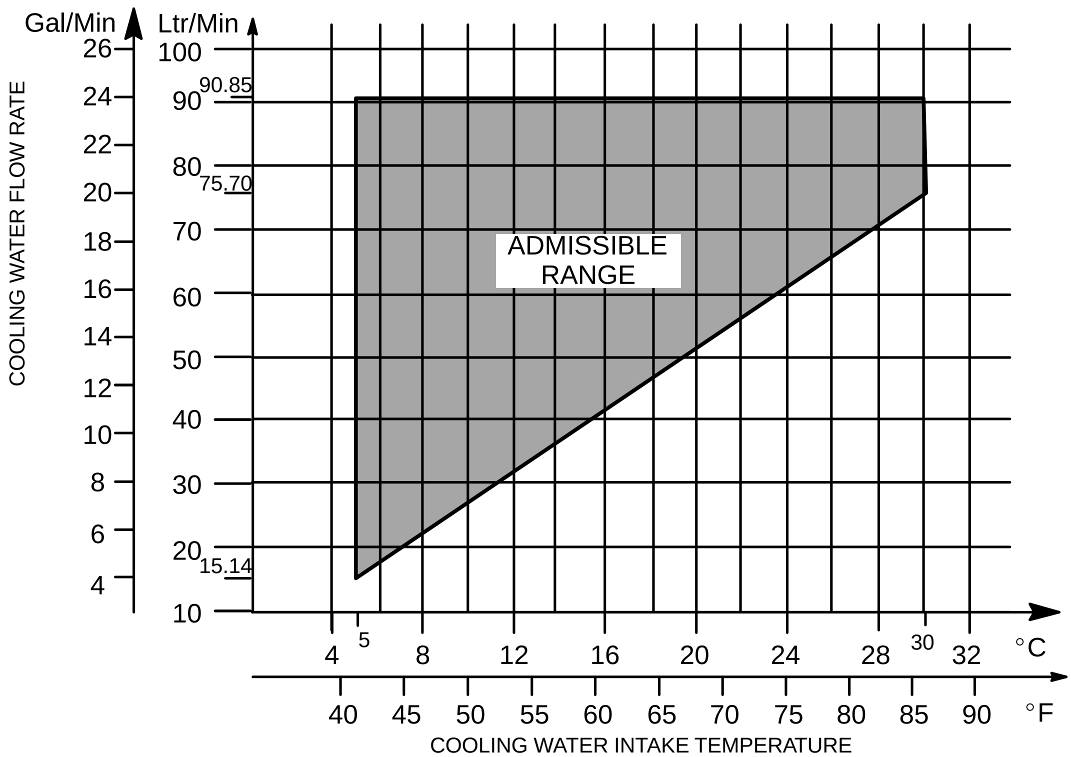
Note

Continuous water cooling is critical for the TSCC and Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week / 365 days per year to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

TABLE 4–6
WATER COOLED TSCC COOLING WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de–ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TSCC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TSCC filter at cabinet inlet.
Anti–freeze	Propylene Glycol (50% maximum concentration in water)	
Inlet Pressure	150 psi maximum	
Pressure Drop	1.6 psi (11 kPa) at minimum flowrate of 4 gpm (15.14 L/min) 5.6 psi (38.5 kPa) at maximum flowrate of 20 gpm (75.70 L/min)	
Maximum Heat Output	100,000 BTU/Hr 29,307 Watts to water 12,000 BTU/Hr 3517 Watts to air	Refer also to **Section 4–3 AIR COOLING REQUIREMENTS**

4–4–3 Water Cooled TSCC Requirements (Continued)



TSCC COOLING WATER FLOWRATE/TEMPERATURE REQUIREMENT
ILLUSTRATION 4–2

4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

The Shield/Cryo Cooler Compressor customer provided water cooling system must be closed loop design. The water cooling design can utilize open loop city water, with required filtering, only as temporary backup during loss of closed loop water cooling system. Open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching to take place which will eventually contribute to failure. Water system capacity must be selected to insure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Compressor.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week / 365 days per year to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

The Shield/Cryo Cooler Compressor closed loop system may be shared with other equipment in the MR suite. The number of sharing systems should be kept to as low as possible in order to minimize contamination and reliability problems. Flow gauges and valves are recommended at all branch lines to control distribution and allow servicing of equipment.

Refer to Tables 4–4 and 4–5 for water cooling specifications.

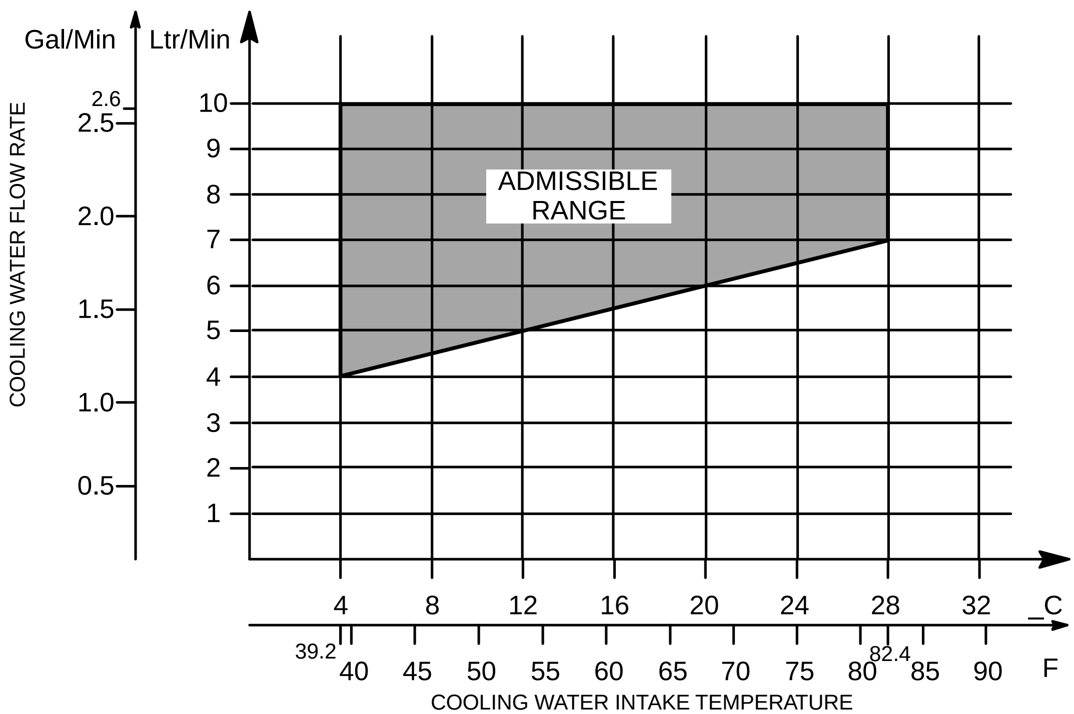
Note

The **Shield/Cryo Cooler Compressor** requires customer supplied flexible hose for mating with water cooling. Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

4–4–4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC (Continued)

TABLE 4–7
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de–ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.
Anti–freeze	Preferred minimum 25% by Volume **Maximum 50%** by volume	Use stabilized product to reduce corrosion and organic growth. Preferred minimum value to minimize organic growth.



SHIELD/CRYO COOLER COMPRESSOR COOLING WATER REQUIREMENT
ILLUSTRATION 4–3

4–4–4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TWGC (Continued)

TABLE 4–8
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPER-ATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOMMENDED FLOW RATE gal/min (liter/min) See Notes 3, 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERA-TURE RISE Δ°F (Δ°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28) See Note 9	Minimum 29 (200) Maximum 100 (690)	Minimum 1.1 (4) See Notes 2, 3 & 9	7.5 (0.5) [52] at minimum flow rate See Notes 2, 5, & 9	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation 20,728 (6100) for 50 Hz operation	28,320* (8300) for 60 Hz operation 23222* (6700) for 50 Hz operation
			Maximum 2.6 (10) See Notes 2, 3 & 9	47 (3.2) [324] at maximum flow rate See Notes 2, 5, & 9	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation		

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The cooling system design must have allowances for pressure/temperature changes due to distance from the Shield/Cryo Cooler Compressor.

1 Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.

2 Water Flowmeter Kit (46–294052G1) is available to check/monitor flow rate for Shield/Cryo Cooler Compressor. Add 2 psi to total system pressure drop if flowmeter is permanently installed in system.

3 Recommend a flowmeter be permanently installed in system, include flowmeter drop in total system pressure drop.

4 Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4–1 for graphic water temperature and flow rate admissible range.

5 Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.

6 Water flow rate and temperature rise value are based on water. Laboratory grade Ethylene Glycol pr Propylene Glycol anti–freeze may be used (do not mix Ethylene Glycol with Propylene Glycol). Preferred concentration is 65% water and 35% Ethylene Glycol to minimize organic growth. Concentration of 50/50 is acceptable with a derate of 0.8 in specific heat calculations and a 20% increase in flow.

7 Pressure drop values based on new system, may rise due to calcification.

8 Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.

9 Water Cooling Circuit typical values:
– water inlet flow 1.8 to 2.1 gal/minute (7 to 8 liter/minute)
– water inlet temperature 53.6 to 59°F (12 to 15°C)
There is a risk of damaging the Shield/Cryo Cooler Compressor with water inlet low temperature and low flow range.

4–4–5 Water Cooled TGWC Requirements

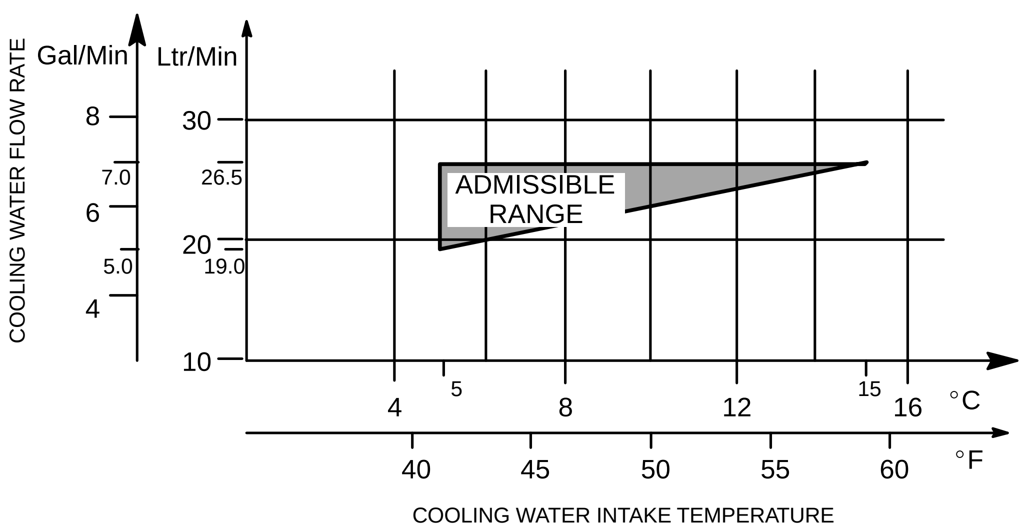
The Water Cooled configuration of the Twin Gradient Water Cooler (TGWC) contains a dedicated, closed loop, liquid–to–liquid heat exchanger system providing water cooling for the Gradient Coil ONLY. The Water Cooled TGWC configuration requires customer provided water cooling, refer to Table 4–9 for cooling water requirements. The water cooled TGWC has 0.5 inch (12.7 mm) female thread connections at the lower left side of the unit for customer water cooling connections. Refer to Illustration 2–24 for water supply and return connection location details.

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor. Refer to **Section 4–4–4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC** for customer provided water cooling requirements.

TABLE 4–9
WATER COOLED TGWC WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
Composition	Water	Maximum of 50% glycol in water mixture
pH level	6.5–8.2	GE recommends the use of de–ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TGWC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TGWC filter at cabinet inlet.
Maximum Heat Output	37,000 BTU/Hr 10,843 Watts	Refer also to Section 4–3 AIR COOLING REQUIREMENTS
Pressure Drop	9.02 PSIG (62.2 kPa) at minimum flowrate of 5.0 gpm (19.0 liters/minutes) 17.86 PSIG (123.1 kPa) at maximum flowrate of 7.0 gpm (26.5 liters/minutes)	These pressure drop values are for pure water is used. Multiply the values by 1.55 if 50/50 Propylene Glycol / water is used.
Inlet Pressure	150 psi maximum	



TGWC COOLING WATER FLOWRATE/TEMPERATURE REQUIREMENT
ILLUSTRATION 4–4

4–5 ALTITUDE

100 ft (30.5 m) below sea level to 11,808 ft (3600 m) above sea level.

4–6 LIGHTING

Direct Current (DC) lighting is recommended in the Magnet Room. Alternating Current (AC) lighting, other than fluorescent, may be used but the magnetic field of the system will significantly reduce the operating life of incandescent bulbs in AC lighting. An optional DC LIghting product is available which provides for two circuits of DC power and adjustable light levels in the Magnet Room. Refer to Table 4–10 for additional lighting requirements.

Note

Fluorescent lighting is not allowed in the Magnet Room due to the RF noise generated by the fluorescent light tubes.

TABLE 4–10
ROOM LIGHTING REQUIREMENTS

AREA	LIGHTING TYPE	LUMINOUS INTENSITY	NOTES
Magnet Room	Direct Current (DC) Incandescent or Quartz (See Notes)	• Minimum 300 lux around the front of the magnet for patient access. • Need provision to provide 300 lux above the magnet (non–magnetic, portable lighting is acceptable).	• Direct Current (DC) lighting is recommended in the Magnet Room to minimize light bulb life issues. • Electrical lines for AC lighting must meet filtering and grounding requirements in Section 7–4 **ELECTRICAL**. • Fluorescent lighting is not allowed in the Magnet Room to avoid RF noise. • Short filament length is recommended, linear lamps are not recommended because of the filament length and high incidence of filament failure. • The alternating current (AC) ripple from the DC power should be no greater than 5%. • Dimmers in the Magnet Room are not acceptable. If a low and high light level is desired, the different levels must be selectable by a switch.

4–7 ACOUSTICS

The following acoustic information is provided for site planning and architectural design activities to address acoustics to meet local regulations and customer requirements. For more information about recommended safety procedures regarding patient exposure to MR–generated acoustic levels, see the system operator manual.

4–7–1 Background

A typical MR suite has two types of acoustic noise issues. The first is the acoustics of the room where patients and technicians are impacted by the noise of the MR system as the gradients are pulsed. The second is noise transmitted to other spaces via airborne and structure borne paths.

Airborne

The airborne transmission path entails the excitation of air within the magnet room; the resonator module consisting of the magnet, RF coil, and gradient coil generates acoustic noise similar to an intense loud speaker. The airborne noise passes through walls via any openings, i.e. small holes, cracks, HVAC ducts, and waveguides, into surrounding spaces within and possibly beyond the confinements of the building. Acoustic energy can transmit across distances of significant length.

Examples of airborne acoustics issues may include the following (not limited to only these):

- MR Operator exposure at Operator Workstation
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In–house library facilities

Structureborne

The structureborne transmission path is the result of mechanical excitation of the floor/building structure causing the building to vibrate. The vibration of the surfaces at surrounding spaces then radiates as acoustic noise. Acoustic energy can transmit across distances of significant length.

Note

Less than 5% of installed base sites have experienced structureborne acoustic issues.

Examples of structureborne acoustics issues may include the following (not limited to only these):

- Areas directly above or below the Magnet Room, may not always be an issue
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In–house library facilities

4–7–2 System Acoustic Noise Levels

Any GE factory–installed protocol can be modified by operators, which can increase or decrease acoustic SPL (Sound Pressure Level); or operators may create their own protocol which could produce a higher or lower acoustic SPL as stated under **Operating Conditions** Condition 1 below. Typical scans generate acoustic levels as stated under **Operating Conditions** Condition 2 below. In addition, the exposure times are completely under operator control. Consequently, hearing protection is required for all people in the Magnet Room during scans to prevent hearing impairment, acoustic levels may exceed 99dBA. Again, for more information about recommended safety procedures regarding patient exposure to MR–generated acoustic noise, see the MR system Operator Manual.

Ambient Conditions

To reduce any background noise due to cabinet blowers, etc., acoustical ceilings, walls, and floors are recommended. The following are typical noise level readings:

- Operator Area . . . . 55 dBA
- Equipment Room . . . . 75 dBA
- Outdoor TSCC (Twin System Cooling Cabinet) . . . . 75 dBA
- Outdoor TGWC (Twin Gradient Water Cooler) . . . . 75 dBA

Operating Conditions

Condition 1

MR scanners under “worst–case” operating conditions, could generate acoustic levels (as measured at the magnet iso–center) as follows:

Average SPL 115 dBA SPL= Sound Pressure Level
Peak 120 dB
Frequency Range 20 to 20k Hz

Condition 2

MR scanners for many typical clinical scanning scenarios though, generate acoustic levels (as measured at the magnet iso–center) somewhat lower as follows:

Average SPL 85 to 95 dBA
Peak 105 to 112 dB
Frequency Range 20 to 20k Hz

As recent history has shown an evolution towards more powerful (and hence louder) gradient subsystems, architects should consider the acoustic levels stated in the “worst case” Condition 1, mentioned above. Note that high–field Signa systems have the ability to run scanning protocols which can generate acoustic levels over the entire human perceptible frequency range (20 to 20k Hz), therefore attenuation over this entire range must be considered for site design.

4–7–3 Design Guidelines

Magnet Room

Noise generated by the MR system is inherent to the operation of the system, refer to Section 4–7–2 System Acoustic Noise Levels. The sound quality (human perception) within the Magnet Room can be modified by including sound absorbing materials to make the room sound more subdued and less harsh. The measured sound levels via a sound level meter will not change. However, the measured sound levels can be reduced only when the sound level generated by the MR System is reduced.

Sound quality improvements can be achieved by the following:

- Use ceiling tiles with fiberglass panels having a 2 inch (51mm) thickness set into the standard T–bar grid system.
- Adding fiberglass panels to the side walls covering approximately 20% of the side wall surface area. The panels should focus on covering the top half of the side walls. Panels could take many different and decorative shapes to improve the sterile look of the rooms. Typically panels might be on the order of 4ft x 6ft (1.2m x 1.8m) with a thickness of 4 inches (102mm) or equivalent. Panels shape could vary to produce mosaic effects to meet the customer preference. Any decorative materials used to cover the wall panels must be porous so that sound waves can pass through with ease. In principle, a person should be able to breath through the material with ease. Fire retardant cloth should be used. The NRC (Noise Reduction Coefficient) of the panels should be 0.95 or better when mounted against a hard surface such as drywall or concrete.

Inter–Spacial Areas

Acoustic Noise Control to mitigate noise from being transmitted to other spaces often amounts to paying attention to small details while working with ordinary construction materials. The key objectives are to eliminate all cracks and gaps in the wall construction while making sure that the doors, walls, floor, and ceiling have adequate transmission loss via mass or special double wall construction along with good fitting massive doors.

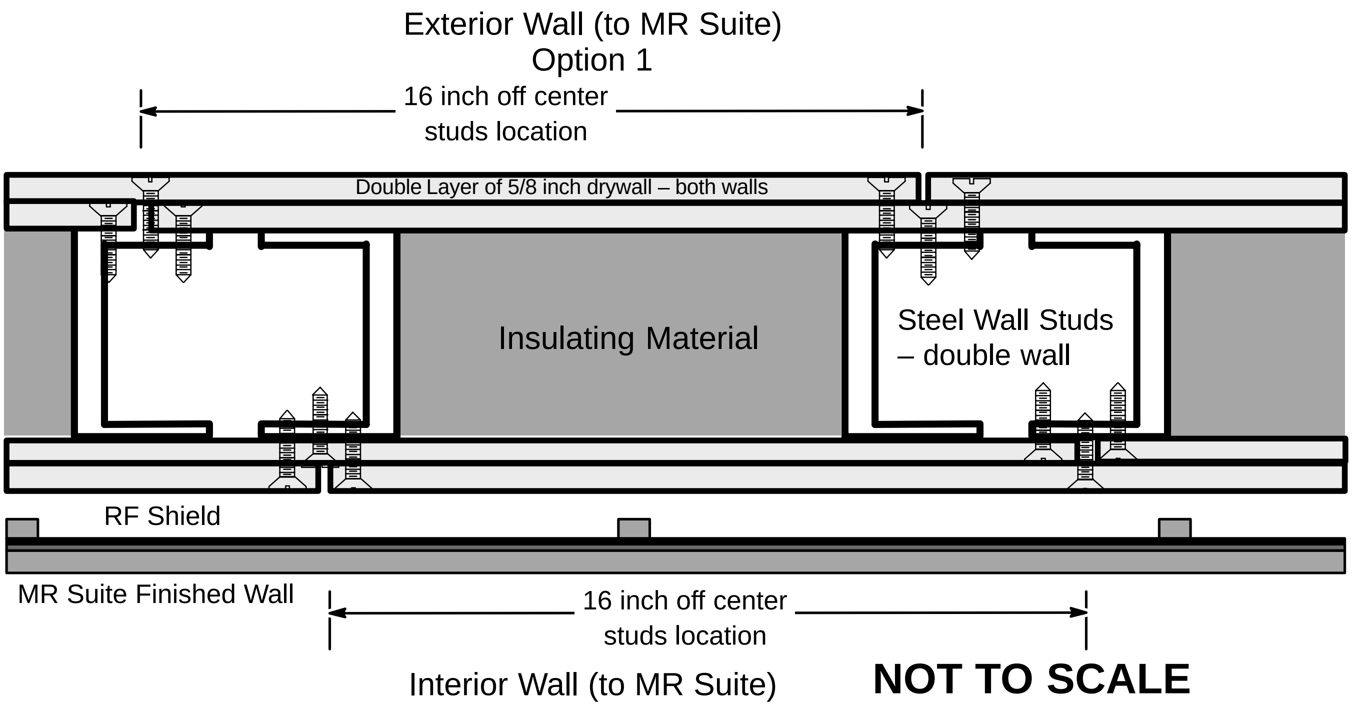
The entire Magnet must be surrounded by walls with substantial mass and/or double wall construction so that noise is contained in the room and not allowed to pass through into nearby spaces. Wall junctions must be sealed with acoustical sealant so that noise waves to do not escape from the room. In principle, if the room were filled with smoke and under a positive pressure, no smoke would leak from the room.

<u>Wall Construction</u>

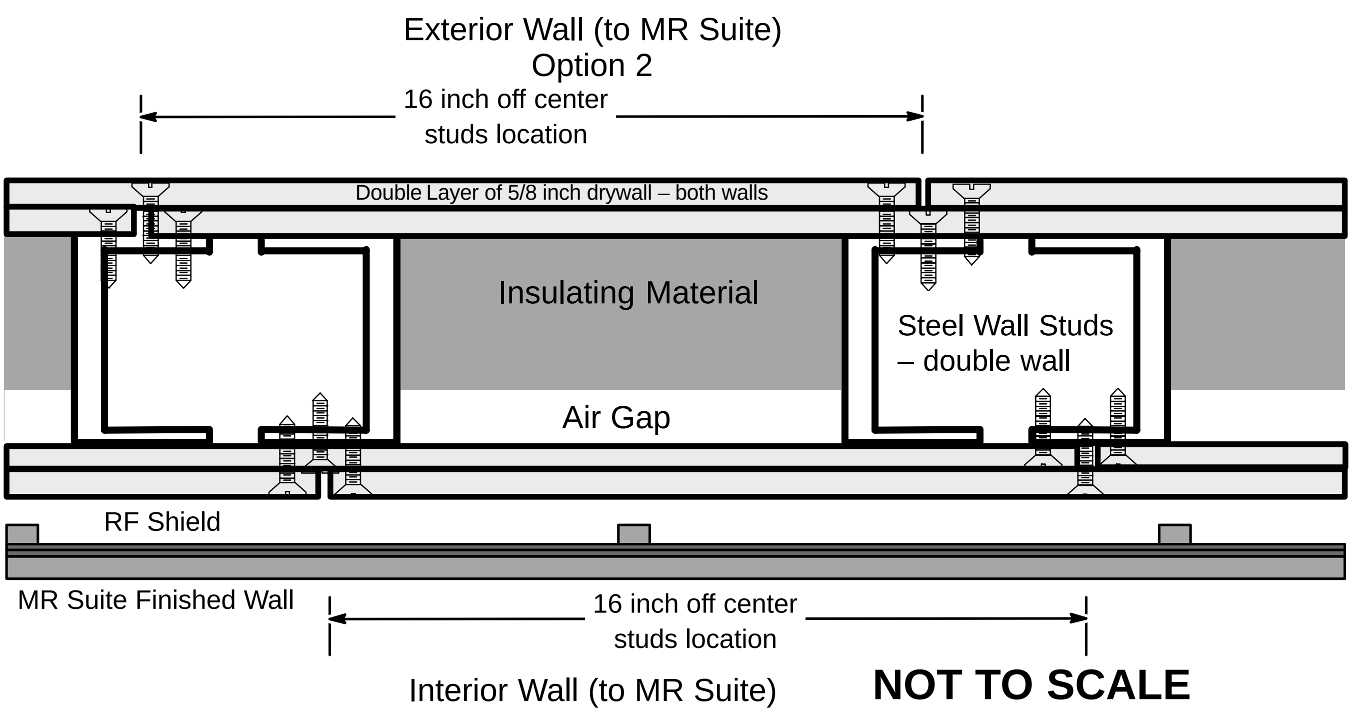
Wall Construction will entail ordinary building materials in a careful configuration.

- The preferred wall would have an ASTM STC 50 construction which entails the use of standard wall construction of steel studs (typically 3–5/8 inch (92 mm)) with 2 layers of Type X drywall (typically 5/8 (16 mm)) on each side totaling 4 layers and fiberglass batt in the stud cavity. All drywall must be overlapped by 6 inches (152 mm) or more. Beads of (USG) Acoustical Caulking (non–hardening) would be used around the entire perimeter of the drywall. Any form of wall penetration should be avoided. Any necessary wall penetrations must be sealed using combination of Acoustical Caulking (non–hardening) and fiberglass batt material. See examples of wall construction shown in Illustrations 4–5 and 4–6.
- The top of the wall must join the ceiling/floor above so that no cracks or gaps occur. If metal pan is used on the ceiling/floor (above), then flute seals would be used to seal the gaps between the drywall and the pan. Alternately drywall can be cut out to fit into the flutes. Acoustical caulking (non–hardening) will be used to seal the remaining cracks and gaps.

4–7–3 Design Guidelines (Continued)



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 1
ILLUSTRATION 4–5



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 2
ILLUSTRATION 4–6

4–7–3 Design Guidelines (Continued)

High Bay RF Room

A high bay RF Room is a self contained RF Room which has open air space between the RF Room ceiling and the building floor above. The air space is an acoustic transmission path. Acoustic energy must be reduced to minimize this transmission of energy through this path.

In cases where the Magnet is to be installed in a high bay, it may be most effective to enclose the RF Room with its own drywall and steel stud room. The key difference being a ceiling assembly that mimics the sidewall construction to contain noise.

- Normal high STC stud walls from above would be used to support a ceiling assembly constructed of structural C channel with two layers of drywall on each side (total of 4 layers) with fiberglass batt in the cavity.
- Penetrations should be avoided via the use of surface mounted lights. HVAC and ducts passing through the ceiling, party wall or side walls would require acoustic noise attenuation in the form of inline silencers. Gaps and cracks would be sealed between the ceiling, party wall or vertical side walls and the cryogen vent plumbing. In essence the Magnet would be enclosed in a drywall “doghouse”.

Miscellaneous Plumbing, RF Windows and RF Doors

Other construction details are equally important to mitigate noise transmission to meet the intended goal.

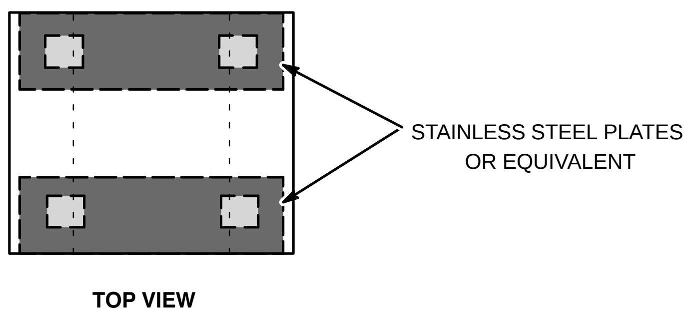
- Pipes (gas or water) and electrical conduit or Magnet Room signal cables must be sealed where they penetrate the walls or ceiling. A heavy mastic material such as Duxseal™ is appropriate.
- RF windows should be purchased as window/frame units with an STC rating obtained from laboratory testing per ASTM standards. STC 50 to 60 windows are needed. The installation must include proper sealing to avoid sound leaks.
- RF doors should be selected to provide an STC 50 to 60 to quell the noise. Contact RF Shield Room supplier for selection of RF doors that meet the local acoustic codes and site acoustic requirements. RF door seals must be selected to prevent small gaps around the door perimeter and at the door threshold. RF door seals would either require periodic replacement or a door seal that would last the life of the Magnet Room.

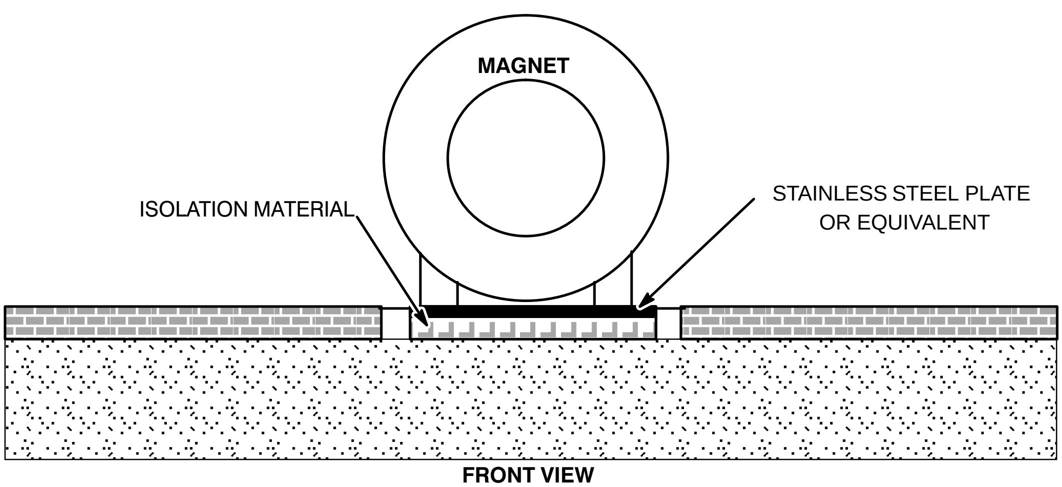
4–7–3 Design Guidelines (Continued)

Structureborne Vibration Control

Upper floor MR installations represent the largest population of sites which may experience structureborne acoustic issues. Typically, an architect, acoustic engineer, or project engineer would suggest placing the magnet on a (soft) vibration isolator. Although this classical solution may reduce the magnet generated vibration from transmitting into the building structure, this solution has produced mixed results ranging from introducing image quality issues with proper vibration attenuation to not providing enough vibration attenuation.

GE has worked with consultants who have designed/installed solution that provide (large) vibration attenuation without compromising image quality. Such solutions would bolt the magnet to stiff plates to meet the magnet bolting requirement (refer to **Section 7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM**). The plate and isolation material would be selected/sized to distribute the vibration over a large surface area and the isolation system would be selected to sufficiently attenuate vibration from the building structure. Additionally the solution natural frequency misaligns with the floor resonance. The solutions used typically had higher natural frequency then the supporting floor. Note: the magnet plate and selected isolation must ensure the magnet vertical, horizontal, and pitch/yaw modes no not compromise the vibration specification or floor levelness requirements, refer to **Section 4–15 VIBRATION** and **Section 7–6–4 Floors**. See Illustration 4–7 for an overview of such a solution concept.





OVERVIEW STRUCTUREBORNE VIBRATION CONTROL SOLUTION CONCEPT
ILLUSTRATION 4–7

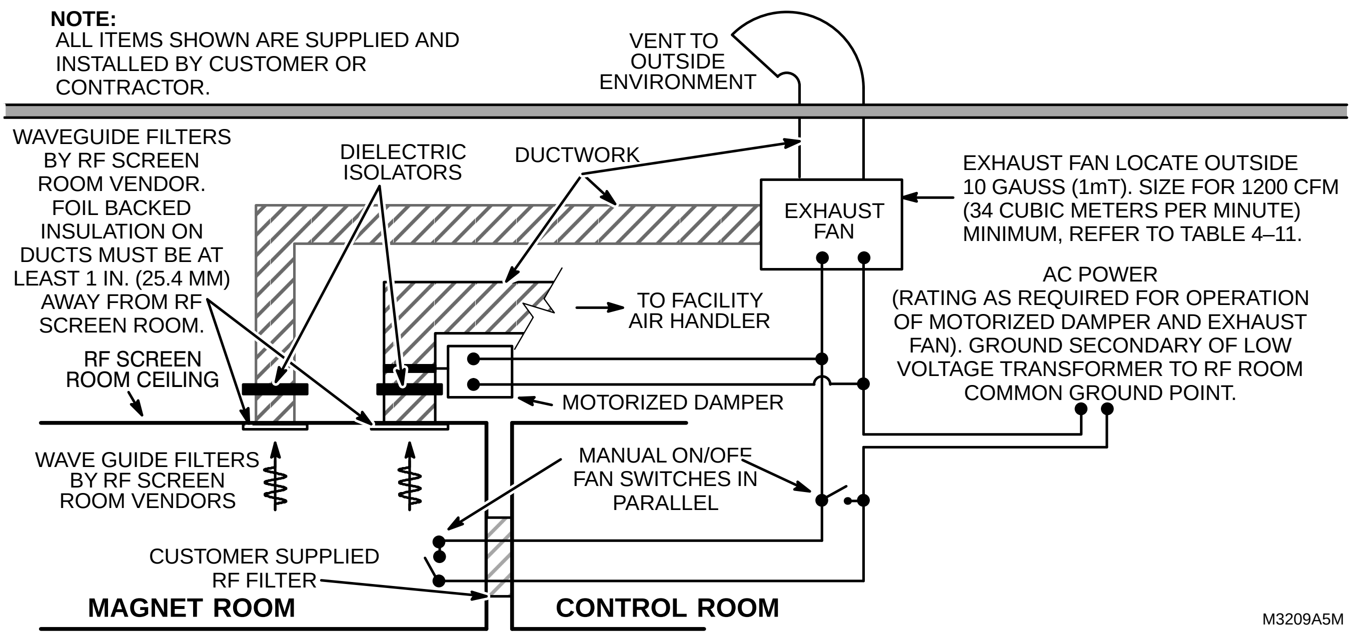
4–8 ROOM VENTILATION

Refer to Table 4–11 for ventilation specifications for the magnet and cryogen storage rooms. Refer to Section 4–12, POLLUTION, for air quality specifications.

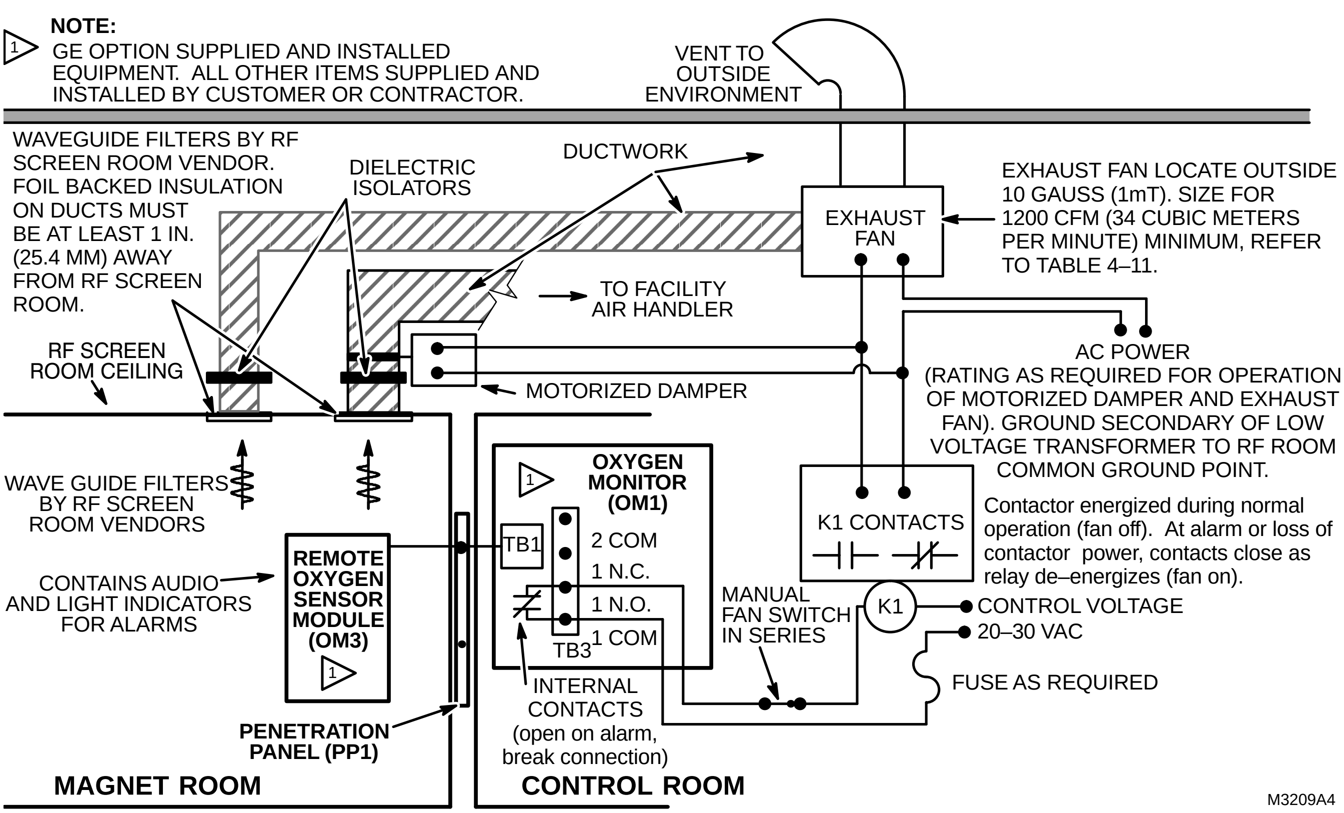
TABLE 4–11
ROOM VENTILATION REQUIREMENTS/RECOMMENDATIONS

ROOM	VENTILATION REQUIREMENTS	VENTILATION RECOMMENDATIONS
MAGNET	• Sufficient air ventilation in the magnet room must be maintained, not only for patient comfort during scans but also to maintain proper oxygen level during cryogen replenishment. • An exhaust fan to be placed above RF shielding with appropriate wave guide filtering for quick removal of helium gas if large amounts of helium disperse into magnet room. Inert gas containers, such as dewars, are not air tight. • **Magnet Room exhaust fan intake vent must be located at the highest ceiling plane near the magnet cryogen vent.** • Exhaust fan to exhaust to safe outside area and be independent of cryogenic venting.	• Minimum 5–7% of outside makeup air to be vented into magnet room. For example, with an air input rate of 1200 cubic feet per minute (CFM) (34 cubic meters per minute), there must be a minimum of 60 CFM (1.7 cubic meters per minute) (5%) of outside makeup air.
	• The exhaust fan and air inlet must be sized for an air flow of 1200 CFM (34 m^3/minute) minimum.	• The exhaust fan and air inlet provide for a minimum of 12 air exchanges per hour or 1200 CFM (34 m^3/minute), which ever is larger. • **Note**: Annual customer inspection and cleaning / maintenance of the exhaust fan system (fan, inlet grill/filter, ducts, etc.) is needed to meet the minimum airflow requirement to an outside area.
	• Two manual exhaust fan controls connected parallel, one to be located near the Operator Console and second control located in the Magnet Room. Refer to Illustration 4–8 for exhaust fan recommended set–up or Illustration 4–9 for recommended set–up with optional Oxygen Monitor. • Exhaust fan (customer supplied) to be installed and operating before magnet is moved into room. • Provide minimum 2 ft x 2 ft (0.61 m x 0.61 m) pressure equalizing waveguide vent in the magnet room **ceiling** to prevent positive or negative pressures from interfering with opening of the magnet room door per IEC 60601–2–33 6.8.3 cc.	
CRYOGEN STORAGE	• Adequate ventilation to ensure acceptable oxygen level per local regulation.	

4–8 ROOM VENTILATION (Continued)



EXHAUST FAN SET–UP
ILLUSTRATION 4–8



EXHAUST FAN SET–UP WITH OPTIONAL OXYGEN MONITOR
ILLUSTRATION 4–9

4–9 CRYOGENIC VENTING

The superconducting magnet used in the MR System contains large amount of liquid helium at 4 Kelvin (K) = –452° Fahrenheit (F) or –269° Celsius (C). During a quench, the magnet quickly boils off 90% of this liquid. Consequently, a very large volume of extremely cold helium gas must be safely vented outside of the building. Failure of the cryogenic vent can result in this cold gas entering the magnet room or another portion of the building and lowering the ambient oxygen supply to an unsafe level. Therefore it is very important that designer/contractors familiar with industrial piping systems be responsible for the vent system design and installation.

When a quench occurs, the helium gas will immediately begin to warm up and expand rapidly. As the gas escapes from the magnet, the helium gas will increase in temperature from 4.5 K to approximately 30 K. As a result, the gas will expand to 10 times of its original volume. Consequently, venting systems must never decrease in size and may increase in size (diameter) as the distance from the magnet increases.

CAUTION

THE CRYOGENIC VENT INSPECTION MUST BE COMPLETED TO FINAL EXIT OUTSIDE OF THE BUILDING INCLUDING RF SHIELD PENETRATION PRIOR TO MOVING THE MAGNET INTO THE MAGNET ROOM.
IN THE SITUATION WHERE THE MAGNET DELIVERY WILL BE THROUGH A ROOF HATCH AND THE CRYOGENIC VENT WILL BE LOCATED IN THE SAME HATCH THEN THE CRYOGENIC VENT MUST BE INSTALLED AND INSPECTED TO FINAL EXIT WITHIN 24 HOURS OF THE MAGNET DELIVERY THROUGH THE HATCH.

Note

To minimize the confusion due to various domestic and foreign venting material sizing systems, the actual dimensions of the vent material are used to describe the required vent sizes in this document. Please use the locally available venting material best matching or exceeding the requirements discussed in this document.

Note

The difference in the American terminologies of pipes and tubes is disregarded in this document.

Note

The vent size is described by the inside diameter in the pressure drop tables, because the pressure drop is calculated with the inside diameter. The vent size is described by the outside diameter for the waveguide requirements, because sealing the Ventglas® and matching the outside diameters are the critical issues.

Note

Portions of this document may not be applicable to the mobile units. Cryogenic vent design for mobile units must be submitted to GE Medical Systems MR Engineering for certification.

4–9–1 Requirements For Outside Magnet Room

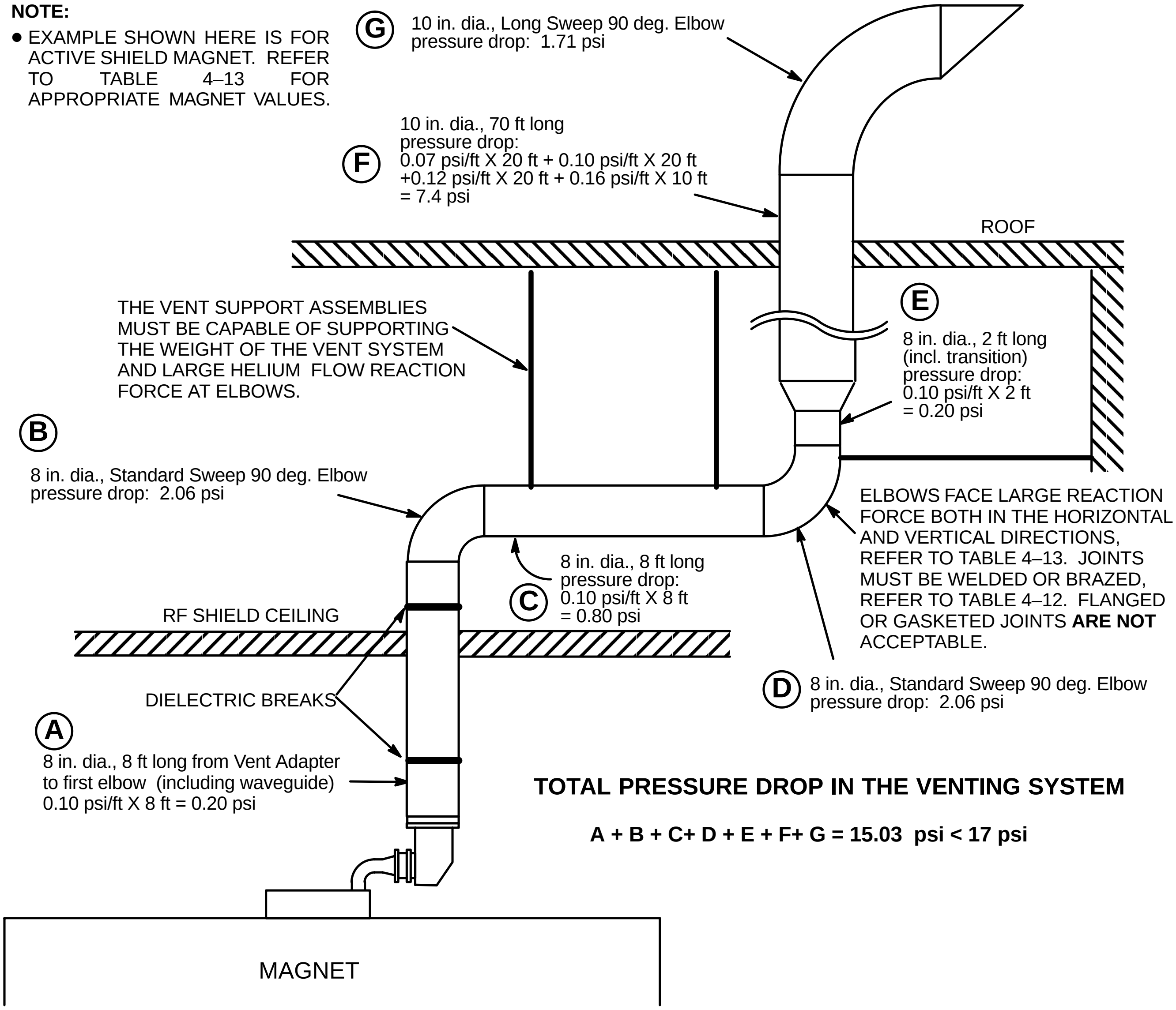
The customer's vent system designer/contractor must design and install the cryogenic vent from the RF shield waveguide to the final exit on the building roof top or outside wall. Table 4–12 contains the requirements which must be strictly adhered to in order to prevent failure of the vent system during a magnet quench.

TABLE 4–12
CRYOGENIC VENTING REQUIREMENTS FOR OUTSIDE MAGNET ROOM

PARAMETER	CRYOGENIC VENTING REQUIREMENTS
VENT SIZE	• The total pressure drop of the entire cryogenic vent system must be less than 17 psi (117.2 KPa). The calculation starts at the magnet vent interface and ends at the termination point outside of the building. Use Table 4–13 to determine proper vent sizes. See Note 1. See Illustration 4–10 for a sample calculation. • The pressure drop of the RF shield waveguide must be included in the overall calculation. Make sure the RF shielded room supplier provides a straight pipe for the waveguide with an outside diameter which matches the tubing from the magnet within $\pm$ 0.125 inch (3 mm). Refer to Section 7, RF SHIELDED ROOM, for location of vent pipe/waveguide. • The vent route must be as direct as possible and designed with a minimum number of elbows. If elbows must be used, they must be standard or long sweep types with K factors as shown in the footnotes of Table 4–13. Use of elbows with higher K factors will result in higher pressure drops than those shown in Table 4–13. Refer to Section 7, RF SHIELDED ROOM, for routing of vent pipe.
VENT CONSTRUC-TION	• Expansion/contraction joints must be provided to account for dimensional changes that will occur during a magnet quench when the vent temperature decreases from ambient to 4.5 K (–451° F or –268° C). • At the waveguide, non–metallic isolation joint must be provided to ensure the integrity of the magnet room RF shield. The joint gap must be 1.0 $\pm$ 0.25 inch (25 $\pm$ 6 mm). • All components must be able to withstand the minimum static pressure as calculated form Table 4–13 and a helium flow reaction force at temperatures from 4.5 K (–451° F or –268° C) and above. • Electro–mechanical fire dampers must not be used in the design of the vent system. Fusible link fire dampers are acceptable and require routine inspection and maintenance. • Appropriate protection must be provided for any portion of the cryogenic vent system which may drip condensation on personnel or ceiling components. • Access must be available for any portions of the cryogenic vent system made of non–metallic materials which require routine inspection and maintenance by the customer.
VENT SUPPORT	• All portions of the cryogenic vent system must be adequately supported. The vent support assemblies must be capable of supporting the weight of the vent system and the helium flow reaction force of 1850 lbs (8229 N) for 1.5T magnet at the vent elbows. • Non–metallic (i.e Ventglas) joints **must not** be used as the support for the vent system.
VENT EXIT AND TERMINATION	**WARNING!** • The exit location of the cryogenic vent system must be chosen to prevent the extremely cold exhaust gas from injuring anyone, including maintenance personnel. Exhaust opening access must be limited by warning signs or other barriers within a distance 20 feet (6.1 m) for 1.5T LCC (Active Shield) magnets. • The exhaust vent must be directed in a manner to prevent the cold gas from injuring personnel or damaging any building components. It must also be directed away from air intake vents. • For a roof top exit, GE recommends that the exhaust flow be directed horizontally using a 90° elbow having minimal pressure drop and the outlet covered with a 0.5 inch (12.7 mm) mesh screen to prevent the entry of foreign material. See Illustration 4–11. Other low pressure drop, high flow rate roof caps are acceptable. • The bottom of the 90° elbow must be at least 3 feet (0.9 meters) above the roof deck. This dimension must be higher if the location of the vent exhaust is susceptible to being blocked by drifting snow.
Note: 1 The customer's designer is responsible for selecting materials and hardware capable of safely handling the pressures and cold temperature generated within the vent at each MRI site.	

4–9–1 Requirements For Outside Magnet Room (Continued)

NOTE:
- EXAMPLE SHOWN HERE IS FOR ACTIVE SHIELD MAGNET. REFER TO TABLE 4–13 FOR APPROPRIATE MAGNET VALUES.



SAMPLE PRESSURE DROP CALCULATION OF CRYOGENIC VENTING
ILLUSTRATION 4–10

4–9–1 Requirements For Outside Magnet Room (Continued)

TABLE 4–13
CRYOGENIC VENT SYSTEM PRESSURE DROP MATRIX FOR 1.5 T LCC (ACTIVE SHIELD) MAGNET
(THIS TABLE MUST BE USED FOR CRYOGENIC VENT SYSTEM DESIGN)

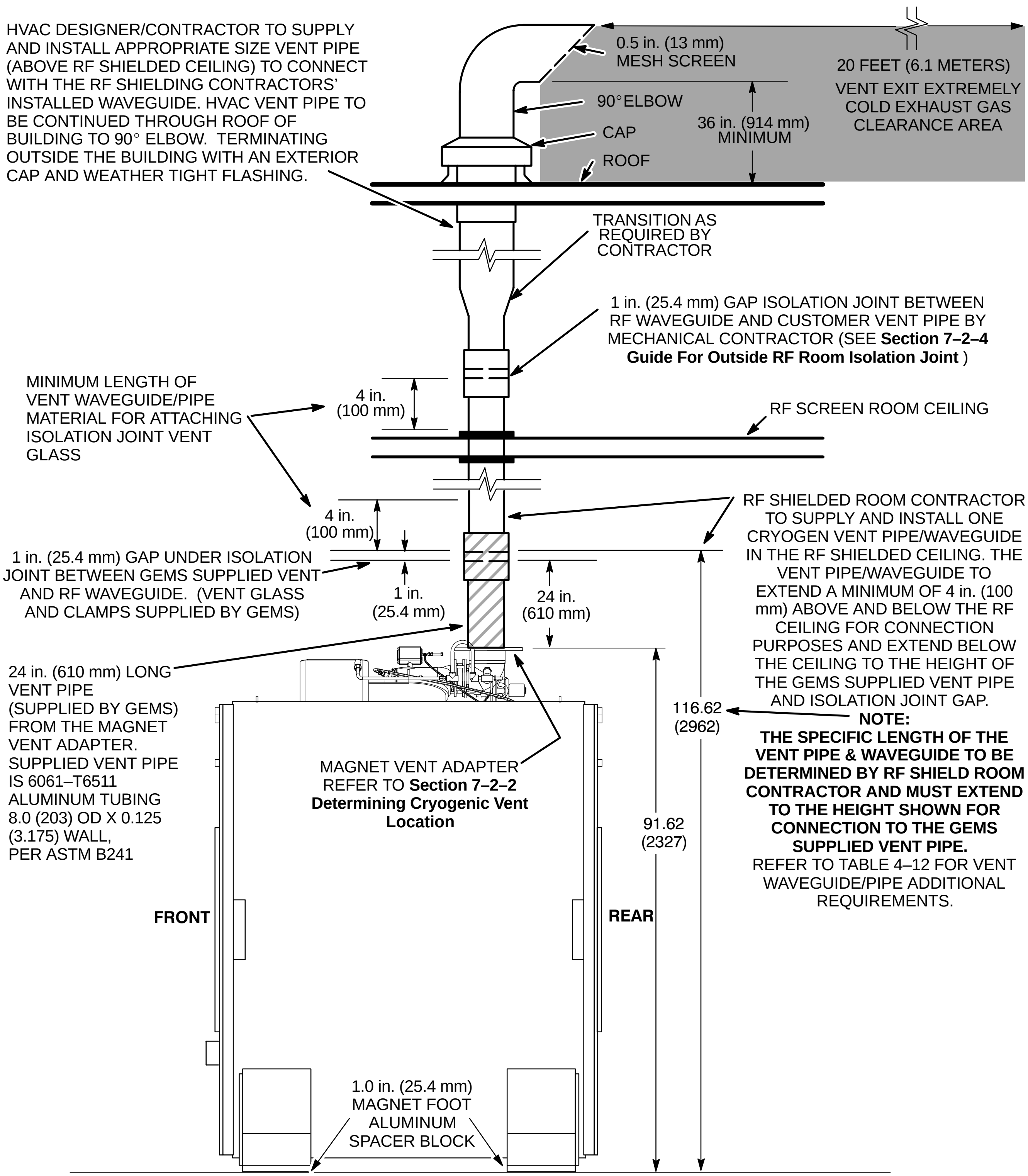
CRYOGENIC VENT SYSTEM PRESSURE DROP MATRIX FOR A 1.5 TESLA LCC (ACTIVE SHIELD) MAGNET					PRESSURE DROP PER ELBOW USED ANYWHERE WITHIN A 20 FT VENT SEGMENT							
INSIDE DIAMETER OF VENT PIPE	DISTANCE OF VENT SYSTEM COMPONENT FROM MAGNET		PRESSURE DROP FOR STRAIGHT VENT PIPE WITH SMOOTH INSIDE SURFACE		STANDARD SWEEP 45° ELBOW		STANDARD SWEEP 90° ELBOW		LONG SWEEP 45° ELBOW		LONG SWEEP 90° ELBOW	
in. (mm)	ft	(m)	psi/ft	(KPa/m)	psi	(KPa)	psi	(KPa)	psi	(KPa)	psi	(KPa)
8 (203.2)	0–20	(0–6.1)	0.10	(2.26)	1.10	(7.58)	2.06	(14.20)	0.55	(3.79)	1.03	(7.10)
	20–40	(6.1–12.2)	0.21	(4.75)	2.10	(14.48)	3.70	(25.51)	1.03	(7.10)	1.85	(12.76)
	40–60	(12.2–18.3)	0.30	(6.79)	2.88	(19.86)	5.21	(35.92)	1.44	(9.93)	2.60	(17.92)
	60–80	(18.3–24.4)	0.38	(8.60)	3.70	(25.51)	6.71	(46.27)	1.85	(12.76)	3.36	(23.17)
	80–100	(24.4–30.5)	0.47	(10.63)	4.52	(31.17)	8.22	(56.68)	2.26	(15.58)	4.11	(28.34)
10 (254.0)	0–20	(0–6.1)	0.03	(0.68)	0.55	(3.79)	0.82	(5.65)	0.27	(1.86)	0.41	(2.83)
	20–40	(6.1–12.2)	0.07	(1.58)	0.82	(5.65)	1.51	(10.41)	0.41	(2.83)	0.75	(5.17)
	40–60	(12.2–18.3)	0.10	(2.26)	1.23	(8.48)	2.19	(15.10)	0.62	(4.27)	1.10	(7.58)
	60–80	(18.3–24.4)	0.12	(2.71)	1.51	(10.41)	2.74	(18.89)	0.75	(5.17)	1.37	(9.45)
	80–100	(24.4–30.5)	0.16	(3.62)	1.92	(13.24)	3.43	(23.65)	0.96	(6.62)	1.71	(11.79)
12 (304.8)	0–20	(0–6.1)	0.013	(0.29)	0.27	(1.86)	0.41	(2.83)	0.14	(0.97)	0.21	(1.45)
	20–40	(6.1–12.2)	0.027	(0.61)	0.41	(2.83)	0.82	(5.65)	0.21	(1.45)	0.41	(2.83)
	40–60	(12.2–18.3)	0.041	(0.93)	0.55	(3.79)	1.10	(7.58)	0.27	(1.86)	0.55	(3.79)
	60–80	(18.3–24.4)	0.054	(1.22)	0.69	(4.76)	1.37	(9.45)	0.34	(2.34)	0.69	(4.76)
	80–100	(24.4–30.5)	0.069	(1.56)	0.96	(6.62)	1.51	(10.41)	0.48	(3.31)	0.75	(5.17)
	100–120	(30.5–36.6)	0.08	(1.81)	1.09	(7.52)	1.77	(12.20)	0.55	(3.79)	0.88	(6.07)
	120–140	(36.6–42.7)	0.10	(2.26)	1.27	(8.76)	2.07	(14.27)	0.63	(4.34)	1.04	(7.17)
	140–160	(42.7–48.8)	0.11	(2.49)	1.43	(9.86)	2.36	(16.27)	0.72	(4.96)	1.19	(8.20)
	160–180	(48.8–54.9)	0.12	(2.71)	1.60	(11.03)	2.53	(17.44)	0.80	(5.52)	1.27	(8.76)
	180–200	(54.9–61.0)	0.17	(3.85)	1.75	(12.07)	2.93	(20.20)	0.88	(6.07)	1.47	(10.14)
14 (355.6)	0–20	(0–6.1)	0.008	(0.055)	0.20	(1.38)	0.301	(2.08)	0.102	(0.70)	0.15	(1.03)
	20–40	(6.1–12.2)	0.017	(0.12)	0.30	(2.07)	0.602	(4.15)	0.154	(1.06)	0.30	(2.07)
	40–60	(12.2–18.3)	0.026	(0.18)	0.40	(2.76)	0.808	(5.57)	0.198	(1.37)	0.40	(2.76)
	60–80	(18.3–24.4)	0.034	(0.23)	0.51	(3.52)	1.01	(6.96)	0.250	(1.72)	0.51	(3.52)
	80–100	(24.4–30.5)	0.043	(0.30)	0.71	(4.90)	1.11	(7.65)	0.353	(2.43)	0.55	(3.79)
	100–120	(30.5–36.6)	0.050	(0.34)	0.80	(5.52)	1.30	(8.96)	0.40	(2.76)	0.64	(4.41)
	120–140	(36.6–42.7)	0.063	(0.43)	0.933	(6.43)	1.52	(10.48)	0.46	(3.17)	0.76	(5.24)
	140–160	(42.7–48.8)	0.069	(0.48)	1.05	(7.24)	1.73	(11.93)	0.52	(3.59)	0.87	(6.00)
	160–180	(48.8–54.9)	0.076	(0.52)	1.18	(8.14)	1.85	(12.76)	0.59	(4.07)	0.93	(6.41)
	180–200	(54.9–61.0)	0.11	(0.76)	1.29	(8.89)	2.15	(14.82)	0.64	(4.41)	1.08	(7.45)
16 (406.4)	0–20	(0–6.1)	0.0053	(0.037)	0.153	(1.05)	0.230	(1.59)	0.078	(0.54)	0.115	(0.79)
	20–40	(6.1–12.2)	0.013	(0.09)	0.229	(1.58)	0.460	(3.17)	0.118	(0.81)	0.229	(1.58)
	40–60	(12.2–18.3)	0.020	(0.14)	0.306	(2.11)	0.618	(4.26)	0.152	(1.05)	0.306	(2.11)
	60–80	(18.3–24.4)	0.026	(0.18)	0.390	(2.69)	0.773	(5.33)	0.191	(1.32)	0.390	(2.69)
	80–100	(24.4–30.5)	0.033	(0.23)	0.543	(3.74)	0.850	(5.86)	0.270	(1.86)	0.421	(2.90)
	100–120	(30.5–36.6)	0.038	(0.26)	0.613	(4.23)	0.995	(6.86)	0.310	(2.14)	0.490	(3.38)
	120–140	(36.6–42.7)	0.048	(0.33)	0.714	(4.92)	1.16	(8.00)	0.352	(2.43)	0.581	(4.01)
	140–160	(42.7–48.8)	0.052	(0.36)	0.803	(5.54)	1.32	(9.10)	0.398	(2.74)	0.666	(4.59)
	160–180	(48.8–54.9)	0.058	(0.40)	0.903	(6.23)	1.42	(9.79)	0.451	(3.11)	0.712	(4.91)
	180–200	(54.9–61.0)	0.084	(0.56)	0.987	(6.81)	1.64	(11.31)	0.490	(3.38)	0.826	(5.70)

Note 1 : Elbows with angles greater than 90° **must not** be used.

Note 2 : The table data is based on the following:

a. Initial flow conditions at magnet interface
b. Gas temperature starting at 4.5 Kelvin (–452° F or –268° C).
c. Helium gas flow rate of 2,737 cubic feet per minute (77.5 cubic meters per minute)
d. 45° standard sweep elbow $K = 15 F_t$
e. 90° standard sweep elbow $K = 30 F_t$
f. 45° long sweep elbow $K = 7.5 F_t$
g. 90° long sweep elbow $K = 15 F_t$

4–9–1 Requirements For Outside Magnet Room (Continued)



TYPICAL CRYOGENIC VENT DETAIL
ILLUSTRATION 4–11

4–9–2 Requirements For Inside Magnet Room

GE provides and installs a vertical 8 in. (203.2 mm) outside diameter 24 in. (610 mm) long cryogenic vent tube within the magnet room straight up from the magnet in line with the waveguide in the RF Shield. The customer is responsible to provide additional vent tube within the magnet room to meet requirements defined in Table 4–14. The Vent diameter is determined by magnet, refer to Section 7–2–2, Determining Cryogenic Vent Location. For other vent configurations (i.e. offset ceiling exits, wall exits and geodesic domes) the customer's contractor is responsible for the design and installation of the cryogenic vent system and vent supports within the magnet room. In these cases, a complete description of the vent size, materials/properties and routing must be sent to GE Medical Systems MR Siting & Shielding Group for final review. Table 4–14 contains the GE requirements for vent design within the magnet room.

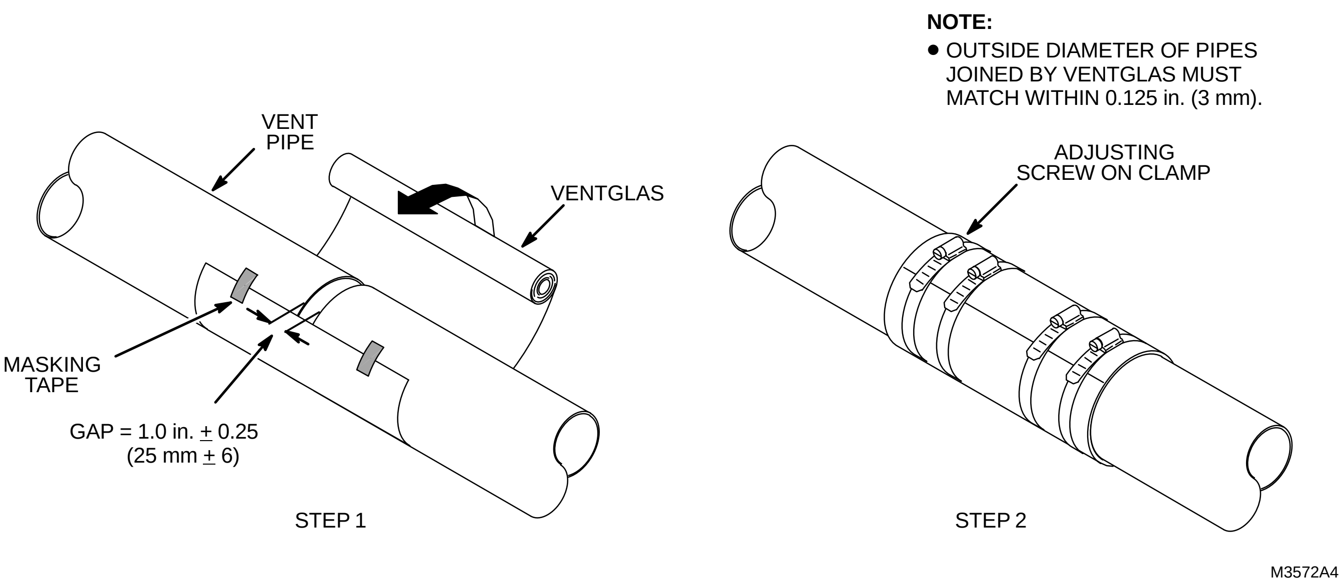
TABLE 4–14
CRYOGENIC VENTING REQUIREMENTS FOR INSIDE MAGNET ROOM

PARAMETER	CRYOGENIC VENTING REQUIREMENTS
VENT SIZE	• The total pressure drop of the entire cryogenic vent system must be less than 17 psi (117.2 KPa). The calculation starts at the magnet vent interface and ends at the termination point outside of the building. Use Table 4–13 to determine proper vent sizes. • The pressure drop of the RF shield waveguide must be included in the overall calculation. Make sure the RF shielded room supplier provides for 1.5T LCC (Active Shield) magnets 8 inch (203.2 mm) outside diameter straight pipe for the waveguide. Refer to Section 7, RF SHIELDED ROOM, for location of vent pipe/waveguide.
VENT MATERIAL	• The vent material must be Stainless Steel 304, Aluminium 6061–T6 or Copper type DWV, M or L. • Either tubes or pipes meeting the requirements may be used for venting. The vent pipe must be either seamless or have welded seams. Corrugated pipe **must not** be used. If necessary, bellows pipe of length less than 1 ft (30 cm) is allowed for thermal expansion joint. • For 1.5T magnets, the vent pipe must be capable of withstanding pressures up to 35 psi (241.4 KPa). • For the waveguide, the vent material must match the outside diameter of the magnet's vent. Refer to Table 4–15. • Venting wall thickness: SS 304 Minimum 0.035 in. (0.89 mm) Maximum 0.125 in. (3.18 mm) AL 6061–T6 Minimum 0.083 in. (2.11 mm) Maximum 0.125 in. (3.18 mm) CU DWV, M or L Minimum 0.083 in. (2.11 mm) Maximum 0.140 in. (3.56 mm)
VENT SUPPORT	• All portions of the cryogenic vent system must be adequately supported. The vent support assemblies must be capable of supporting the weight of the vent system. To ensure the integrity of the RF shield for the magnet room, electrically isolate any support assemblies which are used to support sections of venting between the magnet interface and the isolation joint at the waveguide. • For 1.5T magnets, the vent support assemblies must be capable of supporting the weight of the vent system and 1850 lb. (8229 N) helium flow reaction force at vent elbows. • Non–metallic (i.e. Ventglas) joint **must not** be used as the support for the vent system.
VENT CONSTRUC-TION	• One dielectric break in the vent system (ie. Ventglas) is required within the Magnet Room. A non–metallic isolation joint must be provided between the GEMS supplied 24 in. (610 mm) long vent pipe and the cryogen vent system ensure the integrity of the RF shield for the Magnet Room. • For vent systems with offsets, a thermal expansion joint may be necessary to allow for the thermal expansion and contraction of the vent system. • All joints except 1 non–metallic isolation (ie. Ventglas) must be welded or brazed. No clamped, sealed flanges permitted. • All isolation/thermal expansion joints except 1 non–metallic isolation (ie. Ventglas) must be able to withstand temperatures from 4.5 K (–451° F or 268° C) and above. For 1.5T magnets, these joints must also withstand pressures up to 35 psi (241.4 KPa). • GE requires Ventglas material to be used for 1 isolation joint, which may serve as a non–metallic thermal expansion joint. Rubber soil pipe couplings are not acceptable for this application. When using Ventglas material to join 2 pipes together make sure the gap between the pipes is 1.0 $\pm$ 0.25 inch (25.4 $\pm$ 6 mm) and use a continuous wrap technique during installation. See Illustration 4–12. Use 2 hose clamps on each side of the joint for securing the Ventglas to the vent pipe. • To prevent condensation during magnet ramping, the vent must be insulated with 1.5 inch (38 mm) thick flexible unicellular insulation. For appearances, all exposed insulation should be covered with white PVC jackets. **Note** Access must be available to the non–metallic isolation joint located within 24 in. (610 mm) of the Magnet Vent Adapter for annual inspection and/or maintenance by the customer.

4–9–2 Requirements For Inside Magnet Room (Continued)

TABLE 4–15
MAGNET TYPE MATRIX

MAGNET TYPES	**DIMENSIONS LxWxH inches (mm)**	**HELIUM VOLUME** gallons (liters)	**PEAK HELIUM FLOW DURING QUENCH** ft^3 per min (m^3 per min)	**MAGNET VENT OD** inches (mm)	**VENT LOCATION ON MAGNET** SEE SECTION 7–2–2
1.5T LCC (Active Shield) Magnet	69.7x81.15x91.54 (1770x2061x2325)	520.0 (1970)	2737 (77.50)	8 (203.2) Floating Flange	Illustration 7–1



VENT GLASS CONTINUOUS WRAP
ILLUSTRATION 4–12

4–10 ALARM DEVICES, WATER SENSORS AND THERMOSTATS

4–10–1 System Cabinet

For Signa Twin*Speed* with EXCITE Technology the System Cabinet has 40°C and 50°C temperature sensors. When the temperature in the System Cabinet reaches 40°C a Warning Message will be displayed on the Operator Workspace. When the temperature in the System Cabinet reaches 50°C or above a Warning Message will be displayed on the Operator Workspace and the system scanning operation will be inhibited. Normal scanning operation will be allowed when the System Cabinet temperature returns to normal. Any external alarm device for room temperature monitoring must be supplied by the customer.

For Signa Twin*Speed* the System Cabinet has one temperature sensor and one control module which sounds an alarm located in the System Cabinet when temperature reaches 94°F (34.4°C). After 3 minutes of alarm condition PDU will revert to full off condition. Any external alarm device other than mentioned above must be supplied by the customer.

4–10–2 Water Sensor Alarm and Floor Drain

It is recommended that customer supplied water sensor alarms and floor drain be located on floors where water cooled cabinets are positioned, especially under raised flooring.

4–10–3 Pneumatic Patient Alert

The Pneumatic Patient Alert Control Box provides an audible and visual alarm near the operator when the patient depresses the hand held squeeze bulb. The control box is to be mounted with consideration for ease of use by operator, remaining in sight of operator, and remaining within 5 ft. (1.5 m) of an electrical outlet. Note, an outlet on the Operator Workspace may be used. Options for control box location include mounting box vertically (on a wall or other vertical surface), horizontally (place box on a counter top, desk top, or other horizontal surface), or under a shelf within sight of operator.

4–11 AMBIENT RADIO FREQUENCY INTERFERENCE (RFI)

The MR System utilizes spatially encoded radio frequency information to create the MR image. Therefore, it is sensitive to ambient RFI. To protect the MR from ambient RFI (as well as the local environment from Magnetic Resonance RF), all sites require a 100 dB RF Shield, refer to Section 7, RF SHIELDED ROOM, for exact requirements. It is very unlikely that local signals will affect an MR System with a properly designed and installed RF Shield. During the site evaluation visit, GE notes the location of nearby sources of RFI and will advise if further information or on–site testing is required. Most sites do not require on–site testing. Listed in Table 4–16 are the recommended centerband and bandwidth frequencies to be used when measuring radio frequency interference. This table includes those frequency bands which are important for both proton imaging and spectroscopy.

TABLE 4–16
RADIO FREQUENCY SURVEY SPECIFICATIONS

ISOTOPE	1.5T SYSTEM	
	BANDCENTER MHz	BANDWIDTH Hz
^{1}H	63.86	916,138
^{19}F	60.12	981,882
^{31}P	25.88	390,296
^{23}Na	16.90	242,773
^{13}C	16.06	233,925

When required, RFI site surveys are to be performed by cycling through the preceding frequency bands and a broad band range from 10MHz–100MHz. Special emphasis, however, should be placed on the 1H band since this is used in proton imaging. The RFI site survey should be performed for a length of time necessary to determine, within a reasonable degree of certainty, that the RFI noise at the site will not exceed the 100 db attenuation provided by the RF shielded room. Note that any RFI site survey no matter how thorough, will not preclude the possibility of future or unmeasured RFI caused by new or intermittent sources.

The ambient RF noise measured should be less than 100 millivolt per meter (100 dB microvolt per meter). When a RFI site survey is required, it must be completed before the purchase and installation of the RF shielded room.

To ensure that 100 millivolt (or greater) RF noise peaks outside the bandwidths specified above do not actually extend into these bandwidths and exceed the 100 millivolt limit, adjust the resolution of the test equipment (spectrum analyzer) according to the equation:

BW (resolution) = f_0 /50

where: BW = Bandwidth (resolution)
f_0 = Center frequency (for 1H: at 1.5 Tesla 63.86 MHz)

4–12 POLLUTION

The site must be clean prior to delivery of the equipment. Although individual components have filters for optimum air filtration, care should be taken to keep air pollution to a minimum.

Since static discharge can cause system failures or affect its operation, carpeting should be of the anti–static type or treated with an anti–static solution.

When cleaning tile floors, do not use steel wool which could enter cabinet enclosures and cause internal shorts.

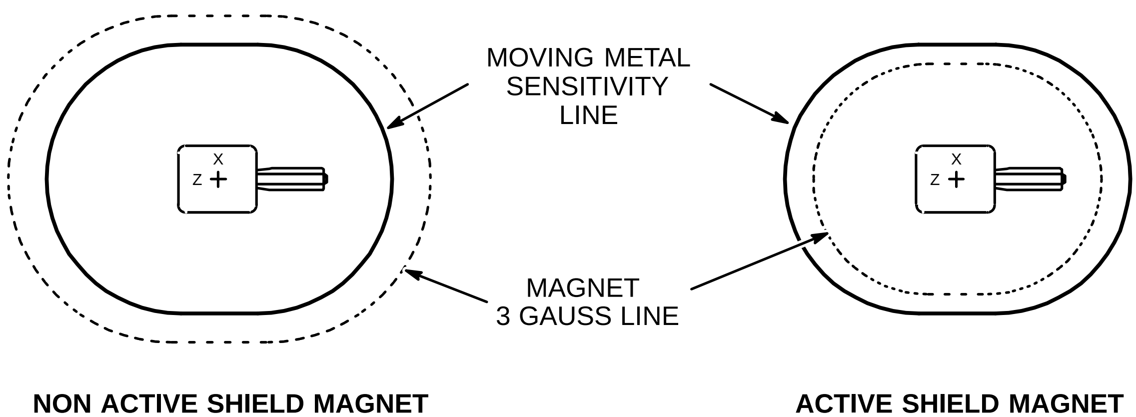
The computer/equipment area requires that the air be filtered to remove 90 percent of all particles down to 10 microns and 80 percent of all particles from 10 to 5 microns in size.

4–13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS

Note
Refer also to Table 2–1, PROXIMITY LIMITS, for additional proximity limitations.

Explanation Of Active Shield Changing Magnetic Environment (Moving Metal Phenomenon)

Prior to the introduction of Active Shield magnets it was necessary to keep moving metal objects outside of the magnet 3 gauss line so the object would not interact with the magnet's field. With Active Shield magnets having very small 3 gauss line plots, a new parameter became the limiting factor for moving metal objects proximity to the magnet. In this case, now the object's magnetic moment interacts with the magnet's field. See Illustration 4–13.



MOVING METAL SENSITIVITY LINE COMPARISON
ILLUSTRATION 4–13

Definition Of Moving Metal

Moving metal means metal objects that move inside of the moving metal sensitivity line during system scans. For example, cars being driven inside the moving metal sensitivity line are moving metal. However, if a car or a dumpster is within the moving metal sensitivity line and **does not** move during scans, then it is not an issue. Note, the 3 gauss line proximity limit for metal objects still applies to Active Shield magnets.

4–13–1 Steel Objects Categories And Requirements

Refer to Table 4–17 for LCC (Active Shield) magnet moving metal requirements and the appropriate illustrations:

- For 1.5T LCC (Active Shield) magnet see Illustrations 4–16 & 4–17.

TABLE 4–17
LCC (ACTIVE SHIELD) MAGNET MOVING METAL REQUIREMENTS

STEEL OBJECTS CATEGORY	DEFINITION OF DISTANCE LOCATION	MINIMUM DISTANCE RADIAL X AXIAL ft (m) See Note 1
Objects 100 – 400 lbs	distance from isocenter radial x axial (See Note 1)	3 Gauss line 7.87 x 13.45 (2.4 x 4.1)
CARS, MINIVANS, VANS, PICKUP TRUCKS, AMBULANCES	distance from isocenter measured to center of driving or parking lane radial x axial (See Note 1)	15.5 x 21 (4.72 x 6.40)
BUS, TRUCKS (UTILITY, DUMP, SEMI)	distance from isocenter measured to center of driving or parking lane radial x axial (See Note 1)	18.1 x 24.5 (5.52 x 7.47)
Objects > 400 lbs, ELEVATORS, TRAINS, SUBWAYS	Place a directional probe (e.g. flux gate sensor) at isocenter of proposed magnet location aligned along the Z–axis. Measure p–p magnetic field change (dc).	See Illustration 4–14 and See Example in Note 2

Note 1 Radial distances are magnet X and Y axis. Axial distances are magnet Z axis.

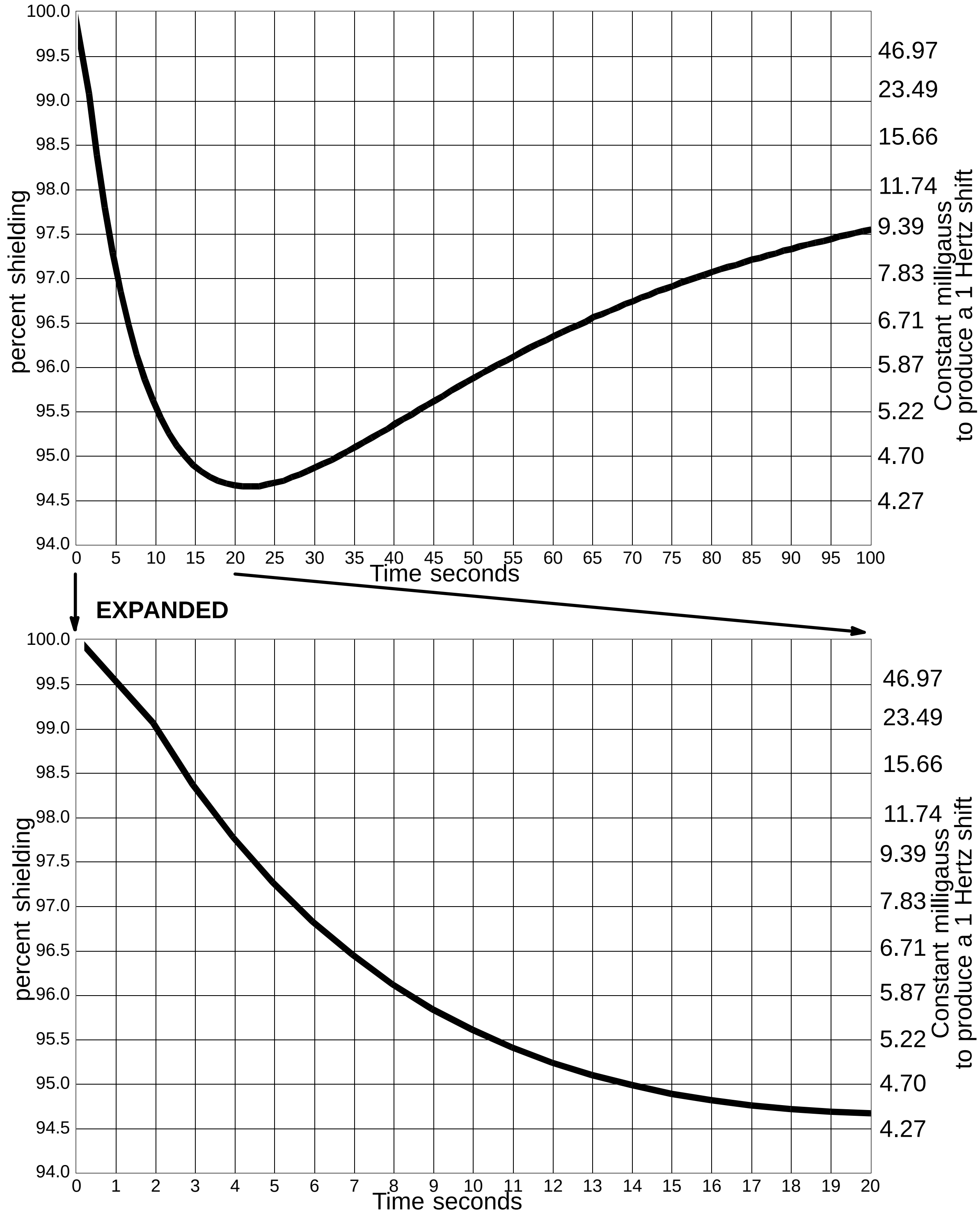
2 EXAMPLE: For Moving Metal Requirements of objects > 400 lbs category you can use the time history of the occurrence to determine what milligauss level to use.

A. If the site has elevators/counter weights near the magnet and the elevator can stop on the floors for longer than 20 seconds (which is usually the case), you need to use a shielding factor of 94.7% and a resulting peak–to–peak milligauss reading of 4.43.

B. If the site has a subway nearby and the field disturbance is less than 5 seconds, you can use a shielding factor of 97.3% and a resulting peak–to–peak milligauss reading of 8.39.

C. Therefore, be conservative and use 4.43 milligauss peak–to–peak.

4–13–1 Steel Objects Categories And Requirements (Continued)



ACTUAL AXIAL SHIELDING PERFORMANCE FOR LCC (ACTIVE SHIELD) MAGNET
ILLUSTRATION 4–14

4–13–2 Distances For AC Power Lines, Transformers And Electric Motors

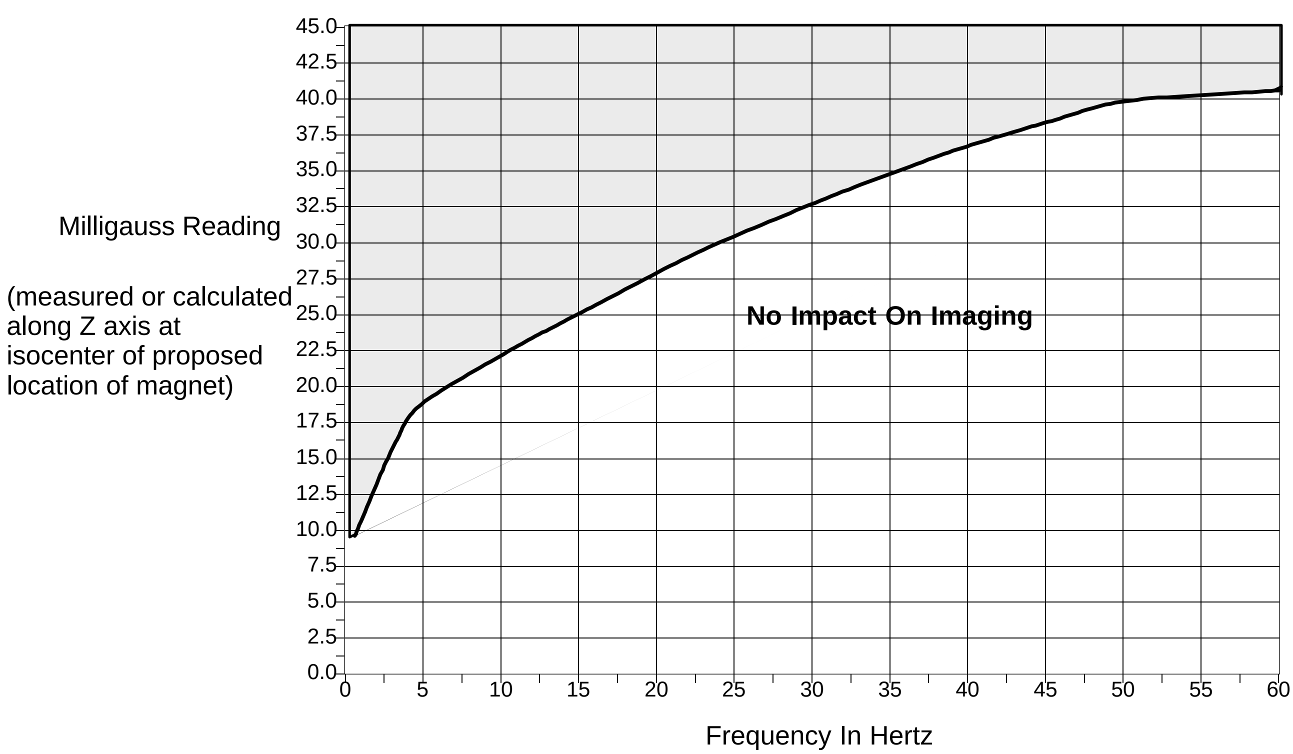
In general most AC equipment in sites is not an issue if it kept outside the 5 gauss line. If a site has large AC equipment (building mains, substations, electric trains, or subways) see Illustration 4–15 and calculate or measure the field along the Z axis at the magnet isocenter.

Electrical currents flowing in high voltage power lines, transformers, and large generators or motors near the magnet can affect the magnetic field homogeneity that is essential to the proper performance of the MR System. Although it is highly unlikely that induced magnetic fields will be a problem, possible sources of AC interference are identified by GE during the site evaluation visit. GE will analyze this information and advise if further shielding or site rearrangement are necessary.

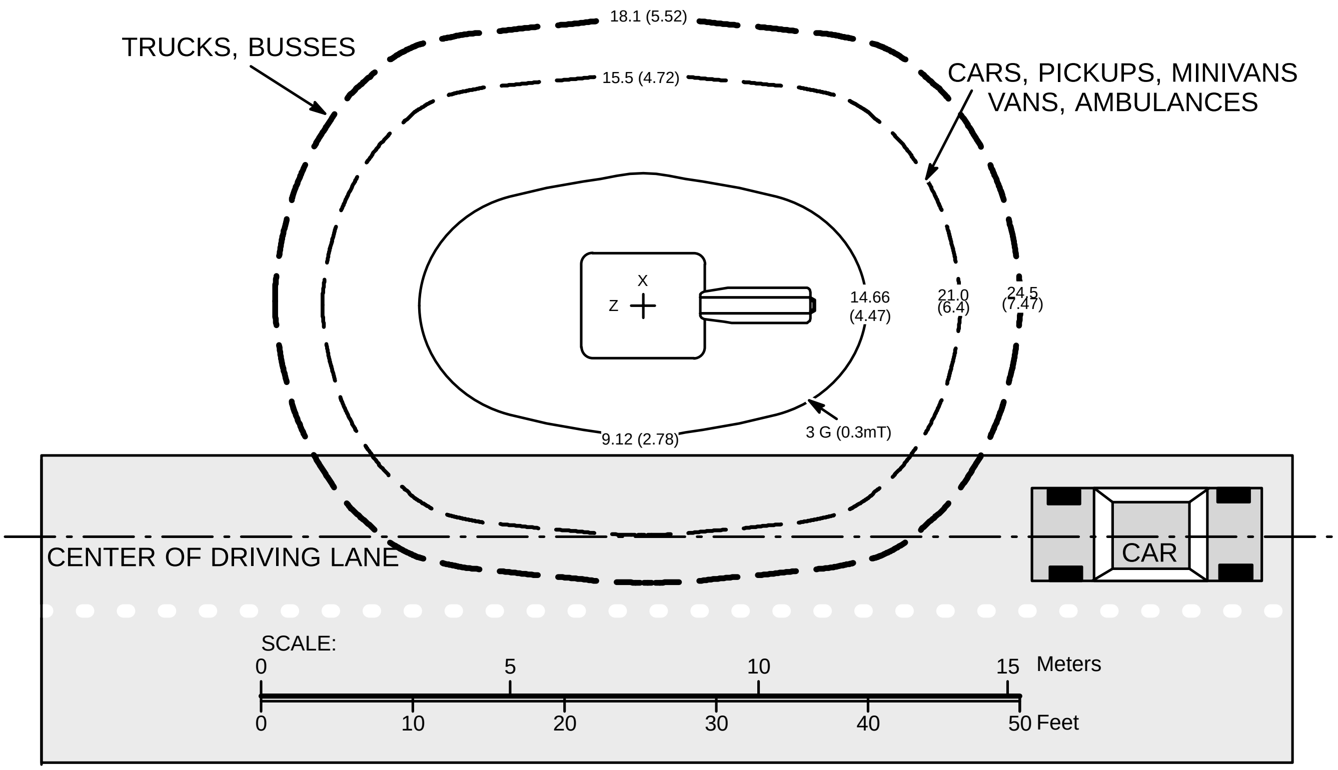
Magnetic field interference at 50 or 60 Hz must not exceed 40 milligauss RMS at the magnet location. The following equation can be used as a general guide in determining allowable current in feeder lines at a given distance from the magnet isocenter.

$$I = \frac{20X^2}{S}$$

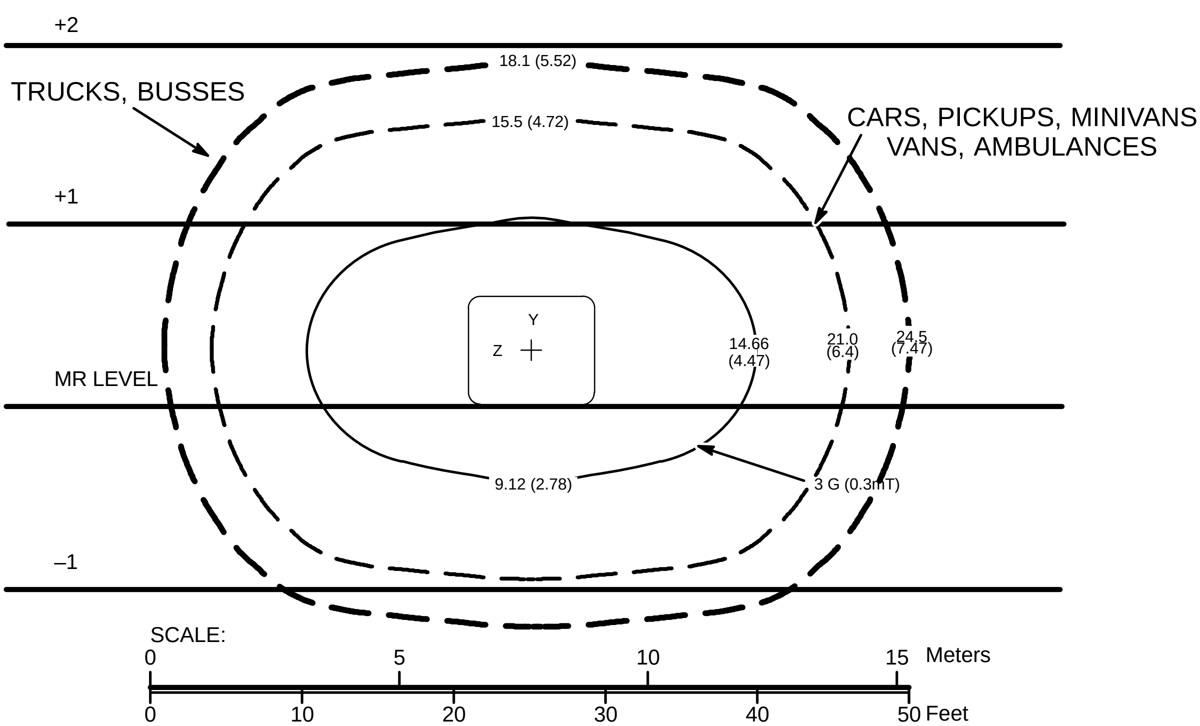
where: I = Maximum allowable RMS single phase current (in amps) or maximum allowable RMS line current (in amps) in three phase feeder lines
S = Separation (in meters) between single phase conductors or greatest separation between three phase conductors
X = Minimum distance (in meters) from the feeder lines to isocenter of the magnet



ALLOWABLE MILLIGAUSS VS LINE FREQUENCY FOR AC EQUIPMENT
ILLUSTRATION 4–15



1.5 TESLA LCC (ACTIVE SHIELD) MOVING METAL SENSITIVITY LINE PLOT (MR CENTER LEVEL)
ILLUSTRATION 4–16



1.5 TESLA LCC (ACTIVE SHIELD) MOVING METAL SENSITIVITY LINE PLOT (VERTICAL VIEW)
ILLUSTRATION 4–17

4–14 CONSTRUCTION MATERIALS

The following recommendations are for maintaining field homogeneity of the magnet. All construction must comply with local and national building codes.

Note

When welding in an MR room with system equipment installed, the return path for the welding must be in very close proximity to the welding. The close proximity is needed to make sure the welding currents do not cause damage to the system. Never use the building structure as a return path for welding.

4–14–1 Floors

For the 1.5T Active Shield magnet, the magnet room floor should be poured slab on grade with polypropelene fiber impregnated or epoxy reinforced concrete. Non–magnetic stainless steel rebar or fiberglass rebar may also be used as a reinforcing material. Steel reinforcing rods or corrugated iron sheets should be avoided especially within the 50 gauss for the 1.5T Active Shield magnets. If these materials exist at the site, or if installation of these materials is contemplated, they must be taken into account in the structural steel evaluation of the site. Refer to Section 3, MAGNETIC FIELD CONSIDERATIONS, for more information. If necessary, systems with the 1.5T Active Shield magnet can correct for *some* steel in the floor. This includes steel rebars and other steel building components within a 10 foot by 10 foot (3.1 meter by 3.1 meter) area directly below the magnet. Table 4–18 illustrates the varying limits of mass of steel in close proximity to the magnet isocenter when using normal system shimming techniques. The data is based on a square area located directly beneath the magnet and calculating an equivalent density from the total volume of existing structural steel. In most cases, an I–Beam located directly beneath the magnet is prohibited. For the 1.5T Active Shielded magnets, a single I–Beam larger than W8x40 (beam 8 inches wide and 40 lbs/ft weight) or metric HE200B (200 mm wide wings steel beam and 61 kg/linear meter), should be kept a minimum of 52 inches (1321 mm) from the magnet isocenter. In any case, the values listed Table 4–18 should not be exceeded.

TABLE 4–18
STEEL MASS PROXIMITY TO MAGNET ISOCENTER

MAGNET TYPE	DISTANCE FROM MAGNET ISOCENTER in. (mm)	DISTANCE BELOW TOP SURFACE OF FLOOR in. (mm)	LIMITS OF STEEL MASS lbs/ft^2 (kg/m^2)
1.5T LCC (Active Shield) See Note 1	42 (1067)	0 (0)	0 (0)
	45 (1143)	3 (76)	2 (9.8)
	47 (1194)	5 (127)	3 (14.7)
	52 (1321)	10 (254)	8 (39.2)
	55 (1397)	13 (330)	20 (98.0)

Note 1 If any of the steel mass proximity to magnet isocenter limits are exceeded for the LCC Magnet then the 1000 Amp Magnet Ramp Power Supply must be used to ramp the LCC magnet to field. Refer to **Section 10–5 Installation Equipment**. When the 1000 Amp Magnet Ramp Power Supply is needed, notify the GE Field Service Engineer to arrange for Power Supply delivery.

Steel rebar must not be positioned in such a manner as to interfere with anchor bolt locations for the magnet or magnet room equipment, refer to **Section 7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** and **Section 7–8 MAGNET ROOM EQUIPMENT MOUNTING**.

4–14–2 Walls, Ceilings, and Fixtures

General

Standard steel nails, screws, and other hardware are acceptable if properly secured. Any loose steel objects can be violently accelerated into the bore of the magnet. Careful thought should be given to the selection of light fixtures, cabinets, wall decoration, etc. to minimize this potential hazard. For safety, all **removable** items within the magnet room such as switch box cover plates, light fixture components, mounting screws, etc. must be non–magnetic. If you have a specific question about material, bring it to the attention of your GE Installation Specialist.

1.5T LCC (Active Shield) Magnets

Non–movable steel such as wall studs or HVAC components will produce negligible effect on the magnet.

4–14–3 Electrical conduits

Electrical conduit within the magnet room may be steel provided it is inside walls and ceilings. Note, conduit for a receptacle must be metallic. Ferromagnetic material inside the magnet room could inadvertently become a projectile.

4–14–4 Plumbing pipes and drains

Pipes and drains within the magnet room may be iron, if desired, without significant effect on the magnet homogeneity. For safety, any removable items such as faucet handles, drain covers, etc. must be non–magnetic material such as PVC, copper, or brass. Any magnetic material inside the magnet room could inadvertently become a projectile. Refer to Section 4–9, CRYOGENIC VENTING, for cryogenic vent materials requirements.

4–15 VIBRATION

4–15–1 Definition

Improvements in MR imaging technology have increased imaging capabilities. Certain MR procedures require a stable environment to achieve high resolution image quality. **Environmental site vibration has the ability to affect the magnet phase stability and ultimately image quality.** The vibration effects on image quality can be minimized early in the site planning of the MR suite.

The magnet may be sensitive to vibration in the frequency range of 0.5 to 45 Hz depending on the amplitude of the vibration. In the physical area where the MR system is located, every precaution must be taken to ensure that the vibration is minimized. In the magnet siting area, the structural stability and behavioral characteristics can be assessed. **The vibration tests outlined in Appendix A, MR SITE VIBRATION TEST GUIDELINES can be used to assess the vibration environment. Sites which currently pass the vibration stability criteria may proceed with the installation. Sites which have marginal vibration stability require source isolation or structural modifications.** Then it is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specified limits. With the vibration consultant present, local GE Field Service and/or Installation Specialist must verify the elimination/reduction of all identified sources do improve the vibration environment. GE can assist in interpreting marginal site test results and predicting the impact on system performance. However it is ultimately the customer/architect/engineer responsibility to design site solution.

To minimize the interference, the magnet should be placed on a solid floor, located as far as possible from the following vibration sources:

- parking lots
- roadways
- subways
- trains
- hallways
- hospital physical plants containing pumps, motors, air handling equipment, air conditioning units
- elevators
- heliports.

Note

Please note that other items not listed could also be potential sources of vibration.

Note

Vibration isolation is highly recommended at floor connection points of the air conditioning unit(s) to be installed for the purpose of cooling the MR Suite.

Note

Isolation of the MR magnet is not a recommended solution for reducing environmental vibration.

Vibration measurements should be made when the proposed site is located near any of the sources listed here. Measurements should be made using a spectrum analyzer capable of performing the test guidelines detailed in Appendix A, MR SITE VIBRATION TEST GUIDELINES.

4–15–2 Types of Vibration Image Quality Issues

Vibration capable of affecting MR image quality can be classified as either steady state or transient vibration. Steady state vibration typically refers to disturbances caused by rotating machinery. Examples of machinery known to have previously generated vibration image quality problems are exhaust fans, air conditioning blower units, compressors, pumps (air and water), etc. Steady state vibration would be measurable during long time intervals. Transient vibrations are typically a function of the building structure or the building foundation. Transient vibration would typically be associated with vehicular traffic, pedestrian motion, patient transport, door slamming, etc. Transient vibration would typically occur during short periods of time. The transient event would typically decay from a high vibration amplitude to lower levels in short periods of time.

Magnet Siting Requirement

Improper mechanical coupling of the magnet to the floor can in some circumstances create unique vibration characteristics. Magnets improperly mechanically coupled to the floor may experience steady state and/or transient vibration system stability issues. When the system stability tests do not meet acceptance criteria the vibration testing defined in Appendix A will help to identify if the magnet is properly mounted to the floor. Refer to Section 7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM for magnet mounting requirements.

4–15–3 Specifications For 1.5T LCC (Active Shield) Magnets

Note

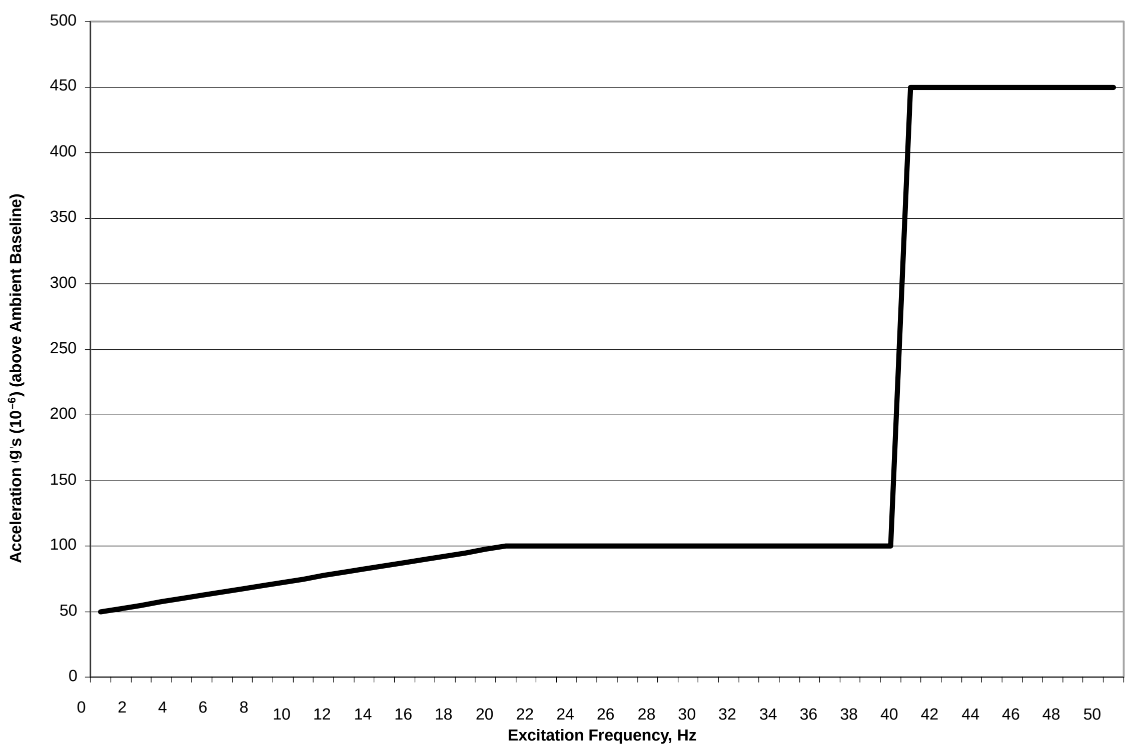
Refer to Appendix A, MR SITE VIBRATION TEST GUIDELINES for definition of ambient baseline.

Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed the following maximum single frequency components **above ambient baseline** (refer to Illustration 4–18):

- **5 x 10^{-5} g rms at 0 Hz ramping to 10 x 10^{-5} g rms at 20 Hz**
- **10 x 10^{-5} g rms 20–40 Hz**
- **45 x 10^{-5} g rms 40–50 Hz**

In order to ensure that any discrete signal represents a real mechanical vibration source, the signal must have a bandwidth that typifies dynamic system response.



LCC MAGNET STEADY STATE VIBRATION SPECIFICATIONS
ILLUSTRATION 4–18

Refer to **Appendix A – MR SITE VIBRATION TEST GUIDELINES** Illustration A–2 for an example applying the steady state specifications with ambient baseline.

4–15–4 Specifications For 1.5T LCC (Active Shield) Magnets

Transient Vibration

Note

Transient vibration analysis requires the elimination of all steady state vibration so as not to mask the transient signal.

Time history transient levels exceeding **500 micro–g, zero to peak** must be fully analyzed to assess impact to the building structure. The building (spectral) response immediately following the **500 micro–g** trigger level must not cause the site environment to exceed the **Steady State Vibration** levels defined in preceding subsection and Illustration 4–18. That is, the vibration consultant must measure and report the transient disturbance of concern. The consultant must determine (assess) the frequency, amplitude, and duration of the transient, then determine whether the disturbance will vibrate the building structure, MR RF Shielded Room, plus the magnet to amplitudes that would exceed the **Steady State Vibration** specifications, refer to preceding subsection and Illustration 4–18.

As an example; a consultant might find transient events with pulse width greater than 5 milliseconds with vibration response (decay) longer than 100 milliseconds which may suggest a need for further analysis. The vibration consultant should document in the report any transient event that may excite the building resonance (including the MR system installed weight).

SECTION 5 – POWER REQUIREMENTS

TABLE OF CONTENTS

5–1 INTRODUCTION

The MR system includes a Power Distribution module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3) which distributes power to most MR system components. Refer to Table 5–1 for required customer power specifics. Refer to Section 5–2, CRITICAL POWER REQUIREMENTS, for specifications of required facility input to the MR System.

WARNING!

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM WITH ACGD.

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Customers should carefully consider the advantages and disadvantages of raised flooring, conduits, floor ducts, and surface raceways for running cables in accordance with local codes. If used, conduits should be large enough to pass any cable and its connector through with all other cables in the conduit.

To reduce voltage regulation problems and wiring costs, minimize the cable length between the primary power source and the Power Distribution Unit. When routing cables, keep all phase conductors and ground for a circuit in the same trough. Whenever possible, keep power cables away from signal and data cables. Use separate trough or dividers in duct.

5–1 INTRODUCTION (Continued)

TABLE 5–1
REQUIRED CUSTOMER POWER

MR COMPONENT	VOLTAGE (VAC)	FREQUENCY	PHASE	MAX. AMPS	COMMENTS
GE Main Disconnect Panel (MDP) See Notes 1, 2 & 3 See Note 4 if customer MDP	480Y/277 VAC ±10% or 400Y/230 VAC ±10%	60 Hz 50 Hz	(3+GND) See Com- ments	See Note 5	Recommend input configuration: 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to PDU or inside the Main Disconnect Panel and not brought to the Power Cabinet. (See Note 6) Optional input configuration: 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Magnet Rundown Unit	100–120 or 200–240	50/60 Hz	1	1.0	Hard wired in unit.
Pneumatic Patient Alert Control Box	115 or 230	50/60 Hz	1	0.2	An outlet on the Operator Workspace Table equipment can be used.
Service Receptacle in Magnet Room	110–120 See Comments	50/60 Hz	1	2.0	Receptacle required for small power tools. Local voltage and portable transformers for voltages values.
*O_2 Monitor	110–120 or 200–240	50/60 Hz	1	3.0	Hard wired in monitor
*MRI Music System	120	60 Hz	1	1.2	Receptacle required in Control Room
*Advantage Windows	Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation.*				

Note * Optional equipment.

1 Power phase conductors, neutral (if present), and ground conductor must be routed inside the same raceway, cable tray, trench cable. or cord per National Electric Code (NEC) 2002 Articles 250.134, 300.3, 517.13 or 1999 Articles 250–134, 300–3, 517–13.

2 Signa Twin*Speed* MDP controls power to the following system equipment:
– Power Distribution Unit
– System Cooling Cabinet
– Shield/Cryo Cooler Compressor
– Magnet Monitor equipment including the Magnet Monitor, Modem, Uninterruptible Power Supply (UPS*) (optional) for Magnet Monitor, Multiplexer Box (optional).

3 MDP power circuits for Twin System Cooling Cabinet and Shield/Cryo Cooler Compressor Cabinet, and cooling for these units, are required immediately upon magnet arrival. If permanent site power is not ready, temporary power drop line and cooling must be made available. If site voltage is not any of the voltages listed above, customer must provide transformer and secondary circuit breaker to provide correct voltage and/or configuration. Refer to **Section 1–4 FACILITY OPTIONS** for listing of step up transformers options.

4 If customer provided MDP has been selected based on meeting Illustration 1–1 flowchart requisites then customer provided MDP MUST meet all MDP requirements.

5 Maximum amps dependent on voltage selected. Refer to **Section 5–2, CRITICAL POWER REQUIREMENTS** for configuration.

6 PDU Module is located in the lower portion of the ACGD/PDU Cabinet (MR3).

5–2 CRITICAL POWER REQUIREMENTS

The system Main Disconnect Panel is a Low Voltage Low Energy local control unit, with multi–point remote control capability, connected to the feeder lines that supply input power to the following:

- Power Distribution Unit (PD1)
- Shield/Cryo Cooler Compressor Cabinet
- System cooling equipment: Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC)
- Magnet Monitor equipment.

The Main Disconnect Panel is provided as part of the system per Illustration 1–1 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.

WARNING!

IF CUSTOMER PROVIDED MDP HAS BEEN SELECTED BASED ON MEETING ILLUSTRATION 1–1 FLOWCHART REQUISITES THEN CUSTOMER PROVIDED MDP MUST MEET ALL MDP REQUIREMENTS.

All work is to be done in accordance with national and local electrical codes. Refer to Section 5–3, POWER DISTRIBUTION SYSTEM, for Main Disconnect Control capability and set up.

WARNING!

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM WITH ACGD.

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

5–2 CRITICAL POWER REQUIREMENTS (Continued)

Configuration:

- Recommend input configuration 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to or inside the Main Disconnect Panel and not brought to the ACGD/PDU Cabinet.
- Optional input configuration 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.

Frequency: 50 ± 3 Hz or 60 ± 3 Hz

Regulation: 4% maximum at system maximum power demand (averaged over 5 seconds) from source to PDU (i.e. includes all feeders and transformer to utility).

Phase Balance: Difference between the highest phase line–to–line voltage and the lowest phase line–to–line voltage must not exceed 2%.

Daily Voltage Variation: ±10% from nominal under worst case line and load regulation.

PDU Voltage: 200/208/380/400/415/480 VAC ±10%

Shield/Cryo Cooler Compressor Voltage: 380/400/415 VAC 50 Hz or 460/480 VAC 60Hz

TSCC or TGWC (excludes Water Cooled TGWC powered from PDU) Voltage: 400 VAC ±10% 50 Hz or 480 VAC ±10% 60Hz

Magnet Monitor equipment Voltage: 100/120 or 200/220 VAC

Voltage Transients: Phase–to–phase voltages must be within 2% of the lowest phase–to–phase voltage. Maximum allowable transient voltage above or below nominal waveshape not to exceed 200 V at a maximum duration of 1 cycle and frequency of 10 times per hour.

Facility Zero Voltage Reference Ground:

- Main facility ground wire to be minimum 1/0 AWG copper or same size as power feeder wire, which ever is larger.
- Main facility ground wire to be insulated.
- Ground impedance to earth at power source to be 2 ohms or less.
- Main facility ground wire to be bonded at every distribution box in an approved grounding block.

5–2 CRITICAL POWER REQUIREMENTS (Continued)

Maximum Momentary Demand: The power demands for the Signa 1.5T Twin*Speed* system with ACGD are specified as a function of the duration of the power demand. Table 5–2 lists points on the curve.

The power system feeding the Signa system with ACGD must be designed to meet the specifications of less than 4% regulation when loaded at 92.4 KVA (corresponds to 5.0 second allowable consumption). For short intervals the Signa system with ACGD power demands can exceed 56.2 KVA and the line voltage delivered to the system will sag below the 4% regulation. The Signa system with ACGD is designed to tolerate these **short** voltage sags.

TABLE 5–2
SYSTEM PEAK POWER DEMAND

COOLING EQUIPMENT	**TSCC *See Note 1***			**TGWC *See Note 2***		
SYSTEM EQUIPMENT	**OUTDOOR TSCC**	**INDOOR WATER COOLED TSCC**	**INDOOR AIR COOLED TSCC**	**OUTDOOR TGWC**	**INDOOR WATER COOLED TGWC**	**INDOOR AIR COOLED TGWC**
PDU draw for 5.0 sec	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA
PDU draw for 1.0 sec	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
PDU draw for 0.1 sec	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
PDU draw for 0.04 sec ***See Note 3***	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
Magnet Monitor ***See Note 4***	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	23 KVA	20 KVA	22.7 KVA	16 KVA	0 KVA ***See Note 5***	11.6 KVA
TOTAL for 5.0 sec	~92.7 KVA	~89.7 KVA	~92.4 KVA	~85.7 KVA	~69.7 KVA	~81.3 KVA
TOTAL for 1.0 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA
TOTAL for 0.1 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA
TOTAL for 0.04 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA

Note
1 TSCC provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil.
2 TGWC provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the TGWC values.
3 The PDU draw on the line will not exceed list values. The ACGD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA.
4 The Magnet Monitor equipment power is 1.5 KVA 1 phase on an unbalanced leg of 3 phase input (4.5 KVA 3 phase equivalent).
5 The Indoor Water Cooled TGWC is powered from the PDU and therefore included in the PDU draw value.

5–2 CRITICAL POWER REQUIREMENTS (Continued)

Average (while scanning) Power Demand: Refer to Table 5–3

TABLE 5–3
SYSTEM AVERAGE (CONTINUOUS) SCANNING POWER DEMAND

COOLING EQUIPMENT	**TSCC** *See Note 1*			**TGWC** *See Note 2*		
SYSTEM EQUIPMENT	**OUTDOOR TSCC**	**INDOOR WATER COOLED TSCC**	**INDOOR AIR COOLED TSCC**	**OUTDOOR TGWC**	**INDOOR WATER COOLED TGWC**	**INDOOR AIR COOLED TGWC**
PDU draw *See Note 3*	43.1 KVA	43.1 KVA	43.1 KVA	43.1 KVA	44.5 KVA	43.1 KVA
Magnet Monitor	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	18.3 KVA	14 KVA	13.0 KVA	15 KVA	0 KVA *See Note 4*	8.8 KVA
TOTAL *See Note 5*	74.9 KVA	70.6 KVA	69.6 KVA	71.6 KVA	58.0 KVA	65.4 KVA

Note

1 TSCC Provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil.

2 TGWC Provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the TGWC values.

3 The PDU is rated for 45 KVA continuous power.

4 The Water Cooled TGWC is powered from the PDU and therefore included in the PDU draw value.

5 GE Main Disconnect Panel (MDP) is rated continuous power draw is 77 KVA but MDP continuous draw does not exceed list demands.

Standby (no scan) Power Demand: 26.9 KVA at 0.9 lagging Power Factor including 4.4 KVA for PDU, 9 KVA for system cooling equipment, 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet, and 1.5 KVA 1 phase for Magnet Monitor equipment (4.5 KVA 3 phase equivalent).

5–3 POWER DISTRIBUTION SYSTEM

5–3–1 Main Disconnect Panel (MDP)

WARNING!

If customer provided MDP has been selected based on meeting Illustration 1–1 flowchart requisites then customer provided MDP MUST meet all MDP requirements.

The Signa Twin*Speed* system includes a Main Disconnect Panel (based on Illustration 1–1 flowchart) with multi–point remote control capability which is shown in Illustration 5–1. The Main Disconnect Panel consists of the following:

- A three–pole circuit breaker rated for the current of the PDU circuit. The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A three–pole circuit breaker rated for the current of the System Cooling Cabinet circuit. The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A three–pole circuit breaker rated for the current of the Shield/Cryo Cooler Compressor Cabinet circuit. The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A circuit to provide 120VAC single phase power to the Magnet Monitor, Modem, UPS for Magnet Monitor (optional), and Multiplexer Box (optional). The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate available fault current. The MDP includes a single phase step down transformer for 120VAC loads such as Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box).
- The MDP Panel has receptacles inside the panel enclosure for connections of the UPS for Magnet Monitor input and output, Multiplexer Box, Magnet Monitor, and modem. The enclosure has provision for these cables to enter through the access panels in the bottom left side of the enclosure. Mounting of the panel must allow for 5–6 inch (127–152 mm) of free space to allow for cable bending and installation. Strain relief bushings are provided with the individual equipment for each of these cables, not provided with the MDP.

Main Disconnect Panel is to be located so the top of the upper circuit breaker handle when in the ON position does not exceed 79 inches (2000 mm) from the floor and visible to Power Distribution Unit (PD1), System Cooling Cabinet (SCC), Shield/Cryo Cooler Compressor Cabinet, and the service personnel. The optional UPS for the Magnet Monitor may be located below the MDP if sufficient space is available or adjacent if sufficient space is not available, refer to **Section 2–8–7 Magnet Monitor**.

5–3–1 Main Disconnect Panel (MDP) (Continued)

Note

The MDP circuits for the system cooling equipment (TSCC or TGWC), the Shield/Cryo Cooler Compressor Cabinet, and the single phase transformer for Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box) have auto restart upon return of normal power after a time delay of 3 to 30 seconds (field adjustable) to minimize cryogen consumption of the system. The MDP Emergency Off circuit turns off power to all branch circuits including the Magnet Monitor UPS option output and turns off the auto restart function.

The PDU circuit has low voltage release feature which disconnects power from the PDU upon the first loss of power. Power to the PDU is not restored automatically after a power interruption. Emergency Off operation disconnects power from all circuits including the PDU.

For the system cooling equipment (TSCC or TGWC), the Shield/Cryo Cooler Compressor Cabinet, and Magnet Monitor equipment the auto restart function is disabled by the Emergency Off pushbuttons. Extended power outages of 2 days or more will result in the auto restart function loss of battery backup and will require pressing of the Main Power ON push button manual restart after restoration of normal power.

The circuit breakers or fuses ahead of the MDP must be capable of handling the magnetizing inrush currents of the System Cooling Cabinet, Shield/Cryo Cooler Compressor, Magnet Monitor equipment, and transformer of the PDU module (PD1) in the ACGD/PDU Cabinet (MR3). If fuses are used time delayed fuses are recommended.

Check local and national codes to determine if an interlock to the air–conditioning unit in the Computer/Equipment Room is required in the protective disconnect set–up.

The GE MDP Option provides two Emergency Off buttons to be connected to the MDP to disable the power to all system equipment in emergency situations. Two Emergency Off buttons must be provided by the customer if GE MDP Option is not used. The Emergency Off buttons are to be mounted near each exit 60 in. (1524 mm) from the floor in the Magnet Room and Equipment Room The Emergency Off buttons are clearly labeled “Emergency Off” and visible to personnel. It is important the buttons are labeled “off” and not “stop” since there exists an “Emergency Stop” button in the Signa system which powers down only a portion of system equipment for patient safety.

Note

The emergency off circuit disconnects power to the PDU, Twin System Cooling Cabinet, Shield/Cryo Cooler Compressor Cabinet, and from the single phase transformer for Magnet Monitor equipment. Power can be restored to the MDP outputs by pressing the MAIN POWER ON pushbutton on the MDP for the TSCC, Shield/Cryo Cooler Compressor Cabinet, Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box). Power to the PDU is restored by pressing the PDU POWER ON pushbutton and also requires pressing the EMO Reset button on the PDU.

5–3–1 Main Disconnect Panel (MDP) (Continued)

The Main Disconnect Panel (MDP) is lockable to meet OSHA requirements for power Lockout/Tagout requirements. The MDP provides for the disconnection of the facility power to the PDU, Twin System Cooling Cabinet, and Shield/Cryo Cooler Compressor Cabinet. Individual branch circuits for the PDU, Magnet Monitor equipment, Twin System Cooling Cabinet, and Shield/Cryo Cooler Compressor Cabinet consisting of lockable GE Spectra circuit breakers. The MDP also has electrical contacts for an interlock to the air–conditioning units in the Equipment Room. Check local and national codes to determine if an interlock to the air–conditioning unit in the Computer/Equipment Room is required in the protective disconnect set–up.

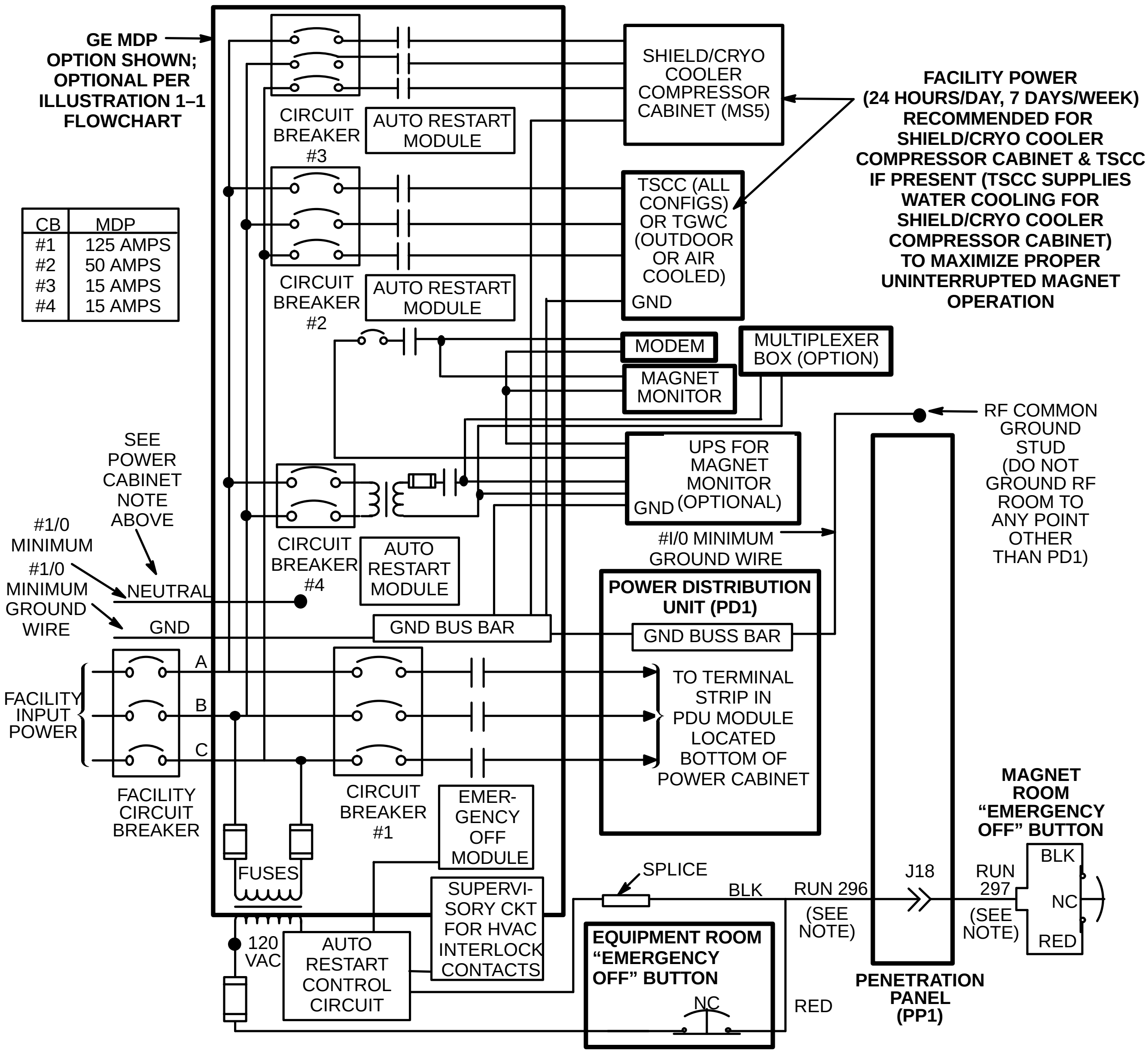
The Main Disconnect Control Panel is UL labeled in accordance with 2002 National Electric Code (NEC) Article 110.2 or 1999 NEC Article 110–2.

Note

The maximum conductor the MDP can accept is #3/0 AWG (83 mm^2). For feeders larger than 3/0 AWG (83 mm^2) the wires must be reduced (ie. splice, junction box, etc.) to 3/0 AWG (83 mm^2) within 10 feet (3 meters) of MDP. It is important to note the maximum cable wire from the MDP to the PDU must not be larger than 2/0 AWG (70 mm^2).

5–3–1 Main Disconnect Control (MDC) (Continued)

NOTE:
- RUNS 296 AND 297, & POWER CORDS FOR SCC & MAGNET MONITOR EQUIPMENT (MAGNET MONITOR, UPS INPUT & OUTPUT, MODEM, OPTIONAL MULTIPLEXER) ARE GE SUPPLIED CABLES. **ALL OTHER WIRING IS CUSTOMER SUPPLIED.**
- TWO REMOTE EMERGENCY “OFF” BUTTONS ARE SUPPLIED WITH GE MDP OPTION, **EMERGENCY OFF BUTTONS ARE CUSTOMER SUPPLIED IF GE MDP OPTION NOT USED** .
- CIRCUIT BREAKERS ARE PROVIDED FOR PDU, TSCC, SHIELD/CRYO COOLER COMPRESSOR CABINET, MAGNET MONITOR EQUIPMENT CIRCUITS.
- ALL BRANCH CIRCUITS DROP OUT ON LOSS OF POWER. TSCC, SHIELD/CRYO COOLER COMPRESSOR CABINET, & MAGNET MONITOR EQUIPMENT AUTOMATICALLY RESTART AFTER 3 SEC TIME DELAY UPON RESTORATION OF POWER. EMERGENCY OFF LOCKS OUT ALL CONTACTORS.
- IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT USED THEN NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE POWER CABINET
- SUPERVISORY CIRCUIT FOR HVAC INTERLOCK CONTACTS OPEN ON LOSS OF DC POWER OR EMERGENCY OFF OPERATION.



PROTECTIVE DISCONNECT SET–UP
ILLUSTRATION 5–1

5–3–2 System Power Distribution Unit

The PDU Module in the lower portion of ACGD/PDU Cabinet has an integrated filter for a level of power conditioning. The largest allowable phase conductor the PDU will accept is 2/0 AWG (83 mm^2). Larger feeder wires can be connected to the MDP with 2/0 AWG (83 mm^2) between the MDP and PDU.

Note

The ground conductor shall be minimum size of 1/0 AWG copper or the same size as the feeder wire, which ever is larger. Lug connector for the ground wire is to be provided by the contractor, recommended Amp Inc. number 36919 lug.

Note

Neutral, if present, must be terminated prior to or inside the Main Disconnect Panel and not brought to the PDU Module in the lower portion of ACGD/PDU Cabinet (MR3).

Note

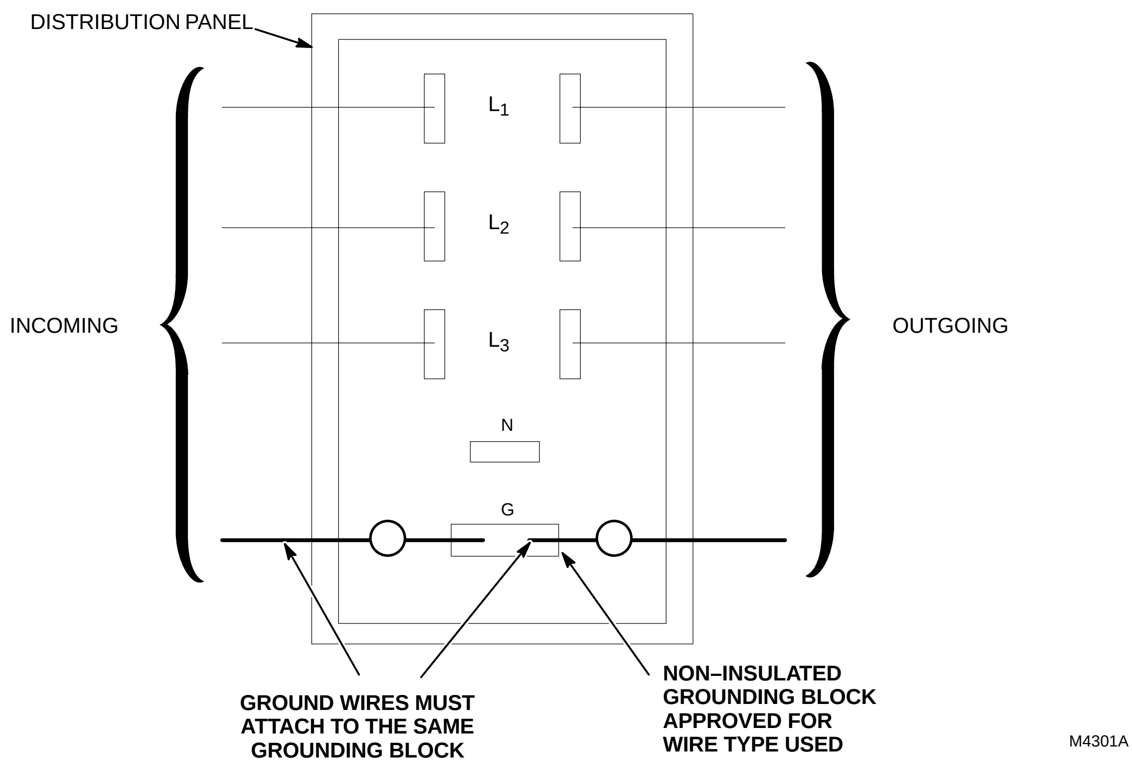
When the full MR system UPS option [Signature 5000 Series 3 UPS 100KVA (E4502FB)] is installed the feeder wiring from the UPS to the PDU Module must be sized to maintain voltage regulation of <5% at 100KVA.

5–4 GROUNDING

5–4–1 Facility Ground

The ground for the MR system shall originate at the system power source, ie. transformer or first access point of power into a facility, and be continuous to the MR system power disconnect in the room. This ground can be spliced with "High Compression Fittings" and should be terminated at each distribution panel it passes through. When it is broken for a connection to a panel, it shall be connected into an approved non–insulated grounding block with the incoming and outgoing ground in this same grounding block, which is then connected to the steel panel, never using the steel or other material of the panel as the block. See Illustration 5–2.

The connection at the power source shall be at the grounding point of the "Neutral – Ground" if a "Wye" transformer is used, or typical grounding points of separately derived system. In the case of an external facility, it shall be bonded to the facility ground point at the service entrance.



GROUND CONNECTION AT DISTRIBUTION PANEL
ILLUSTRATION 5–2

Ground Wire

The ground wire shall be copper wire with a minimum of AWG 1/0 or the same size as the power feeders whichever is larger. This means that if there is a primary feeder to a distribution panel of 500 MCM with a secondary feeder to the MR system of AWG 2 wire, the ground to the distribution panel shall be 500 MCM with an AWG 1/0 to the MR system. The ground wire impedance from the MR system disconnect, including the ground rod, shall not have an impedance greater than 2 ohms to earth as measured by one of the applicable techniques described in Section 4 of ANSI/IEEE Standard 142 – 1982 which can be accomplished using 3–point Fall Of Potential (3 point measurement) method or Clamp–On Ground Resistance measurement which requires a ground measurement device such as AEMC 3730.

5–4–2 System Ground

The MR system is designed with minimum ground loops to prevent noise currents and natural disturbances from flowing through the low–level signal reference path.

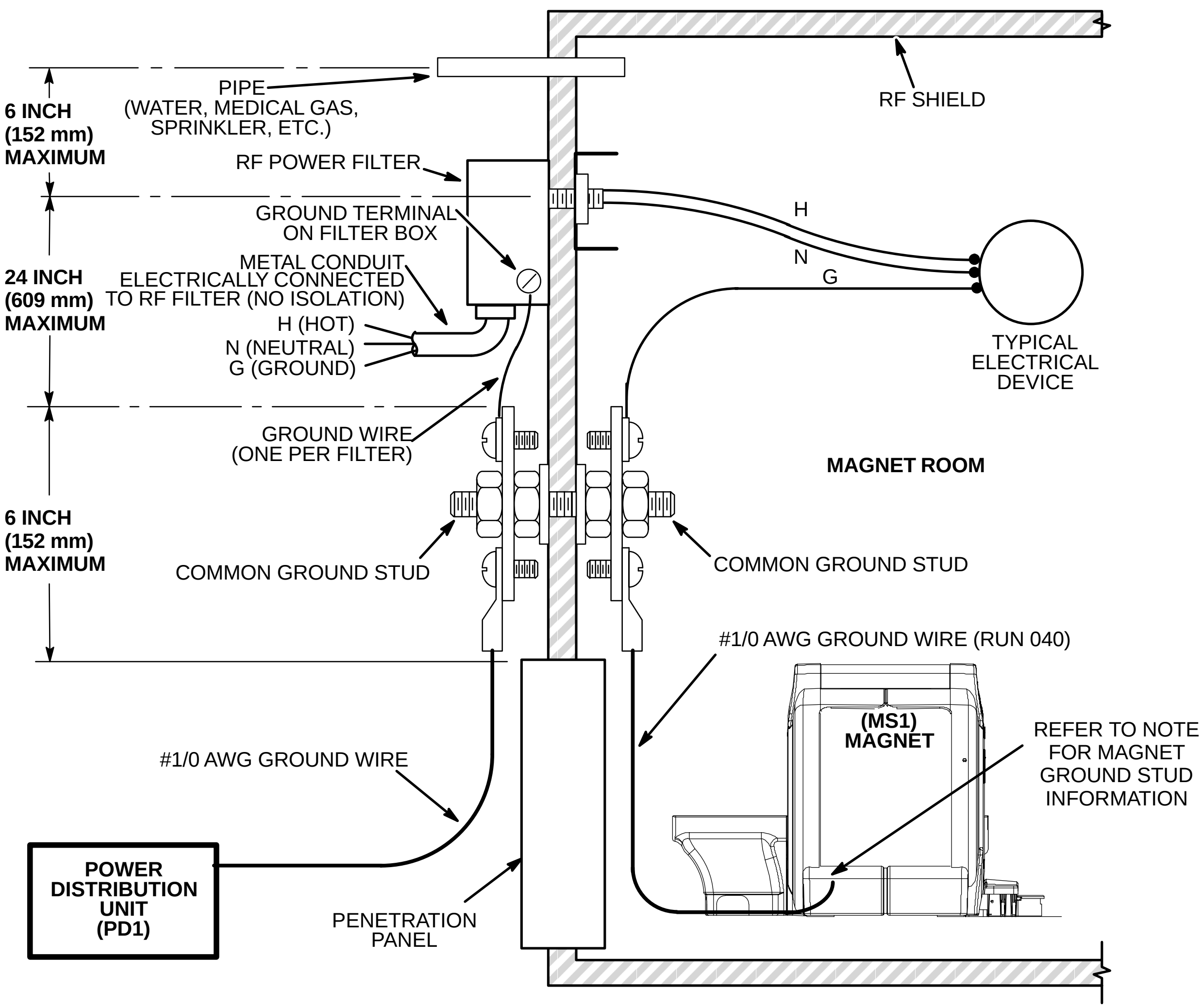
The three major grounding points in the MR system are: the system ground point (bus) in the System PDU (PD1), the enclosure ground points (ground studs located in each cabinet or enclosure), and the RF shielded room common ground point. **This RF shielded room common ground point is to be located within 6 in. (152 mm) of the GE supplied Penetration Panel.** Refer to Section 7, RF SHIELDED ROOM, for a further description of the RF shielded room common ground point.

To ensure patient safety and system performance, the conditions defined in Illustration 5–3 must be met when running power lines into the Magnet Room.

Any modifications or non–MR equipment grounds added to the MR ground system must be approved by your GE Service Representative in order to ensure safety and performance.

5–4–2 System Ground (Continued)

NOTE:
- ALL ITEMS SHOWN ARE CUSTOMER SUPPLIED EXCEPT POWER DISTRIBUTION UNIT, MAGNET, AND ONE #1/0 AWG GROUND WIRE BETWEEN MAGNET GROUND STUD AND RF COMMON GROUND POINT.
- **RESISTANCE BETWEEN ANY TWO GROUNDED DEVICES *MUST NOT EXCEED 0.1 OHM* TO ENSURE EQUAL POTENTIAL GROUND SYSTEM WITHIN MAGNET ROOM.**
- **LOCATE FILTERS WITHIN 2 FEET (600 mm) OF RF COMMON GROUND STUD WHICH MUST BE LOCATED WITHIN 6 INCHES (152 mm) OF PENETRATION PANEL.**
- ALL EXTERNAL CONDUIT MUST BE **METAL AND ELECTRICALLY CONNECTED** TO THE RF POWER FILTERS (REGARDLESS OF FILTER VOLTAGE) PER NEC 2002 ARTICLE 250.110.
- RF POWER FILTERS OF 30 VOLTS OR LESS MAY BE LOCATED ANYWHERE ON THE RF SHIELD **PROVIDED THE INCOMING CONDUIT IS METALLIC** PER NEC 2002 ARTICLE 725.21**, THESE FILTERS MUST ALSO BE LOCATED WITHIN 24 INCHES (609 mm) OF THE RF COMMON GROUND STUD .**
- ALL CONDUITS IN THE RF ROOM MUST BE **METAL**. STEEL IS ACCEPTABLE PROVIDED IT IS ADEQUATELY ANCHORED.
- ALL ELECTRICAL DEVICES (IE. OUTLETS, LIGHT FIXTURES, ETC.) MUST HAVE A GROUND WIRE FROM ITS POWER SOURCE AND BE GROUNDED TO RF ROOM SHIELD AT THE RF COMMON GROUND STUD AS SHOWN BELOW.
- ALL METALLIC PIPES ENTERING THE RF ROOM, EXCLUDING CRYOGENIC VENT AND FLOOR DRAINS, MUST BE LOCATED WITHIN 30 INCHES (762 MM) OF THE RF COMMON GROUND.
- LCC MAGNET HAS 2 GROUND STUDS, ONE ON FRONT FOOT AND ONE ON REAR FOOT (COLDHEAD SIDE OF MAGNET). HOWEVER, THERE IS ONLY ONE #1/0 AWG GROUND WIRE TO BE CONNECTED TO ONLY ONE OF THE GROUND STUDS.



MR MAGNET ROOM GROUNDING REQUIREMENTS AND TYPICAL DIAGRAM
ILLUSTRATION 5–3

5–5 GROUND FAULT PROTECTION

MR suites and radiology departments are considered health care facilities pursuant to National Electric Code (NEC) 2002 Article 517.2 definitions or NEC1999 Article 517–3 definitions and as such must be powered from sources that comply with the ground fault requirements of NEC 2002 Article 517.17 or NEC 1999 Article 517–17. NEC 2002 Article 517.17(A) or NEC 1999 Article 517–17(a) states "Where ground fault is required for the operation of the service disconnecting means or feeder disconnecting means as specified in NEC 230.95 or 215.10, an additional step of ground fault protection shall be provided in the next level of feeder disconnecting means downstream towards the load."

NEC 2002 Article 230.95 or 215.10 or NEC 1999 Article 230–95 or 215–10 requires ground fault protection on service disconnecting means rated 1000 Amps or more on solidly grounded WYE services over 150 volts to ground but not over 600 volts phase to phase.

The two or more levels of ground fault shall be coordinated to provide selectivity between each level of ground fault such that a ground fault on the load side of the feeder would cause the feeder and not the service disconnect to open on a ground fault. Six cycles of separation between the different levels of ground fault tripping is required for the system to be considered selective in accordance with NEC 2002 Article 517.17(B) or NEC 1999 Article 517–17(b).

Check national and local electrical codes.

5–6 POWER SOURCE MONITORING

The facility input power for the proposed system should be checked using a power line disturbance monitor for average line voltage, surges–sags, impulses, and frequency. Some of the recommended line analyzers which are designed for unattended monitoring are the Dranetz Models 656A or 658 and RPM Models 1651, 1656, or 1658.

Analysis should span a period to include two weekends so as to cover several days of normal use. The possibility of "brown–out" conditions which may be experienced in summer must be considered. Any existing power problems with large power consuming systems (x–ray units, CT scanners, etc.) or other computer installations at the proposed site should be reviewed as they may affect the MR system. Results of this analysis should be reviewed with your GE representative to determine if line conditioning is needed.

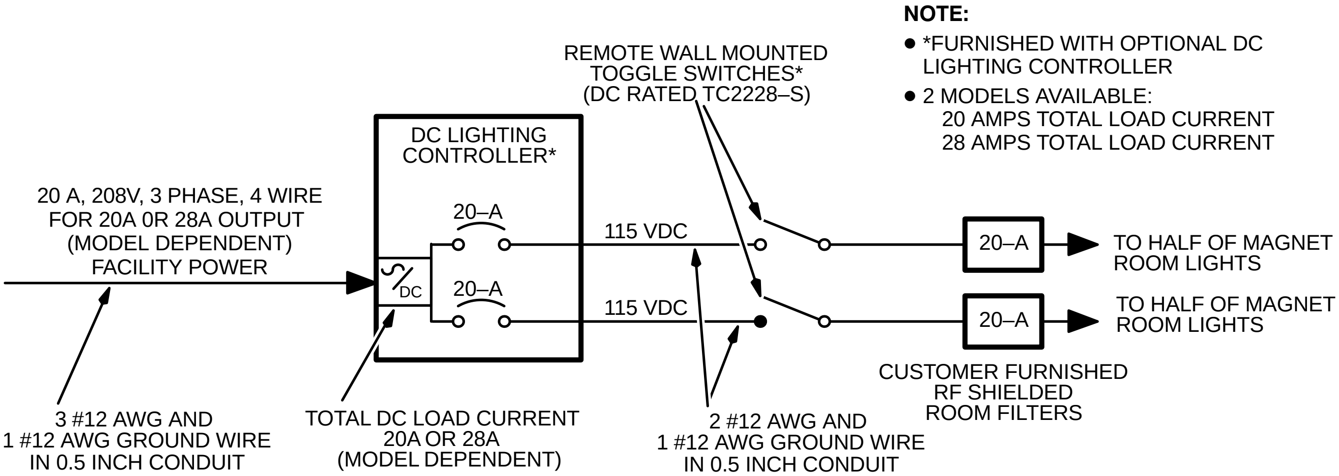
5–7 EMERGENCY POWER

Primary power should be distributed from the customer's emergency life–safety power branch to an emergency lighting source in the Magnet Room. All input power lines must be filtered upon entrance into the RF shielded room (Magnet Room) and grounded according to the requirements listed in Section 5–4–2, SYSTEM GROUND. Always check national and local codes for other emergency power requirements.

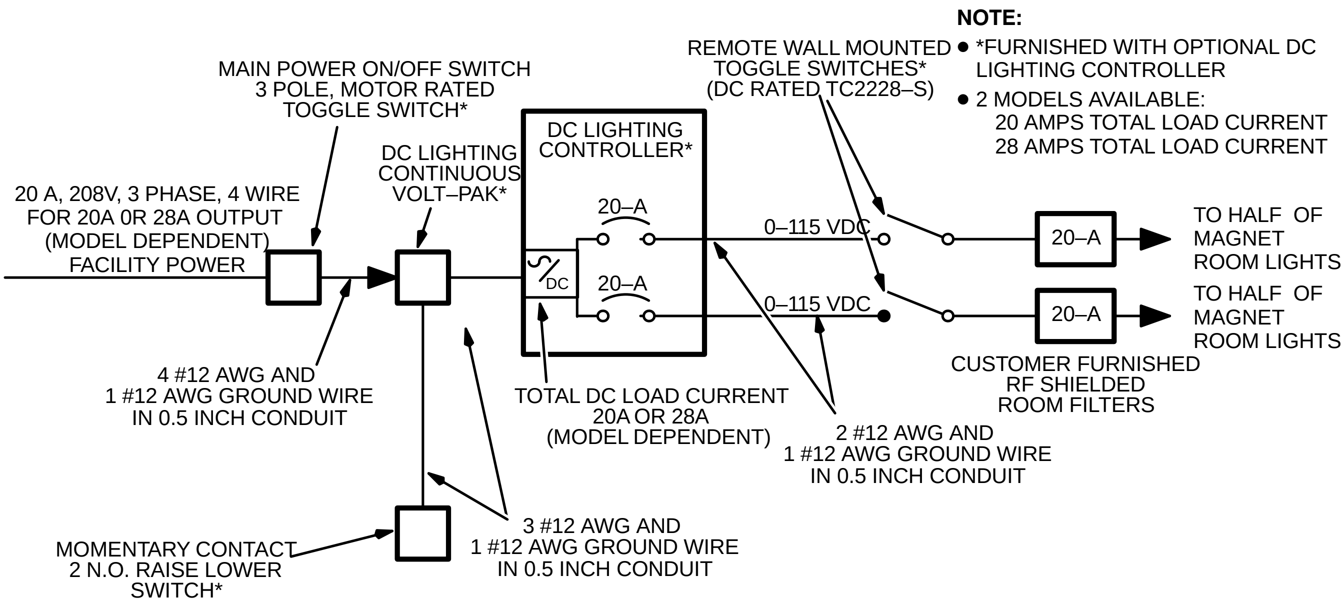
5–8 DC LIGHTING CONTROLLER (Facility Option)

Direct current (DC) powered lighting is recommended in the Magnet Room per Section 4–6 LIGHTING. A DC light controller is available from GE as well as a variable DC lighting controller system. The wiring diagrams for these units are shown in Illustrations 5–4 and 5–5. The input power, interconnect cabling, RF shielded room filters, and conduit are customer furnished.

5–8 DC LIGHTING CONTROLLER (Facility Option) (Continued)



DC LIGHTING CONTROLLER (Facility Option) WIRING DIAGRAM
ILLUSTRATION 5–4



VARIABLE DC LIGHTING CONTROLLER (Facility Option) WIRING DIAGRAM
ILLUSTRATION 5–5

SECTION 6 – INTERCONNECT DATA

TABLE OF CONTENTS

6–1 INTRODUCTION

Installation must be in accordance with local and national code.

Section 6, INTERCONNECT DATA, addresses cable interconnections and customer furnished components for the system. It's subsections are broken down as follows:

- 6–1 INTRODUCTION
  - overall system interconnects, component designations
- 6–2 POWER INTERCONNECTS
  - cable connections to the system Main Disconnect Panel and Power Distribution Unit, subsystem power distribution
- 6–3 EMERGENCY OFF WIRING
  - wiring to main disconnect for emergency off
- 6–4 SYSTEM INTERCONNECTS
  - cable interconnects for the system
- 6–5 TSCC ADDITIONAL INTERCONNECTS
  - additional system cable interconnects for each of the TSCC configurations
- 6–6 TGWC ADDITIONAL INTERCONNECTS
  - additional system cable interconnects for each of the TGWC configurations
- 6–7 CONTRACTOR FURNISHED COMPONENTS
  - miscellaneous components typically provided by a contractor
- 6–8 SPECTROSCOPY INTERCONNECTS
  - cable connections for the 1.5T spectroscopy option
- 6–9 OXYGEN MONITOR OPTION INTERCONNECTS
  - cable connections for the optional Oxygen Monitor
- 6–10 BRAINWAVE OPTION INTERCONNECTS
  - cable connections for the BrainWave option

Note

Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation* for Advantage Workstation information. The BrainWave option offering utilizes an Advantage Workstation.

6–1–1 Component Designators

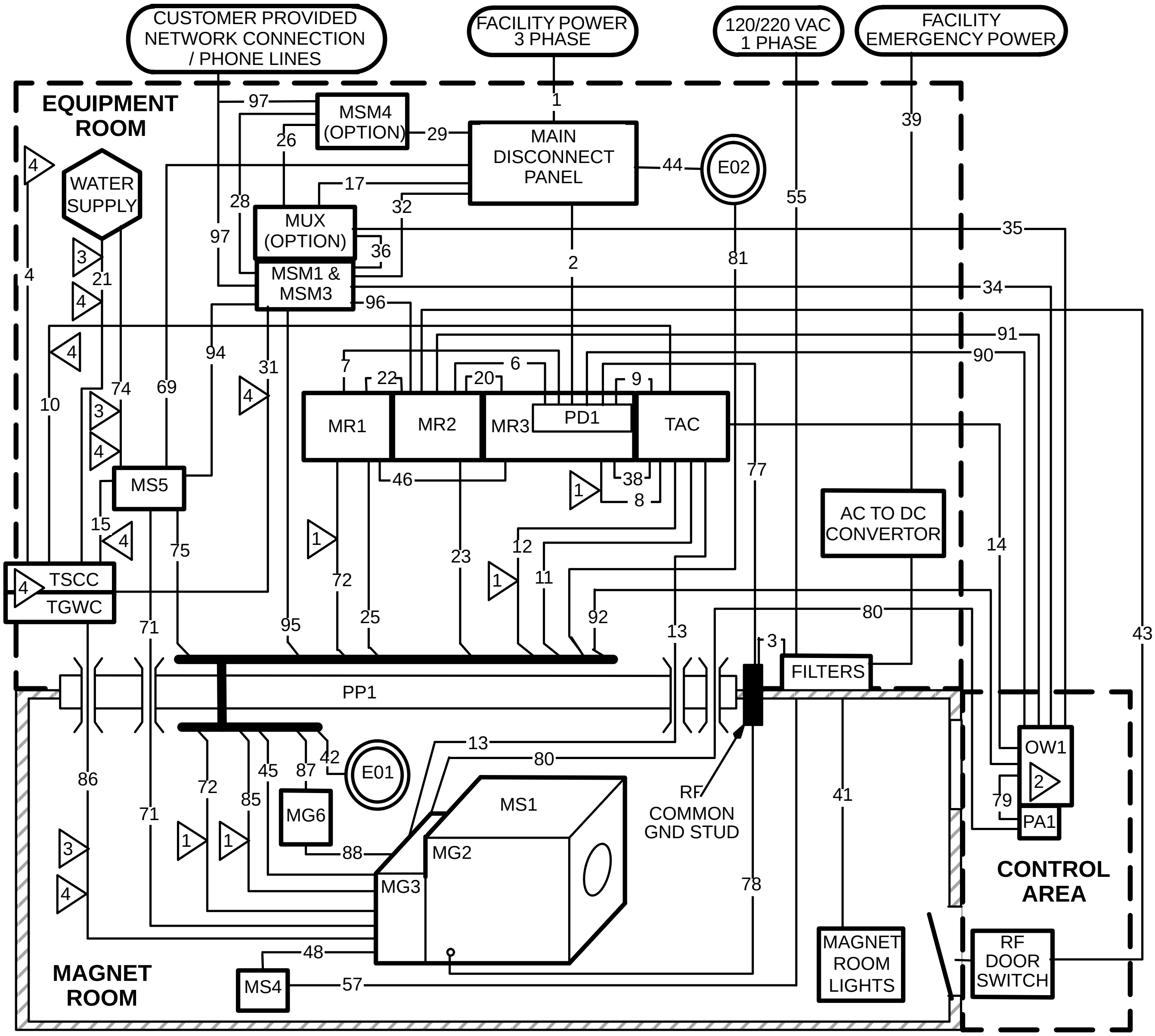
GE uses a Component Designator System as a means of identifying system components in a consistent manner. All subsystem cabinets and other components are referred to by their component designators in the diagrams and tables of this section. For example, the System Cabinet is referred to as MR2. Refer to Table 6–1 for all component designators.

TABLE 6–1
COMPONENT DESIGNATION

BASIC SYSTEM OR OPTION	COMPONENT DESIGNATOR	DESCRIPTION
Basic System	EO1/EO2	Emergency Off Buttons
	MDP	Main Disconnect Panel
	MG2	Magnet Enclosure
	MG3	Magnet Rear Pedestal
	MG6	Blower Box
	MR1	1.5T SRFD II Cabinet
	MR2	System Control Cabinet
	MR3	ACGD/PDU Cabinet has PD1 in lower portion cabinet
	MS1	Superconducting Magnet
	MS4	Magnet Rundown Unit
	MS5	Shield/Cryo Cooler Compressor Cabinet
	MSM1	Magnet Monitor
	MSM3	Modem for Magnet Monitor
	OW1	Operator Workspace
	PD1	Power Distribution Unit (PDU) is a module in lower portion of ACGD/PDU Cabinet
	PA1	Pneumatic Patient Alert Control Box
	PP1	Penetration Panel
	PT1	Patient Transport Table
	RCP	Remote Control Box for Outdoor TSCC, Outdoor TGWC, & Water Cooled TGWC
	TAC	Twin Accessory Cabinet
	TSCC	Twin System Cooling Cabinet (provides water cooling for Shield/Cryo Cooler Compressor & Gradient Coil)
	TGWC	Twin Gradient Water Cooler (provides water cooling for Gradient Coil ONLY; Customer supplied water cooling is required for Shield/Cryo Cooler Compressor Cabinet)
System Options	AW	Advantage Workstation
	AUDIO	BrainWave Audio Console
	BW CABINET	BrainWave Cabinet
	BW CART	BrainWave Cart
	MNS	MNS Cabinet
	MSM4	UPS for Magnet Monitor
	MUX	Phone Line Multiplexer for Magnet Monitor
	OM1	Oxygen Monitor
	OM3	Remote Oxygen Sensor Module

6–1–2 Group Interconnects

Illustration 6–1 shows the Group Interconnect Diagram for 1.5T Signa Twin*Speed* system. Each group contains one or more cables. This diagram should be referred to when using the tables in this section.



1 MUST BE CUT AND CONNECTED TO PP1 AT SITE.
NOTE: IMPEDANCE IS NOT CRITICAL SO EXCESS CABLE SHOULD BE CUT OFF.

2 OPERATOR WORKSPACE (OW1) SUBSYSTEM EQUIPMENT IS PROVIDED WITH MAXIMUM LENGTH CABLES POSSIBLE. SEVERAL OW ASSEMBLIES ARE MOUNTED TO OW TABLE & OW INTERCONNECTS ARE ROUTED THROUGH TABLE CABLE TRAY. FOR REFERENCE USE ONLY THE OW1 RUN NUMBERS ARE 049, 792, 793, 794, 795, 796, 797, 798, 799, 806, & 807.

3 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

4 FOR GROUPS 4, 10, 15, 21, 31, 49, 50, 51, 53, 74, & 86 REFER TO **SECTION 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **SECTION 6–6 TGWC ADDITIONAL INTERCONNECTS** FOR CONFIGURATION APPROPRIATE INTERCONNECTS AND DETAILS.

SIGNA 1.5T TWIN*SPEED* SYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–1

6–1–3 Definition of Terms

The definition of terms used in Tables 6–2 and 6–3 are:

- Group Number: identifying number referenced to bundles (i.e. groups) of cables as shown in Illustrations 6–1
- Between Units (From/To): component designators as found in Table 6–1
- Area: cross–sectional area of the combined cables in a group

Note

The group area was found by adding up the circular cross–sectional areas of all individual cables within a group. It does not take any fill factors or space between cables into account. Adhere to applicable electrical codes for fill factors.

- Usable Length: total length of a cable MINUS any required take up within cabinets
- Run Number: unique number assigned to each GE–supplied cable

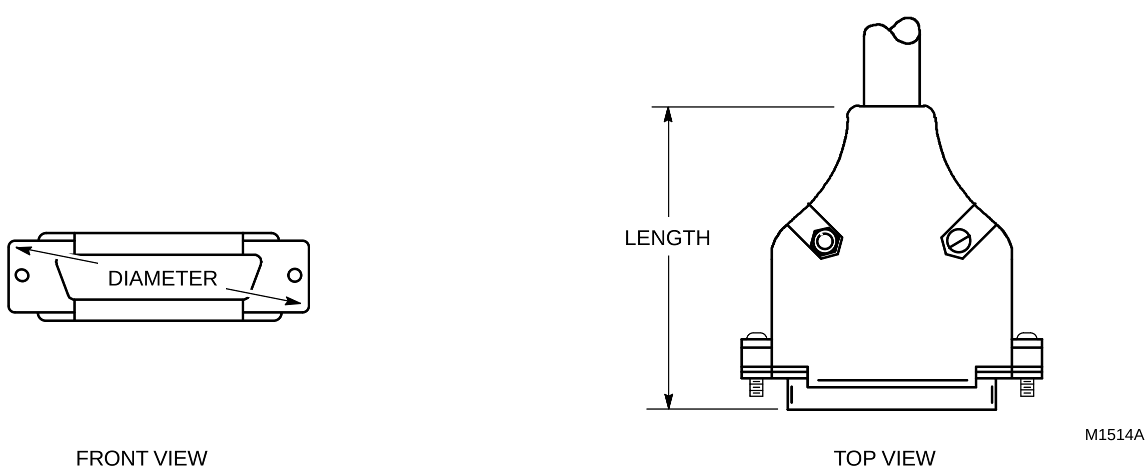
Note

The run number must be used when making special cable orders.

- Cable Diameter: diameter of an individual cable
- Plug Pulling Diameter x Length: cable plug dimensions as shown in Illustration 6–2

Note

In some cases, a cable has more than one connector on an end. These cables will have the number of connectors following the english dimensions of the plug pulling diameter times length (e.g. '2.0x3.25 –x2' means there are 2 connectors with dimensions of 2.0 in. diameter and 3.25 in. length). Of course, the same number of connectors apply to the metric dimensions as well.



SUBMINIATURE–D CONNECTOR PLUG PULLING DIMENSIONS
ILLUSTRATION 6–2

6–2 POWER INTERCONNECTS

The interconnects for the MDP include:

- Facility Main power; refer to **Section 5 – POWER REQUIREMENTS** for detailed information on main power connections.
- Main power to the system Main Disconnect Panel (MDP) and Power Distribution Unit (PD1).
- Emergency off wiring; refer to **Section 6–3 EMERGENCY OFF WIRING**, for information on the emergency off circuit interfacing.
- System Cooling Cabinet
- Shield/Cryo Cooler Compressor Cabinet (MS5)
- Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, Modem, and optional Multiplexer Box)

Note

The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

The interconnects for the PD1 include:

- Power cables GE–supplied subsystems for Operator Workspace, System Cabinet, Twin Accessory Cabinet, Patient Comfort Compressor.
- 3 auxiliary ground cables to the PD1.
- 2 control cables to the PD1

Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. However, there may be instances in which conduit is used for power cables. In those cases, cables may be pulled by the lug terminal ends so the connector pulling dimensions on the plug ends will not be a factor for power cables.

Note

The power cables will probably need to be terminated at PD1 located in the lower portion of the Power Cabinet if conduit is used.

Unless otherwise specified, cables and components listed in Table 6–2 are supplied by GE.

6–2 POWER INTERCONNECTS (Continued)

TABLE 6–2
POWER CABLES FROM MDP & PD1

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
1	—	Facility Power	MDP	—	—	—	—	—	Customer Furnished. **See Note 2.**
2	—	MDP	PD1 See Note 1	—	—	—	—	—	Customer Furnished. Refer to **Section 5–3–2 System Power Distribution Unit** for wire size information.
4	—	MDP/ PDU See Notes	TSCC/ TGWC See Notes	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** or **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM & TO connections.
6	0.803 (518) ◆◆ 0.905 (584) ◆◆	PD1 **See Note 1**	MR2	22 (6.70)	037	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
					971	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	
					972◆◆	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					968	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
7	1.36 (878)	PD1 **See Note 1**	MR1	22 (6.70)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
					966	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
					967	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
9	0.43 (278)	PD1 **See Note 1**	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	#10 AWG / 4 wire Power cable
17	0.05 (32.3)	MDP	MUX	7 (2.1)	—	0.25 (6.35)	1.00x2.50 (25.4x63.5)	1.00x2.50 (25.4x63.5)	MUX is an option, refer to **Section 2–8–8**

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

2 If low Voltage Step–Up Transformer Option (R4500AW or R4500BE) is used then customer furnished interconnects are required between facility power, transformer and MDP.

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–2 POWER INTERCONNECTS (Continued)

TABLE 6–2 (Continued)
POWER CABLES FROM MDP & PD1

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
29	0.22 (142)	MDP	MSM4	6 (1.83) minus takeup at each end	— 939	0.375 (9.53) 0.375 (9.53)	1.00x2.50 (25.4x63.5) 1.00x2.50 (25.4x63.5)	Hard Wired 1.00x2.50 (25.4x63.5)	UPS Power IN (for Magnet Monitor UPS option) UPS Power OUT (for Magnet Monitor UPS option)
32	0.159 (103)	MDP	MSM1 & MSM3	6 (1.8) minus takeup at each end	— —	0.375 (9.5) 0.25 (6.35)	1.2x3.00 (30.5x76.2) 2.00x2.50 (50.8x63.5)	1.25x2.5 (31.8x63.5) 2.00x2.50 (50.8x63.5)	MSM1 power cable Modem for MSM1 power cable. Modem located on top of MSM1. Refer to **Section 2–8–7**.
44	—	MDP	EO2	—	—	—	—	—	Customer Furnished. Refer to **Section 6–3**.
69	1.31 (842)	MDP	MS5	29.8 (9.1) minus takeup at MDP	—	1.29 (32.75)	Hard Wired	Hard Wired	Both ends have liquid tight connector which wires pass through for hard wire connections.
77	0.27 (172)	PD1 **See Note 1**	RF COM-MON GND STUD	—	—	0.584 (14.8)	Ring Terminal	Ring Terminal	Customer supplied ground cable, refer to **Section 5–4–2**.
81	0.460 (297.3)	EO2	PP1	40 (12.2)	296	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	Refer to **Section 6–3**.
90	0.413 (266)	PD1 **See Note 1**	OW1	51 (15.5) **See Note 3**	048 969	0.188 (4.78) 0.70 (17.78)	Hard Wired Hard Wired	Hard Wired Hard Wired	#10 AWG / 1 wire Ground cable #10 AWG / 4 wire power cable

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).

3 M1090ZP, M1090ZN, M3044TC, M3044TA can be ordered to increase usable length of Group 90 to 84 ft (25.6 m).

6–3 EMERGENCY OFF WIRING

6–3–1 Introduction

This section addresses wiring for the Emergency Off circuit (also known as protective disconnect circuit). Refer to **Section 5–3 RECOMMENDED POWER DISTRIBUTION SYSTEM** for information on the recommended protective disconnect device and emergency off button locations and mounting.

The emergency off wiring for the MR system is unique because the wiring into the magnet room must be RF tight.

6–3–2 Main Disconnect Panel Connections

The emergency off circuit is shown in **Section 5–3 POWER DISTRIBUTION SYSTEM** Illustration 5–1. The circuit utilizes the normally closed series loop shown. The MDP provides 2 emergency off buttons.

6–3–3 Magnet Room Wiring

GE provides two cables for routing the emergency off circuit through the Penetration Panel and into the magnet room (Runs 296 and 297). Alternate wiring may be used by the customer; however, the use of these cables ensures that the emergency off wiring will be RF tight.

In Illustration 5–1 black and red wires are used for connections on the ends of Runs 296 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. (Runs 296 and 297 are actually nine wire cables.)

6–4 SYSTEM INTERCONNECTS

Table 6–3 contains information on interconnects between all system components. Cables found in Tables 6–2 are not repeated here, although their groups are referenced for completeness. Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. Unless otherwise specified, cables and components listed in Table 6–3 are supplied by GE.

TABLE 6–3
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
1	—	Facility Power	MDP	—	—	—	—	—	Refer to Table 6–2.
2	—	MDP	PD1 **See Note 1**	—	—	—	—	—	Refer to Table 6–2.
3	—	Facility Emerg Power Filter	PP1	—	—	—	—	—	Customer Furnished Ground.
4	—	MDP/ PDU	TSCC/ TGWC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM & TO connections.
5	—	—	—	—	—	—	—	—	Not Used
6	—	PD1	MR2	—	—	—	—	—	Refer to Table 6–2.
7	—	PD1	MR1	—	—	—	—	—	Refer to Table 6–2.
8	3.45 (2221)	MR3	TAC	16 (4.88)	976 977 978	1.21 (30.7) 1.21 (30.7) 1.21 (30.7)	Hard Wired Hard Wired Hard Wired	Hard Wired Hard Wired Hard Wired	Runs 976, 977, & 978 are cut and connected to TAC at site. Total length of 65 ft (19.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
9	—	PD1	TAC	—	—	—	—	—	Refer to Table 6–2.
10	—	TSCC/ TGWC	TAC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM unit.

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
11	0.31 (201)	TAC	PP1	21 (6.40)	611	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 611 for system with 2nd Order Resistive Shim Option
12	3.61 (2323)	TAC	PP1	27 (8.2)	979 980 981 988 X 3	1.21 (30.7) 1.21 (30.7) 1.21 (30.7) 0.26 X 3 (6.6 X 3)	Hard Wired Hard Wired Hard Wired Ring Terminal	Ring Terminal Ring Terminal Ring Terminal Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 65 ft (19.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1. A Ground wire for each of Runs 979, 980, & 981
13	1.27 (819)	TAC	MG2/3	80 (24.4)	—	1.27 (32.3)	Flexible Line	Flexible Line	Run is routed through waveguide in PP1, cut to length at site.
14	0.10 (66)	TAC	OW1	46 (14.0) See Note 3	1004◆or 568◆◆	0.36 (9.14)	1.30x2.00 (33.5x50.8)	2.30x2.10 (58.5x53.5)	Used for 2nd Order Resistive Shim Option
15 **See Note 2**	—	MS5	TSCC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS**
16	—	—	—	—	—	—	—	—	Not Used
17	—	MDP	MUX	—	—	—	—	—	Refer to Table 6–2.
18 to 19	—	—	—	—	—	—	—	—	Not Used

Note 2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

3 Run 568 or 1004 can be extended on a 60 Hz system by using M3090RA 60 Hz High Order Shim Cable Extension Kit. This kit includes a 100 ft (30.5 m) extension cable and converters, power supply, etc. A 50 Hz system will need to follow the Special Order Inquiry (SOI) process if extensions are needed.

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
20	0.78 (506)	MR2	MR3	14 (4.27)	1034◆ or 710◆◆	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 or 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
21	—	Facility Water Supply	TSCC	—	—	—	—	—	For Indoor Water Cooled TSCC or TGWC ONLY: **Section 6–5–2 Indoor Water Cooled TSCC System Interconnects** or **Section 6–6–2 Indoor Water Cooled TGWC System Interconnects.**
22	0.83 (537)	MR2	MR1	14 (4.27)	229	0.23 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
					702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					708◆◆	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
					774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
					1033◆	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 1033 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
23	1.54 (997) ◆ 1.06 (686) ◆◆	MR2	PP1	31 (9.45) **See Note 4**	231	0.212 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	**RF Receive Cable**
					489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	**RF Receive Cable**
					490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	**RF Receive Cable**
					499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					711/ 712◆◆	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
					1025◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1026◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1027◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1028◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1029◆	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
					1030◆	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
					1045◆	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).

Note 4 M3044TC or M3044TA can be ordered to increase usable length of Group 23 to 56 ft (17.1 m).

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
24	—	—	—	—	—	—	—	—	Not Used
25	1.50 (966) ◆ 1.76 (1134) ◆◆	MR1	PP1	21 (6.40) **See Note 5**	044	0.20 (5.1)	Ring Terminal	Ring Terminal	Ground cable
					486◆◆	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					487◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					726◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
					769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
					773	0.34 (8.64)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	

Note 5 M3044TC or M3044TA can be ordered to increase usable length of Group 25 to 51 ft (15.5 m).

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
26	—	MSM4	MUX	—	—	—	—	—	ONLY USED WITH UPS FOR MAGNET MONITOR: Customer provided phone line routed through UPS for transient protection.
27	—	—	—	—	—	—	—	—	Not Used
28	0.07 (45.2)	MSM1 & MSM3	MSM4	6 (1.8) **minus** takeup at each end	—	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	ONLY USED WITH UPS FOR MAGNET MONITOR: Customer provided phone line, cable diameter & plug pull information are estimates.
29	—	MSM4	MDP	—	—	—	—	—	Refer to Table 6–2.
30	—	—	—	—	—	—	—	—	Not Used
31	—	MSM1	TSCC/ TGWC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM connection.
32	—	MDP	MSM1 & Modem	—	—	—	—	—	Refer to Table 6–2.
33	—	—	—	—	—	—	—	—	Not Used
34	0.09 (58)	MSM1	OW1	70 (21.3) **minus** takeup at MSM1	942	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
(Continued)									

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
35	—	MUX	OW1 InSite Modem	—	—	—	—	—	Customer provided phone line.
36	0.09 (58)	MSM3	MUX	6 (1.8) **minus** takeup at each end	—	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	Modem–MUX phone line
37	—	—	—	—	—	—	—	—	Not Used
38	0.10 (66)	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
39	—	Facility Emerg Power	Filter	—	—	—	—	—	Refer to **Section 5–8** for DC Lighting Controller cabling.
40	—	—	—	—	—	—	—	—	Not Used
41	—	Filter	Magnet Room Lights	—	—	—	—	—	Refer to **Sections 5–7** and **5–8**.
42	0.096 (61.94)	PP1	E01	84 (25.6) **minus** takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	Refer to **Section 6–3**.
43	0.096 (61.94)	MR2	RF Door Switch	67 (20.4) **minus** takeup at RF Door Switch **See Note 6**	701	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	RF Door Switch provided by RF Screen Room vendor.
44	—	MDP	EO2	—	—	—	—	—	Refer to Table 6–2.
Note 6 M3044TC or M3044TA can be ordered to increase usable length of Group 43 to 97 ft (29.6 m) **minus** takeup at RF Door Switch.									
(Continued)									

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
45	2.61 (1681) ◆ 3.02 (1943) ◆◆	PP1	MG2/3	30 (9.15)	262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					282	0.464 (11.8)	2.30x2.00 (57.2x50.8	2.30x2.00 (57.2x50.8	
					300	0.44 (11.18)	2.30x2.00 (57.2x50.8)	Ring Terminals	
					318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					385	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
					387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					494◆◆	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					495◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					612	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 612 for system with 2nd Order Resistive Shim Option
					624	0.26 (6.6)	Ring Terminals	1.00x1.38 (25.4x35.1)	
					700◆◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
					715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
45 (contin- ue)	2.61 (1681) ◆ 3.02 (1943) ◆◆	PP1	MG2/3	30 (9.15)	716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					727◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					828	0.31 (7.75)	2.30x2.00 (57.2x50.8	2.30x2.00 (57.2x50.8	
					829	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					1036◆	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					1037◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1038◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1039◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1040◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1041◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					1042◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
46	0.541 (349)	MR1	MR3	14 (4.27) **See Note 7**	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is flexible conduit containing fiber optic cables.
47	—	—	—	—	—	—	—	—	Not Used
48	0.071 (45.6)	MS4	MS1	86 (26.2)	606	0.30 (7.6)	0.65x1.85 (16.5x47.0)	0.65x1.85 (16.5x47.0)	
49	—	Out-door TSCC	See Notes	—	—	—	—	—	Refer to **Section 6–5–1 Outdoor TSCC System Interconnects**
50	—	Out-door TSCC/ TGWC	See Notes	—	—	—	—	—	Refer to **Section 6–5–1 Outdoor TSCC System Interconnects** or **Section 6–6–1 Outdoor TGWC System Interconnects** for details and to determine FROM and TO units.
51	—	Out-door TSCC/ TGWC	RCP	—	—	—	—	—	Refer to **Section 6–5–1 Outdoor TSCC System Interconnects** or **Section 6–6–1 Outdoor TGWC System Interconnects** for details and to determine FROM unit.
52	—	—	—	—	—	—	—	—	Not Used
53	—	Indoor Water Cooled TGWC	RCP	—	—	—	—	—	Refer to **Section 6–6–2 Indoor Water Cooled TGWC**.
54	—	—	—	—	—	—	—	—	Not Used
55	—	Facility Power	Filter	—	—	—	—	—	Customer Furnished Magnet Room power (refer to **Sections 5–1** and **7–4**).
56	—	—	—	—	—	—	—	—	Not Used
57	—	Filter	MS4	—	—	—	—	—	Customer Furnished (refer to **Sections 5–1**).
58 to 68	—	—	—	—	—	—	—	—	Not Used
69	—	MDP	MS5	—	—	—	—	—	Refer to Table 6–2.
70	—	—	—	—	—	—	—	—	Not Used

Note 7 M3044TC or M3044TA can be ordered to increase usable length of Group 46 to 59 ft (18.0 m).

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
71	4.28 (2758)	MS5	MS1	38.5 (11.7)	621 622	1.65 (41.9) 1.65 (41.9)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	Runs 621 and 622 are continuous helium lines routed through PP1. Bend radius of Runs 621 and 622 is 8 in. (203.2 mm). Cable diameter includes foam insulation installed on lines.
72	1.27 (820)	MR1	MG2/3	68 (20.7) **See Note 8**	235 & 257 236 & 258	0.87 (22.1) 0.59 (15.0)	1.36x2.12 (34.5x53.9) 1.5x2.50 (38.1x63.5)	1.36x2.40 (34.5x61.0) 1.5x2.50 (38.1x63.5)	**These runs are RF Transmit Cables.** Runs 235/257 ans 236/258 must be cut to length and connected via PP1 at site. Usable length provided for routing in Equipment Room and Magnet Room.
73	—	—	—	—	—	—	—	—	Not Used
74	—	Facility Water Supply	See NOTES	—	—	—	—	—	Refer to **Section 6–5–2 Indoor Water Cooled TSCC System Interconnects** or **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details and to determine FROM and TO units.
75	0.035 (22)	MS5	PP1	42 (12.8)	623	0.21 (5.28)	1.00x1.38 (25.4x35.1)	Ring Terminals	
76	—	—	—	—	—	—	—	—	Not Used
77	—	PD1 **See Note 1**	RF COM-MON GND STUD	—	– –	—	—	—	Refer to Table 6–2.

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

8 M3044TC or M3044TA can be ordered to increase usable length of Group 72. These catalogs provide the following:
– For Runs 235 & 257: 88 ft (26.8 m) usable length Heliax to be cut to length and connected via PP1 at site for routing in Equipment and Magnet Room. Note this Heliax has a minimum bend radius of 10 in. (254 mm)
– For Runs 236 & 258: One cable with total length 50 ft (15.24 m) and one cable with total length 55 ft (16.76 m) to be used for Runs 236 & 258. Run 236 From MR1 to PP1 requires takeup of 4 ft (1.2 m) at PP1. Run 258 From PP1 to MG2 requires takeup of 4 ft (1.2 m) at PP1 and 4 ft (1.2 m) at MG2.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
78	0.338 (218)	RF COM-MON GND STUD	MS1 GND STUD	60 (18.29) **minus** takeup at RF Common GND Stud **See Note 9**	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
79	0.013 (8.0)	OW1	PA1	5 (1.5) **minus** takeup at PA1	—	0.13 (3.2)	3.00x3.00 (76.2x76.2)	0.38x1.75 (9.6x44.5)	
80**	0.049 (32.1)	PA1	MG2	97 (29.6) **minus** takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
81	—	E02	PP1	—	—		—	—	Refer to Table 6–2.
82 to 84	—	—	—	—	—	—	—	—	Not Used
85	6.899 (4441)	PP1	MG3	40 (12.2)	982	1.21 (30.7)	Ring Terminals	Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.
					983	1.21 (30.7)	Ring Terminals	Hard Wired	
					984	1.21 (30.7)	Ring Terminals	Hard Wired	
					985	1.21 (30.7)	Ring Terminals	Hard Wired	
					986	1.21 (30.7)	Ring Terminals	Hard Wired	
					987	1.21 (30.7)	Ring Terminals	Hard Wired	

Note ** If installation requires greater than 97 feet (29.6 meters) of pneumatic tubing between the squeeze bulb, located on the front of the Magnet Enclosure, and the Patient Alert Control Box (PA1), located near the Operator's Console or Operator Workspace, an Extender Kit (46–317758P2) must be ordered. The Extender Kit consists of a small Extender Box (to be mounted in Equipment Room) and 95 feet (29.0 meter) of pneumatic tubing.

9 M3044TC or M3044TA can be ordered to increase usable length of Group 78 to 91 ft (27.7 m) **minus** takeup at RF Common GND Stud.

10 M3044TC or M3044TA can be ordered to increase usable length of Group 87 to 71 ft (21.6 m).

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
86 **See Note 2**	—	TSCC/ TGWC See Notes	MG2	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM connection.
87	0.352 (228)	PP1	MG6	32 (9.76) minus takeup at MG6, **See row Notes & Note 10**	045 784	0.195 (4.95) 0.64 (16.3)	Ring Terminal Ring Terminals	Ring Terminal 2.28x3.85 (57.9x97.8)	The usable length is based on MG6 being mounted on floor and floor routing of cables.
88	15.9 (8544)	MG6	MG3	13 (3.95) minus takeup	—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.
89	—	—	—	—	—	—	—	—	Not Used
90	—	PD1	OW1	—	—	—	—	—	Refer to Table 6–2.
91	0.55 (355) ◆ 0.54 (302) ◆◆	MR2	OW1	51 (15.5) **See Note 11**	789 791◆◆ 818◆◆ 836◆◆ 1031◆ 1032◆ 1044◆	0.44 (11.2) 0.44 (11.2) 0.44 (11.2) 0.34 (8.64) 0.34 (8.64) 0.525 (13.3) 0.34 (8.64)	1.30x2.00 (33.5x50.8) 1.30x2.00 (33.5x50.8) 1.30x2.00 (33.5x50.8) 0.50x0.75 (12.7x19.1) 0.50x0.75 (12.7x19.1) 2.30x2.00 (57.2x50.8) 0.50x0.75 (12.7x19.1)	1.30x2.00 (33.5x50.8) 1.30x2.00 (33.5x50.8) 1.30x2.00 (33.5x50.8) 0.50x0.75 (12.7x19.1) 0.50x0.75 (12.7x19.1) 2.30x2.00 (57.2x50.8) 0.50x0.75 (12.7x19.1)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8). Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

11 M1090ZP, M1090ZN, M3044TC, M3044TA can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
92	0.223 (144) ◆ 0.152 (99) ◆◆	PP1	OW1	53 (16.1) **See Note 12**	788 838◆	0.44 (11.2) 0.3 (7.6)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.0x50.8)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.0x50.8)	
93	—	—	—	—	—	—	—	—	Not Used
94	0.09 (58.6)	MSM1	MS5	54 (16.4) minus takeup at MSM	826	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
95	0.22 (144)	MSM1	PP1	54 (16.4) **minus** takeup at MSM1	824 825	0.44 (11.2) 0.30 (7.62)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.5x50.8)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.5x50.8)	
96	0.09 (58.6)	MSM1	MR2	54 (16.4) **minus** takeup at MSM1	823	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
97	—	Phone Line Con-nection	MSM1 or MSM4 (Op-tion)	—	—	—	—	–	Refer to Section **Section 2–8–8** for additional customer network and/or phone line information. WITH UPS FOR MAGNET MONITOR OPTION: Customer provided phone line routed through UPS for transient protection, refer to Group 26.
Note 12 M1090ZP, M1090ZN, M3044TC, M3044TA can be ordered to increase usable length of Group 92 to 88 ft (26.8 m).									

6–5 TSCC ADDITIONAL INTERCONNECTS

The following sections show the system interconnects required for the indicated Twin System Cooling Cabinet (TSCC) configurations in addition to the interconnects in **Section 6–4 SYSTEM INTERCONNECTS**.

6–5–1 Outdoor TSCC System Interconnects

Note

Refer to **Section 2–7–1 TSCC Configurations Siting Considerations** Table 2–5 for listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 470 Pre installation Installation Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for the Outdoor TSCC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Outdoor TSCC must be in accordance with local and national codes**.

Vertical Separation Requirements

- Outdoor TSCC must not exceed 100 ft (30.5 m) vertical separation from the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements

Customer must provide water supply and return lines between the Outdoor TSCC and the GE supplied hoses for the Shield/Cryo Cooler Compressor.

- Ellis And Watts provides two 0.75 in. (19 mm) FNPT fittings at the Outdoor TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2–14 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 0.75 in. (19 mm) MNPT connections and 0.75 in. (19 mm) ID copper lines.
- Ellis And Watts provides two 0.5 inch (12.7 mm) ID rubber hoses of 60 ft (18.3 m) total length [usable length of 55 ft (16.7 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Shield/Cryo Cooler Compressor Cabinet. Site layout for the Shield/Cryo Cooler Compressor water cooling lines and hoses must meet the following:
  - Total line length from the Outdoor TSCC connection to the Shield/Cryo Cooler Compressor connection MUST NOT EXCEED 160 ft (48.8 m). Ellis And Watts supplied rubber hoses for Shield/Cryo Cooler Compressor is a maximum length of 60 ft (18.3 m) total length [55 ft (16.8 m) usable length].
  - If the 0.75 in. (19 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
  - Copper lines must be insulated outdoors.

6–5–1 Outdoor TSCC System Interconnects (Continued)

Gradient Coil Water Cooling Lines & Hoses Requirements

- Ellis And Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2–14 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
  - Total line length from the Outdoor TSCC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TSCC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
  - If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).

RCP Data Cables Requirements

- Ellis & Watts provides three data cables that connect between the RCP and the Outdoor TSCC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TSCC and RCP placement [total length minus height of connection at Outdoor TSCC and height of connection at RCP, no cable takeup inside of TSCC or RCP].

Note

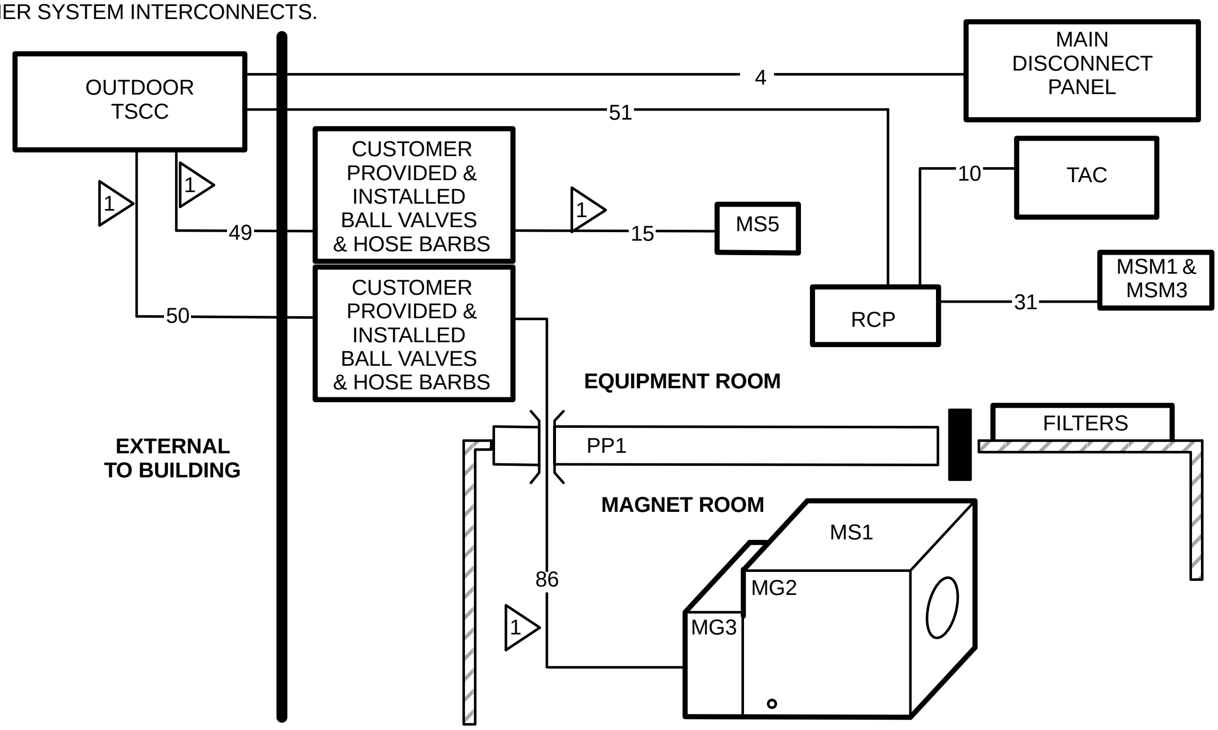
Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

System Additional Interconnects

Illustration 6–3 shows the Outdoor TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–4 for specifics.

6–5–1 Outdoor TSCC System Interconnects (Continued)

NOTE:
- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

OUTDOOR TSCC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–3

6–5–1 Outdoor TSCC System Interconnects (Continued)

TABLE 6–4
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	Out-door TSCC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	RCP	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	Cus-tomer pro-vided ball valves & hose barbs for Shield/ Cryo Cooler	MS5	55 (16.7) minus takeup at hose barbs See Notes	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	Group 15 & 49 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements**
31	0.10 (66)	MSM1	RCP	50 (15.24) **minus takeup at both ends**	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
49 SEE NOTE 1	—	Out-door TSCC	Cus-tomer ball valves & hose barbs for Shield/ Cryo Cooler	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	• Customer to provide & install 0.75 (19 mm) ID copper lines for Shield/Cryo Cooler Compressor water cooling supply & return. • Customer to provide & install in Equipment Room in end of each copper line a 0.75 in. (19 mm) ball valve & 0.5 in. (12.7 mm) hose barb. • Groups 49 & 15 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements**

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

(Continued)

6–5–1 Outdoor TSCC System Interconnects (Continued)

TABLE 6–4 (Continued)
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
50 SEE NOTE 1	—	Out-door TSCC	Cus-tomer ball valves & hose barbs for Gradi-ent Coil	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	• Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. • Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. • Group 50 and 86 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements**
51	1.097 (613)	Out-door TSCC	RCP	See NOTES	— — —	0.63 (16.0) 0.65 (16.5) 0.76 (15.9)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TSCC and height of connection at RCP. No cable takeup inside of TSCC or RCP.
86 SEE NOTE 1	1.666 (1078)	Cus-tomer pro-vided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	• Group 86 and 50 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements** • These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

6–5–2 Indoor Water Cooled TSCC System Interconnects

Note

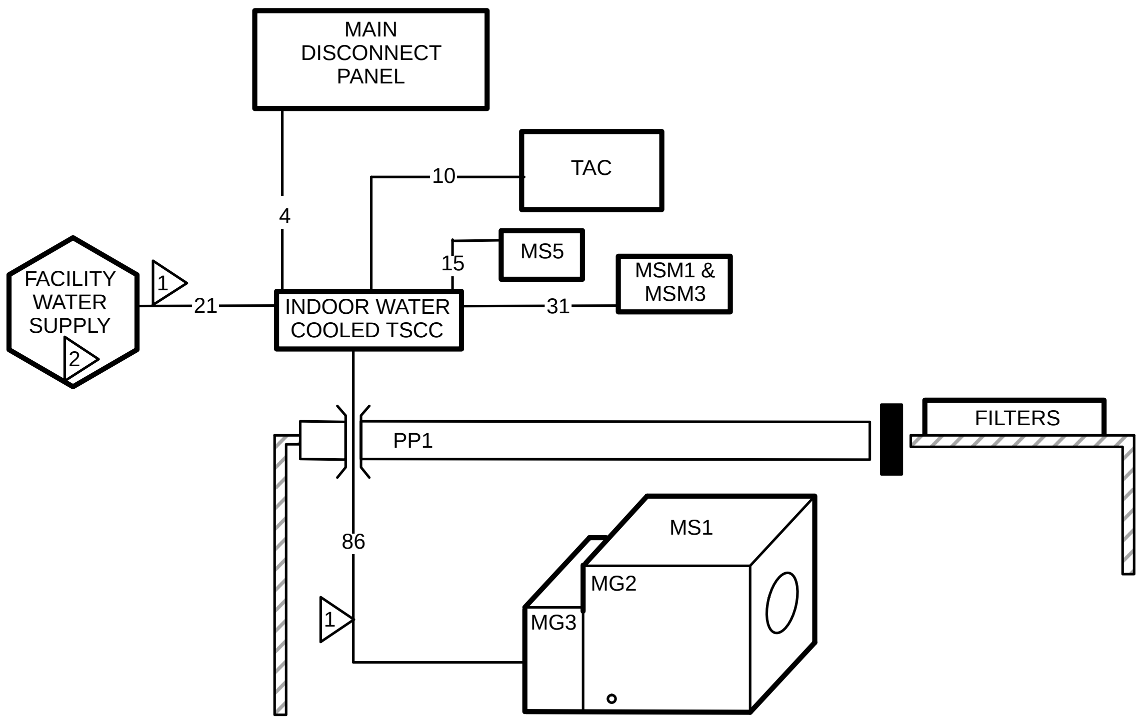
Refer to **Section 2–7–1 TSCC Configurations Siting Considerations** Table 2–5 listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 471 Pre installation Installation Operating LC 18MT/WC Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TSCC must be in accordance with local and national codes.

Illustration 6–4 shows the Indoor Water Cooled TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–5 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
2. FOR INDOOR WATER COOLED TSCC ONLY. REFER TO **Section 4–4–3 Water Cooled System Cooling Cabinet Configuration Requirements**

INDOOR WATER COOLED TSCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–4

6–5–2 Indoor Water Cooled TSCC (Continued)

TABLE 6–5
INDOOR WATER COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	rubber hose rubber hose	rubber hose rubber hose	See Notes 2 & 3 for vertical separation limits & TSCC hoses/connections.
21 SEE NOTE 1	—	Facility Water Supply	TSCC	—	—	—	—	—	For Indoor Water Cooled TSCC ONLY: Customer furnished, refer to **Section 4–4–3 Water Cooled System Cooling Cabinet Configuration Requirements** & refer to ***Note 5***
31	0.10 (66)	MSM1	TSCC	38 (11.6) **minus** takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1	1.666 (1078)	TSCC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 2 & 4 for vertical separation limits & TSCC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 The Indoor Water Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

3 The Indoor Water Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2–17 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor.

4 The Indoor Water Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2–17 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.

5 Facility chilled water connections at TSCC are 1.5 inch (38.1 mm) FNPT. Customer to connect with 1.5 inch (38.1 mm) MNPT.

6–5–3 Indoor Air Cooled TSCC System Interconnects

Note

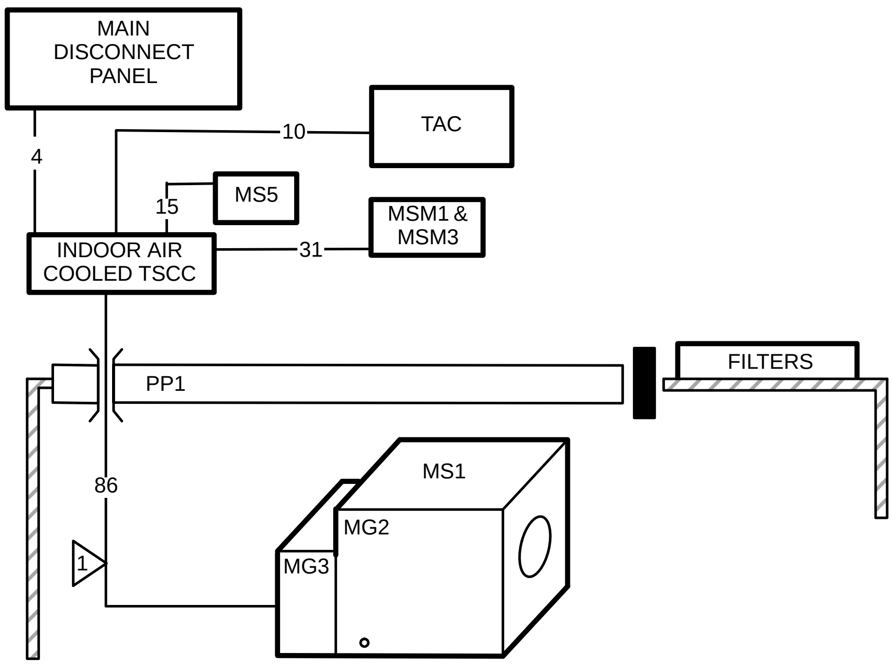
Refer to **Section 2–7–1 TSCC Configurations Siting Considerations** Table 2–5 for listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 472 Pre installation Installation Operating LC 20MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Air Cooled TSCC must be in accordance with local and national codes.

Illustration 6–4 shows the Indoor Air Cooled TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–5 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

INDOOR AIR COOLED TSCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–5

6–5–3 Indoor Air Cooled TSCC System Interconnects (Continued)

TABLE 6–6
INDOOR AIR COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	See Notes 2 & 3 for vertical separation limits & TSCC hoses/connections.
31	0.10 (66)	MSM1	TSCC	40 (12.2) **minus** takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1	1.666 (1078)	TSCC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 2 & 4 for vertical separation limits & TSCC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 The Indoor Air Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

3 The Indoor Air Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2–19 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor.

4 The Indoor Air Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2–19 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.

6–6 TGWC ADDITIONAL INTERCONNECTS

The following sections show the system interconnects required for the Twin Gradient Water Cooler (TGWC) configurations in addition to the interconnects in **Section 6–4 SYSTEM INTERCONNECTS**.

6–6–1 Outdoor TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2–7–2 TGWC Configurations Siting Considerations** Tables 2–6 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 474 Pre installation Installation Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for Outdoor TGWC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Loop Outdoor TGWC must be in accordance with local and national codes.**

Vertical Separation Requirements

- Single Loop Outdoor TGWC must not exceed 100 ft (30.5 m) vertical separation from the Gradient Coil.

Gradient Coil Water Cooling Lines & Hoses Requirements

- Ellis & Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2–14 for location of fittings on TGWC. The Outdoor TGWC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
  – Total line length from the Outdoor TGWC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TGWC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
  – If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduces 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
  – Copper lines must be insulated outdoors.

RCP Data Cables Requirements

- Ellis & Watts provides two data cables that connect between the RCP and the Outdoor TGWC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TGWC and RCP placement [total length minus height of connection at Outdoor TGWC and height of connection at RCP, no cable takeup inside of TGWC or RCP].

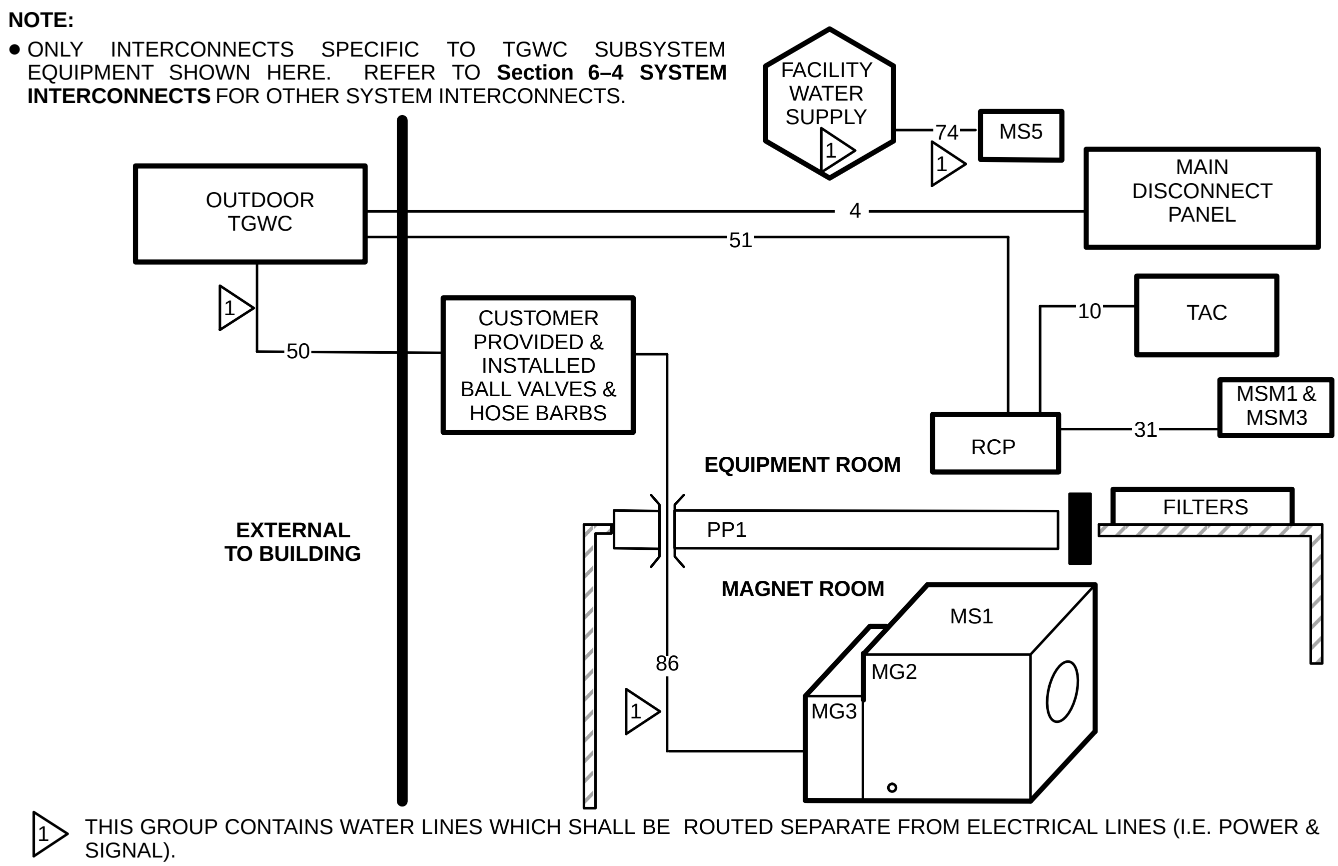
Note

Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

6–6–1 Outdoor TGWC System Interconnects (Continued)

System Additional Interconnects

Illustration 6–6 shows the Outdoor TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–7 for specifics.



OUTDOOR TGWC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–6

6–6–1 Outdoor TGWC System Interconnects (Continued)

TABLE 6–7
OUTDOOR TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
4	—	MDP	Out-door TGWC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.10 (66)	MSM1	RCP	60 (18.3) **minus takeup at both ends**	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
50 SEE NOTE 1	—	Out-door TGWC	Cus-tomer ball valves & hose barbs for Gradi-ent Coil	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	• Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. • Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. • Group 50 and 86 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements**
51	4.791 (3091)	Out-door TGWC	RCP	See NOTES	— —	1.59 (40.4) 1.89 (48.0)	1.97x2.25 (50.0x57.2) 1.75x2.25 (44.5x57.2)	1.97x2.25 (50.0x57.2) 1.75x2.25 (44.5x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TGWC and height of connection at RCP. No cable takeup inside of TGWC or RCP.
74 SEE NOTE 1	—	Water	MS5	—	—	—	—	—	Customer Furnished water lines (refer **to Sections 4–4–4 and 5–1**).
86 SEE NOTE 1	1.666 (1078)	Cus-tomer pro-vided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	• Group 86 and 50 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements** • These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

6–6–2 Indoor Water Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2–7–2 TGWC Configurations Siting Considerations** Tables 2–6 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 469 Pre installation Installation Operating LTL–10 Chilled Water System for GE Twin Magnet MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TGWC must be in accordance with local and national codes.

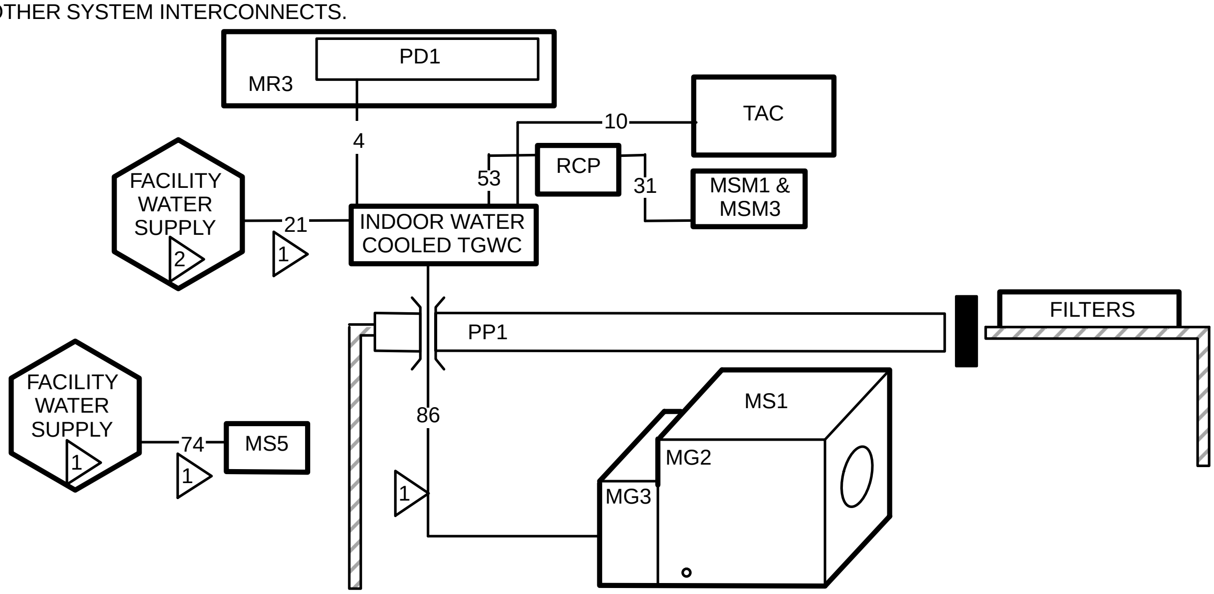
Illustration 6–7 shows the Indoor Water Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–8 for specifics.

Note

The Indoor Water Cooled TGWC and Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) of water lines. The TGWC vertical separation must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TGWC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

2 FOR INDOOR WATER COOLED TGWC ONLY. REFER TO Section 4–4–3 **Water Cooled System Cooling Cabinet Configuration Requirements**

INDOOR WATER COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–7

6–6–2 Indoor Water Cooled TGWC (Continued)

TABLE 6–8
INDOOR WATER COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	0.159 (103)	PD1 **See Note 2**	TGWC	23 (7.0)	—	0.45 (11.43)	Hard Wired	1.53x2.78 (38.9x70.6)	Ellis & Watts provided cable
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
21 SEE NOTE 1	—	Facility Water Supply	TGWC	—	—	—	—	—	For Indoor Water Cooled TGWC ONLY: Customer furnished, refer to **Section 4–4–5 Water Cooled Twin Gradient Water Cooler (TGWC) Requirements**
31	0.10 (66)	MSM1	RCP	50 (15.24) **minus** takeup at booth ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
53	0.157 (103)	Water Cooled TGWC	RCP	28 (8.53) **minus** takeup at RCP See Notes	— — —	0.258 (6.6) 0.258 (6.6) 0.258 (6.6)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	No cable takeup inside of RCP.
74 SEE NOTE 1	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to **Sections 4–4–4 and 5–1**).
86 SEE NOTE 1	1.666 (1078)	TGWC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See **Notes 3 & 4** for vertical separation limits & TGWC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

3 Indoor Water Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil.

4 The Indoor Water Cooled TGWC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2–24 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.

6–6–3 Indoor Air Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

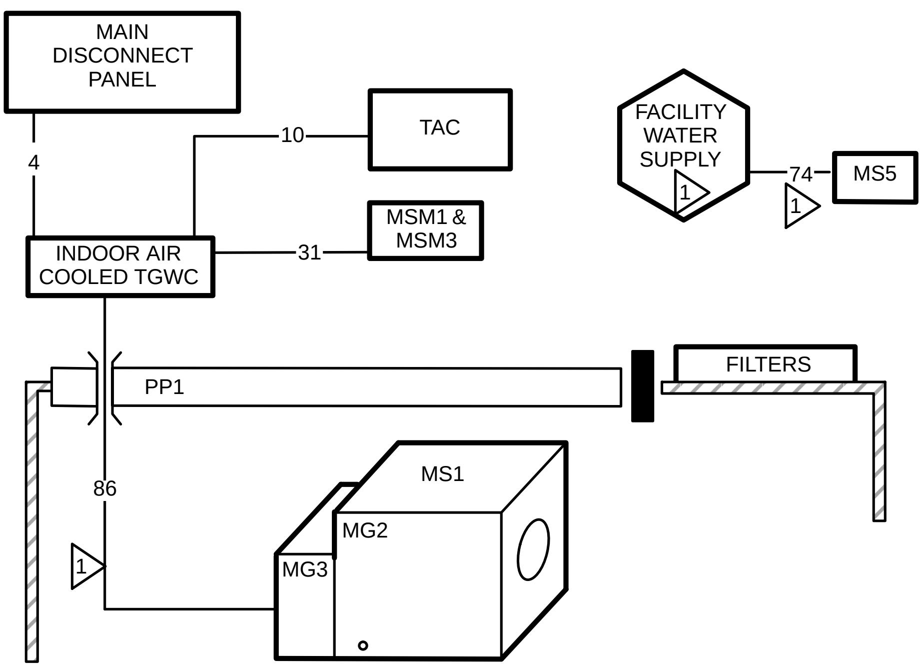
Refer to **Section 2–7–2 TGWC Configurations Siting Considerations** Tables 2–6 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 473 Pre installation Installation Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Air Cooled TGWC must be in accordance with local and national codes.

Illustration 6–8 shows the Indoor Air Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–9 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

INDOOR AIR COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–8

6–6–3 Indoor Air Cooled TGWC (Continued)

TABLE 6–9
INDOOR AIR COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TGWC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.16 (103)	MSM1	TGWC	48 (14.6) **minus** takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
74 SEE NOTE 1	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to **Sections 4–4–4 and 5–1**).
86 SEE NOTE 1	1.666 (1078)	TGWC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 2 & 3 for vertical separation limits & TGWC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 Indoor Air Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil.

3 The Indoor Air Cooled TGWC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2–26 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.

6–7 CONTRACTOR FURNISHED COMPONENTS

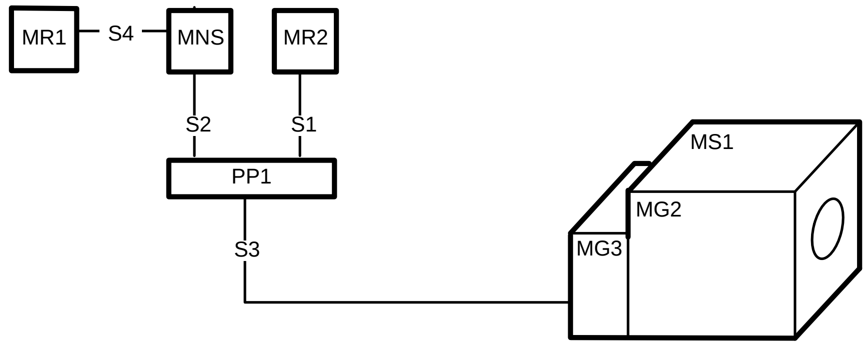
Table 6–10 lists contractor furnished components and details for connections to the system.

TABLE 6–10
CONTRACTOR FURNISHED COMPONENTS

ASSOCIATED EQUIPMENT	MATERIAL/LABOR PROVIDED BY CUSTOMER CONTRACTOR
POWER IN MAGNET ROOM	Provide and install power and wall duct for magnet rundown unit. (See **Section 5, POWER REQUIREMENT** for power specifications.)
SYSTEM GROUND	Provide ground cable between RF shielded room common ground point and Power Distribution Unit (PD1). (See **Section 5–4 GROUNDING** for cable specifications.)
EQUIPMENT POWER • Main Disconnect Panel (MDP) • Magnet Rundown Unit • MDP to Power Distribution Unit (PDU) • MDP to Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC) • Service Outlet in Magnet Room • * Oxygen Monitor • * Advantage Workstation	Provide and install power, duct work, receptacle, and coverplate for each item listed. (See **Section 5 POWER REQUIREMENTS** for power specifications.)
PLUMBING	Provide and install all water cooling equipment, see **Section 4 SITE ENVIRONMENT**, for customer supplied components & requirements.
CRYOGENIC VENTING	Provide and install cryogenic vent system. (See **Section 4–9 CRYOGENIC VENTING**, for vent specifications.)
ROOM VENTILATION	Provide and install all room ventilation equipment (e.g. Magnet Room exhaust fan) **Section 4–8 ROOM VENTILATION** for room ventilation specifications.
PENETRATION PANEL MOUNTING HARDWARE	RF shielded room vendor to provide appropriate mounting hardware for GE supplied penetration panel. (See **Section 7 RF SHIELDED ROOM**.)
RF DOOR SWITCH AND CABLING	RF shielded room vendor to provide and install RF door switches on all RF shielded room doors. All switches must be wired in series. GE supplies a 100 ft (30.5 m) cable from System Cabinet which is terminated with 2 leads. These leads are connected to the set of switches. Switches must be in the open position when RF door is open but closed when door is closed. (See **Section 7 RF SHIELDED ROOM**.)
Note * Optional Equipment. ** The Pneumatic Patient Alert Control Box can be powered from an outlet on the Operator Workspace.	

6–8 SPECTROSCOPY INTERCONNECTS

Illustration 6–9 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure. Table 6–11 contains the cable data for these interconnects.



SPECTROSCOPY SYSTEM INTERCONNECTS
ILLUSTRATION 6–9

TABLE 6–11
SPECTROSCOPY 1.5T SYSTEM INTERCONNECT LIST

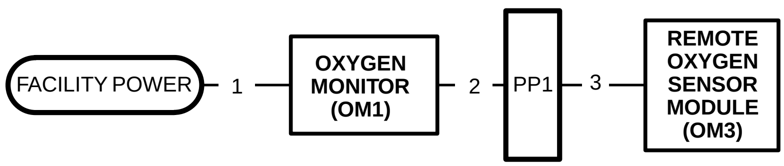
GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6–3 Group 23.
S2	0.41 (267)	PP1	MNS	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6–3 Group 25.
S3	6.824 (4403)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6–3 Group 45.
					472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6–3 Group 45.
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6–9 OXYGEN MONITOR OPTION INTERCONNECTS

The Oxygen Monitor option consists of the following items:

- Oxygen Monitor
- Remote Oxygen Sensor Module
- Interconnect cables

Illustration 6–10 shows the Interconnect Diagram. Table 6–12 contains the cable data.



DISTANT REMOTE CONSOLE CABLING
ILLUSTRATION 6–10

TABLE 6–12
OXYGEN MONITOR INTERCONNECT LIST

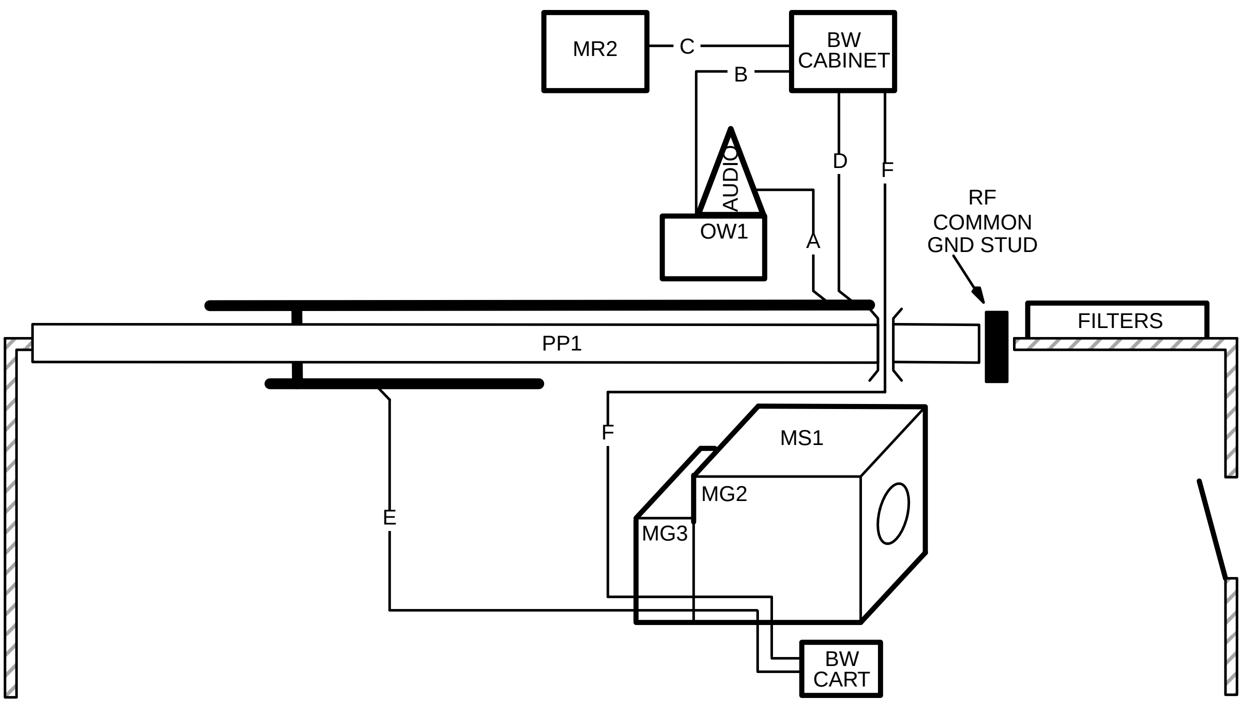
GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
1	—	Facility Power	OM1	—	—	—	—	—	Customer Furnished recommended power source for OM1 (refer to **Section 5–1**).
2	0.096 (61.94)	OM1	PP1	94 (28.7) **minus** takeup at OM1	457	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	
3	0.096 (61.94)	PP1	OM3	84 (25.6) **minus** takeup at OM3	458	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	

6–10 BRAINWAVE OPTION INTERCONNECTS

The BrainWave option consists of the following items:

- BrainWave Audio Console
- BrainWave Cabinet
- BrainWave Cart
- Interconnect cables

Illustration 6–11 shows the Interconnect Diagram. Table 6–13 contains the cable data.



BRAINWAVE OPTION GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–11

6–10 BRAINWAVE OPTION INTERCONNECTS (Continued)

TABLE 6–13
BRAINWAVE OPTION INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER (Vendor number)	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
A	0.113 (74)	Audio on OW1	PP1	92 (28.0)	758620	0.38 (9.7)	0.60x1.90 (15.2x48.3)	1.26x1.75 (32.0x44.5)	Audio power / signal cable
B	0.081 (53)	OW1	BW Cabinet	66 (20.1)	560213	0.20 (5.1)	1.66x0.38 (42.2x9.7)	1.55x0.95 (39.4x24.1)	560213 connects to Audio Console located on OW1
					758616	0.25 (6.4)	0.50x0.88 (12.7x22.3)	0.50x0.88 (12.7x22.3)	Network Cable
C	0.194 (125)	MR2	BW Cabinet	10.0 (3.0)	520423	0.38 (9.7)	1.45x2.5 (36.8x63.5)	1.45x2.5 (36.8x63.5)	Trigger Extension Cable
					560202	0.32 (8.1)	1.28x3.0 (32.5x76.2)	Hardwired	AC Power
D	0.227 (148)	BW Cabinet	PP1	60 (18.3)	979153	0.38 (9.7)	0.72x2.43 (18.3x61.7)	1.26x1.75 (32.0x44.5)	DC Power – response
					760065	0.38 (9.7)	0.6x1.9 (15.2x48.3)	2.2x2.5 (55.9x63.5)	DC power – video
E	0.340 (222)	PP1	BW Cart routed through MG2/3	53 (16.2) 53 (16.2)	770065	0.38 (9.7)	2.2x2.5 (55.9x63.5)	0.75x2.0 (19.1x50.8)	DC Power – vidoe. This run is routed through MG2/3 and out to the BW Cart.
					758619	0.38 (9.7)	1.26x1.75 (32.0x44.5)	0.60x1.90 (15.2x48.3)	Audio power/signal cabler. This run is routed through MG2/3 and out to the BW Cart.
					979152	0.38 (9.7)	1.26x1.75 (32.0x44.5)	0.72x2.43 (18.3x61.7)	DC power – Response This run contains a connector at MG2 with a separate cable routed out to the BW Cart.
F	0.239 (156)	BW Cabinet	BW Cart	106 (32.3)	758615	0.40 (10.2)	1.75x2.5 (44.5x63.5)	1.75x2.5 (44.5x63.5)	Video fiber optic
					781100	0.38 (9.7)	air tubing	air tubing	These interconnects are routed through PP1 Waveguide, through MG3, & partially through MG2 (exits at midpoint of Enclosure side). Usable length allows 6 ft (1.8 m) from MG2 to BW Cart.

SECTION 7 – RF SHIELDED ROOM

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TABLE OF CONTENTS

7–1 RF SHIELDED ROOM SPECIFICATION

Every GE MR system requires that the Magnet Room be RF shielded. Table 7–1 contains the RF Shielded Room specifications.

Note

It is advisable to check the integrity of any RF Shielded (screen) Room in proximity to the MR system. Poor RF Shield integrity can contribute to system image quality artifacts i.e. zippers.

TABLE 7–1
RF SHIELDED ROOM REQUIREMENTS

PARAMETER	REQUIREMENTS
RF ATTENUATION	• 100dB (10MHz – 100MHz) **planewave** with RF vendor supplied blank panel, refer to TESTING requirements in this table.
GROUND ISOLATION	• 1,000 ohms or greater Refer to Section 7–7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM for additional information on electrical isolation requirement.
MATERIALS	• The choice of material is the responsibility of the customer's architect and RF vendor. Normally, copper–brass or treated aluminum is used because these materials are non–magnetic and will not affect homogeneity. However, RF Shielding has also been fabricated from galvanized steel or by modifying steel magnetic shielding to produce the required RF attenuation. Any steel RF enclosure will affect the magnet's homogeneity and must be reviewed by GE Medical Systems MR Siting and Shielding Group. The door or any other moving or non–rigid parts must not be fabricated from magnetic materials.
CONSTRUCTION	• The design of the shield support system is the responsibility of the customer's architect and RF vendor. If magnetic steel panels are used, these materials must be rigidly supported to prevent any slight movement, from air pressure changes or other reasons, that could degrade magnet homogeneity and system performance. For safety reasons, magnetic steel material must also be well anchored; loose steel components can become dangerous projectiles and accelerated into the magnet. • It is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems.
TESTING	• The customer's architect and RF vendor are responsible for conducting RF attenuation and ground isolation tests to verify that the shield meets GE specifications. The RF shielded room verification test is to be performed in the presence of a GE representative. • When to test: The RF shielded room must be 100% finished, magnet installed, all penetrations complete including floor mounting bolts. The test must be conducted with an RF vendor supplied blank penetration panel and the same mounting hardware which is used with the GE penetration panel. • How to test: The FINAL RF Shielded room acceptance test shall be performed in accordance with Appendix B, RF SHIELDED ENCLOSURE TEST GUIDELINE.
MAINTENANCE	• Follow RF vendor's recommended maintenance. Alert GE Service Representative of any RF shielded room maintenance issues since there may be system performance impacts.
ACOUSTIC	• RF Screen Room including all openings (i.e. windows, doors, vents, etc.) need acoustic properties to meet local regulations and customer requirements. **Note** RF Screen Room doors with <55db acoustic attenuation have caused customer acoustics issues. Refer to Section 4–7 ACOUSTICS for additional information.

7–2 VENTS

7–2–1 Cryogenic

Due to normal boil–off of liquid helium and the possibility of a quench with superconducting magnets, outside cryogenic venting is required. RF shielded room contractor is to provide one straight pipe with maximum 0.125 in. (3.175 mm) wall thickness for the cryogenic vent pipe/waveguide. The vent pipe/waveguide is to be made of non–magnetic material which is grounded to the RF room and electrically isolated from any other grounds. The vent pipe/waveguide must extend inside and outside of the RF shielded room to allow for non–metallic isolation joint connections. The HVAC (heating, ventilation, and air conditioning) contractor is to make cryogenic vent connections to vent pipe/waveguide outside of the RF shield and GE will make the normal connection in the Magnet Room. Refer to Section 4–9–2, Requirements For Inside Magnet Room, for exceptions.

7–2–2 Determining Cryogenic Vent Location

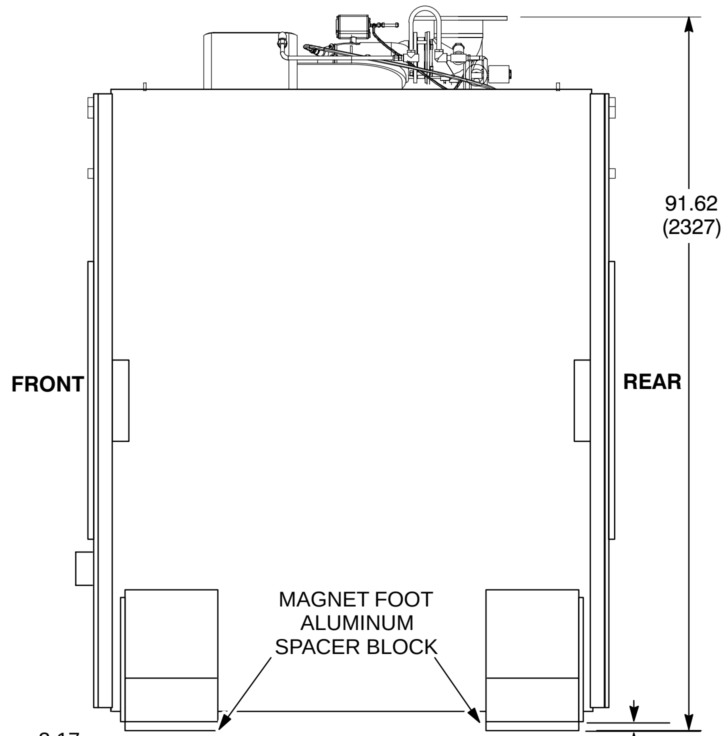
Shown in Illustration 7–1 is the LCC magnet vent location. It is important that the 0.25 in. (6.35 mm) tolerance for vent opening be met. Included with the 1.5T LCC (Active Shield) magnets is a vertical helium vent tube made with with tubing of 6061–T6511 aluminum tubing 8 in. (203.2 mm) OD x 0.125 in. (3.175 mm) wall, per ASTM B241. A typical route for this vent tube is shown in Illustration 7–2. However, if the tolerances shown in Illustration 7–1 can not be met, the customer's contractor is responsible for the design and installation of a cryogenic vent system which meets the requirements in Section 4–9, CRYOGENIC VENTING.

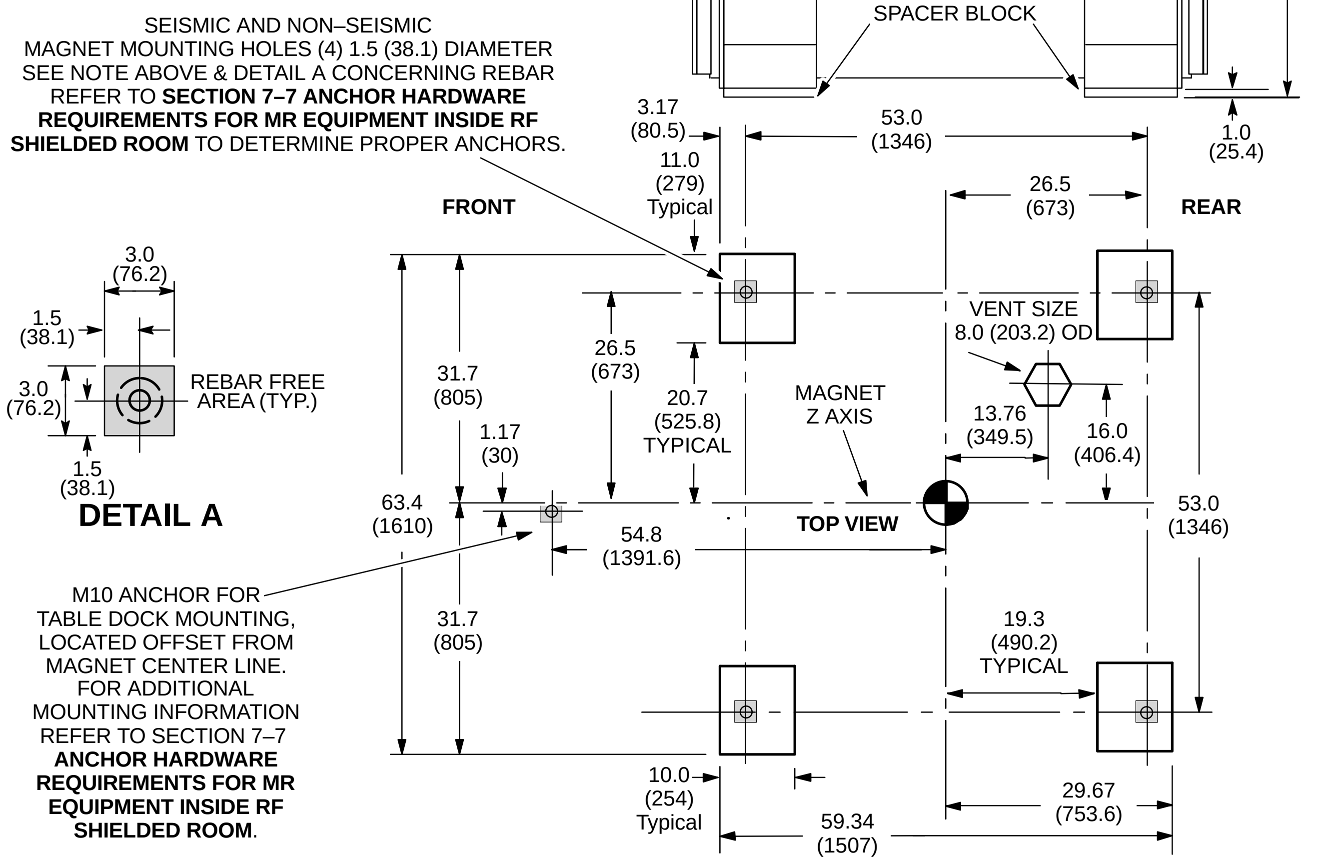
GE provides a Floating Flange Vent Adaptor Kit with all of the 1.5T Active Shield magnet. The kit increases the vent adapter radial adjustability in any direction to one inch (25.4 mm). The adjustability helps absorb some mismatch (1 inch (25.4 mm) in any horizontal direction) of the locations of the Vent Pipe from the magnet and the Ceiling Vent Pipe, installed by the RF Room contractor. See Illustration 7–3. The GE provided Vent Pipe is 24 inches (610 mm) long with a wall thickness of 0.125 inch (3.175 mm) and may be cut short to create 1 $\pm$ 0.25 inch (25 $\pm$ 6 mm) gap between the Vent Pipe and the Ceiling Waveguide for dielectric isolation to ensure the integrity of the RF shield room.

7–2–2 Determining Cryogenic Vent Location (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8–1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH TRM 12,826 lbs (5818 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **SECTION 8 SHIPPING AND DELIVERY DATA**

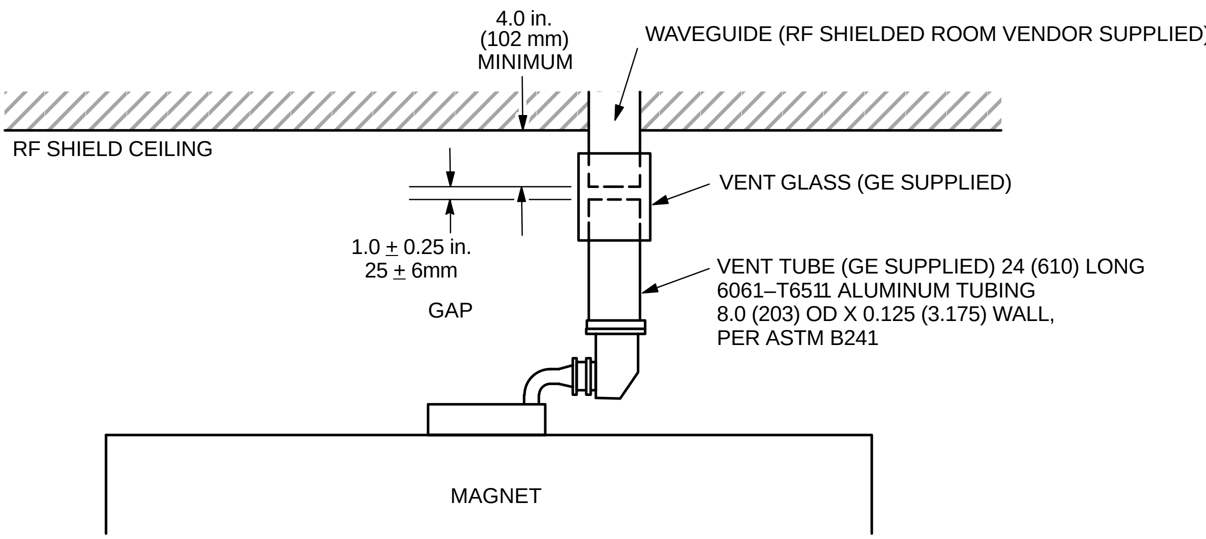




LCC MAGNET CRYOGENIC MOUNTING & VENT LOCATION
ILLUSTRATION 7–1

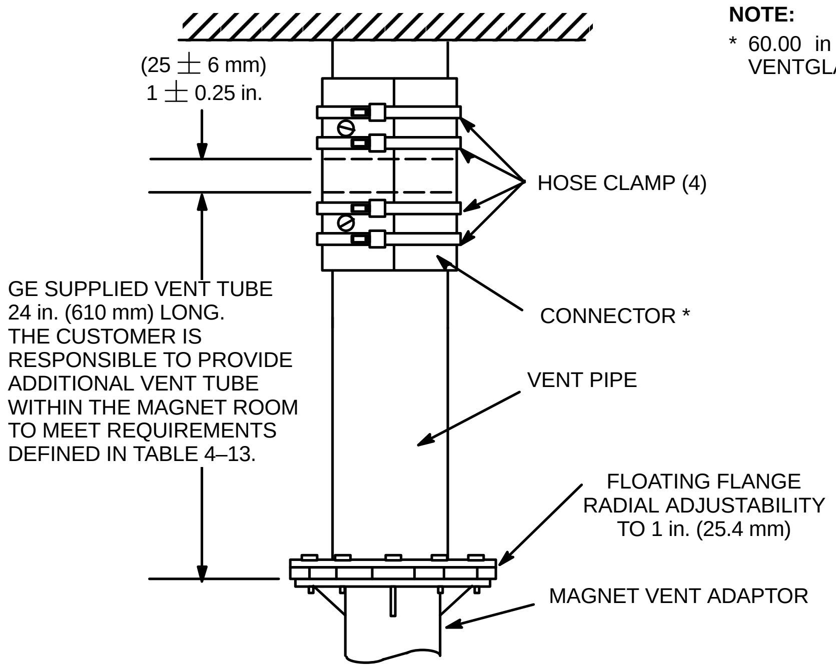
7–2–2 Determining Cryogenic Vent Location (Continued)

NOTE:
- ALL DIMENSIONS ARE IN INCHES
  ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.



CRYOGENIC VENT ROUTING
ILLUSTRATION 7–2

NOTE:
* 60.00 in X 8.00 in (152 cm X 20 cm) VENTGLAS CONTINUOUS WRAP SLEEVE



FLOATING FLANGE
ILLUSTRATION 7–3

7–2–3 Waveguide

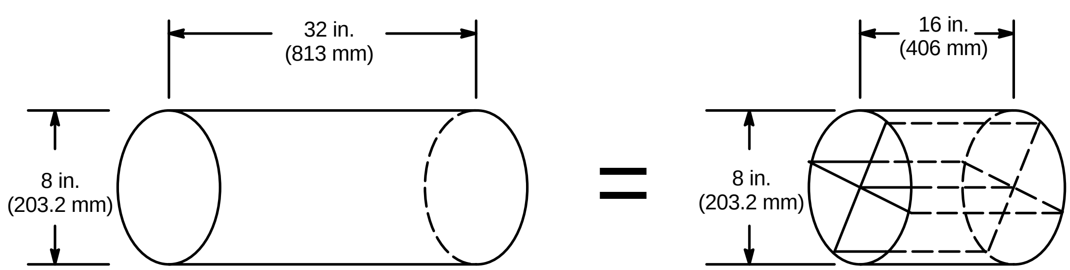
RF shield room contractor/designer is responsible for choosing and installing a RF shield waveguide . The generally accepted length of the waveguide is four times the outside diameter of the tube. Therefore, an 8 in. (200 mm) OD waveguide should be 32 in. (800 mm) long. Refer to Table 7–2 for list of GE requirements for the waveguide.

TABLE 7–2
WAVEGUIDE REQUIREMENTS

PARAMETER	REQUIREMENTS
WAVEGUIDE SIZE	• If the provided GE vent tube is to be connected directly to the waveguide via the GE Ventglas method, the outside diameter of the waveguide must match the outside diameter of the GE vent tube within $\pm$ 0.125 in. (3 mm). Larger mismatches in diameters will result in unacceptable leakage during a quench. • The generally accepted length of the waveguide is four times the inside diameter of the tube. Therefore, an 8 in. (203.2 mm) OD waveguide should be 32 in. (813 mm) long.
WAVEGUIDE MATERIAL	• The Waveguide must be constructed from one of the GE accepted materials (i.e. stainless steel, aluminium or copper). Typically, the waveguide is made from the same material as the RF shielded enclosure to avoid dissimilar metal interfaces (i.e. galvanic reaction).
WAVEGUIDE CONSTRUCTION	• The waveguide does not have to be positioned equally on either side of the RF ceiling (or wall). For example, 6 in. (152.4 mm) of a 32 in. (813 mm) long waveguide may hang down below the RF ceiling with the remaining 26 in. (660 mm) extending above. • If a 90 degree elbow is required to avoid ceiling structures, the elbow can be a part of the waveguide and contribute to its overall length. Waveguides do not have to be completely straight. • If a full length waveguide is not possible due to structural interferences, a shorter waveguide can be fabricated by dividing the inside volume into no more than four chambers. For example, if an 8 in. (203.2 mm) OD waveguide is divided into four equal chambers, as shown in ILLUSTRATION 7–4, the length of the waveguide may be decreased from 32 in. (813 mm) to 16 in. (406 mm). If this technique is used, 1 psig must be added to the pressure drop calculation to account for the pressure drop of the four chambered waveguide. (Section 4–9, CRYOGENIC VENTING) • Flat, honeycomb type waveguide is not acceptable.

NOTE:

- 1 psig MUST BE ADDED TO THE PRESSURE DROP CALCULATION TO ACCOUNT FOR THE PRESSURE DROP OF THE FOUR CHAMBERED WAVEGUIDE.
- IN A CASE OF WAVEGUIDE LENGTH RESTRICTION, A HALF LENGTH WAVEGUIDE WITH FOUR CHAMBERS MAY BE USED.



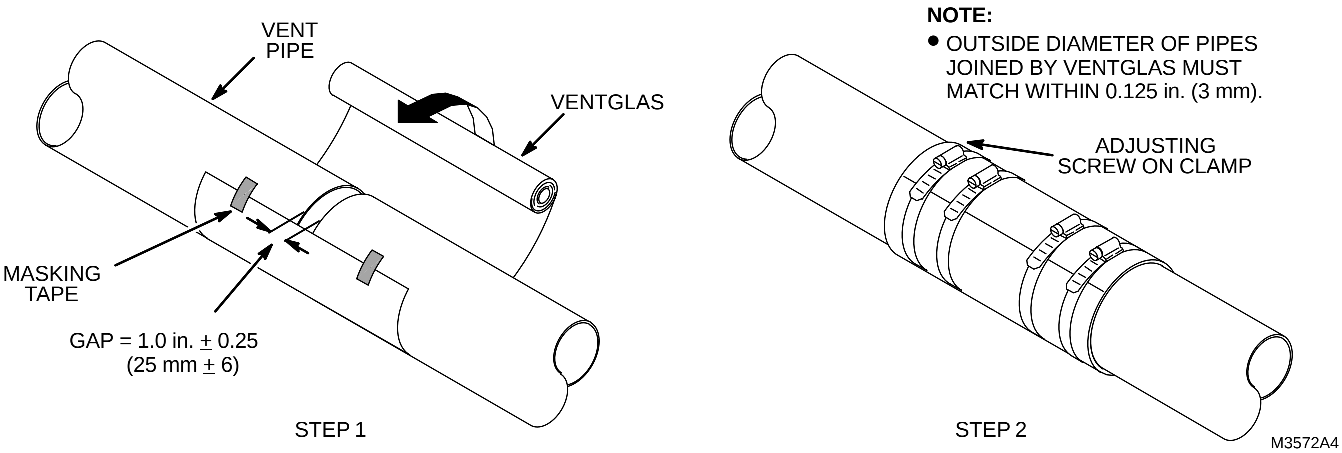
4 CHAMBER WAVEGUIDE
ILLUSTRATION 7–4

7–2–4 Guide For Outside RF Room Isolation Joint

The RF shielded room contractor/designer is responsible for choosing and installing an isolation joint outside of the RF shielded room as shown in Section 4–9–1, Requirements For Outside Magnet Room. This isolation joint is required to maintain the single point ground concept for the RF shielded room. Table 7–3 contains suggestions for the RF room isolation joint.

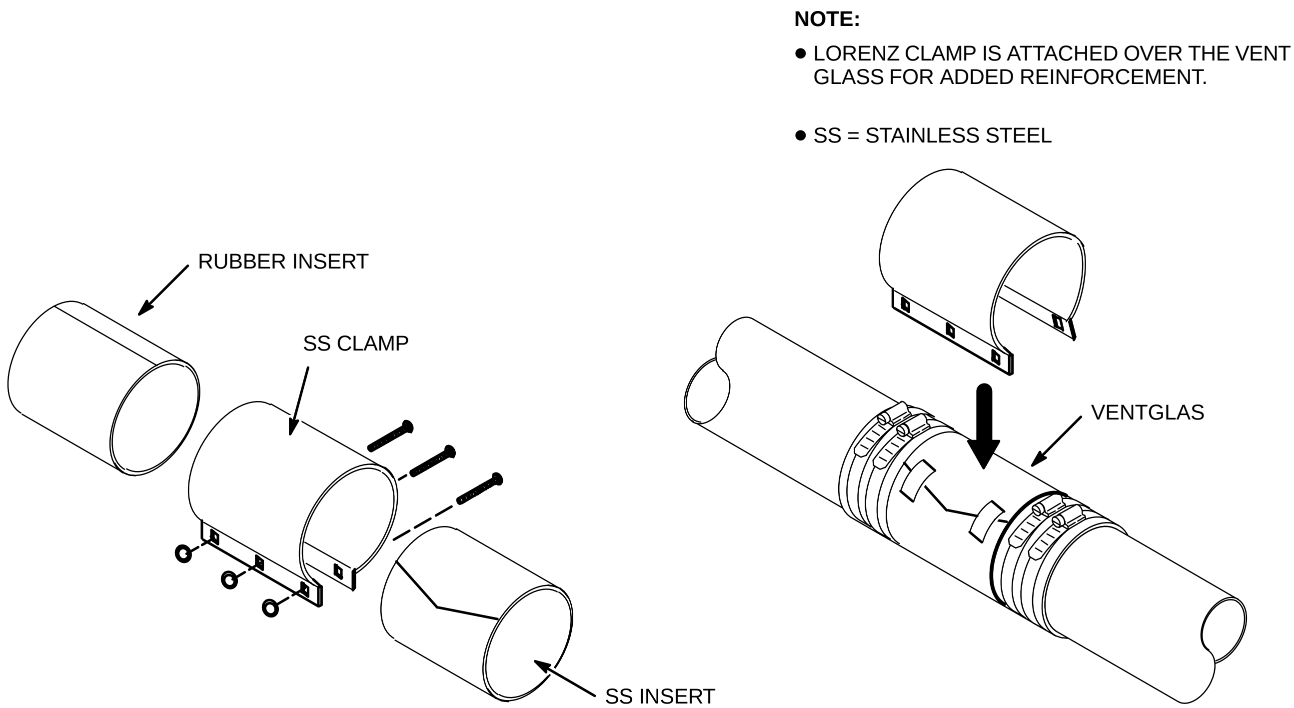
TABLE 7–3
OUTSIDE RF ROOM ISOLATION JOINT SUGGESTIONS

PARAMETER	ISOLATION JOINT SUGGESTIONS
ISOLATION JOINT MATERIAL	• PVC, rubber or soil pipes **must not** be used to construct the isolation joint. • Ventglas and Lorenz clamp are two GE recommended methods of achieving the isolation.
ISOLATION JOINT CONSTRUCTION	• Ventglas connection method is shown in Illustration 7–5 • Lorenz clamp connection is shown in Illustration 7–6 • The mating diameters must match within ± 0.125 in. (3 mm). • The Ventglas **must not** be used for structural support.
INFORMATION	• Ventglas information may be obtained from: Industrial Machine & Fabricating Inc. e–mail address indmachfab@aol.com 2808 E Sammys Lane Florence, SC 29506–3841 USA (843) 667–4582 Vent Fabrics Inc. web site address www.ventfabrics.com 5520 N Lynch Avenue Chicago, IL 60630–1418 USA (800) 621–1207 or (773) 775–4477 • Lorenz clamp information may be obtained from: Lorenz and Son Mfg. Co. LTD. P.O. Box 1002 Cobourg, Ontario, Canada K9A4W4 (905) 372–2240, fax (905) 372–4456



VENT GLASS CONTINUOUS WRAP
ILLUSTRATION 7–5

7–2–4 Guide For Outside RF Room Isolation Joint (Continued)



LORENZ CLAMP
ILLUSTRATION 7–6

7–2–5 HVAC

RF shielded room contractor is to install HVAC waveguides (open pipes or honeycomb–type) which penetrate room and to ensure waveguides are non–magnetic and electrically isolated. HVAC contractor is to determine size and number of vents, consistent with local codes.

An exhaust fan placed outside the RF shielding with appropriate wave guide filtering is required for quick removal of helium gas in the event large amounts of helium disperse into the Magnet Room. The exhaust fan can be connected to the output relay of the optional oxygen monitor. The fan will then be activated in the event the room oxygen level is less than 18%. Refer to Section 4–8, ROOM VENTILATION, for other exhaust fan requirements.

7–3 PLUMBING

All metallic pipes entering the RF Room, excluding cryogenic vent and floor drains, must be located within 30 inches (762 mm) of the RF common ground.

Note

When welding in an MR room with system equipment installed, the return path for the welding must be in very close proximity to the welding. The close proximity is needed to make sure the welding currents do not cause damage to the system. Never use the building structure as a return path for welding.

7–3–1 Water

All pipe waveguides must meet the 100 dB requirements. If a floor drain is to be installed in the Magnet Room, it must be electrically isolated and meet the 100 dB requirements. All plumbing must be consistent with local codes.

7–3–2 Medical Gases

The customer should consider if medical gases are to be piped into the Magnet Room along with suction service for patient life support. Remember, all non–electrical entries into the Magnet Room must use appropriate waveguide. Special precaution must be taken to ensure that ferromagnetic medical gas cylinders are not brought into the Magnet Room.

7–3–3 Sprinklers

If using sprinklers in the Magnet Room, dry pipe systems have the advantage of reducing ground problems. However, all decisions regarding fire protection systems are the customer’s responsibility. If wet–type sprinkler system is used, pipe penetration should be limited to one location.

7–4 ELECTRICAL

The entry of any electrical lines into the RF shielded room must be filtered to ensure that the RF shielded room meets the minimum attenuation levels. The RF shielded room vendor must supply filters for all penetrations of the RF shielding excluding the lines entering through the GE supplied RF penetration panel. All filters (for electrical lines) must be located outside the 200 gauss line.

Note

AC outlets must have ground wires firmly attached per electrical codes to avoid intermittent grounding which can cause undesirable emi (electro magnetic interference) issues within the RF shielded room.

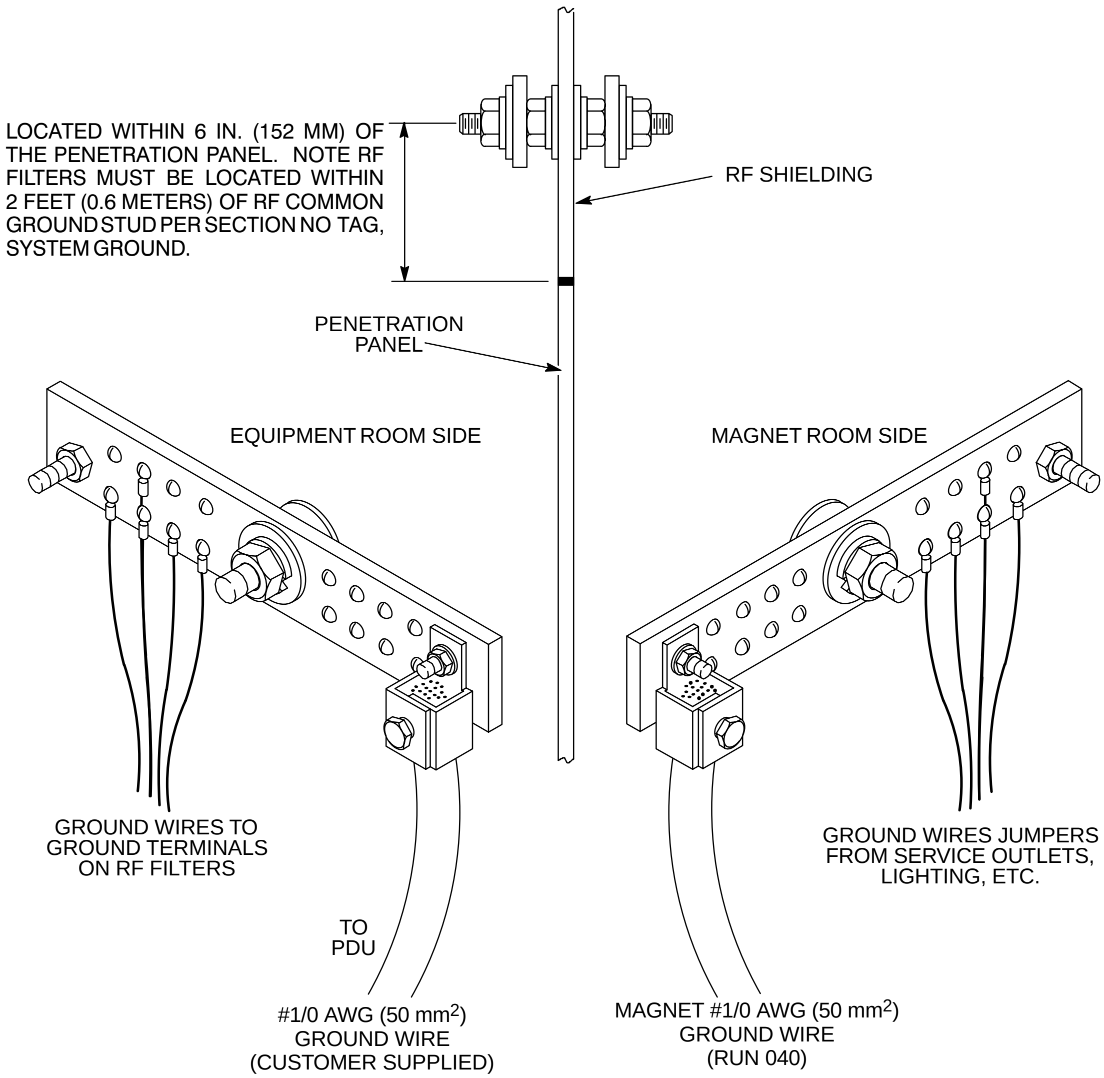
RF shielded room vendor should review with the electrical contractor the number of incoming power lines to the Magnet Room in determining the number of filters needed for electrical requirements.

The power ground lines from customer supplied power filters must be grounded to the RF shield common ground point and be located as close as possible to the RF penetration panel. See Section 5, POWER REQUIREMENTS, for power and grounding requirements of all incoming power lines to the RF shielded room.

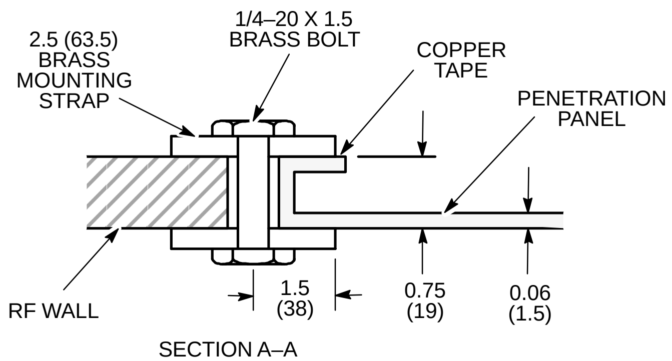
All lighting in the RF shielded room must be DC/incandescent. Dimmer switches must not be used; however, a selectable switch may be used to change the light intensity. For additional Magnet Room lighting information refer to Section 4, SITE ENVIRONMENT, Section 5, POWER REQUIREMENTS, and Section 6, INTERCONNECT DATA.

Common ground connection for shielded room must be located within 6 in. (152 mm) of the RF shielded room Penetration Panel with RF filters located within 2 feet (0.6 meters) of the RF common ground. RF shielded room vendor to provide this common ground connection on both sides of shielded room by means of a stud extending through the shielded room (see Illustration 7–7). RF Common ground stud must be accessible for servicing purposes on both sides of shield room. For aesthetic purposes, it is recommended that the stud be positioned above the Penetration Panel so it is concealed behind the penetration panel cover (see Illustration 7–10).

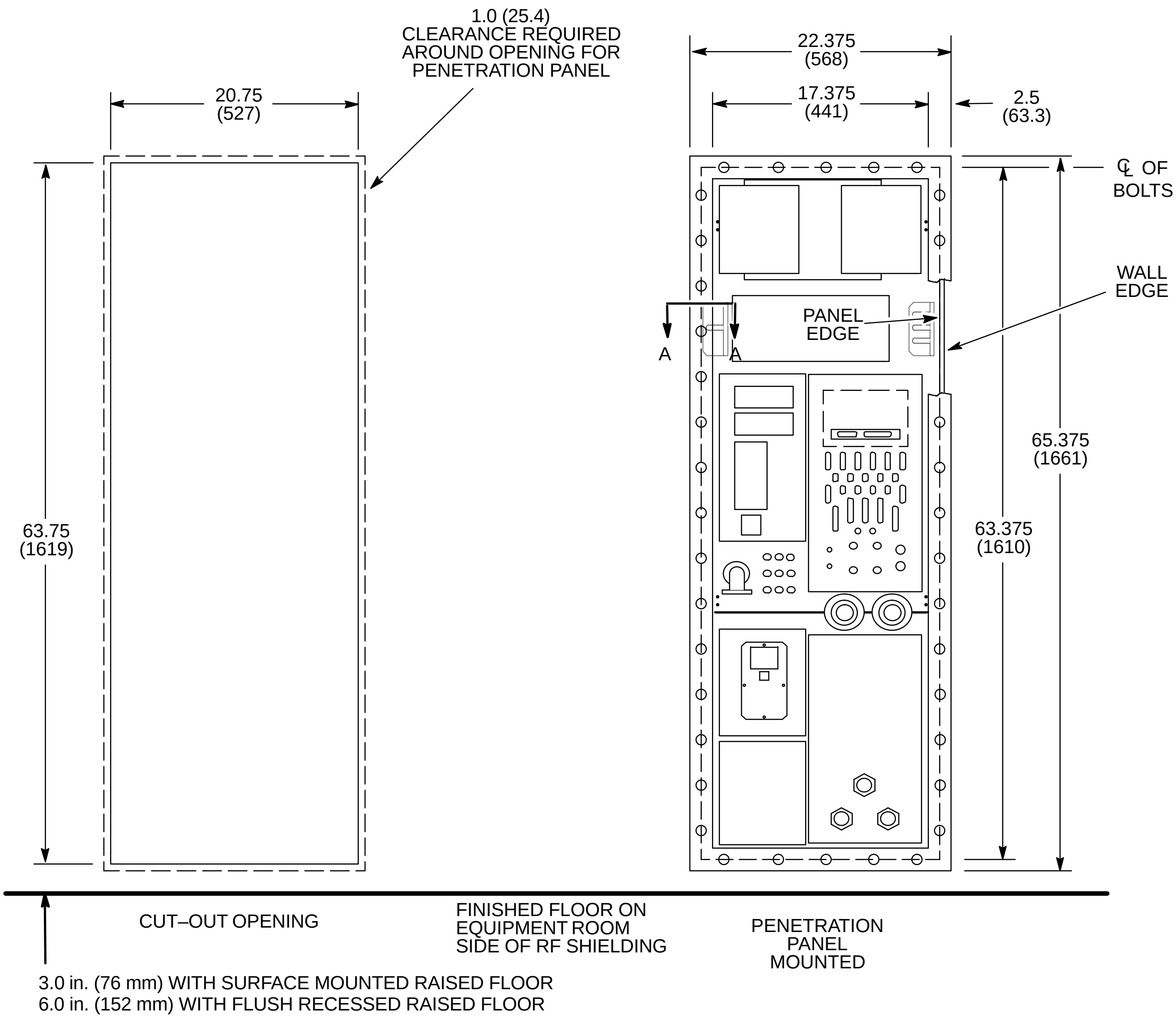
NOTE:
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS



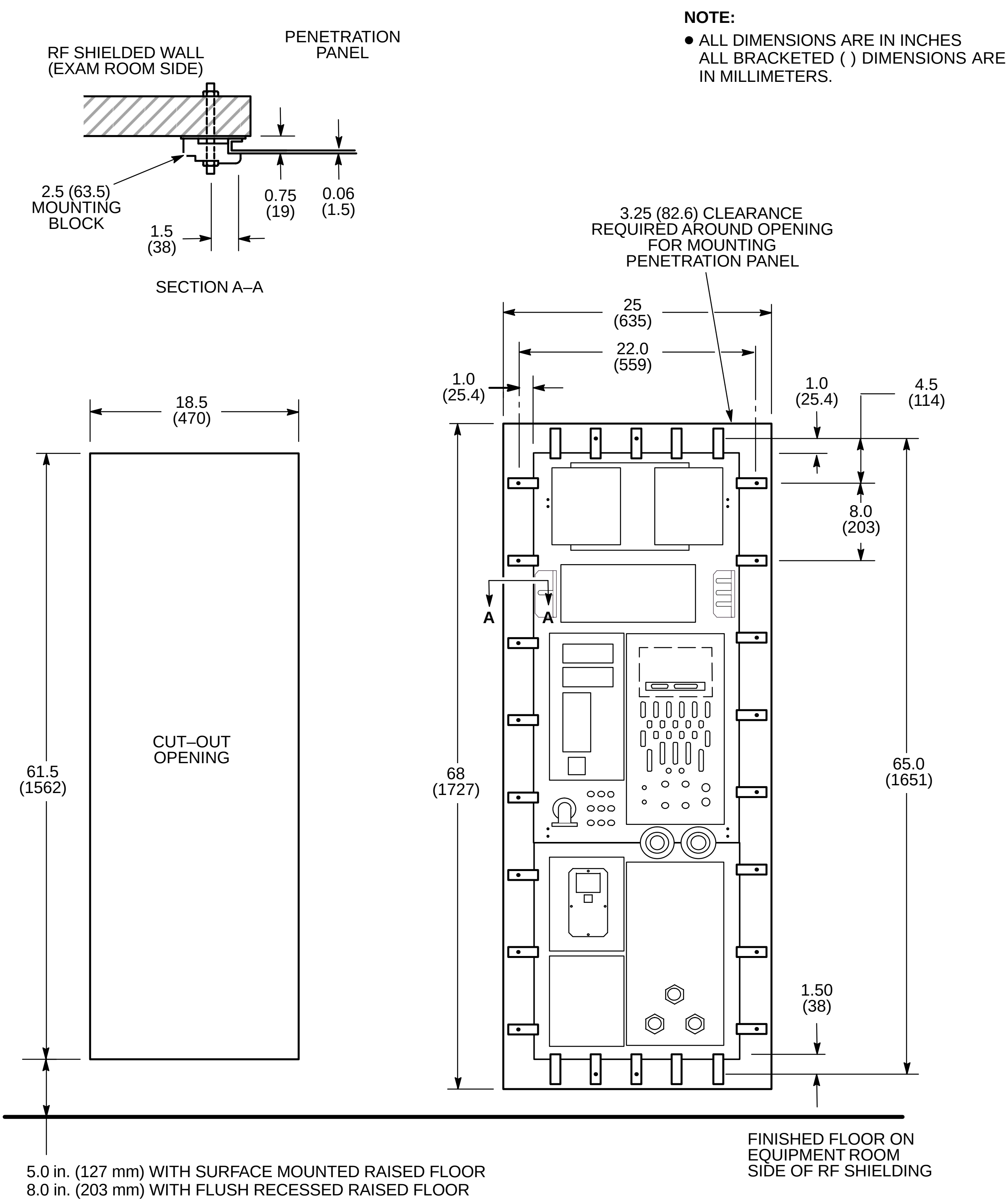
RF COMMON GROUND PENETRATION STUD
ILLUSTRATION 7–7



NOTE:
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- ALL DIMENSIONS ARE ± 0.0625 (1.6)



PENETRATION PANEL CUT OUT FOR 0.75 INCH (19 MM) THICK RF WALL
ILLUSTRATION 7–8



PENETRATION PANEL CUT OUT FOR RF WALL THICKNESS VARYING FROM SMALL TO LARGE
ILLUSTRATION 7–9

7–5 RF PENETRATION PANEL

Illustrations 7–8 and 7–9 show two different methods for mounting the GE MR penetration panel. Either method may be used depending on RF shielded room wall thickness. Ensure if the mounting method in Illustration 7–8 is used, the RF wall thickness is 0.75 in. (19 mm) $\pm$ 0.0625 in. (1.6 mm). Check with RF shielded room vendor to determine appropriate mounting method. The penetration panel must be covered on both sides for safety. If GE supplied adjustable covers are not used, customer must furnish covers or enclosures with key or tool required for opening to limit access to the panel.

The penetration panel is to be mounted above the finished floor on the penetration cabinet side of the RF shielded room. GE supplies only the penetration panel as shown in Illustrations 7–8 and 7–9. The mounting and clearance dimensions for the penetration panel covering are shown in Illustration 7–10.

RF shielded room vendor must provide the opening in the RF shielding and appropriate mounting hardware for the GE penetration panel. The RF shielded room acceptance test must be performed after the opening is cut in the RF shielding for the GE penetration panel. This acceptance test must be conducted with vendor supplied blank panel and the same mounting hardware to be used with the GE penetration panel. It is the facility's responsibility to ensure that the RF shielded room vendor testing meets the attenuation specifications listed in Section 7–1, RF SHIELDED ROOM INTRODUCTION.

7–6 PHYSICAL CONSIDERATIONS

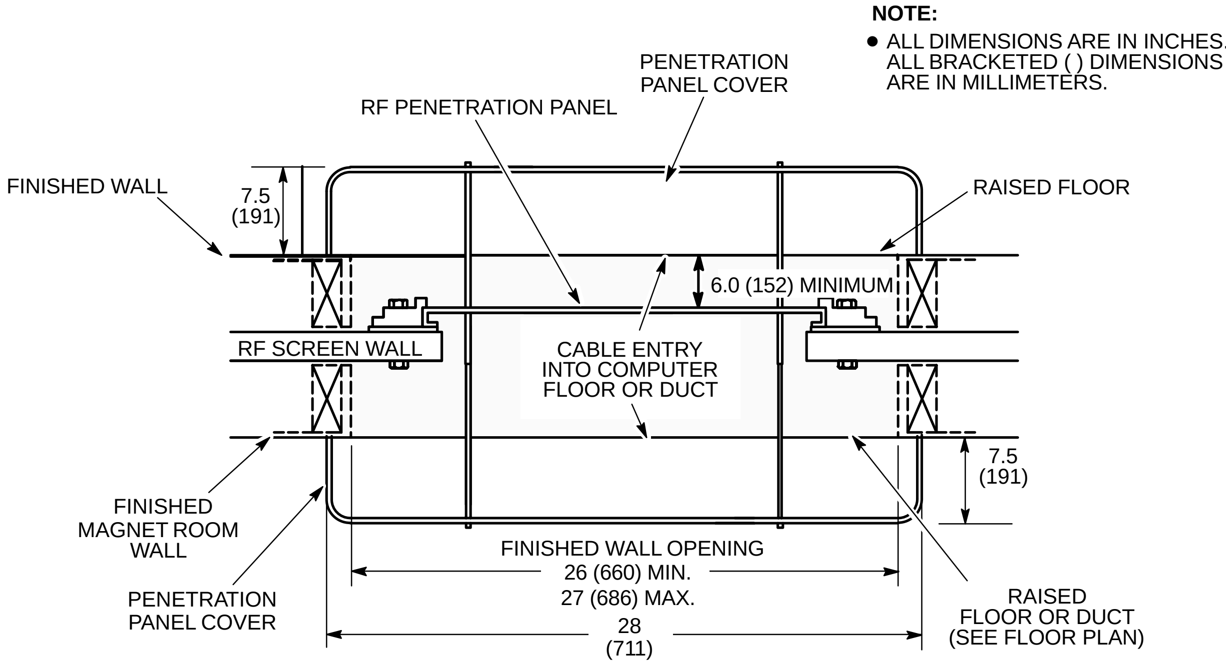
The RF shielded room can be either a free standing shielded structure or a shielded room within an existing room. Either style must be electrically isolated from earth ground by 1000 ohms or greater. Magnetic shields that are used as RF shield must also be isolated from ground by the same specification.

7–6–1 Doors and Other Openings

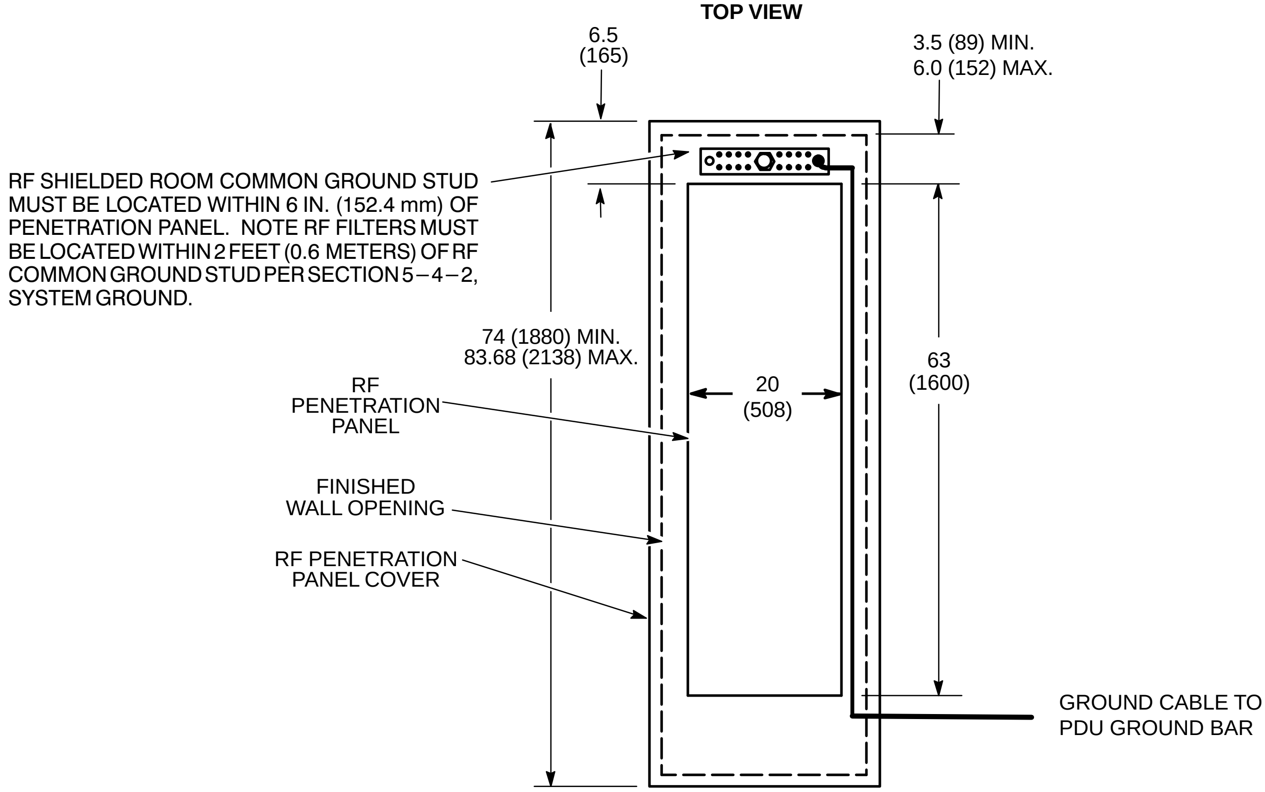
Shielded room doors are a major source of RFI leaks and should be kept to a minimum.

The main door requires a minimum finished opening of 43 in. (1092 mm) to allow for helium dewars and patient tables to pass through the opening. However, a 48 in. (1219 mm) wide door is recommended for easy maneuvering of the patient table. Maximum door sill height is 1 in. (25 mm) with a 10 degree maximum threshold inclination.

For moving the standard magnet into the room, a 9 ft wide by 10 ft high removable panel wall or 9 ft by 10 ft roof hatch is required. This normally consists of several panels that are bolted together to ensure a good RF shield when opening is not in use. Since future access may be required for magnet entrance/exit through the removable panels, avoid permanently sealing the removable panels. Note the room removable wall panel or roof hatch opening may need to be larger to achieve the required opening defined above in this section. Consideration for clear opening dimensions is especially applicable to sites requiring magnetic shielding.



TOP VIEW



PENETRATION PANEL/COVER
(FRONT VIEW)

PENETRATION PANEL/COVERING MOUNTING REQUIREMENTS
ILLUSTRATION 7–10

7–6–2 Ceiling Height

Table 7–4 lists the Magnet Room absolute minimum ceiling height required for servicing the listed magnets. This height is required for the area directly above the magnet. GE Medical Systems, Modality Installation Planning Service group must be notified of any ceiling dimensions less than ceiling heights stated in Table 7–4 and shown in Illustration 7–11. Modality Installation Planning Service group can be reached at (262) 548–4500.

TABLE 7–4
MAGNET SERVICING CEILING HEIGHT REQUIREMENTS

MAGNET TYPE	**ABSOLUTE MINIMUM CEILING HEIGHT IN. (MM) See Note 1**	**COMMENTS**
1.5T LCC (Active Shield)	105 (2667)	Magnet servicing is performed from a platform ladder which is positioned at the Coldhead side of the Magnet. Ceiling height allows for clearance for fill line stinger insertion.
1.5T LCC (Active Shield) with Low Ceiling Height Siting Option (M1060SR) installed	98.5 (2500)	Magnet servicing is performed from a platform ladder which is positioned at the Coldhead side of the Magnet. Ceiling height allows for clearance for insertion of fill line with 7 inch (178 mm) stinger insertion.
Note * Magnet servicing is performed on a platform mounted on top of the Magnet. Ceiling height includes 39 inches (991 mm) of clearance for Service Personnel. 1 Absolute minimum ceiling height values are from magnet room finished floor to fixed ceiling.		

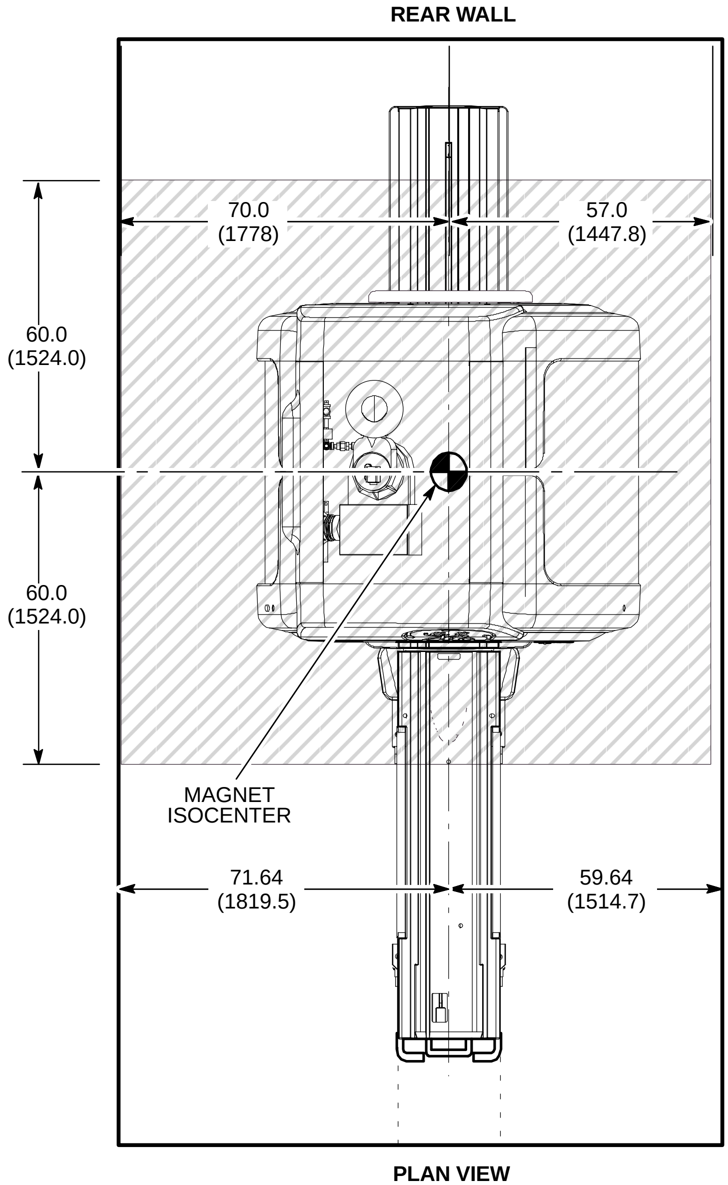
Use of a standard valved helium transfill line and a 250 liter dewar (not more than 70 in. (1778 mm) high) requires a ceiling height of 135.5 in. (3442 mm) for inserting transfill line into the dewar. Note that this need only be a 24 in. (610 mm) square ceiling recess located either in the Magnet Room or in an accessible area near the Magnet Room where the transfill line can be inserted into the dewar. A 500 liter dewar (not more than 73 in. (1854 mm) high) requires a ceiling height of 138 in. (3505 mm) for the same process.

If the helium transfill requirements cannot be satisfied in or near the Magnet Room, consider a location outside the building or on a loading dock. The standard valved transfill line, after insertion into either a 500 or 250 liter dewar, will fit through 79 in. (2007 mm) high doorways and hallways. Provide free access from the dewar location to the magnet. If elevators are to be used along cryogen delivery route, verify that elevator dimensions and weight capacity is sufficient to handle the cryogen dewars. Also, elevator must be dedicated with restricted access during cryogen transport (will not allow stops between initial start and final floor destination).

7–6–2 Ceiling Height (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
  ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- SHADED AREA ROOM CEILING MUST NOT BE BELOW TABLE 7–4 REQUIREMENTS.



MINIMUM CEILING HEIGHT 1.5T LCC (ACTIVE SHIELD) MAGNET
ILLUSTRATION 7–11

7–6–3 Walls

It is recommended that walls be covered to protect RF material and to add to the aesthetics of the room for patient comfort. Fire retarding material must be used per building codes. Consult RF shield room vendor for RF shielding service requirements prior to covering RF walls. Removable wall covering may be needed if periodic RF shield servicing is required to maintain RF integrity.

The recommended patient viewing window dimensions are 48 in. wide by 42 in. high (1219 mm x 1067 mm). The location of the window is dependent on the position of Operator Workspace position.

Note
The operator at the Operator Workspace must be able to view the patient during a scan.

7–6–4 Floors

The Magnet Room floor levelness requirement is important for accurate patient table docking. Floor levelness in the Magnet Room must not be greater than 0.3125 in. (8 mm) between depression and high spots over any 120 in. (3048 mm) distance within the area of the magnet enclosure and the area in front of the enclosure, see shaded area shown in Illustration 7–12.

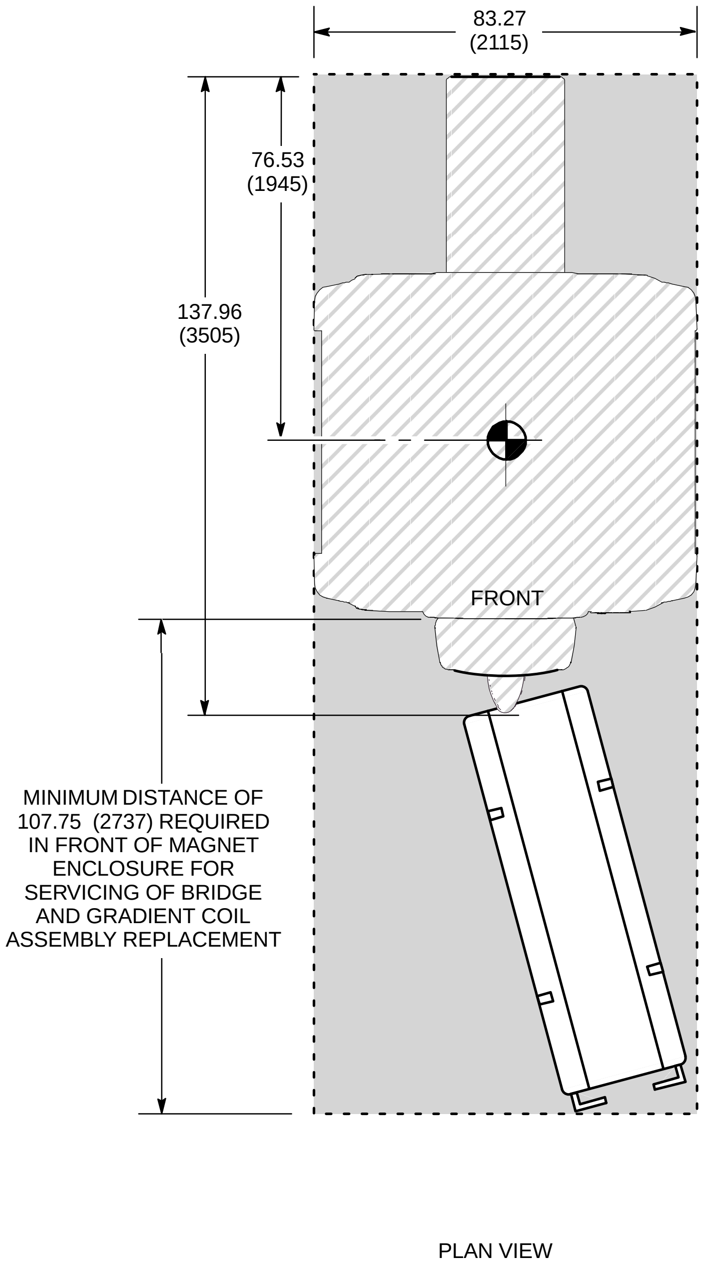
The finished floor is to be designed to insure the finished floor to the center of the magnet dimension of 42.125 in. $\pm$0.25 in. (1070 mm $\pm$6.35 mm) is maintained. This will allow the Patient Transport to properly dock to the enclosure. Also the material used for the finished floor directly under the magnet must not allow the magnet to migrate into the material therefore requiring constant leveling (e.g. vinyl composition tile over a 0.5 inch self leveling top coat).

The finished floor in the Magnet Room should be waterproof and be a conductive type flooring to reduce the possibility of a static discharge. The floor covering should be hard–surfaced with the seams sealed to protect the RF flooring against possible water damage. This floor should also be easily replaceable to repair tiles that are damaged from occasional cryogen spillage. Removable access flooring in the Magnet Room must be aluminium or non magnetic for safety purposes.

Note
The Magnet Room floor must be designed to support TRM Gradient Coil replacement. Refer to **Section 2–4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY** for weight and dimensions of the TRM Gradient Coil on cradle/cart utilized for replacement.

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED ( ) DIMENSIONS ARE IN MILLIMETERS.
- ROOM FLOOR MUST BE CLEAR AND LEVEL FLOOR TO THE ROOM DOOR TO ALLOW FOR PATIENT TABLE MOVEMENT.



MAGNET FLOOR LEVELNESS AREA
ILLUSTRATION 7–12

7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM

Overview & Background Information

All GE MR systems require an RF enclosure which is commonly called an RF Shielded Room or an RF Screen Room. RF signals from sources outside of the RF Shielded Room are attenuated so they do not interfere with the MR system. Likewise, the RF signals produced by the MR scanner are kept from interfering with other RF devices. This RF quiet environment is necessary for the MR system to produce quality images.

RF Shield Rooms come in a variety of shapes and sizes but the feature they all have in common is a total RF shield produced by one continuous ground plane. This is achieved by making the walls, floor, doors and ceiling out of an electrically conductive material such as copper, aluminum, brass, or steel. All the room components (walls, floor, ceiling, etc.) must be electrically bonded together to form one solid, common ground plane. Once this is established the ground plane is then tied to earth ground to create the RF shield. This ground point is known as the PRIMARY ground. The primary grounding technique works well for any battery powered devices that may be operated within the RF Shield Room. Devices requiring facility power or the introduction of facility power into the Screen Room requires a change in the primary grounding technique for the RF Shield Room. The RF Shield Room must now be grounded back to the facility power ground.

The addition of water systems or other grounds required by the national electrical code will cause the RF Shield Room ground impedance to be ZERO ohms. These additional grounds that connect the outside of the RF Shield Room to earth grounds are called SECONDARY grounds. It is the secondary grounding that needs to be controlled. If the secondary ground introduces any current to the RF Shield, indicating the RF Shield is a better ground path than the secondary ground, then the current can set up electrical fields on the surface of the RF Shield which may cause image artifact.

The equipment anchors are generally installed into the grade below the RF Shield Room because the floor thickness within the room does not provide the proper embedment depth. With this in mind the hardware used to anchor the equipment will be either a bolt or a stud. This hardware must penetrate the RF Shield and be in full contact with the shield thereby preventing RF leaks at the point of penetration as specified in **Section 7–7–2 RF Shield Integrity**. In this case, the anchoring hardware may be in direct electrical contact to the anchored equipment. The anchoring device that is set below the RF Shield Room floor must be isolated from ground. If it should come in contact with rebar or wire mesh and this in turn is contacting building steel then a secondary ground has been established and needs to be corrected.

RF Shield Rooms that have the RF shield far enough below the finished floor in the room to allow the anchor to be installed without penetrating the shield have no issues with secondary grounding.

Following are the detailed roles and responsibilities of the Customer, RF Shield Room vendor and GE Medical Systems.

The Customer is responsible for the following tasks:

- Coordinate equipment anchor methods and anchor location with the contracted RF shield room vendor, structural engineer, and architect to prevent RF leaks and secondary grounding problems.
- Coordinate with the contracted RF shield room vendor, structural engineer, and architect the design of the equipment anchor hardware per GE specifications, refer to Section 7–7–6, Example – Select Magnet Anchor Size.
- Obtain any and all approvals necessary for the construction of equipment support and seismic anchoring. Provide copy of building inspector’s (inspection) report and approval on anchor method to GE Medical Systems local office.

7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM (Continued)

The RF Shield Room vendor is responsible for the following tasks:

- Assist Customer structural engineer to determine stud length due to floor and room shield thickness variability.

WARNING!

THE RF SHIELD ROOM VENDOR MUST SUPPLY TORQUE SPECIFICATIONS FOR ALL ANCHORS, STUDS, AND FASTENING HARDWARE TO MEET THE CLAMPING FORCE SPECIFIED EACH PIECE OF EQUIPMENT LISTED IN TABLE 7–5.

- Procurement of commercially available anchors, studs, and fastening hardware required for equipment listed Table 7–5.
- Layout and installation of the equipment anchors (create own template from GE supplied information) Coordination with Building Contractor/Architect may be necessary to prevent interference with rebar or structural steel that would cause a secondary ground path through the anchor.
- GE Service to assist RF Shield Room vendor by locating magnet isocenter during layout of equipment anchors. RF Shield Room vendor to be present when GE Service identifies magnet isocenter location to maximize the accuracy of the location.
- Anchors for Magnet shall be installed prior to magnet delivery to allow time to address any issues that may arise.
- Acnhor for Dock Assembly shall be installed during installation process of the Dock Assembly to the Magnet.
- RF Shield Room Vendor to perform pull test on all anchors and indicate the torque required to achieve the specified clamping force (tension).
- Perform ground impedance test on installed anchors prior to connection to RF shield
- Perform RF integrity test on room after anchors and studs are installed and connected to RF shield
- Provide copy of ground impedance and RF room integrity tests to GE Medical Systems local office
- Required to use a electrically conductive fiberous or equivalent washer to electrically connect anchors and studs to the RF Shield. Flat washers will not maintain RF Shield integrity. Flat washer may introduce image artifacts.

7–7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM (Continued)

GE Medical Systems is responsible for the following tasks:

- GE Installation Specialist to provide equipment dimensions drawings showing equipment mounting locations to the RF Shield Room vendor and review the anchor method prior to anchor installation.
- GE Service to assist RF Shield Room vendor by locating magnet isocenter during layout of equipment anchors. RF Shield Room vendor to be present when GE Service identifies magnet isocenter location to maximize the accuracy of the location.
- GE Service to inspect and verify the anchor location is correct and obtain the anchoring hardware from the RF Shield Room vendor prior to equipment delivery. Carefully inspect the electrical connectivity of the anchor/stud to the RF Shield to make sure the fiberous washer or equivalent device is in place.
- GE Service to work with riggers to secure Magnet to anchors at time of delivery and installation.
- GE Service to attach/align Dock Assembly to Magnet and mark anchor location for RF Shield Room Vendor.

Refer to Table 7–5 for equipment type and seismic anchor characteristics.

7–7–1 Clamping Force (Tension) And Pull Test

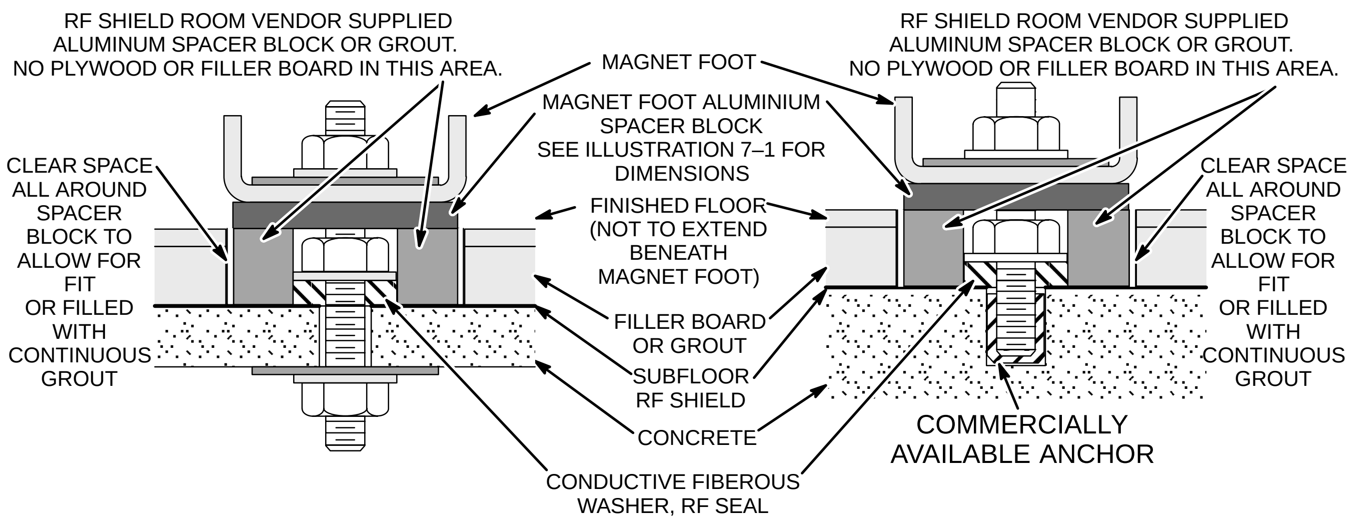
1. The type of anchor to be used for mounting the equipment will be dependent on the building construction (refer to Illustration 7–13) and the clamping force (tension) requirement for the equipment to be anchored (refer to Table 7–5 for equipment clamping force (tension) requirements).

2. To understand how to properly select the anchor diameter and embedment refer to Section 7–7–6, Example – Select Anchor Size.

3. Each anchor must meet the clamping force (tension) requirement.

4. The equipment must be clamped directly to the floor and the entire equipment base must maintain full contact to the floor.

5. A pull test equal to the clamping force (tension) must be performed by the RF Shield Room Vendor prior to the equipment delivery. The test results indicating the torque required to achieve the specified clamping force (tension) must be recorded by RF Shield Room Vendor and the information forwarded to GE Medical Systems for the customers record file.



Note: For sites with RF Shield on top of subfloor, the RF Shield needs to be recessed to the concrete level to provide a proper RF Seal.

RF SHIELD ROOM ANCHOR DETAILS
ILLUSTRATION 7–13

7–7–2 RF Shield Integrity

The anchor hardware must maintain RF shield integrity. This is accomplished by electrically sealing the stud at the penetration point on the RF shield. The method by which the electrical contact is made must take into account any stretch in the stud resulting from the applied clamping force (tension). A fiberous washer or equivalent will provide a proper RF seal where a solid flat washer could produce an RF leak and introduce artifacts into the MR images. The RF room test should result in a specific attenuation at the operating frequency of the system under the following conditions:

1. Blank Penetration Panel installed
2. Anchor hardware installed
3. Electrical connection made between the anchor stud and the RF shield.

Refer to APPENDIX B – RF SHIELDED ENCLOSURE TEST GUIDELINE.

TABLE 7–5
EQUIPMENT CLAMPING FORCE (TENSION) REQUIREMENTS

EQUIPMENT TYPE	CLAMPING FORCE (TENSION) TO THE FLOOR APPLIED TO EACH ANCHOR lbs (N) See Note 1
1.5T LCC (Active Shield) Magnet (See Note 2)	2,500 ± 200 (11,100 ± 900)
Dock Assembly for LCC (Active Shield) Magnet	600 ± 100 (2700 ± 450)
Blower Box (See Note 3)	100 ± 10 (450 ± 45)

Note 1 The RF Screen Room Vendor must perform a pull test on each anchor, equal to the clamping force (tension) required for the equipment, prior to equipment delivery. A copy of the pull test results, anchor ground impedance measurements and building inspection certification must be given to GE Medical Systems for customer files.
A floor made of compressible material may require the anchor hardware to be tightened several times to achieve the clamping force (tension). The clamping force (tension) must be maintained for the equipment over the life of the product.

2 All four feet of the magnet must be anchored to the floor. The bolt hole openings in the magnet base are to be used to anchor the magnet.

3 The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or a access support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room. The Blower Box may be mounted to a raised floor section within the magnet room. If the Blower Box is secured to the finished floor and the anchor hardware penetrates the RF shield then the RF Shield Vendor must install the anchors prior to system equipment delivery. The need for Blower Box anchors must be planned in advance to allow the RF Shield Room Vendor to include the cost in their bid.

7–7–3 Electrical Isolation

The anchor hardware must not provide a secondary ground path for the RF Shield Room. A secondary ground may occur by having the anchor come in contact with steel rebar, wire mesh or structural steel in the floor. Ideally the ground impedance between the anchor and earth ground should be greater than 1000 ohms. This may not be possible due to moisture conditions in the concrete or soil beneath the concrete. Therefore the following requirements for ground isolation shall be observed:

1. A ground isolation test performed on each electrically conductive anchor and stud.
2. If the result is greater than1000 ohms on each stud then record the resulting measurement and give documentation to GE Medical Systems.
3. If the result is less than1000 ohms but greater than100 ohms, contact the power and grounding support personnel at GE Medical Systems and review process and site conditions.
4. If the result is less than100 ohms then it is very likely the anchor has made contact to steel rebar or wire mesh. In this case the steel in the floor will need to be removed or the anchors will need to be relocated. In either case GE Medical Systems must be notified and a retest performed after the corrective action is taken.
5. The test results must be recorded by RF Shield Room Vendor and the information forwarded to GE Medical Systems for the customers record file.

Electrical Isolation Measurement Method

Prior to attaching the primary ground and installing any power outlets, room lights, water supplies, etc. into the RF Shield Room, a ground isolation test is performed between the room's ground plane and earth ground.

- This measurement should be greater than 1000 ohms DC. However, the results of this measurement may appear lower for RF Shield Rooms located below or at grade level. This is caused by a capacitive voltage, setup by the measurement meter's DC voltage, between the RF Shield Room and earth ground.
- For RF Screen Rooms with a resistance reading less than 1000 ohms the following method should be used to determine if the low resistance reading is caused by this capacitive effect.
  - Use a Meggar Insulation Tester capable of reading < 1000 ohms to most accurately make the measurement. An analog meter with a D'Arsonval meter movement may be used if a Meggar is not available.
  - After making the measurement reverse the leads and watch for the measurement to start high and decrease to a lower resistance value. This change in measurement verifies the capacitive effect.
  - In this case the peak measurement is the approximate resistance between the RF Shield Room and earth ground.

7–7–4 Physical Characteristics

Stud Size

The following factors will contribute to the stud size:

- Magnet anchor/stud diameter: minimum 0.625 in. (M16) and maximum 1.25 in. (M32).
- clamping force (tension) requirement for the equipment, refer to Table 7–5
- Seismic codes (affect the length and diameter of the stud and anchor size)
- Embedment depth (affects the length of the stud)
- Variation in floor thickness (RF Shield Room floor and finished floor)
- Equipment base thickness and any spacers required under the base
- Equipment base clearance for protrusion of stud inside the base
- Size of the hole in the equipment base (affects the diameter of the stud)

Refer to Table 7–6 for Equipment Characteristics.

TABLE 7–6
EQUIPMENT CHARACTERISTICS

EQUIPMENT TYPE	EQUIPMENT MOUNTING BASE THICKNESS in. (mm)	CLEARANCE HOLE IN EQUIPMENT BASE in. (mm)	MAXIMUM CLEARANCE ABOVE EQUIPMENT BASE in. (mm) See Note 1	SEISMIC REQUIREMENT PRE–APPROVED BY OSHPD FOR BOLT OR STUD DIAMETER in. (mm) See Note 2	EQUIPMENT MOUNTING ILLUSTRATION
1.5T LCC (Active Shield) Magnet See Note 3	1.75 (44) See Note 4	1.5 (38)	2.5 (64)	1.0 (M24)	7–1
Dock Assembly for LCC Magnet	0.75 (20)	0.43 (11)	2.0 (50)	Pre–approval not required. Select anchor/stud to clamping force (tension).	7–1
Blower Box See Note 5	0.25 (6)	0.25 (6)	0.5 (13)	Pre–approval not required. Select anchor/stud to clamping force (tension).	2–27

Note 1 Maximum Clearance Above Equipment Base is the dimension for protrusion of the stud inside the equipment base including clearance for tools to tighten hardware to meet specifications in Section 7–7–1 Clamping Force (Tension) And Pull Test.

2 Seismic codes do not allow for any gap between the stud and clearance hole in the base of the equipment. In California, USA all anchor hardware designs must be submitted to the Office of Statewide Health Planning and Development (OSHPD) for pre–approval. For all other states, providences or countries, plans must be submitted to the local planning authority (where seismic codes apply).

3 All four feet of the magnet must be bolted to the floor. The bolt hole openings in the magnet base are to be used to anchor the magnet.

4 LCC Magnet mounting base thickness includes the 1 inch (25 mm) foot block required to maintain the Magnet Center Line Height dimension as defined in Illustration 2–9.

5 The Blower Box for the Signa Horizon system can be mounted to either the wall or the floor. The floor mounting could be made onto a raised floor section within the magnet room. If the Blower Box is secured to the finished floor and the anchor hardware penetrates the RF shield then the RF Shield Vendor must install the anchors prior to system equipment delivery. The need for Blower Box anchors must be planned in advance to allow the screen room vendor to include the cost in their bid.

7–7–4 Physical Characteristics (Continued)

Material Properties For The Stud And Anchor

The stud must be made of a conductive material to allow it to be electrically connected to the Room's RF Shield at the point of penetration. The material properties should be compatible with the material properties of the RF Shield and not produce galvanic corrosion due to dis–similar metals. The stud does not require electrical isolation from the equipment base.

The embedment anchor must comply with local building codes especially for seismic designated areas. An epoxy type anchor may be used if it meets the following requirements:

1. Able to achieve clamping force (tension) requirement for the equipment to be anchored.
2. Is approved by the local building inspector.

7–7–5 Installation Location

The exact location for installing the Magnet anchors is determined by dimensional footprint drawings for the MR equipment to be installed. The Installation Services Group at GE Medical Systems will provide to the RF Shield Room Vendor the dimensional drawing showing all anchor locations. The drawing can be issued in either hard copy or electronic format. The RF Shield Room Vendor is responsible for supplying their own template to precisely mark the Magnet anchor locations within the room.

GE Service to attach/align Dock Assembly to Magnet and mark anchor location for RF Shield Room Vendor.

Coordination between the RF Shield Room Vendor and Building Contractor/Architect may be necessary to mark the location of the anchors to prevent interference with rebar or structural steel. A re–arrangement of the room may be necessary to ensure ground isolation.

Refer to Table 7–6, equipment dimensional illustration references.

7–7–6 Example – Select Magnet Anchor Size

The following is an example of to illustration the selecting of proper anchor to install an LCC Magnet into a building structure with 2000 psi (13.8 MPa) concrete. For this example the area is not under seismic requirements.

1. Determine magnet clamping force (tension) by referring to requirements in Table 7–5.

2500 lbs + 200 lbs = 2700 lbs	11,100 N + 900 N = 12,000 N

2. Refer to Tables 7–7 or 7–8 (examples of anchor vendor catalogs) to select anchor diameter and embedment which meets the clamping force (tension) determined in Step 1.

Diameter : Min. 0.625 inch Max. 1.25 inch	Diameter : Min. M16 Max. M32
For 8 inch embedment select 3/4 inch diameter	For 130 mm embedment select M20 diameter
For 4.5 inch embedment select 1 inch diameter	For 114 mm embedment select M24 diameter

3. The vendor instructions and torque to the maximum recommended level for the anchor selected in Step 2 must be provided to the RF Shield Room vendor for proper installation of the anchor and equipment.

7–7–6 Example – Select Magnet Anchor Size (Continued)

TABLE 7–7
EXAMPLE OF ENGLISH UNITS STAINLESS STEEL ANCHOR ALLOWABLE LOADS IN CONCRETE

ANCHOR DIAMETER in. (mm) See Note 1	**EMBEDMENT DEPTH in. (mm)**	**2000 psi (13.8 MPa)**		**3000 psi (20.7 MPa)**		**4000 psi (27.6 MPa)**		**6000 psi (41.4 MPa)**	
		TENSION lb (kN)	**SHEAR lb (kN)**	**TENSION lb (kN)**	**SHEAR lb (kN)**	**TENSION lb (kN)**	**SHEAR lb (kN)**	**TENSION lb (kN)**	**SHEAR lb (kN)**
5/8 (15.9)	2 3/4 (70)	1250 (5.6)	2800 (12.5)	1600 (7.1)	3070 (13.7)	1810 (8.1)	3330 (14.8)	1920 (8.5)	3330 (12.5)
	4 (102)	1870 (8.3)	3330 (14.8)	2400 (10.7)	3330 (14.8)	2930 (13.0)	3330 (14.8)	3200 (14.2)	3330 (12.5)
	7 (178)	2500 (11.2)	3330 (14.8)	3010 (13.4)	3330 (14.8)	3650 (16.2)	3330 (14.8)	3650 (16.2)	3330 (12.5)
3/4 (19.1)	3 1/4 (83)	1550 (6.9)	2880 (12.8)	1950 (8.7)	3310 (14.7)	2350 (10.5)	3730 (16.6)	2610 (11.6)	4800 (21.4)
	4 3/4 (121)	2510 (11.2)	4510 (20.1)	3250 (14.5)	4650 (20.7)	3870 (17.2)	4800 (21.4)	4670 (20.8)	4800 (21.4)
	8 (203)	2930 (13.0)	4800 (21.4)	3870 (17.2)	4800 (21.4)	4530 (20.2)	4800 (21.4)	5120 (22.8)	4800 (21.4)
1 (25.4)	4 1/2 (114)	3120 (13.9)	6080 (27.0)	3870 (17.2)	6770 (30.1)	4610 (20.5)	7470 (33.2)	4800 (21.4)	7470 (33.2)
	6 (152)	4400 (19.6)	7470 (33.2)	6400 (28.5)	7470 (33.2)	7200 (32.0)	7470 (33.2)	7330 (32.6)	7470 (33.2)
	9 (229)	5600 (24.9)	7470 (33.2)	8000 (35.59)	7470 (33.2)	9390 (41.77)	7470 (33.2)	9390 (41.8)	7470 (33.2)
Note 1 All shaded values fail to meet the clamping force (tension), therefore are not acceptable anchors.									

TABLE 7–8
EXAMPLE OF METRIC STAINLESS STEEL ANCHOR ALLOWABLE LOADS IN CONCRETE

ANCHOR DIAMETER See Note 1	**EMBEDMENT DEPTH mm (in.)**	**13.8 MPa (2000 psi)**		**20.7 MPa (3000 psi)**		**27.6 MPa (4000 psi)**		**41.4 MPa (6000 psi)**	
		TENSION kN (lb)	**SHEAR kN (lb)**	**TENSION kN (lb)**	**SHEAR kN (lb)**	**TENSION kN (lb)**	**SHEAR kN (lb)**	**TENSION kN (lb)**	**SHEAR kN (lb)**
M16	105 (4 1/8)	11.2 (2500)	25.1 (5650)	20.9 (4705)	39.9 (8965)	24.2 (5450)	10125 (45.0)	6900 (30.7)	10550 (46.9)
M20	130 (5 1/8)	25.1 (5650)	52.9 (11900)	30.7 (6910)	58.7 (13195)	36.4 (8175)	14490 (64.5)	10005 (44.5)	14490 (64.5)
M24	155 (6 1/8)	30.0 (6735)	61.2 (13760)	36.9 (8300)	70.5 (15855)	43.9 (9860)	29.8 (17950)	57.7 (12980)	95.6 (21490)
Note 1 All shaded values fail to meet the clamping force (tension), therefore are not acceptable anchors.									

7–8 MAGNET ROOM EQUIPMENT MOUNTING

7–8–1 Remote Oxygen Sensor Module (OM3) – Optional

The Remote Oxygen Sensor Module (if option ordered) must be mounted approximately 60 in. (1524 mm) above the Magnet Room floor near the front of the magnet enclosure.

7–8–2 Magnet Rundown Unit (MS4)

The magnet rundown unit should be mounted 60 in. (1524 mm) above the Magnet Room floor near the front of the magnet enclosure but outside the 200 gauss zone.

7–8–3 Emergency Off buttons

Customer supplied emergency off buttons to be located near each room exit including magnet and equipment rooms. These buttons must be clearly labeled, “Emergency Off”. Refer to Section 5, POWER REQUIREMENTS, for emergency off power requirements.

7–9 RF DOOR SWITCH

RF shielded room vendor must supply and install RF door switches on all RF shielded doors. These switches must be wired in series and a GE supplied cable (two loose lead conductors) will attach to one door switch. RF switches must be rated for 24 volts at 750 milliamperes maximum and the switches must be in the open position when the doors are open (switch contacts close when the doors are completely closed).

7–10 EMERGENCY EXIT

Emergency exiting from the Magnet Room is to be specified by the customer’s architect and contractor. Such measures as an out swinging door, emergency door latch release, easily removed window, or other measures must be designed into the room. Emergency exit instructions must be permanently and prominently mounted near the door and/or window.

7–11 ROOM VENTILATION SWITCH

Placement of the room ventilation switch should be near the Magnet Room door and is the responsibility of the architect and mechanical contractor.

SECTION 8 – SHIPPING AND DELIVERY DATA

TABLE OF CONTENTS

8–1 SHIPMENT

Note

The Signa 1.5T Twin*Speed* system is shipped in 3 steps: 1) GE MDP and TSCC or TGWC equipment 2) Magnet with partial Enclosure and accessories 3) remaining system equipment. Refer to **Section 9–8 TYPICAL MR INSTALLATION PROJECT SCHEDULE** for more information.

Domestic transportation for the MR system, excluding the magnet, will be via an air–ride moving van. The magnet will be shipped on an air–ride flat–bed truck. Export transportation for the MR system overseas will be via air shipment in a pressurized cargo hold. See Table 8–2 for the shipping weights and dimensions of the major MR system components. Actual shipping may vary, international shipment may require equipment to be crated. Refer to Table 8–1 for transportation environmental conditions.

TABLE 8–1
TRANSPORTATION ENVIRONMENTAL CONDITIONS

SYSTEM EQUIPMENT	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	RELATIVE HUMIDITY %	HUMIDITY CHANGE (%/Hr)	ATMOSPHERIC PRESSURE hPa
Electronics Cabinets & equipment	–30 to 140 (–34 to 60)	176 (80)	0–90 non–condensing	30	1012 to 525
Magnet	–31 to 122 (–35 to 50)	176 (80)	0–90 non–condensing	30	1012 to 525

The GE magnet utilizes a Shield/Cryo Cooler System to maintain a reduced helium boil–off. However, the Shield Cooler System is not operational during transportation. Therefore the magnet will require liquid helium replenishment if transportation time exceeds two weeks. Contact GE Service for magnet servicing.

The LCC Magnet is filled with liquid helium at initial shipment but can be allowed to warm up during transportation without damage to the support structure.

8–2 STORAGE REQUIREMENTS

If the system is stored before installation, it must be stored in a warehouse protected from weather. The storage temperature should be between –30 and 140 degrees F (–34 and 60 degrees C) and the relative humidity between 0 and 90% (non–condensing).

There are two scenarios for storing the magnet. One is to maintain the magnet cold temperature by connecting and operating the Shield/Cryo Cooler System. Periodic replenishment of liquid helium may be required depending on the storage time. The other scenario is to store the magnet without maintaining the cold temperature. The LCC magnet can be moved cold or warm, refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation* for magnet moving requirements and details. Contact GE Service for necessary servicing before moving the magnet.

8–3 MAGNET CONSIDERATIONS

For domestic, the magnet is shipped covered with plastic (no shipping pallet). For export, the magnet is crated for shipment on a special shipping pallet. Refer to Table 8–2 for the weight and dimensions of the magnet in its cold ship configuration (i.e. with liquid cryogens in vessel within the cryostat) and with the RF/Gradient Coil inside the magnet bore.

The magnet moving dimensions are shown in Illustration 8–1.

Consideration must be given to the delivery route of the magnet to ensure that the floor can support the magnet and any rigging equipment required to move it. A structural analysis should be performed by a professional structural engineer. The magnet must not be tilted more than 30° in any direction when being moved into position.

The customer is to provide and arrange for riggers to move the magnet from the delivery truck to the final site location. Contact local GE Service for a list of recommended rigger companies. The customer's riggers should have an adequate amount of liability insurance to cover any damage to property or MR system that may occur during delivery of the magnet. The GE sales representative or installation specialist can provide customer riggers with the replacement value of the MR system for insurance purpose.

8–4 COLD–SHIPPED MAGNET DELIVERIES

Cold–shipped magnets are those magnets which are shipped with liquid helium in the vessel within the cryostat. Thus, when these magnets arrive at site, a cryogen delivery route must be available for moving cryogen dewars to the magnet for periodic replenishment of liquid helium. Also, means must be provided for venting of the cryogenic gases if the GE supplied venting kit is not yet installed and if the RF shielded room vent opening is not completed.

Note

Power and cooling water for the shield cooler cabinet must be available when the magnet is delivered to minimize cryogens usage. See Section 4–4, WATER COOLING REQUIREMENTS, and Section 5, POWER REQUIREMENTS, for specifications.

TABLE 8–2
SHIPPING DATA

MR COMPONENT	APPROXIMATE W x D x H	APPROXI-MATE WEIGHT	METHOD OF SHIPMENT
LCC Magnet with cryogens, partial Quiet Technology Enclosure installed	93 x 144 x 107 (2362 x 3658 x 2718)	See Note 1	Domestic – Tarped International – crate/pallet
Magnet Accessory Equipment	48 x 48 x 28 (1219 x 1219 x 711)	400 (182)	crate
Shield/Cryo Cooler Compressor Cabinet	26 x 28 x 42 (660 x 711 x 1067)	240 (109)	skid with box cover
Rear Pedestal Assembly with Rear Split Bridge Assembly	24 x 48 x 45 (610 x 1219 x 1143)	60 (27)	box
Enclosure Top	48 x 36 x 36 (1219 x 914 x 914)	30 (14)	box
Enclosure Skirts	40 x 24 x 24 (1016 x 610 x 610)	30 (14)	box
Patient Table	95 x 36 x 50 (2413 x 914 x 1270)	670 (305)	pallet
Patient Blower Box	24 x 30 x 24 (610 x 762 x 610)	30 (14)	box
1.5T SRFD II Cabinet	24 x 38 x 72 (610 x 965 x 1829)	500 (225)	on cabinet casters, wrapped with plastic
ACGD/PDU Cabinet without Fan Module installed	24 x 37 x 75 (610 x 940 x 1905)	2140 (970)	on cabinet casters, wrapped with plastic
Fan Module for ACGD/PDU Cabinet	28 x 38 15 (711 x 965 x 381)	160 (73)	on pallet
System Control Cabinet for Signa Twin*Speed* with EXCITE Technology	24 x 33 x 78 (610 x 838 x 1981)	475 (215)	on cabinet casters, wrapped with plastic
System Control Cabinet for Signa Twin*Speed*	24 x 33 x 78 (610 x 838 x 1981)	500 (227)	on cabinet casters, wrapped with plastic
Twin Accessory Cabinet	23 x 39 x 50 (584 x 991 x 1270)	600 (272)	on cabinet casters, wrapped with plastic
SPT Phantom Set	34 x 32.5 x 60 (864 x 826 x 1524)	350 (159)	on cart casters with box cover
Operator Workspace Cabinet	25 x 38 x 34 (635 x 965 x 864)	200 (91)	skid
Operator Workspace Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	box
Operator Workspace LCD Color Monitor	27 x 33 x 27 (686 x 838 x 686)	125 (57)	skid
Operator Workspace equipment	32 x 32 x 23 (813 x 813 x 584)	100 (45)	box
Operator Workspace Table	45 x 54 x 37 (1143 x 1372 x 940)	180 (82)	box
MNS Cabinet *	24 x 42 x 71 (610 x 1067 x 1803)	430 (195)	on cabinet casters, wrapped with plastic
BrainWave Cabinet *	24 x 23 x 72 (610 x 584 x 1829)	350 (159)	on cabinet casters, wrapped with plastic
BrainWave Option Several Boxes *	40 x 48 x 32 (1016 x 1219 x 813)	165 (75)	all boxes on 1 pallet

Note * Optional Equipment

1 Approximate magnet shipping weight (includes packaging material) and weight of magnet with cryogens, TRM Gradient & RF coils, Enclosure parts installed on magnet, lifting beams, crate, and skid (i.e. minus packaging material): 12,195 lbs (5532 kg). International shipments must add shipping crate/pallet of 2,200 lbs (998 kg).

(Continued)

TABLE 8–2 (Continued)
SHIPPING DATA

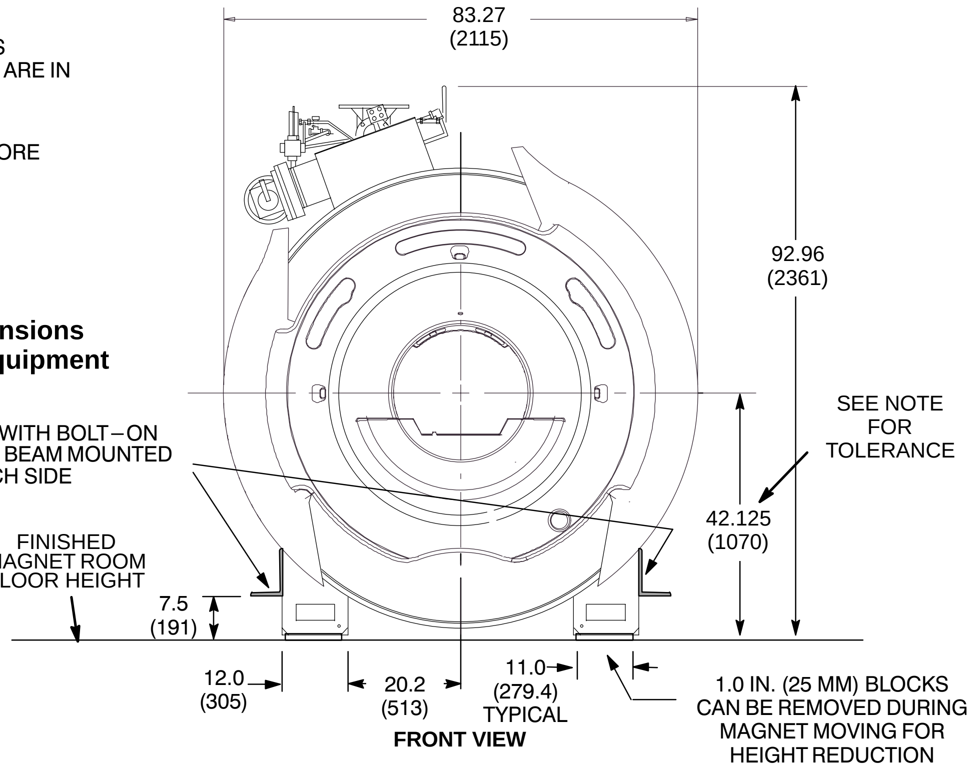
MR COMPONENT	APPROXIMATE W x D x H	APPROXI-MATE WEIGHT	METHOD OF SHIPMENT
TSCC Outdoor Air Cooled equipment			
TSCC unit	52 x 82 x 75 (1321 x 2083 x 1905)	1320 (599)	crate/skid
Remote Control Panel	20 x 10 x 20 (51 x 25 x 51)	33 (15)	box
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	4 boxes
Ship loose items	32 x 32 x 32 (813 x 813 x 813)	50 (23)	box
TSCC Indoor Water Cooled equipment			
TSCC Indoor Water Cooled	58 x 38 x 82 (1473 x 953 x 2083)	1100 (499)	crate/skid
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	3 boxes
Ship loose items	32 x 32 x 32 (813 x 813 x 813)	50 (32)	box
TSCC Indoor Air Cooled equipment			
TSCC Indoor Air Cooled	68 x 36 x 80 (173 x 92 x 203)	1740 (789)	crate/skid
DowFrost HD glycol & ship loose items	30 x 50 x 70 (762 x 1270 x 1778)	300 (136)	box on pallet
TGWC Outdoor Air Cooled equipment			
TGWC unit	55 x 40 x 82 (141 x 102 x 208)	918 (417)	crate/skid
Remote Control Panel	13 x 7 x 15 (330 x 16 x 37)	20 (9)	box
DowFrost HD glycol & ship loose items	24 x 24 x 18 (610 x 610 x 457)	50 (23)	2 boxes
TGWC Indoor Water Cooled equipment			
TGWC Indoor Water Cooled	38 x 24 x 27 (97 x 61 x 69)	200 (91)	box on pallet
Remote Control Panel	12 x 12 x 12 (305 x 305 x 305)	20 (9)	box
DowFrost HD glycol	12 x 12 x 18 (305 x 305 x 457)	50 (23)	2 boxes
Ship loose items	12 x 12 x 12 (305 x 305 x 305)	30 (14)	box
Ship loose items	26 x 6 x 6 (660 x 152 x 152)	15 (7)	box
TGWC Indoor Air Cooled equipment			
TGWC Indoor Air Cooled	34 x 40 x 76 (59 x 76 x 186)	918 (417)	crate/skid
DowFrost HD glycol & ship loose items	24 x 24 x 12 (610 x 610 x 305)	50 (23)	box

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MAGNET WITH CRYOGENS, TRM GRADIENT & RF COILS IN BORE WEIGHT, AND LIFTING BEAMS: 12,195 lbs (5532 kg)

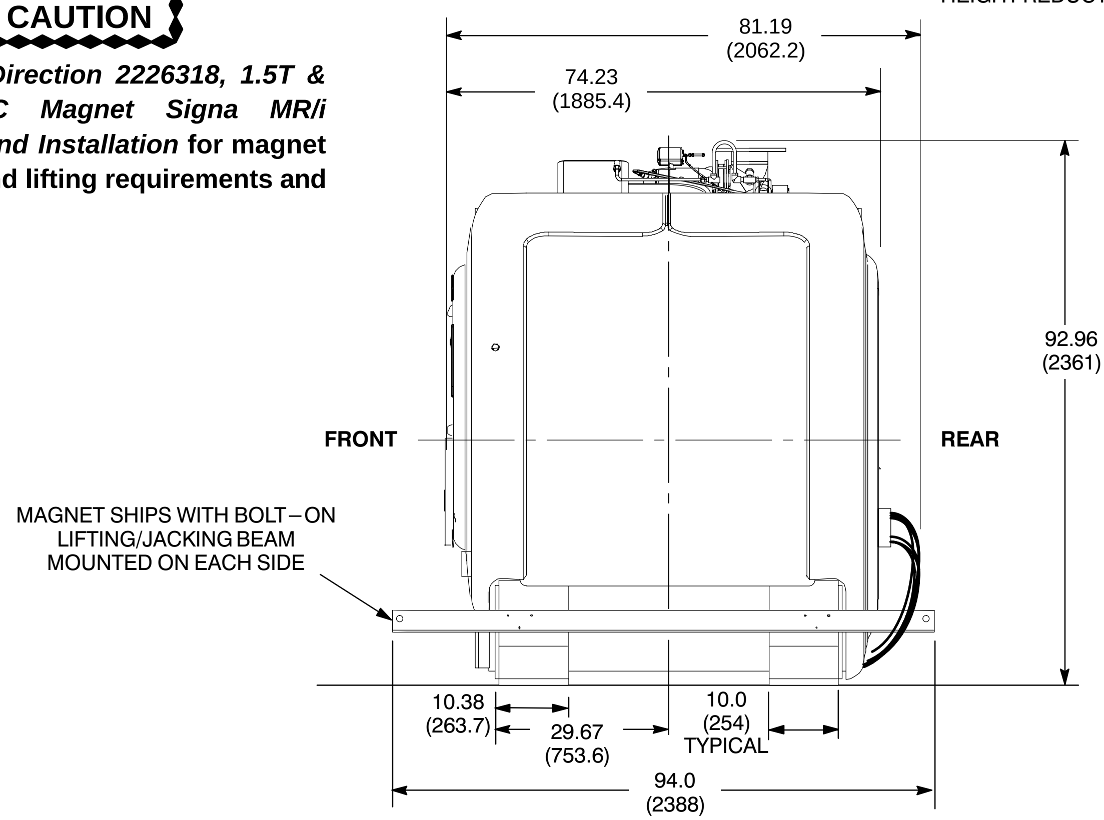
CAUTION

Final magnet moving dimensions are dependent on rigger equipment requirements.



CAUTION

Refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation* for magnet moving and lifting requirements and details.



1.5T LCC (ACTIVE SHIELD) MAGNET MOVING DIMENSIONS
ILLUSTRATION 8–1

SECTION 9 – PREINSTALLATION CHECKLIST

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9–1 INTRODUCTION

“Preinstallation” refers to work necessary to plan and prepare a site for delivery and installation of equipment. Delay, confusion, and waste of manpower can be avoided by completing preinstallation work. It is recommended to have GE Service Representative make on–site inspections during construction.

The purpose of this section is to outline key areas of concern in the preparation of a customer’s site for magnetic resonance. It is intended as a guide for GE‘s Installation Specialists and/or Installation Leaders when making on–site inspections during the construction phase of an MR project. Note that these inspections by GE are intended to aid the overall site preparation process. They do not relieve the customer and project architect from responsibility for the design, engineering, and coordination efforts necessary to ensure a successful MR project.

During the course of the site preparation process, GE‘s Installation Specialists and/or Installation Leaders may observe that the requested arrival date (RAD) for the MR equipment is not realistic. If this is the case, appropriate actions must be taken by the GE Field Sales/Service Team to adjust the RAD accordingly.

All work must be in compliance with national and local safety codes.

9–2 GENERAL PREINSTALLATION REMINDERS

☐ 1. Has the customer’s architect fully reviewed final site construction drawings using the guidelines provided by MR Siting?

☐ 2. Has the vibration study been completed per **Section 4–15, VIBRATION**? It is the customer’s responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specified limits.

☐ 3. If there were vibration issues; has local GE Field Service verified, with the vibration consultant present, the elimination/reduction of all identified sources?

☐ 4. Have vehicle parking arrangements been made for installation personnel?

☐ 5. Is temporary storage space available for use during installation?

☐ 6. What is facility smoking policy?

☐ 7. Have facility arrangements been made for refuse disposal during installation?

9–3 SAFETY

WARNING!

THE FOLLOWING SAFETY ITEMS MUST BE ADHERED TO PRIOR TO MOVING THE MAGNET FROM THE TRUCK INTO THE ROOM. THE MAGNET WILL BE EXHAUSTING HELIUM GAS DURING THIS TIME. GASEOUS HELIUM IS AN INVISIBLE, ODORLESS GAS THAT CAN CAUSE ASPHYXIATION WHEN DEPLETING THE AMBIENT OXYGEN SUPPLY. HELIUM GAS IS LIGHTER THAN AIR AND WILL RISE TO THE CEILING.

☐ 1. Have all of the required room ventilation items been installed and tested to make sure sufficient ventilation is available? Is exhaust fan and fan control installed and functional? The customer is to supply a copy of the Exhaust Fan Test Report to GE Medical Systems for the customers record file. Refer to **Section 4–8 ROOM VENTILATION** for specific requirements.

☐ 2. Are a first aid kit and non–ferrous fire extinguishers available at site?

☐ 3. Are removable chain or strap available for securing cryogen gas cylinders in an upright position to prevent the cylinders from falling which may cause injury or damage?

☐ 4. Are functioning telephones as defined in **Section 2–8–8 System Monitoring & Support Connectivity Requirements** available at site for duration of installation? Phone lines are needed to dial out in case of an emergency.

☐ 5. Has the cryogenic vent been installed complete from the Magnet Room RF shield penetration to the final exit and been inspected to final exit outside of the building? The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customers record file. Refer to **Section 4–9, CRYOGENIC VENTING**.

☐ 6. Have plans been made to connect the cryogenic vent to the magnet after magnet installation in the room? Refer to **Section 4–9, CRYOGENIC VENTING**.

☐ 7. Is there adequate ventilation for helium dewar storage area. Refer to **Section 4–8, ROOM VENTILATION**.

9–4 PREPARATIONS REQUIRED IN ADVANCE OF MAGNET DELIVERY AND MOVING INTO MAGNET ROOM

The following items must be completed prior to magnet delivery. A site inspection by GE Service Representative must be completed prior to magnet delivery to ensure site readiness.

☐ 1. Has Main DIsconnect Panel been installed, electrician wiring complete, MDP operational with power available? Refer to **Table 5–1 REQUIRED CUSTOMER POWER.**

☐ 2. **Is 24 hour/day, 7 days/week power and air cooling available for Main Disconnect Panel (MDP) for powering the Outdoor or Indoor Twin System Cooling Cabinet (TSCC), Shield/Cryo Cooler Compressor cabinet, and Magnet Monitor equipment?**

☐ 3. **If MR system configuration is utilizing water cooled TSCC then is 24 hour/day, 7 days/week water cooling available?**

9–4 PREPARATIONS REQUIRED IN ADVANCE OF MAGNET DELIVERY AND MOVING INTO MAGNET ROOM (Continued)

☐ 4. Are all walls and ceiling in magnet room essentially complete except for removable section?

☐ 5. Has a clear route to the magnet room been defined for magnet installation (refer to **Section 2–4, MINIMUM DOOR SIZES**)?

☐ 6. Is a secure space available to store equipment on site?

☐ 7. Have arrangements been made for the use of special rigging equipment for moving the magnet into the magnet room?

☐ 8. Has the raised floor been installed in the equipment/computer room in preparation for the superconducting and resistive power supplies?

☐ 9. Has work in the magnet room been completed or suspended and the magnet room closed off to provide a dust–free, closed environment?

☐ 10. Has all equipment been removed from the magnet room to allow space for the magnet with rigging equipment and cryogen dewars?

☐ 11. Are equipment anchors installed in floor as required for magnet room equipment? Refer to **Section 7–7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM.**

☐ 12. For all magnet room anchors, has pull test been performed and installation torque value defined? Test results recorded by RF Shield Room Vendor and information forwarded to GE Medical Systems for the customers record file? Refer to **Section 7–7–1 Clamping Force And Pull Test.**

☐ 13. For all magnet room anchors, has ground impedance test been performed? Test results recorded by RF Shield Room Vendor and documentation given to GE Medical Systems for the customers record file? Refer to **Section 7–7–3 Electrical Isolation.**

9–5 PREPARATIONS REQUIRED IN ADVANCE OF CRYOGEN TRANSFER/MAGNET FILL

☐ 1. Have arrangements been made for cryogen delivery and storage?

☐ 2. Does a clear path exist to magnet room for delivery of cryogens?

☐ 3. Is the magnet room adequately vented to allow exhaust of cryogen boil–off?

☐ 4. Is the magnet room clear to provide room for cryogen dewars during filling?

9–6 PREPARATIONS REQUIRED IN ADVANCE OF SYSTEM DELIVERY/INSTALLATION

The following items must be completed prior to system delivery. A site inspection by GE Service Representative must be completed prior to system delivery to ensure site readiness.

☐ 1. Has the equipment/computer room raised floor been installed if not previously completed?

☐ 2. Has area under the raised floor been vacuum cleaned and free of all debris?

☐ 3. Have any necessary conduits or raceway for power cables been installed?

☐ 4. Has delivery route been defined for equipment (refer to **Section 2–4, MINIMUM DOOR SIZES**) including plans for protecting flooring for heavy equipment & carts (ie. RF/Gradient Body Coil & shipping cart)?

☐ 5. Is power available for connection to the Main Disconnect Panel?

☐ 6. Does incoming power have all the specified safety precautions and remote disconnects?

☐ 7. Is the operator's area complete to provide a dust–free environment for installation of the Operator Workspace equipment?

☐ 8. Are the areas for optional equipment complete to provide a dust–free environment for installation of equipment?

☐ 9. Is air conditioning running in equipment room?

☐ 10. Are functioning telephones as defined in **Section 2–8–8 System Monitoring & Support Connectivity Requirements** available at site for duration of installation?

9–7 PREPARATIONS REQUIRED IN ADVANCE OF MAGNET RAMP–UP

☐ 1. Is the crygenic vent connected to the magnet?

☐ 2. Has the magnet room been completely closed (removable section closed up and sealed)?

☐ 3. Has the magnet room been tested to ensure that the RF shielding meets the electrical isolation and the100dB requirement for electric/plane waves and the 90dB requirement for magnetic waves refer to **Section 7–1 RF SHIELDED ROOM INTRODUCTION**? The customer is to supply a copy of RF shielded room vendor test reports.

☐ 4. Have all ferrous metal objects been removed from the magnet room?

☐ 5. Have adequate signs (Safety and Exclusion Zones) been posted to warn personnel about dangers of magnetic field?

☐ 6. Have facility personnel been informed of magnet safety precautions and procedures?

☐ 7. Has power been connected form the Main Disconnect Panel to Power Distribution Unit?

☐ 8. Has the penetration panel been installed?

☐ 9. Is all contractor construction work completed?

☐ 10. Has all contractor equipment that could affect shimming been removed from within the three gauss zone?

☐ 11. Have precautions been taken to prevent movement of large metal objects within the three gauss zone?

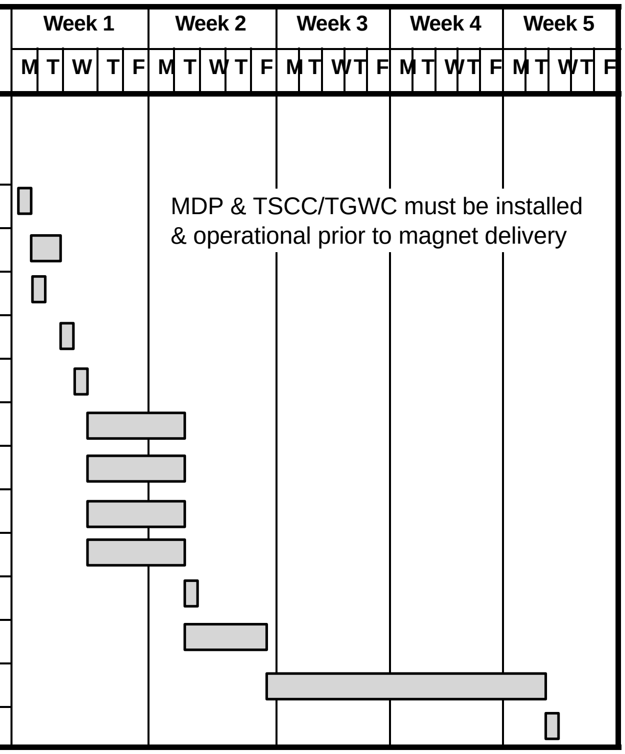
☐ 12. Have local fire department(s) and police department(s) been informed of unique characteristics (e.g. strong magnetic field, cryogens, etc.) of magnet and correct precautions to take in event of emergencies?

9–8 TYPICAL MR INSTALLATION PROJECT SCHEDULE

The MR Project schedule shown below depicts a typical Signa Twin*Speed* installation. The purpose of this project schedule is to provide a guide for your MR project in regards to tasks which need to be completed, duration of each task and who is primarily responsible for completion of the task. This project schedule starts at MDP and TSCC/or magnet delivery which occurs when your site construction is completed, except for magnet room opening. Your GE Installation Specialist can provide more detailed information if needed.

TYPICAL MR INSTALLATION PROJECT SCHEDULE

Typical MR Installation Schedule	Duration	Responsibility
GE MDP & TSCC or TGWC equipment delivered 1 week prior to magnet delivery	0.25 day	Installation Specialist (IS) & Contractor
MDP Installation	0.5 day	IS & Contractor
Outdoor TSCC/TGWC Installation	1 day	IS & Contractor
Indoor TSCC/TGWC Installation	0.5 day	IS & Contractor
Magnet delivered	0.5 day	GE
Magnet mechanical / electrical	0.5 day	GE
Magnet room closure	3 days	Contractor
Shielding is completed and tested		Contractor
Complete magnet room opening		Contractor
Flooring completed		Contractor
MR system delivered	0.5 day	GE
Mechanical Installation of system	3 days	GE
Calibration of system	12 days	GE
MR system available to customer		



SECTION 10 – TOOLS AND TEST EQUIPMENT

TABLE OF CONTENTS

10–1 INTRODUCTION

The following tables lists the tools and test equipment needed to install and calibrate a Signa system.

10–2 MOVING METAL MEASUREMENT EQUIPMENT

TABLE 10–1
MOVING METAL MEASUREMENT EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—–	Magnetometer sensitive to 50 μ gauss Flux gate or equivalent Example: Barrington Instruments Ltd. Model Mag–03MC 500 μT
2	—–	Spectrum analyzer capable of moving metal measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer.
3	—–	Accelerometer capable of measuring from 0.2 Hz beyond 250 Hz.

10–3 RIGGER/CUSTOMER SUPPLIED EQUIPMENT

Note

Table 10–2 contains an overview of equipment for initial planning purposes, refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation Manual* for details of equipment and requirements.

TABLE 10–2
RIGGER/CUSTOMER SUPPLIED EQUIPMENT

SUPPLIED BY	ITEM	DESCRIPTION
Rigger	1	Crane for removing magnet crate from delivery truck: 12 ton (10,909 kg)
	2	Motorized tow vehicle for pushing/pulling magnet when it is on dollies (e.g. an electric fork lift).
	3	Steel floor plates (8) to cover floors while transporting magnet, 36 in. x 12 in. x 0.25 in. (914 mm x 305 mm x 6.4 mm).
	4	Wood blocks, assorted sizes.
	5	Jacks and accessories for lifting magnet: 20,000 lbs (9091 kg) total capacity
	6	Spreader Beam
	7	For LCC magnet: Lifting straps which meet specifications in *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery/Installation Manual*
Customer	8	Equipment for off loading electronics and other miscellaneous components. (e.g. fork lift, hand trucks, straps, etc.)
	9	Panel lifters for computer flooring: Smooth floor Indicon Industries PL2DC; Standard carpeted floor Indicon Industries PL30P; Level loop carpet floor Indicon Industries SCLV1.
Customer/ Contractor	10	Resistance meter: • Megger Insulation Tester **– PREFERRED TEST UNIT** • Analog d'Arsonval Meter (meter must have test source >9 VDC per specification in **Appendix B–6 ENCLOSURE POWER REFERENCE INSOLATION**)

10–4 CRYOGENIC EQUIPMENT

TABLE 10–3
CRYOGENIC EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	46–306734G1	Low Pressure Regulator Kit (Non–magnetic gas cylinder regulator / hose assembly consisting of regulator, hose, and case) (See Note 2)
2	—–	Liquid helium in non–magnetic dewars is needed for refilling the magnet (See Note 2). Refer to appropriate magnet manual for helium volume.
3	46–306717G1	Non–magnetic gaseous helium cylinder cart (See Note 2)
4	46–294705G1	Universal Fill Line Kit (See Note 2)
5	46–294511P1	250 Liter Dewar Stinger (See Note 2)
6	46–294511P2	500 Liter Dewar Stinger (See Note 2)
7	46–294512P1 (12 ft) 46–294512P2 (8 ft)	Transfer Line (See Note 2)
8	46–282336P2	Dewar stinger assembly (High Efficiency) (See Note 2)
9	46–306812G1	Dewar tube ”Thumper” tool (See Note 2)
10	2253802	Burst Disk and Gasket Assembly for LCC Magnet Helium Vessel (See Note 1)
11	46–271136G1	Dewar Adapting Kit including O–ring Kit for Dewar Adapting kit (46–271135P9) (See Note 2)
12	46–271137G1	Cryogen Safety Shield Kit (Includes gloves, face shield, and safety glasses.) (See Note 2)
13	46–294804G1	Non–magnetic Aeroquip wrench Set (See Note 2) includes the following: • 46–294800G1 case for bronze wrench kit • 46–294805P1 bronze 1 $^5/_8$ in. open end wrench • 46–294805P2 bronze 1 $^3/_8$ in. open end wrench • 46–294805P3 bronze 1 $^3/_{16}$ in. open end wrench • 46–294805P4 bronze 1 $^1/_8$ in. open end wrench • 46–294805P5 bronze 1 in. open end wrench
14	46–281088G3	Shield Cooler Installation / Maintenance Kit (See Note 2)
15	46–265273G1	Liquid Helium Level Meter Kit (See Note 2)
16	46–306781G1	Helium Mechanical Gas Flowmeter (See Note 2)
17	46–301477G1	Cryogenic Thermometer, Lakeshore 208 Cryogenic Thermometer Kit or alternate Diode Temperature current source 46–317543G1 (See Note 2)
18	46–252210P1	3 Inch Valve Operator (vacuum break tool) (See Note 1)
19	46–294872G2	SAV–CON / Instrumentation Lead Service Kit (See Note 2)
20	46–318784G2	Shield Cooler Test Kit (See Note 2)
21	46–318696G1	Water Tee for Shield Cooler Water Samples access (See Note 2)
22	46–294052G1	Water Flow Meter Kit (for checking flow of water to Shield Cooler Compressor) (See Note 2)
23	2171219	RUO Temperature Monitor for LCC 1.5T & 1.0T Magnet (See Note 2)
24	46–252805P2 (10 ft) 46–252805P3 (15 ft)	Nitrogen Transfill line (only needed if magnet is warmer than nitrogen temperture)
25	46–260201P1	N2 Precool Syphon (only needed if magnet is warmer than nitrogen temperture)

Note 1 Supplied as part of Signa.

2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.

10–5 INSTALLATION EQUIPMENT

TABLE 10–4
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	——	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	——	Wrecking bar
3	——	Claw hammer, 3/4 lb
4	46–271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5	46–294060G4	Mapping Fixture for field plotting equipment for Cx/LCC magnet Typically not needed to map Cx/LCC Magnet depending on proximity of steel at site, refer to **Section 4–14–1 Floors**.
6	2163971	Field Mapping Fixture Upgrade Kit to make kit compatible with Cx/LCC Magnets Used to upgrade 46–294060G3 to 46–294060G4
7	——	Magnet Log Book
8	——	Installation log book
9	——	4 foot or equivalent carpenter level
10	2230707	For LCC Magnet: Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
11	2134776	Gradient Cable Crimper/Stripper Kit (Note 2) consisting of: • 2134586 Cable stripping tool • 2134586–2 Stripping tool replacement blade • 2134587 Cable slicer • 46–282853P1 Ratcheting crimper • 2135839 1/2 inch terminals • 2135839–2 3/8 inch terminals
12	46–301450G1	Fiber optic connector repair kit (See Note 2)
13	46–260776G3	Magnet Service Tool 750 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
14	46–260776G4	Magnet Service Tool 1000 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
15	46–260777G3	Shim Service Tool Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 290 lbs (132 kg).
16	46–306763G1	Subminiature–D Connector Removal/Re–termination Repair Kit for Robinson–Nugent Sub–D connectors
17	46–251865G2	Field Plotting Kit – Metro Lab Teslameter

Note 1 Supplied as part of Signa.
2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.

(Continued)

10–5 INSTALLATION EQUIPMENT (Continued)

TABLE 10–4 (Continued)
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
18	46–260703G3	Magnet Ramping Equipment Kit
19	46–318833G1	Ramp Cable Holder
20	46–294998G1	Ramping Supply and Equipment kit includes the following: • 46–260703G3 Magnet Ramping Equipment Kit (see Item 18) • 46–260776G3 Magnet Service Tool Power Supply Cabinet (see Item 13)
21	46–320273G3 or G4	Non–Magnetic Tool Kit – Universal (See Note 2) Both metric and inch Non–Magnetic Tool Kits needed. May substitute both of the following kits: • 46–320273G1 Non–Magnetic Tool Kit – Metric • 46–320273G2 Non–Magnetic Tool Kit – Inch
Note	1	Supplied as part of Signa.
	2	Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.

10–6 TEST EQUIPMENT

TABLE 10–5
TEST EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	2284754	Textronic TDS3012 digital scope
2	2284763	Fluke FLK196M scopemeter for medical applications
3	46–194427P226	Dual trace oscilloscope, 100 MHz bandwidth, 2 channel, digital storage, Textronic 2232
4	46–194427P222	Dual trace oscilloscope, 350 MHz bandwidth, 4 channel, Textronic 2465
5	46–194427P284	Battery operated digital multi–meter, Fluke 87, 4.5 digits with frequency counter and capacitance
6	46–208572P9	Clamp–on Ammeter, 1–200A, AC/DC
7	46–317724P14	Attenuator/ load, 200 Watt, 30 dB attenuator
8	46–317724G2	RF Power Measurement Kit with Attenuator/ load, 200 Watt, 30 dB attenuator 46–317724P14
9	2218826	RPM 1650 Power Analyzer Kit
10	46-328143G3	Dranetz 626 analyzer kit with 1 and 3 phase modules
11	Catalog E6320DA or 46–194427P144	Densitometer
12	46–306801G1	Bell Gauss Meter Kit
13	46–194427P248	Microguard or ECOS leakage tester
14	46–306797G1	Fogg (ECG) Simulator and Memory Module
15	46–320433P1	Infrared Scanner 0 – 100° C
16	46–317830G1	Fiber Optic Light Meter Kit
17	46–294047G1	Shield Cooler Vacuum Pump Kit
18	46–251867G1	Main Vacuum Pump Down Kit

10–7 CALIBRATION TOOLS/FIXTURES

TABLE 10–6
CALIBRATION TOOLS/FIXTURES

ITEM	GE PART NUMBER	DESCRIPTION
1	46–307500G1	Longitudinal Drive Force Gauge
2	2133388–6	System Performance Test (SPT) Phantom Set for Systems with Cx/LCC (Active Shield) systems (See Note 1) consisting of the following: • 2125247–10 Nesting Plate • 2125245 Large Volume Shim Phantom Assembly • 2125244 Short Loader Assembly • 2131027–2 Daily Quality Assurance (DQA III) Phantom • 2134213 SPT Shipping/Storage Cart (Not used for Mobile Systems) • 2170481 EPI Foam Positioner • 2141454 Quick Shim Positioner
3	46–287900G3	Head TLT Sphere (46–265826G6), one 100 mm NIckel Chloride sphere (46–317586G1), and Loader (46–287899G1) See Note 1
4	46–255816G1	RF Test Cables Kit
5	46–265434G1	Magnet Rundown Unit Test Box
6	46–307164G4	1.5T / 1.0T GRAFIDY II Kit (See Note 4)
7	2234621	ECMT Kit (See Note 4)
8	46–328021G1	Enmet Oxygen Monitor Calibration consisting of the following: • 1800 PSI cylinder of 20.9% Oxygen in Nitrogen • 1800 PSI cylinder of 17% Oxygen in Nitrogen • Regulators with calibration adapter For use with portable or permanent Oxygen Monitor.
9	2106236	Portable Oxygen Monitor (Connecticut Analytical)
10	2107184	Permanent Oxygen Monitor (Enmet)
11	46–301549G1	TPS RF Test Cable/Adapter Kit
12	46–301927G1	TPS RF Service Interface Kit (See Note 2)
13	46–306712G1	Torque driver kit
14	46–306864G1	Magnet Helium Resistance Box Kit
15	2101360	Power Supply Calibration Kit
16	46–265826G3	Head Signal–to–noise (SNR) sphere (See Note 3)
17	46–265635G4	Body Signal–to–noise (SNR) sphere (See Note 3)
18	46–265635G2	Body T2 sphere (See Note 3)
19	46–328480G1	100 mm Nickel Chloride sphere (46–317586G1) and Universal Phantom Holder (46–328383P1) See Note 2

Note 1 Supplied as part of Signa.
2 Supplied by GE until turnover of system to customer, then available as part of a GE Service Contract.
3 Customer may purchase these items.
4 Either the ECMT Kit or the 1.5T/1.0T GRAFIDY II Kit can be used to calibrate the system.

10–8 TOOL KIT

TABLE 10–7
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
1	—–	Extension cords, with ground conductor
2	—–	Power strip, grounded type, with minimum of five outlets
3	—–	Soldering iron, pencil type with solder
4	—–	Solder sucker
5	—–	Assortment of Brady Quick labels
6	—–	Micro clip leads
7	—–	14 pin and 16 pin DIP clips
8	46–258218P3	Vinyl electrical tape
9	46–258218P4	Copper tape, 3 in. wide
10	46–258218P5	Copper tape, 2 in. wide
11	46–258218P6	Copper tape, 1 in. wide
12	—–	Alcohol cleaning solution
13	—–	Plastic or aluminum flashlight
14	—–	Plastic or aluminum pen light
15	AMP No. 458994–1	Pin extractor, Universal Mate'n'Lock
16	AMP No. 305183–R	Pin extractor, M–series
17	46–237072P1	Pin extractor, Sub–D
18	46–307307G1	Crimping tool for coax cable and BNC connectors Inserts for RG8, 9, 11, 214 46–255841P103 RG58, 223 46–255841P100 RG59 46–255841P101 RG174 46–255841P102 Die Removal Tool 46–255841P201
19	—–	Assorted crimp tools.
20	—–	Non–magnetic level
21	—–	Non–magnetic tape rule, 12 ft
22	—–	Assorted drill bits
23	—–	Inspection mirror
24	—–	Tap set, standard, and tap handle, T–type
25	—–	6 in. rule, standard and metric markings
26	—–	Alignment tool (tweaker)
27	—–	Hex/alignment tool
28	—–	Hemostat, 5 in., curved
29	46–198094P1	Wrist grounding strap
(Continued)		

10–8 TOOL KIT (Continued)

TABLE 10–7 (Continued)
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
30	Xcelite 110CG	Diagonal Cutting Pliers, 4–1/2 in.
31	—–	Screw Starter, aluminum or plastic shaft
32	—–	Hobby and utility knives
33	—–	Spring scales, 0–10 lbs and 0–50 lbs
34	46–313413P1	Extractor for 20 – 100 pin PLCC Chips

APPENDIX A –MR SITE VIBRATION TEST GUIDELINES

A–1 TEST MEASUREMENTS

- Vibration measurements are in the range of 10^{-6} g. Test equipment must have the required sensitivity to these levels.
- All analyses are to be narrowband Fast Fourier Transforms (FFT’s) over the frequency bands listed in Table A–1.

TABLE A–1
FREQUENCY BANDS FOR FFT’S

FREQUENCY BAND	FREQUENCY RESOLUTION
0.2 to 50 Hz	Δf= 0.125 Hz

- Time histories of the vibration must be recorded (i.e. acceleration levels vs. time). The resolution of the time history must be adjusted to clearly capture the transient events. The analyzer set–up will be site dependent.

A–2 EQUIPMENT (SPECTRAL ANALYZER) SET–UP

- Frequency average a minimum of 20 linear averages (do not use peak hold or 1/3 octave analysis)
- Hanning window must be applied to the entire spectra

Spectrum analyzers capable of these measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer. Accelerometers must have the capability to measure from 0.2 Hz beyond 50 Hz. Time histories can be recorded using any of the analyzers mentioned above. Good quality strip chart recorders may also be sufficient. Please note that the equipment mentioned are for example only. It is the responsibility of the Engineering test firm to provide equipment that will follow the test requirements.

A–3 DATA COLLECTION

A–3–1 Ambient Baseline Condition:

All of the measurements defined in Section A–1 and A–2 must be made in a ‘quiet' environment. That is, in areas where excessive traffic, subway trains, etc. exists, a vibration measurement must also be made during periods without traffic or during periods of light traffic. Measurements must define the lowest levels of vibration possible at the site.

The source of any steady state vibration whose levels exceed the specifications in **Section 4–15–3 Specifications For 1.5T LCC (Active Shield) Magnets** must be identified as to the source of the vibration disturbance. A second measurement should be made with all of the identified contributors powered down if possible. In situations where it is not possible to power down equipment, vibration data must be collected to identify specific source of the vibration concern. The majority of steady state vibration problems can be negated by isolating the vibration source.

A–3–2 Normal Condition

All of the measurements listed above must be repeated during periods of 'normal' environmental conditions including the FFT's and time histories. The transient measurements must be provided to define the dynamic disturbances the MR system might be exposed to. This transient disturbance is required for a true assessment of the site.

Transient vibration is very difficult to assess and to eliminate. Environments exceeding the 500 micro–g, zero to peak trigger level, requires analysis of the building's dynamic response to the transient event or selection of an alternative location should be pursued. The building response must remain below the vibration specifications in **Section 4–15–3 Specifications For 1.5T LCC (Active Shield) Magnets.** The dynamic response following the transient event must exceed the specifications for sustained periods of time, i.e. for random dynamic response of duration less than a few seconds will have less impact on system performance than response of longer duration.

Building structures excited by transient events exceeding the specifications in **Section 4–15–3 Specifications For 1.5T LCC (Active Shield) Magnets** must either provide recommendations to reduce levels to meet specifications or an alternative location should be identified. A second vibration measurement should be made to help identify a more stable location.

A–4 PRESENTATION/INTERPRETATION OF RESULTS

The recommended format for site vibration data collection, presentation, and analysis is illustrated in the examples shown in Illustrations A–1 through A–4. Presentation of the data in any other format (linear units only) may result in an incorrect interpretation and diagnosis of the site. Additional data collection or presentation methods is at the option of the vibration testing service.

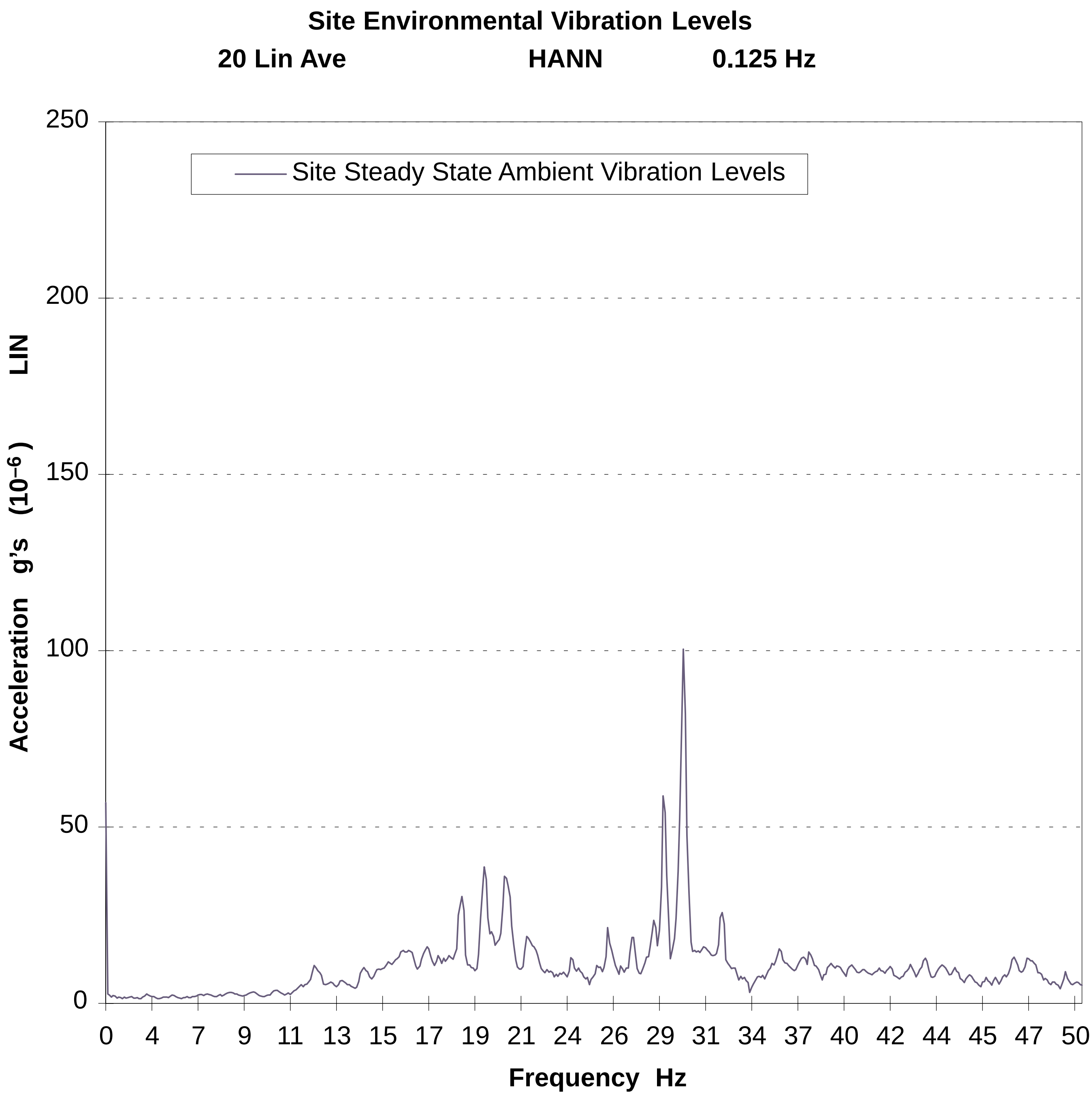
It is the responsibility of the customer's vibration testing service to interpret the results and determine if that site meets GE's specifications. Illustrations A–1 and A–2 are examples provided to assist a test consultant in the use of GE Steady State specifications (vibration specifications above ambient baseline). If the vibration levels are too high, additional data acquisition may be necessary to:

- determine the source of the vibration
- propose a solution to the problem
- find an alternate site location.

Illustrations A–3 and A–4 are examples provided to assist a test consultant in the use of GE Transient specifications. The 500 micro–g, zero to peak trigger level identifies data collection to begin assessment of the site vibration analysis. The response of the transient must be assessed relative to the Steady State vibration specifications in **Section Specifications For 1.5T LCC (Active Shield) Magnets.**

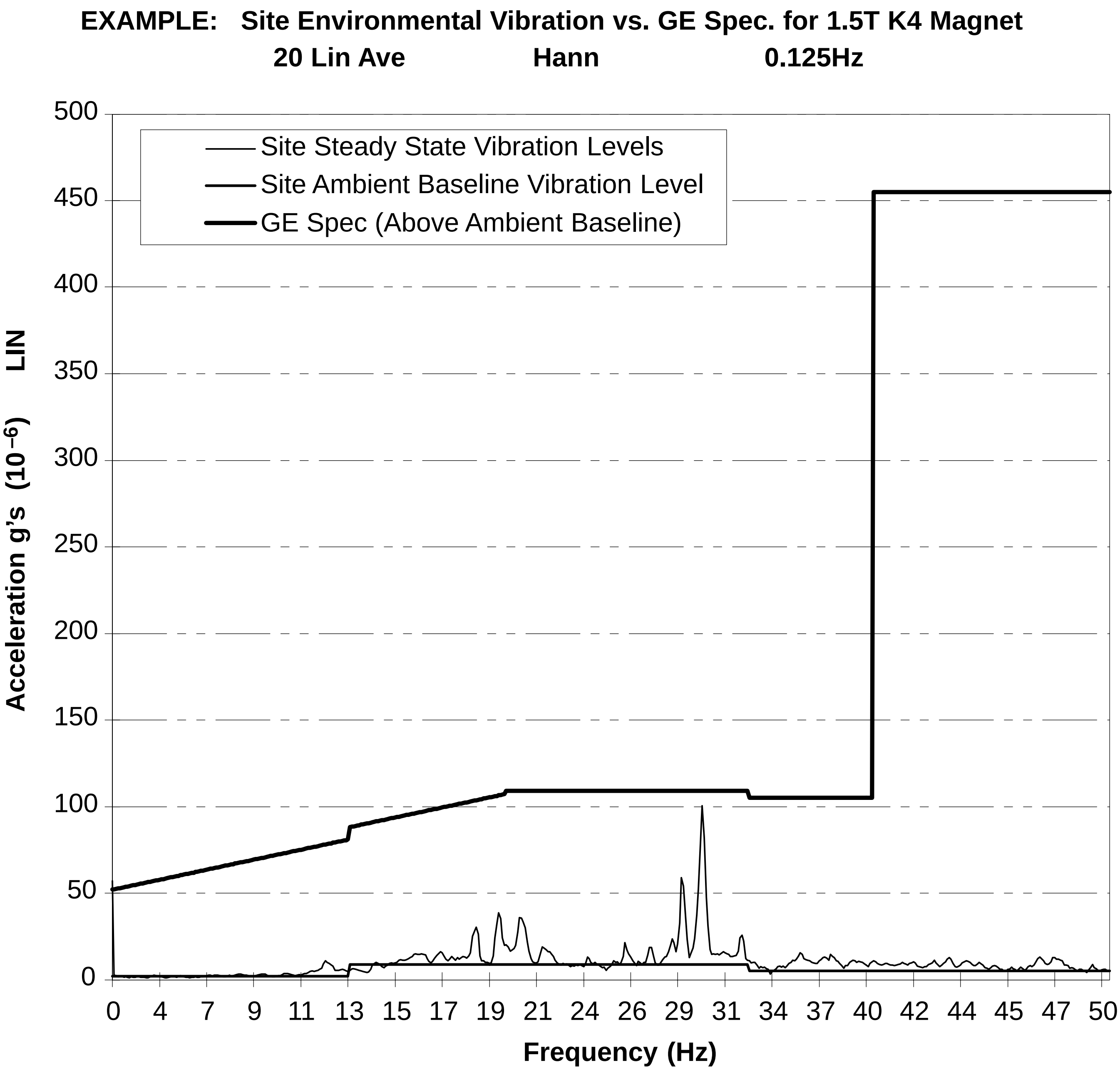
Any questions regarding test equipment requirements, test parameters, or general questions should be discussed with your GE Installation Specialist.

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



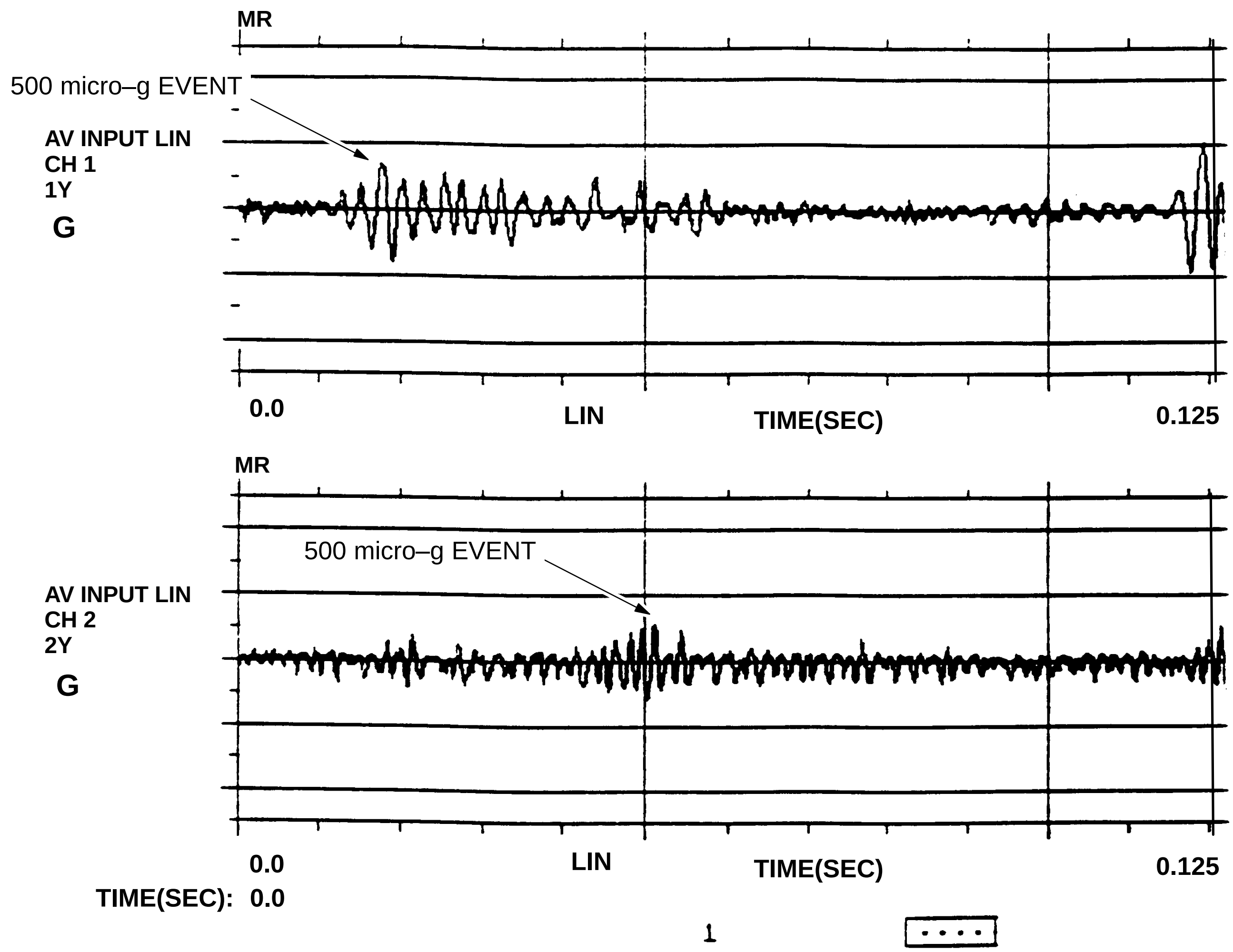
EXAMPLE SITE ENVIRONMENTAL VIBRATION
ILLUSTRATION A–1

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



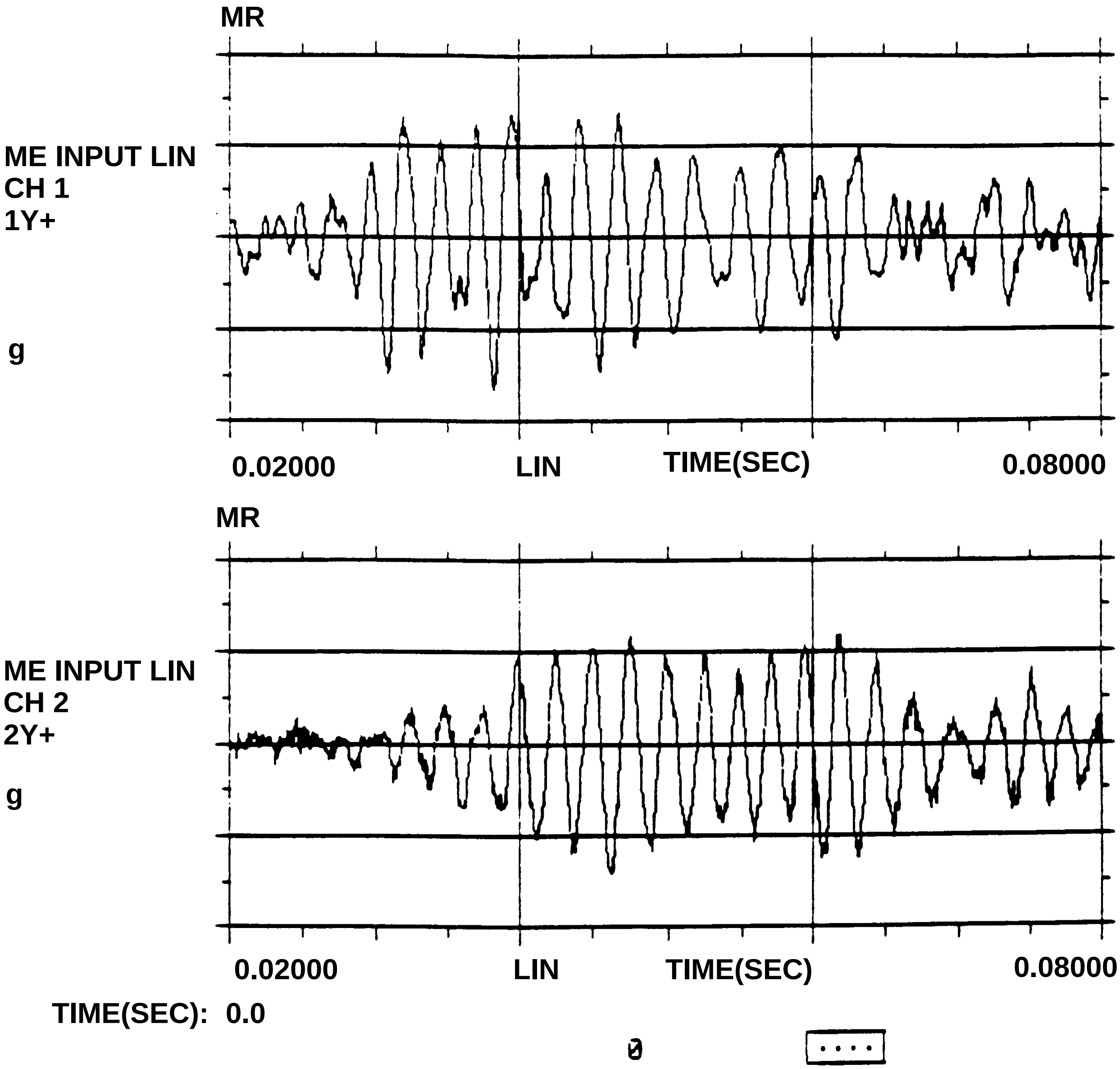
EXAMPLE SITE ENVIRONMENTAL VIBRATION VS. GE SPECIFICATION FOR 1.5T LCC MAGNET
ILLUSTRATION A–2

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY
ILLUSTRATION A–3

A–4 PRESENTATION/INTERPRETATION OF RESULTS (Continued)



ACCELERATION TIME HISTORY (ZOOM IN ON TRANSIENT EVENT)
ILLUSTRATION A–4

APPENDIX B –RF SHIELDED ENCLOSURE TEST GUIDELINE

B–1 INTRODUCTION

This document describes the procedure and methodology of performing an RF shielding effectiveness verification test on enclosures which will house GE Medical Systems (GEMS) Magnetic Resonance Imaging (MRI) equipment. MRI equipment is sensitive to RF energy from sources outside of the shielded enclosure. To ensure proper operation of the MRI equipment, the shielded enclosure must attenuate local RF signals to levels that do not cause interference.

Note

RF Shielding Performance is based on **planewave** measurements. "H" field and "E" field tests are not required, but are allowed as needed for diagnostic purposes.

B–1–1 PURPOSE OF TEST PLAN

The purpose of this test plan is to describe a series of RF shielding effectiveness tests to demonstrate compliance of an MRI shielded enclosure to the requirements of GE Medical Systems.

The test procedure described in this guideline is a modification of MIL–STD–285 and IEEE Std 299–1991. This procedure provides a through evaluation of the shield integrity at the upper end of the frequency range of interest showing any RF leakage which may cause imaging problems. These test guidelines ensure that the electromagnetic environment inside of the enclosure will meet the requirements of General Electric Medical Systems.

B–2 APPLICABLE DOCUMENTS

MIL–STD–285	MILITARY STANDARD ATTENUATION MEASUREMENTS FOR ENCLOSURES, ELECTROMAGNETIC SHIELDING, FOR ELECTRONIC TEST PURPOSES, METHOD OF; 25 June 1956
IEEE Std 299–1991	IEEE STANDARD FOR MEASURING THE EFFECTIVENESS OF ELECTROMAGNETIC SHIELDING ENCLOSURES; 2 July 1991

B–3 TEST SAMPLE SETUP

The shielded enclosure under test shall be set–up in a normal configuration, which consists of the following:

1. Magnet installed including all floor mounting bolts
2. RF shielded door(s)
3. Waveguide penetrations, HVAC, vents, medical gas lines, etc.
4. AC power supplied through low–pass filters
5. Patient view window, skylights, windows, hatches, etc.
6. Blank penetration panel installed, dimensionally equivalent to the GE panel and the same mounting hardware to be used with the GE penetration panel.

For safety reasons, the enclosure shall be electrically grounded during the shielding effectiveness test. Any variances from the normal configuration shall be noted in the certification report.

B–4 SHIELDING EFFECTIVENESS

This test procedure determines the worst case shielding effectiveness based on the lowest test point reading obtained. The lowest reading obtained will be the reading of the room.

B–5 MEASUREMENT PROCEDURE

To simulate the effects of external RF sources, the transmit antenna shall be located outside the enclosure on a plane parallel to the face of the enclosure wall, at a distance of 6 feet (1.8 meters) unless physically constrained to a lesser separation. The areas of least effectiveness are located by searching the inside of enclosure with the antenna connected to the spectrum analyzer.

B–5–1 Test Position

The transmitting antenna will be positioned in front of all critical areas (doors, windows, filters, penetration areas, etc.) and at a minimum of every 20 feet (6.1 meters) of wall. The receiving antenna is scanned over all panel section joints (where accessible), at floor, wall, and ceiling for a minimum of 10 feet (3.05 meters) in all directions from the location of transmitting antenna. The receiving antenna shall be at a minimum of 1 foot (0.3 meters) from the shield. For areas that are inaccessible for direct location of the transmitting antenna, the inside of that area will still be scanned using the receiving antenna with the transmitting antenna positioned in front of the adjacent wall or adjacent test position.

B–5–2 Frequency Range

The standard frequency for shielding measurements shall be 100 MHz $\pm$ 10 MHz. This allows the frequency to be adjusted slightly to avoid interference from local active transmitters and/or RF noise from other sources. Test frequency utilized shall be noted in the certification report.

B–5–3 Free Field Calibration

The incident field, i.e. "free field", is measured by the following procedure:

Position the transmit antenna parallel to the exterior wall of the enclosure at a distance of 6 feet (1.8 meters) (unless physically constrained to a lesser separation, in which case a separate reference will be established and documented at the new test distance), using horizontal polarization. The receive antenna shall be placed between the transmit antenna and 1 foot (0.3 meters) from the exterior wall of the enclosure. The receive antenna will be moved vertically and horizontally to achieve maximum signal strength. The receive antenna shall be placed no closer than 2 inches (51 mm) from the exterior wall of the enclosure and in line with the transmit antenna. The maximum received voltage at the test frequency will be recorded.

B–6 ENCLOSURE POWER REFERENCE INSOLATION

To prevent personnel hazard, it is necessary for the enclosure to be properly grounded.

To minimize common mode currents, the ungrounded enclosure should be isolated from ground with a minimum of 1000 ohms of DC resistance. The isolation measurement is performed by the following procedure:

All power to the enclosure is removed. For safety reasons, an AC voltage measurement will be made to verify that no power is connected. With electrical power and intentional ground disconnected, connect the test instrument between the shielded enclosure and AC power ground. Take a reading and record the value. This test shall be made using either an isolated, current limited high voltage (>150 VDC) DC source and DMM to read drop across the limiting resistor or a megger instrument capable of reading values less than 1000 ohms. Conventional resistance meters employing test sources of 9 VDC or less shall not be used.

B–7 TEST EQUIPMENT

Test equipment shall be selected to provide measuring capabilities as described in this test guideline. The signal source, amplifier, antennas, and receiver or spectrum analyzer shall be such that the difference between the induced reference voltage and the receiver sensitivity shall be at least 6 dB greater than the required attenuation specification.

The signal source and power amplifier shall output a CW signal for a nominal test frequency of 100 MHz. Receiver or spectrum analyzer and preamplifier (if required), shall provide adequate sensitivity to permit attenuation measurements to be made to the specified limits. Dipole antennas and other miscellaneous equipment required to transmit and receive the proper RF fields shall be used.

The absolute performance calibration, of the equipment requiring calibration, shall be performed on an as needed basis in accordance with MIL–STD–45662. The calibration period shall not exceed one year. The test equipment tolerances of at least $\pm$ 2% frequency and $\pm$ 2 dB amplitude shall be met. Equipment certifications shall be traceable to the National Institute of Standards and Technology (NIST). All equipment will be verified for proper operation between and after each series of tests by repeating the reference readings at the specified frequency(s).

B–8 DATA RECORDING AND VERIFICATION

Measurements shall be performed by qualified responsible EMC test personnel. The test must be performed in the presence of a GE representative unless other arrangements have been made by GEMS. All data collected during the course of the tests will be recorded on standardized data sheets. The data sheets will include the test location, frequency, reference level, measured enclosure level, and attenuation level.

B–9 TEST REPORT

A final certification report will be provided after the test is performed. This report will include all recorded data necessary for the evaluation of the shielded enclosure test results and will list any changes pertinent to the test set–up or the shielding effectiveness. The certification report will also include the test procedures and a list of the actual equipment used during the test.

Along with the data sheet, there will be a presentable drawing showing the shape of the enclosure, all test point locations, doors, filters, windows, and existing building walls.

Prerequisites for Scheduled Performance Verification of Chillers For Signa *TwinSpeed*

Revision 02/14/02

The following list indicates items that must be completed before contacting Ellis and Watts to schedule the Factory Authorized Performance Verification of the pertinent *Twin Speed* Cooling Cabinet. When this step is completed, then the PM schedule will be established for the year. The LTL 10 (M3088TE) is started up and serviced by GE, therefore, performance verification by an Ellis and Watts Service Provider is not required.

1. Appropriate line power is connected to the Chiller and correct phasing established.

2. Chiller is permanently mounted as required by GE and / or state / local code requirements.

3. **For units with <u>W</u>ater <u>C</u>ooled condensers**; (LC 18MT/WC, See Ellis and Watts Tech. Pub. 471)
   - Piping has been flushed out to remove any debris before being attached to Chiller
   - Water supply to Chiller condenser has an inline filter or strainer.
   - <u>Full Gradient and Shield Cooler</u> (Usually this happens about a week before turnover to customer.)

4. **For Indoor Air-Cooled Condenser units**; (LC 20MT, See Ellis and Watts Tech. Pub. 472)
   (LC 10MT, See Ellis and Watts Tech. Pub. 473)
   - Ambient temperature and humidity are controlled to meet the equipment room environmental requirements.
   - <u>Full Gradient and Shield Cooler</u> (For Dual Loop) or <u>Full Gradient</u> (For Single Loop) loading is available.(Usually this happens about a week before turnover to customer.)

5. **For Outdoor Condenser unit;** (LC 20MT/O See Ellis and Watts Tech. Pub. 470)
   (LC 10MT/O See Ellis and Watts Tech Pub 474.)
   - All coolant piping connections are complete and leak tested (**Do not use tap water)**.

<u>NOTE:</u> It is recommend that distilled water or nitrogen gas be used to pressurize and leak test.
<u>Full Gradient and Shield Cooler</u> (For Dual Loop) or <u>Full Gradient</u> (For Single Loop) loading is available.(Usually this happens about a week before turnover to customer.)

Performance Verification is usually completed just before the system is turned over to the Customer. Ellis and Watts service department needs a two week notification to schedule the Local Service Provider. Please provide the name and address of the Imaging Center, FE contact name with the best method of contact, requested date of service, and Cryo System ID #. Contact information is listed below.

Ellis and Watts International Inc. is determined to be as helpful as possible with the installation of the chiller equipment that we provide to support the General Electric Medical Systems equipment, but, Ellis and Watts and our Approved Service Providers are not contracted to perform **any** of the installation of this equipment. If you desire for E&W or any of our ASP's to assist with the actual installation of the chiller equipment, then, you will need to contact the Ellis and Watts Service Department at 513-752-9000 x749 or E-mail at service@elliswatts.com to discuss scheduling and payment for this work, or start-up scheduling. Information is free so call any time to discuss questions about installation or operation of our chiller equipment.

Ellis and Watts International Inc.
4400 Glenwillow Lake Lane
Batavia, Oh. 45103

APPENDIX D – REQUIREMENTS FOR EQUIPMENT ROOM SHARED BY MULTIPLE MR SYSTEMS

D–1 INTRODUCTION

When the Equipment Room is shared by more than one MR system of the same field strength there is a potential for cross–talk of RF energy between the MR systems. RF cross–talk may cause noise artifacts in images.

The potential for cross–talk exists when the RF transmit cables and equipment of two or more MR systems are located in the same Equipment Room. For example, when one system is transmitting, the other system could be in receive mode and therefore pick up the RF energy being transmitted resulting in a cross–talk scenario.

Note

The potential for cross–talk exists for RF transmit cables and equipment that produces RF that are part of a non–GE MR System of the same field strength.

The following subsections provide requirements for shared Equipment Room design, layout, and installation which reduce the potential for RF cross–talk.

D–2 EQUIPMENT CABINETS RELATIVE LOCATIONS

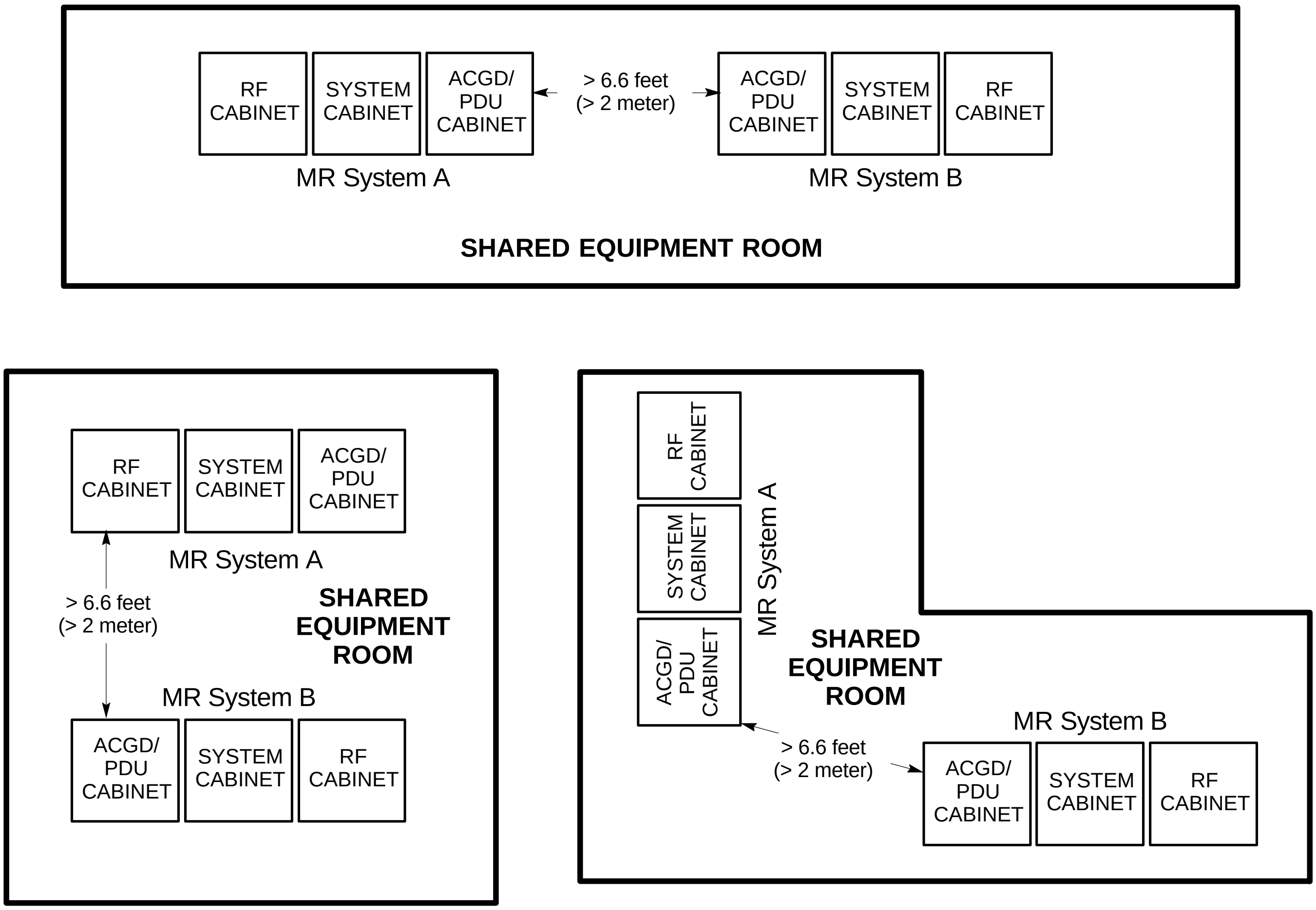
The following are requirements for locating equipment cabinet of one MR system relative to the other MR system equipment cabinets.

- Maximize separation distance between the RF Cabinet of one MR system and the System Cabinet of the other MR system.
- RF, System, or Gradient Cabinets of one MR system and RF, System, or Gradient Cabinets of the other MR system must be separated by a minimum of > 6.6 feet (2 meters) in all directions, see Illustration C–1.

Note

Relative placement should not be an issue for chillers, compressors, and other equipment which does not produce RF.

D–2 EQUIPMENT CABINETS RELATIVE LOCATIONS (Continued)

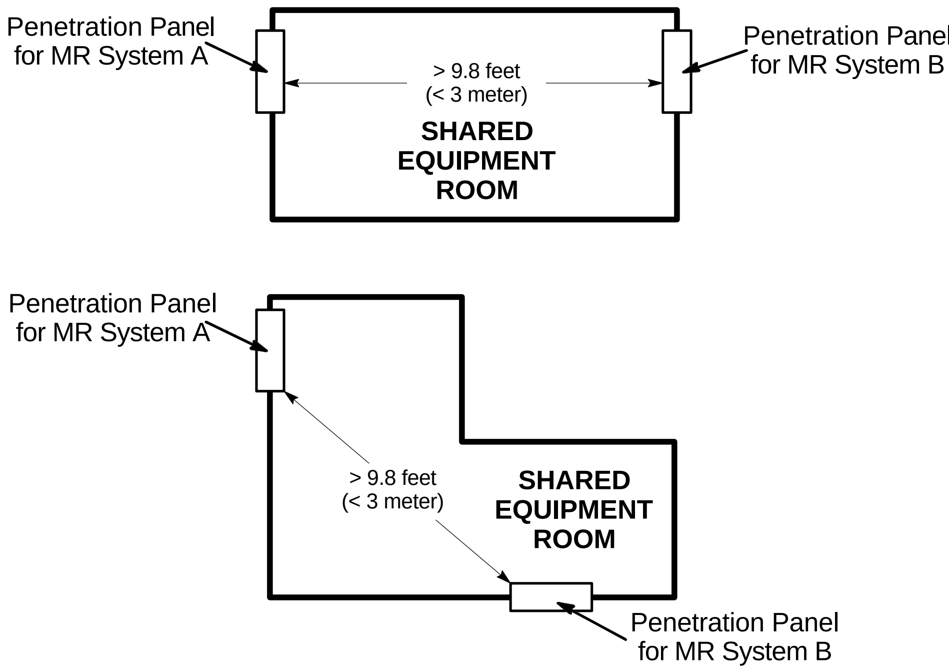


MULTIPLE MR SYSTEMS ELECTRONICS CABINETS SPACING
ILLUSTRATION C–1

D–3 PENETRATION PANELS LOCATION

The following are requirements for locating the RF Shielded Room Penetration Panel of one MR system relative to the other MR system RF Shielded Room Penetration Panel.

- There must be > 9.8 feet (3 meters) separation between the Penetration Panels of each system sharing the Equipment Room space, see Illustration C–2.

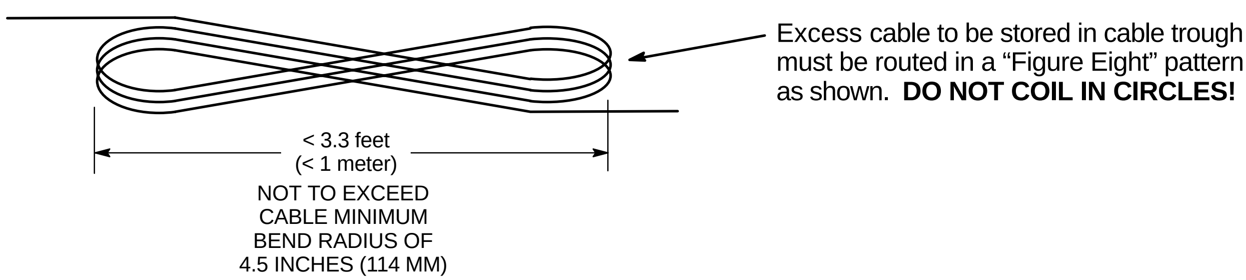


MULTIPLE MR SYSTEMS PENETRATION PANEL SPACING
ILLUSTRATION C–2

D–4 SYSTEM CABLES REQUIREMENTS

The following are requirements for locating and managing excess RF Receive and Transmit cables of the MR systems sharing the Equipment Room.

- In the shared Equipment Room there must be > 6.6 feet (2 meters) separation between the RF receive cables of one system and the RF transmit cable of the other system. Refer to **Section 6 SYSTEM INTERCONNECTS** for Receive and Transmit cable Run Numbers.
- There must be > 6.6 feet (2 meters) separation between the system interconnect cables of each system sharing the Equipment Room space.
- Transmit cables in the Equipment Room must be cut to length to minimize excess cable length reducing the potential for signal coupling with other cables. No excess transmit cable can be stored in the Equipment Room.
- Receive cables excess length must be stored in a “figure 8” with overall dimension of < 3.3 feet (1 meter), see Illustration C–3.



PROPER STORAGE OF EXCESS RECEIVE CABLES
ILLUSTRATION C–3